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(54) **MULTIFUNCTIONAL BISPECIFIC FUSION
POLYPEPTIDE**

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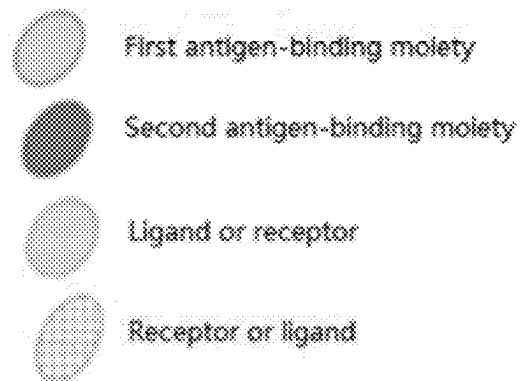
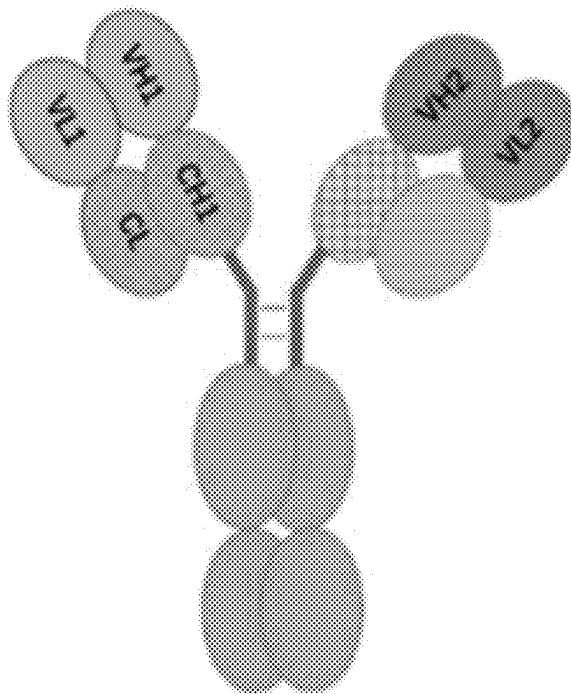
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(57)

ABSTRACT

The present application relates to the technical field of biomedicines, and particularly to a bispecific fusion polypeptide. The bispecific fusion polypeptide comprises an antigen-binding moiety. The antigen-binding moiety comprises a first antigen-binding moiety comprising a first polypeptide comprising a first heavy chain variable domain VH1 of a first antibody from the N terminus to the C terminus, which is operably linked to a first conjugate fragment; and a second polypeptide comprising a first light chain variable domain VL1 of the first antibody from the N terminus to the C terminus, which is operably linked to a second conjugate fragment, wherein the first conjugate fragment and the second conjugate fragment are capable of specifically binding; and the first conjugate fragment is a receptor and the second conjugate fragment is a ligand; or the first conjugate fragment is a ligand and the second conjugate fragment is a receptor.

Specification includes a Sequence Listing.



FiBody

Fig. 1

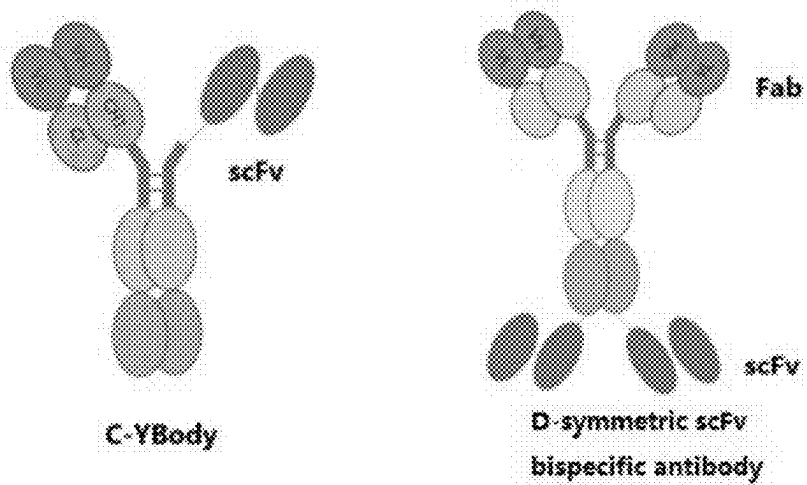
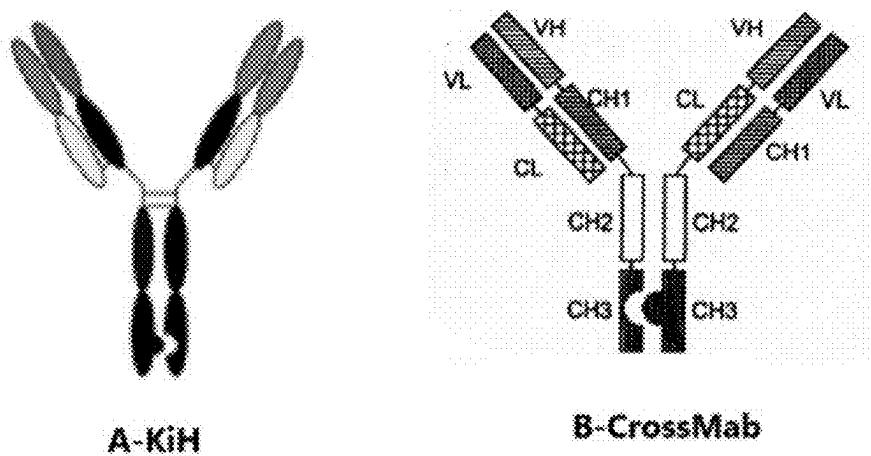


Fig. 2

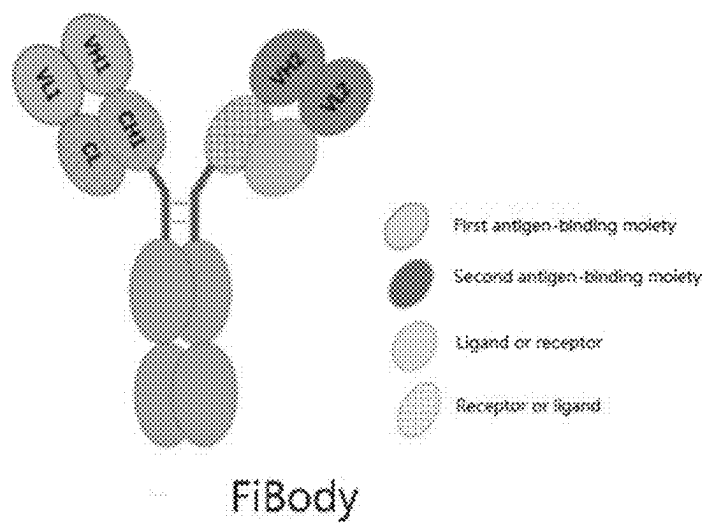


Fig. 3

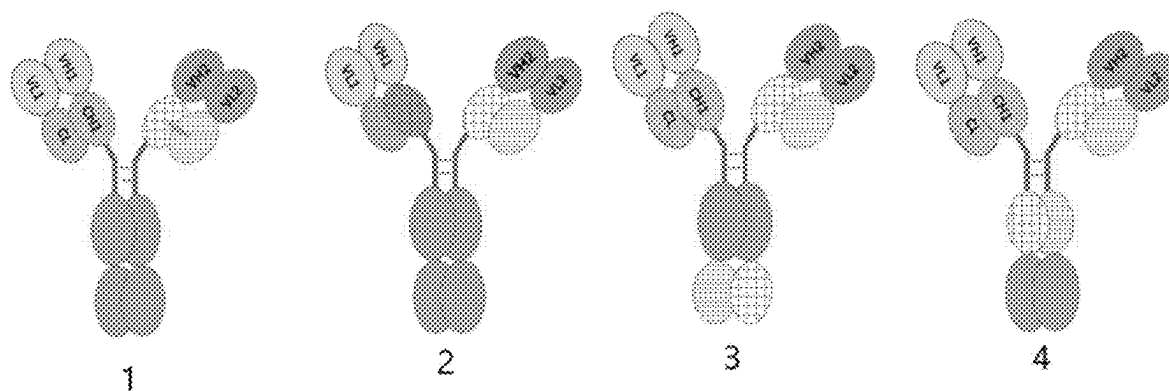


Fig. 4

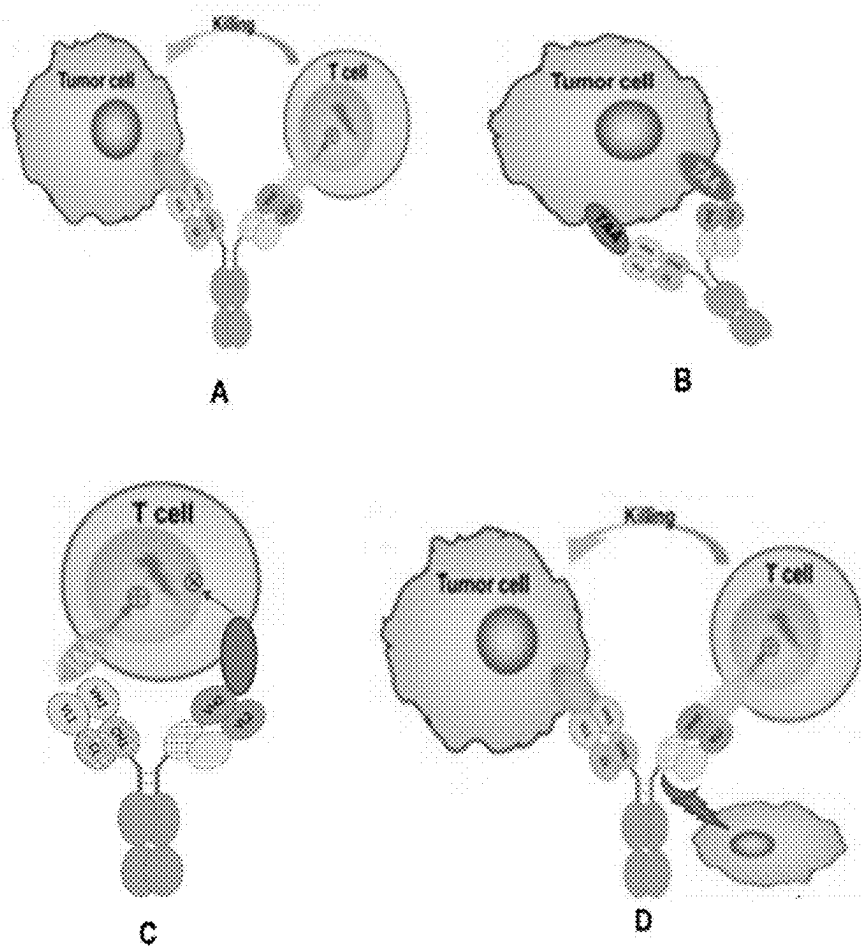


Fig. 5

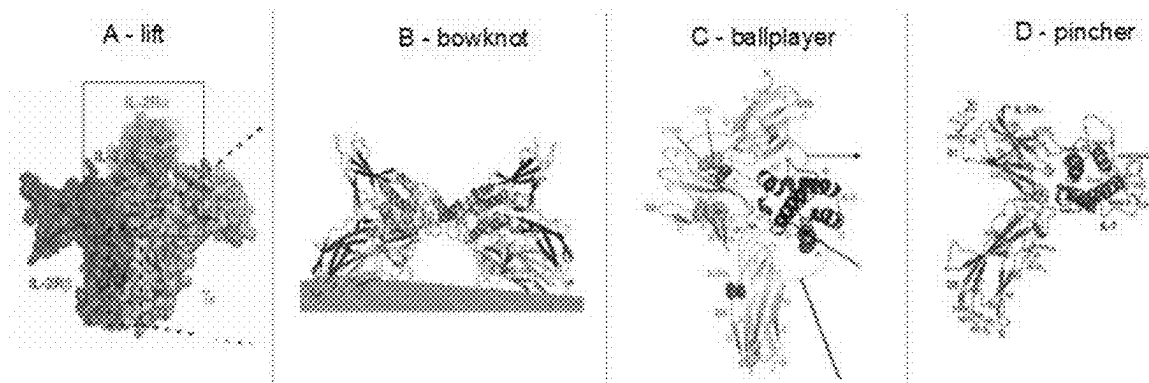


Fig. 6

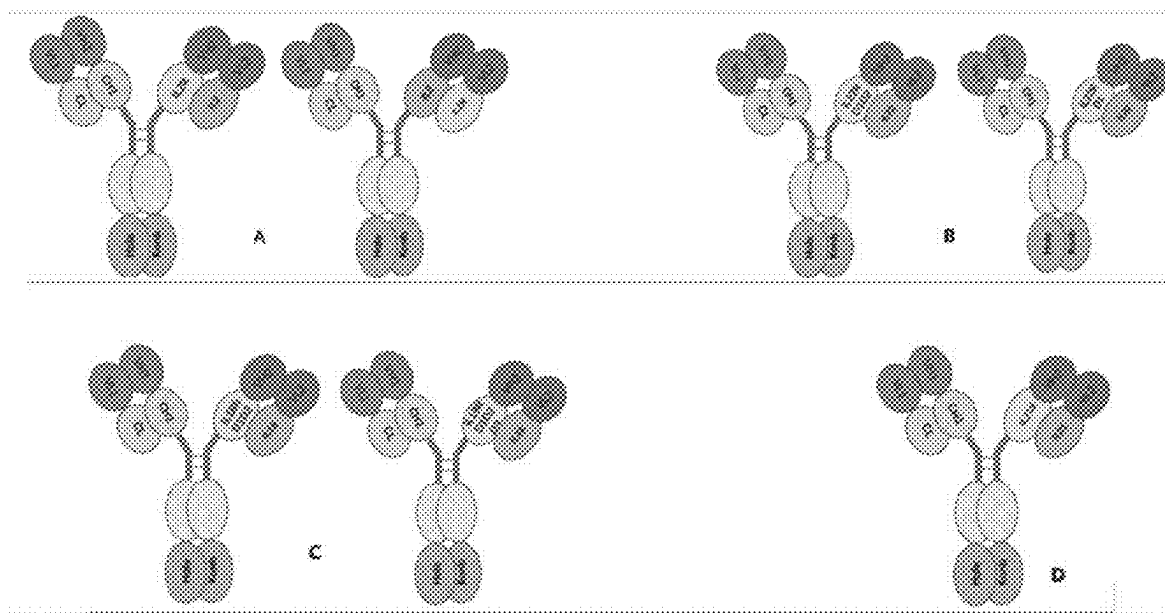


Fig. 7

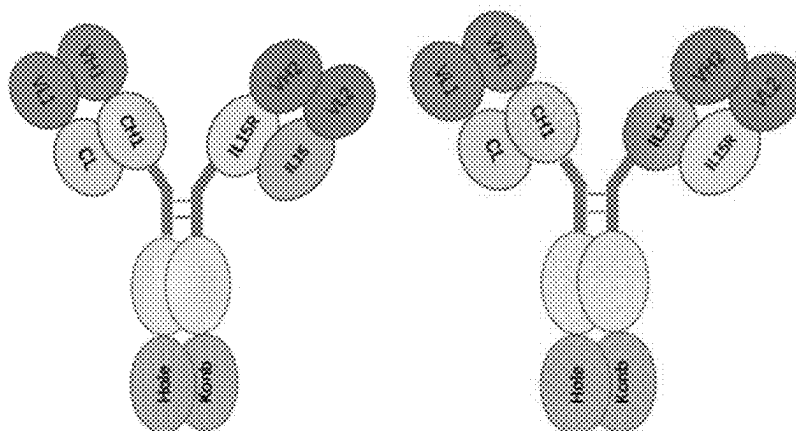


Fig. 8

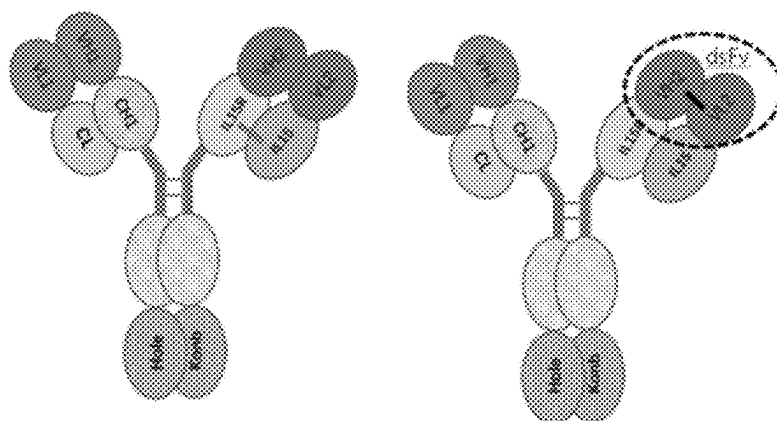


Fig. 9

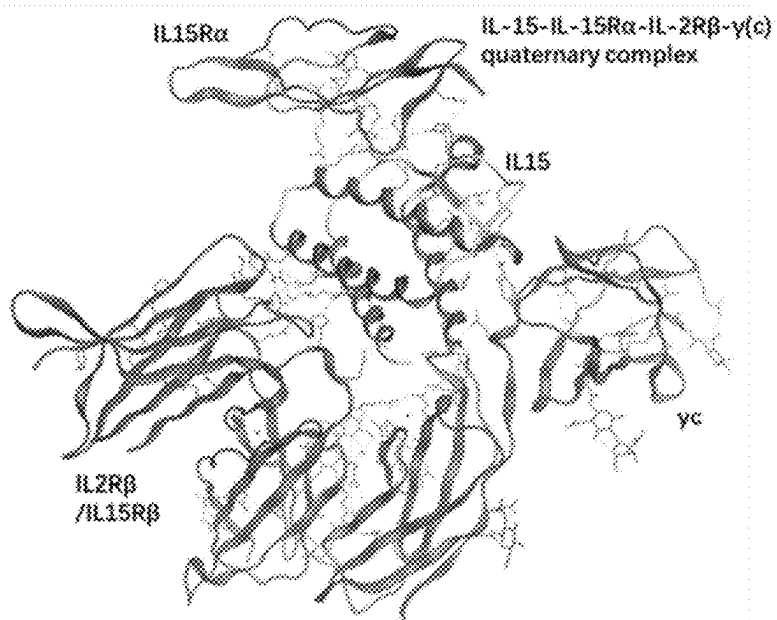


Fig. 10

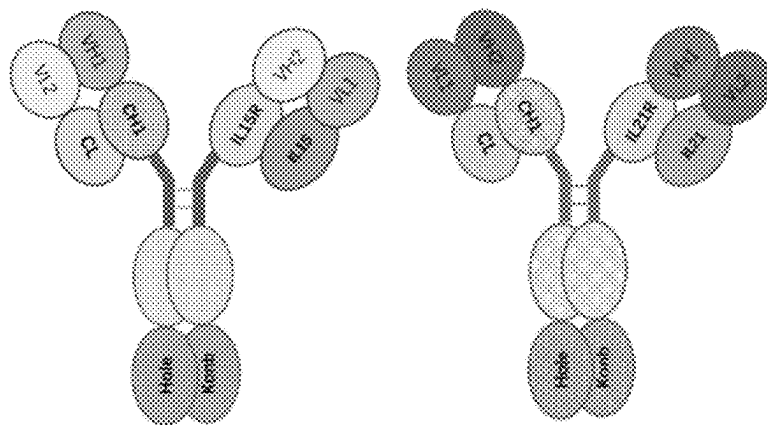


Fig. 11

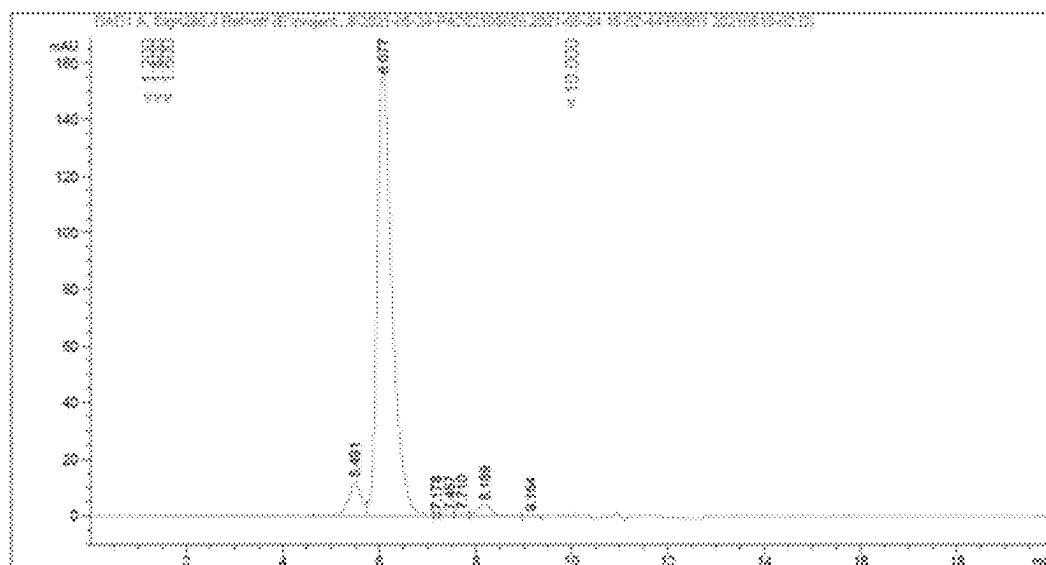


Fig. 12

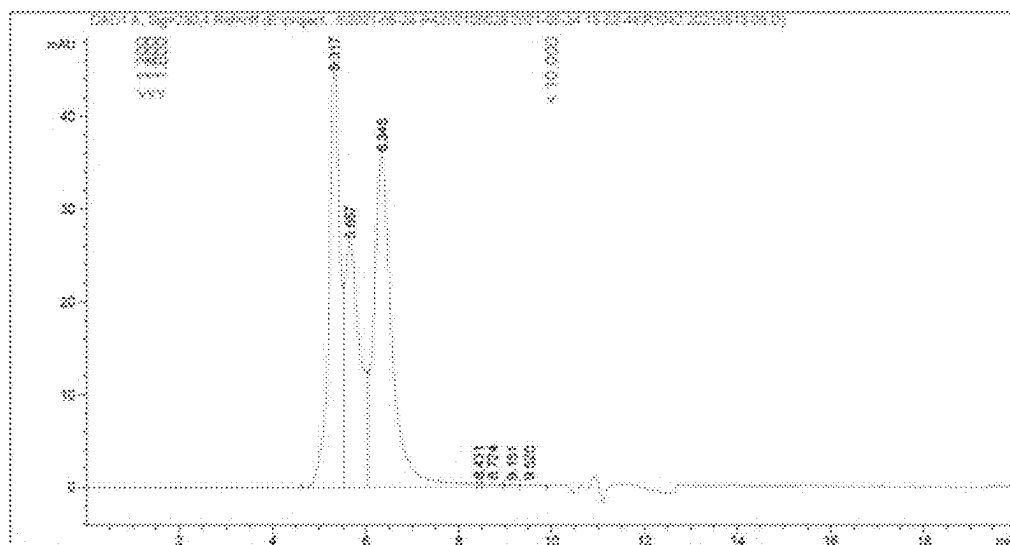


Fig. 13

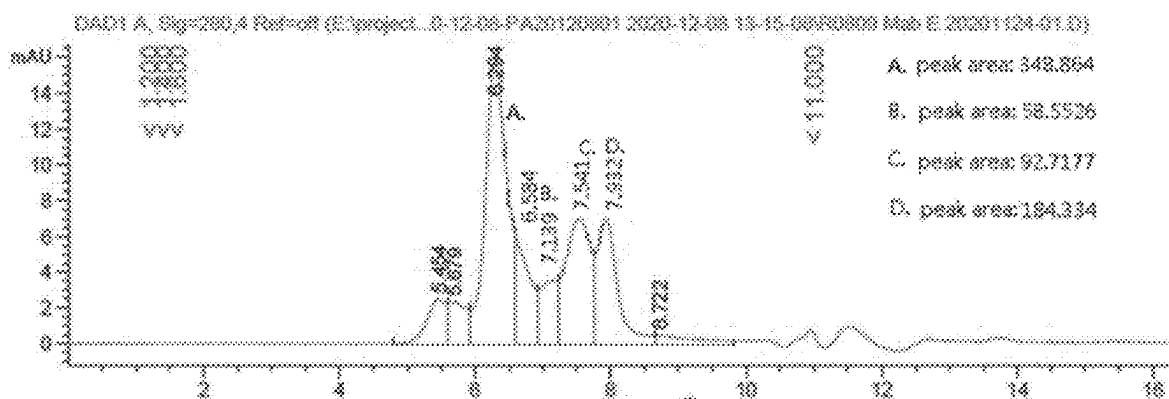


Fig. 14

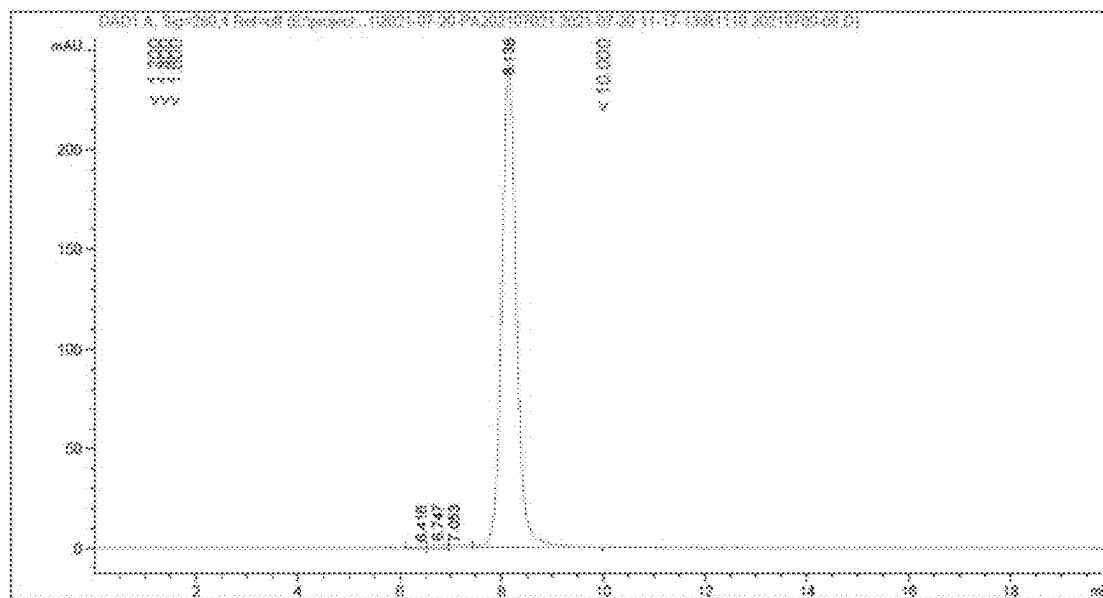


Fig. 15

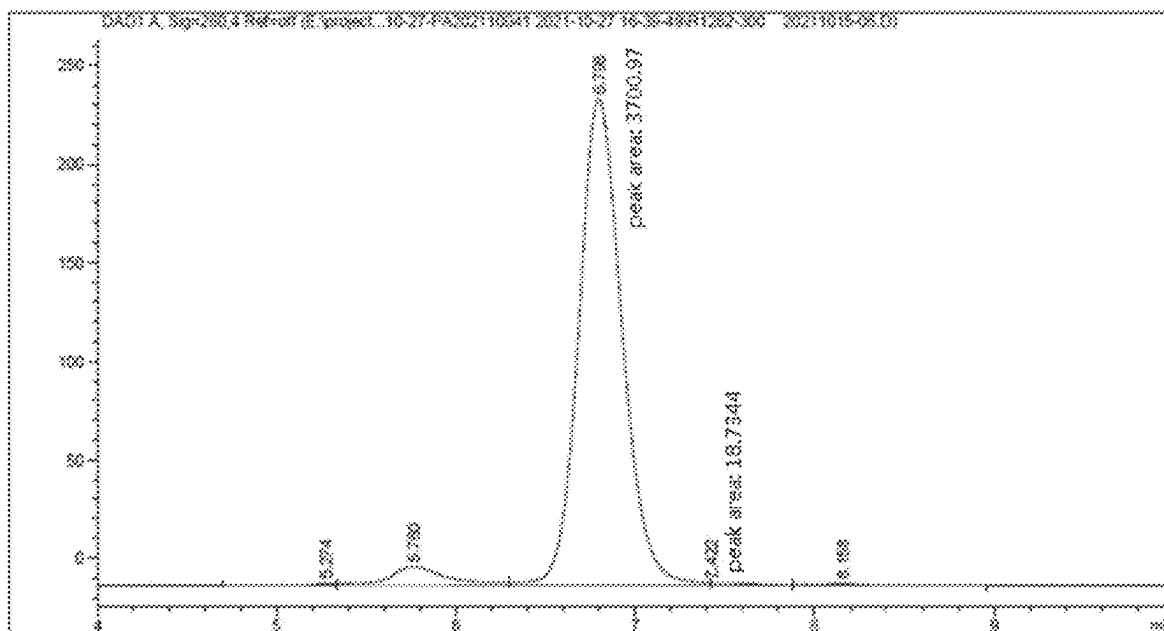


Fig. 16

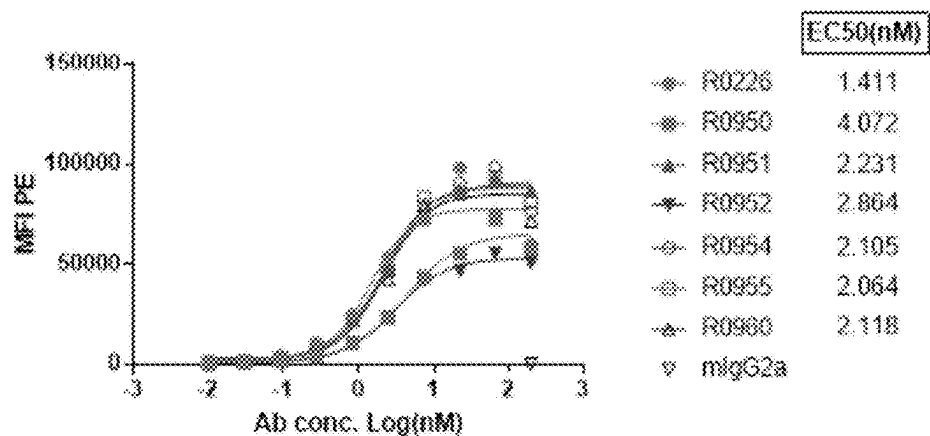


Fig. 17

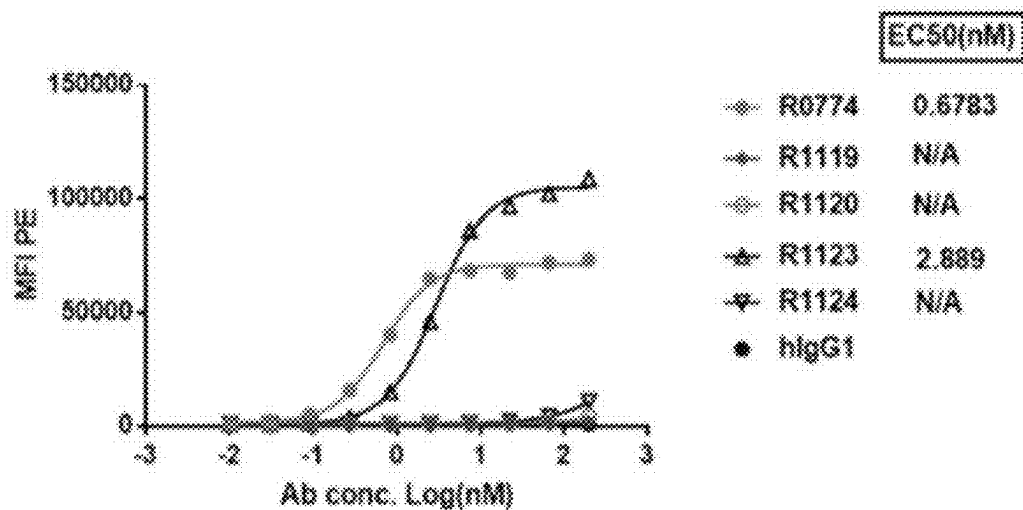


Fig. 18

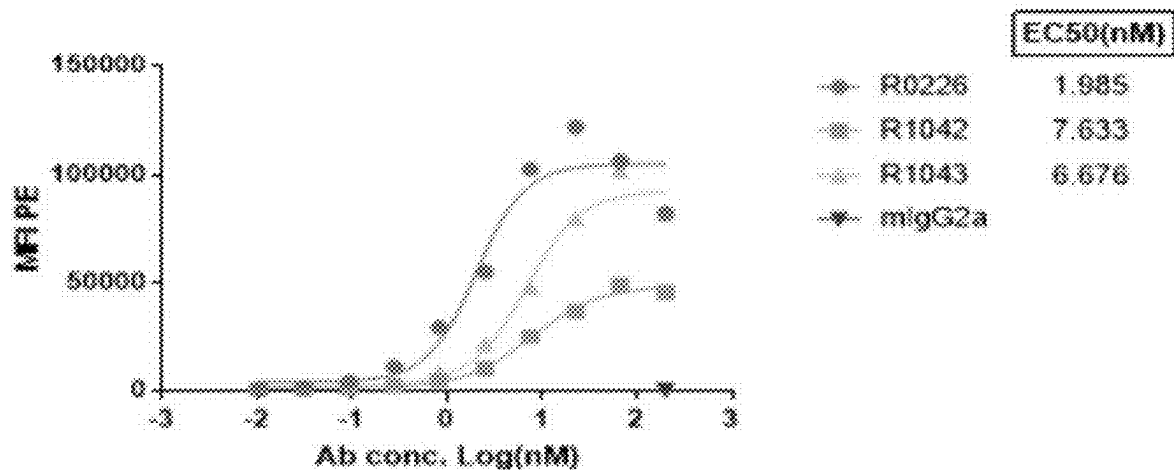


Fig. 19

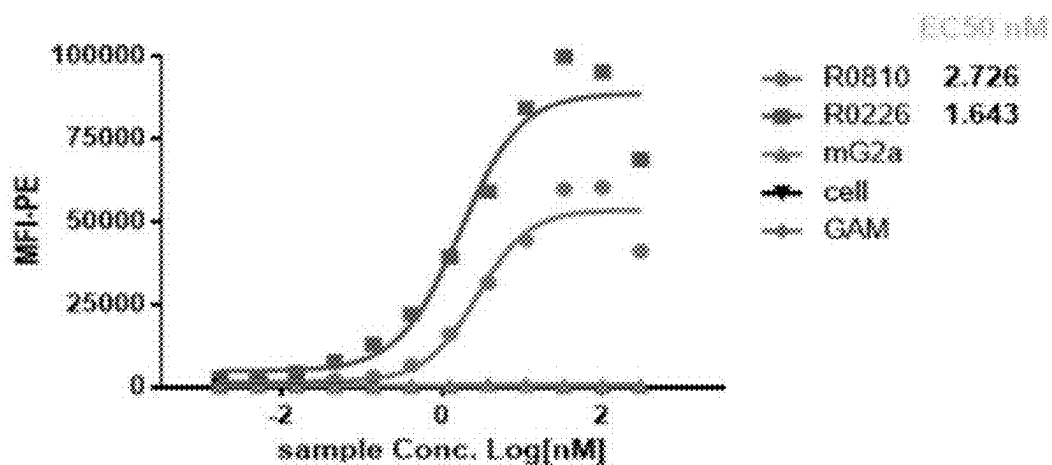


Fig. 20

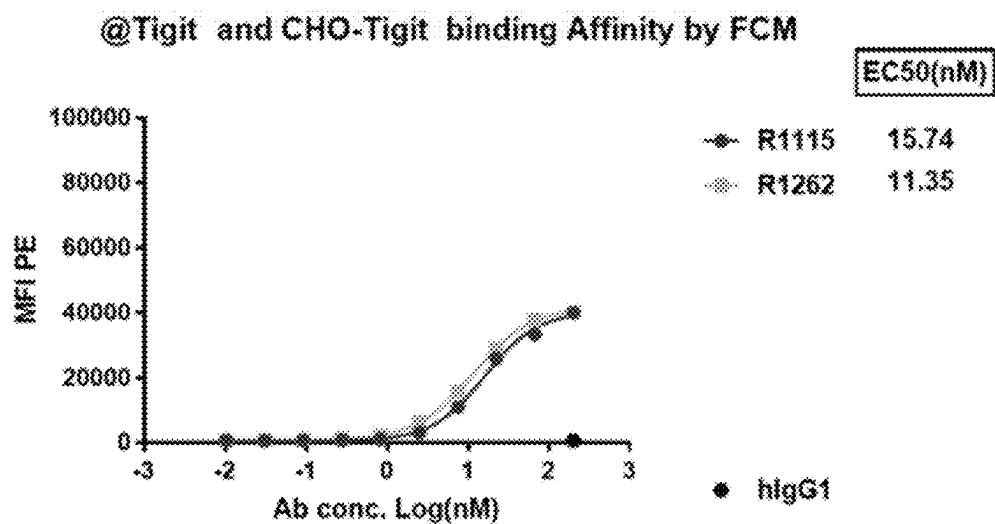
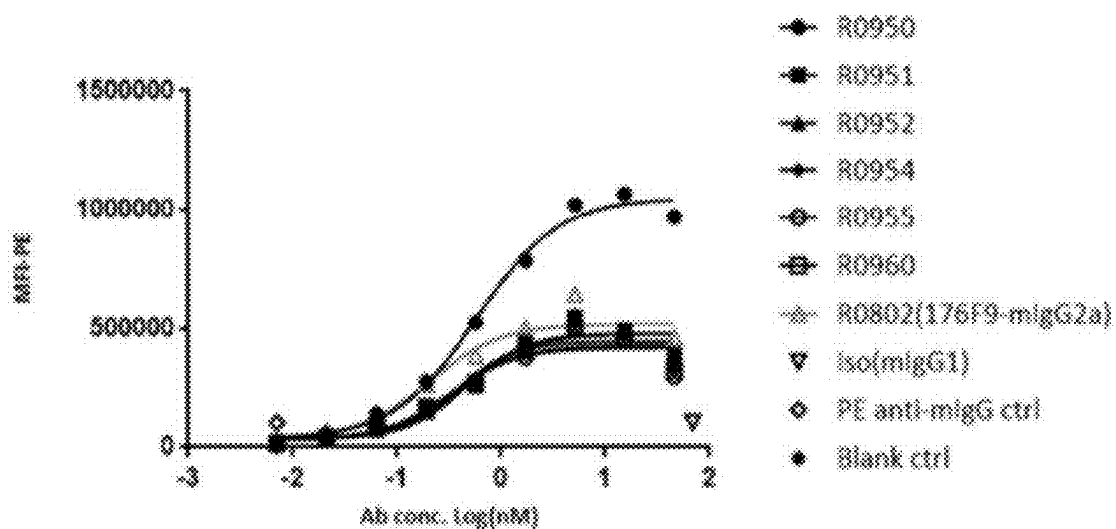
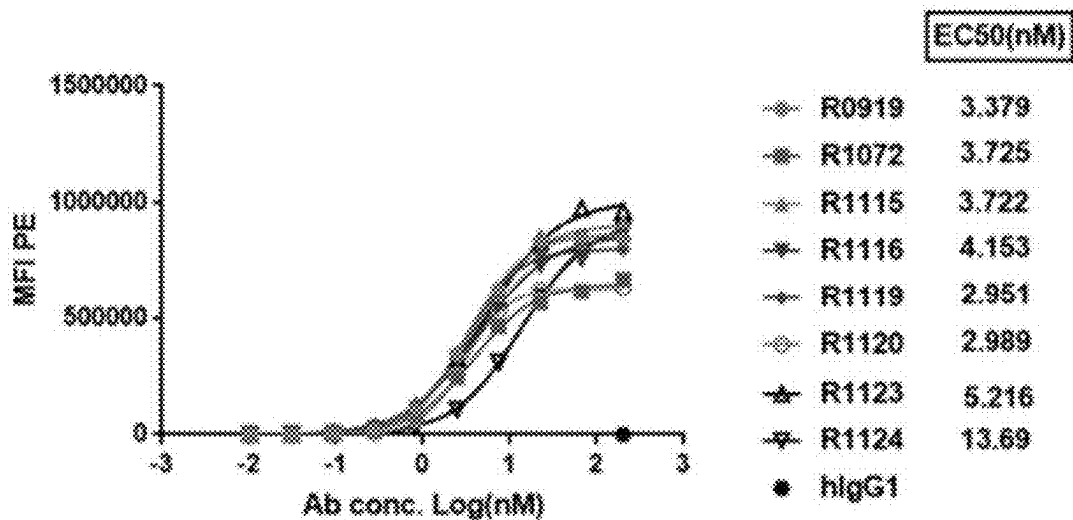


Fig. 21



	R0950	R0951	R0952	R0954	R0955	R0960	R0802	Iso(mIgG1)
EC50	0.5712	0.4596	0.4303	0.373	0.33	0.3257	0.2016	

Fig. 22



EC50(nM)

R0919	3.379
R1072	3.725
R1115	3.722
R1116	4.153
R1119	2.951
R1120	2.989
R1123	5.216
R1124	13.69
hlgG1	

Fig. 23

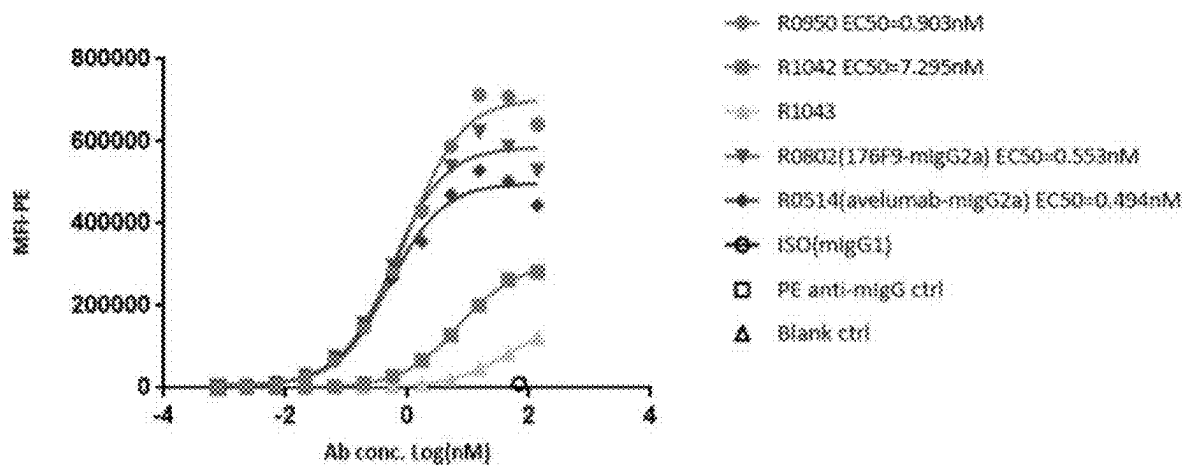
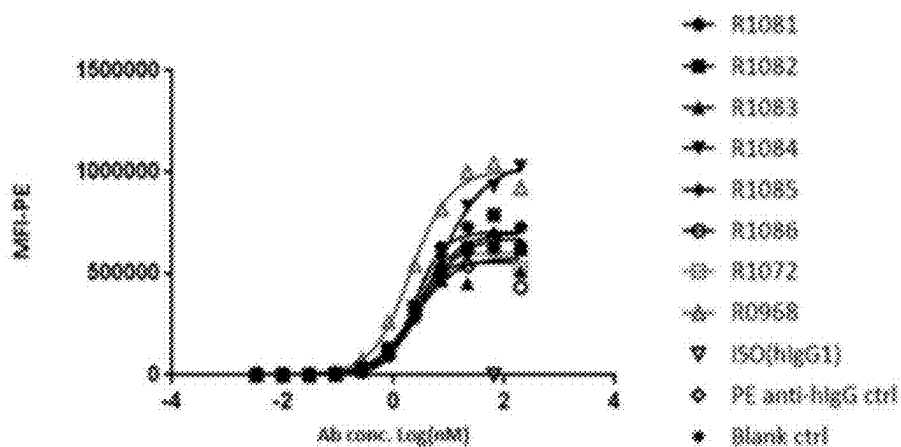


Fig. 24



	R1081	R1082	R1083	R1084	R1085	R1086	R1072	R0968
EC50	2.37	2.784	2.287	5.853	3.198	2.015	2.003	2.013

Fig. 25

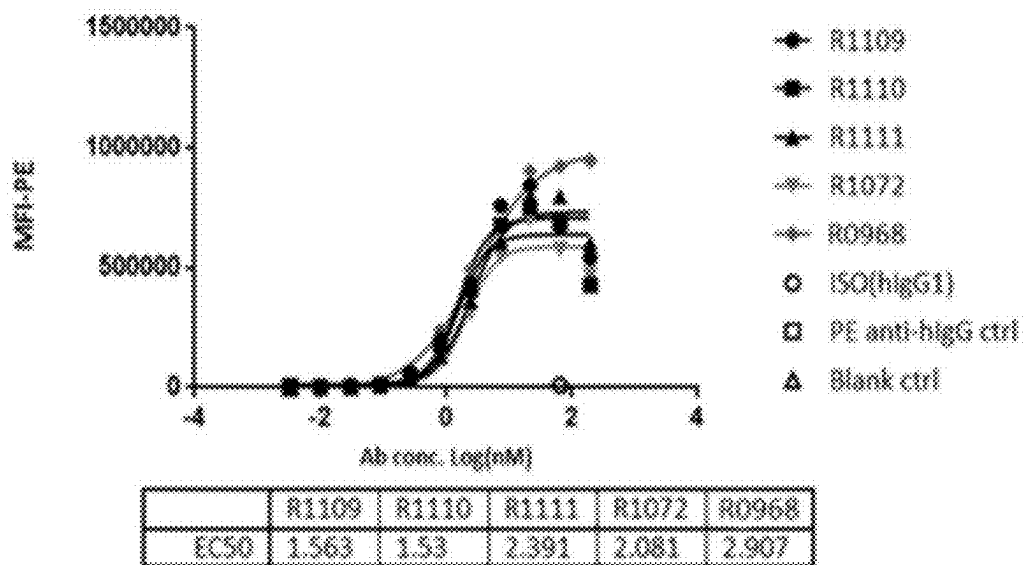


Fig. 26

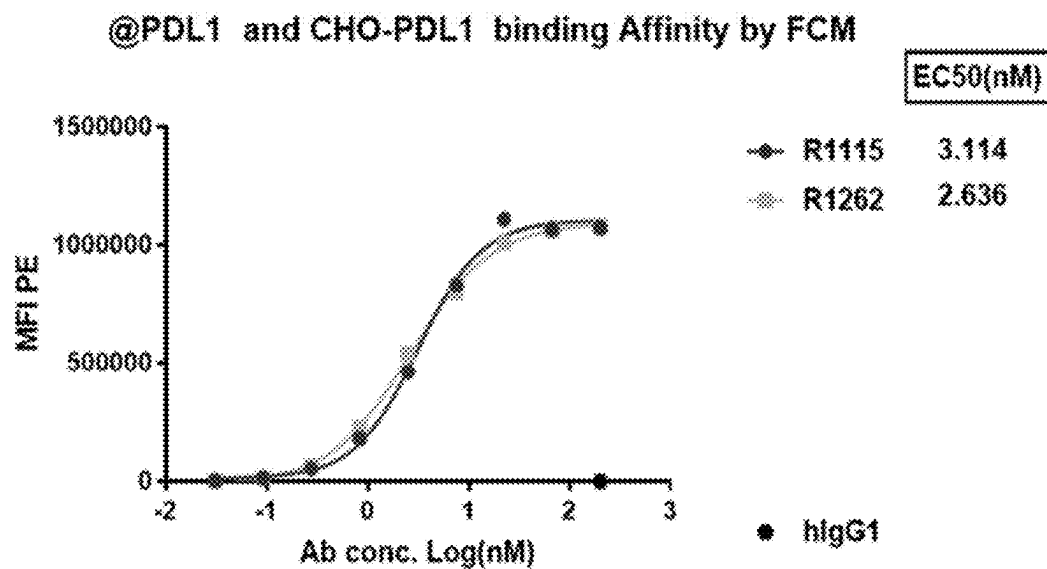


Fig. 27

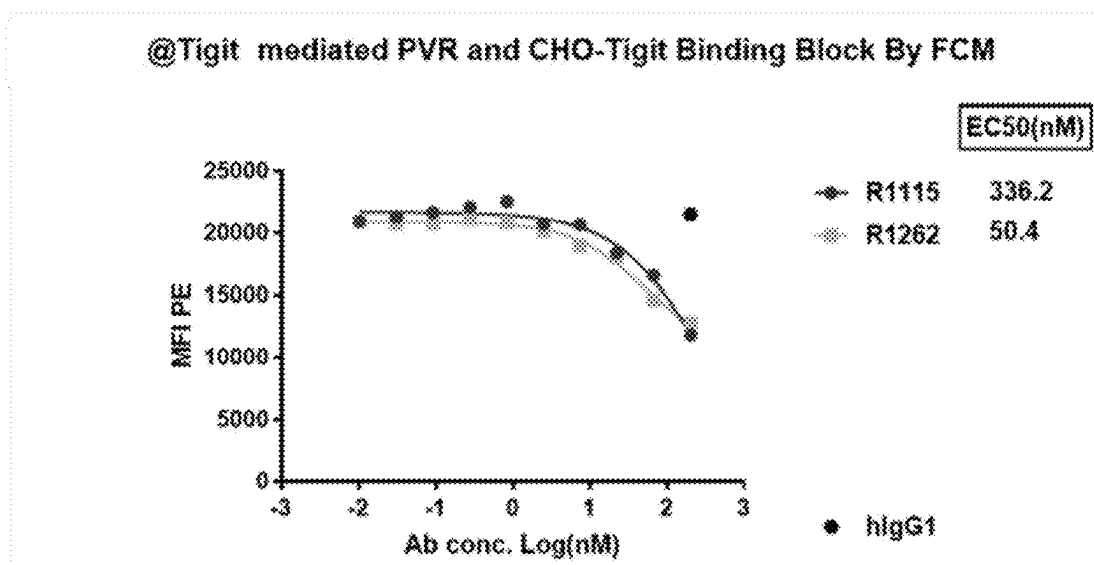


Fig. 28

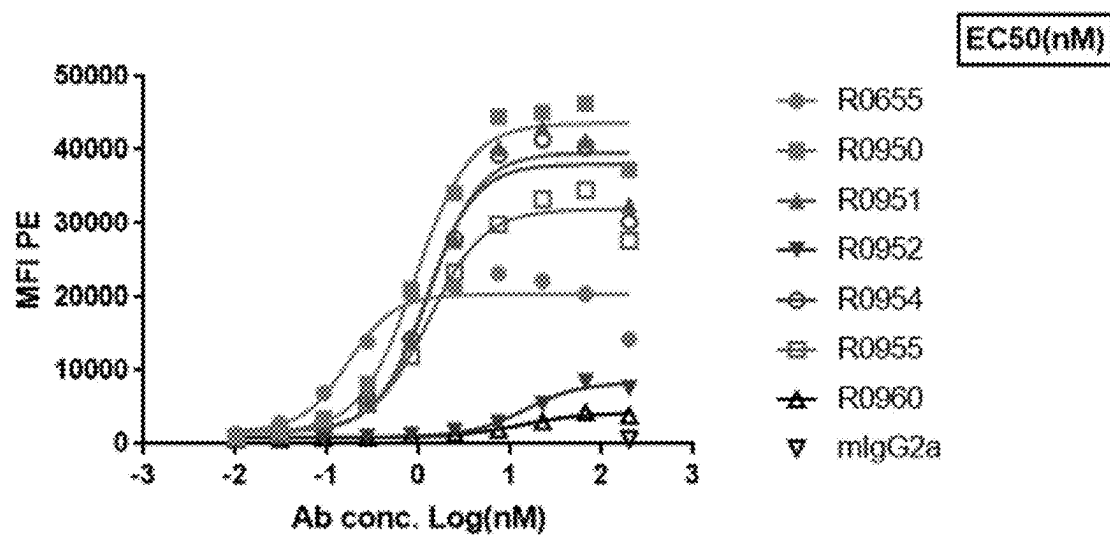


Fig. 29

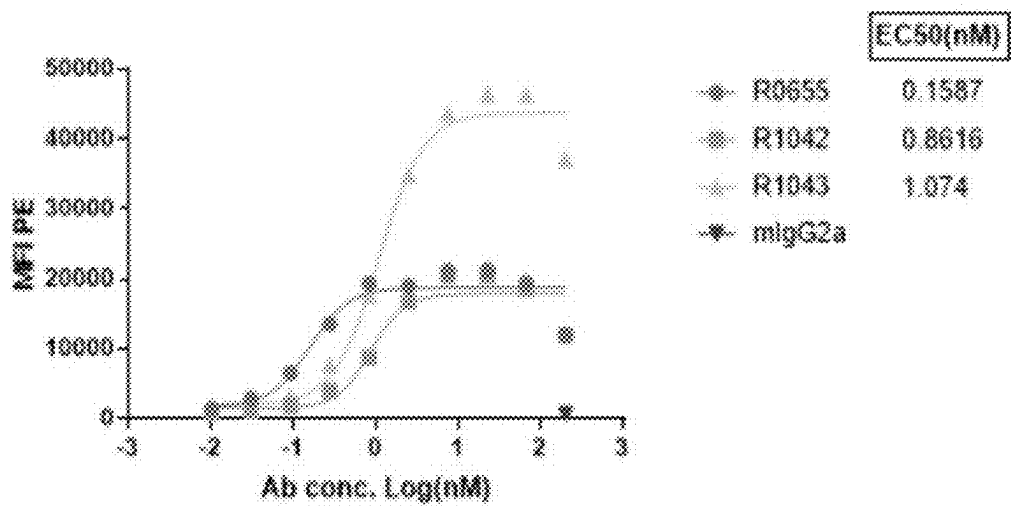


Fig. 30

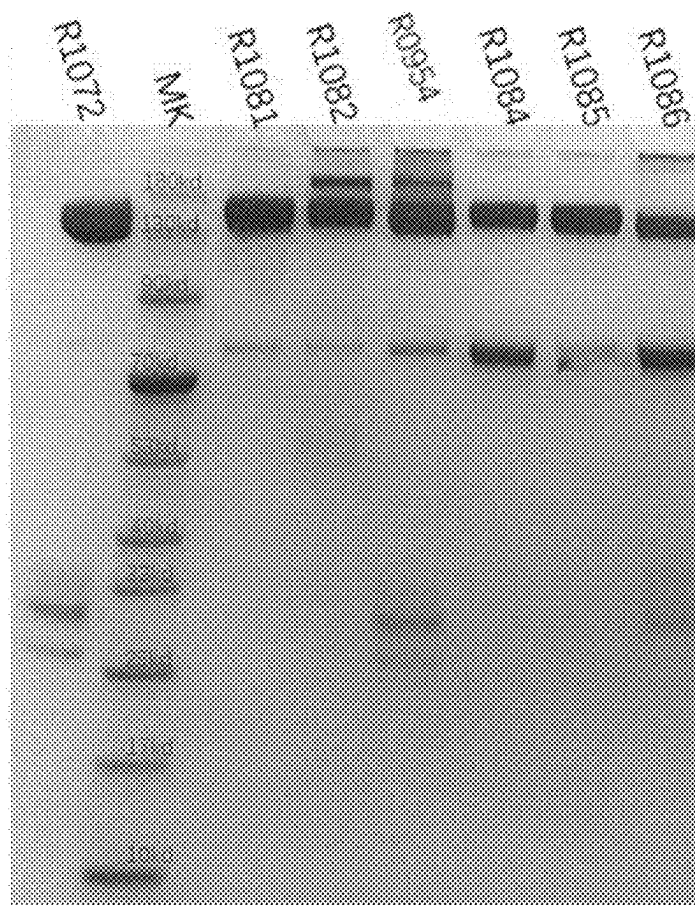


Fig. 31

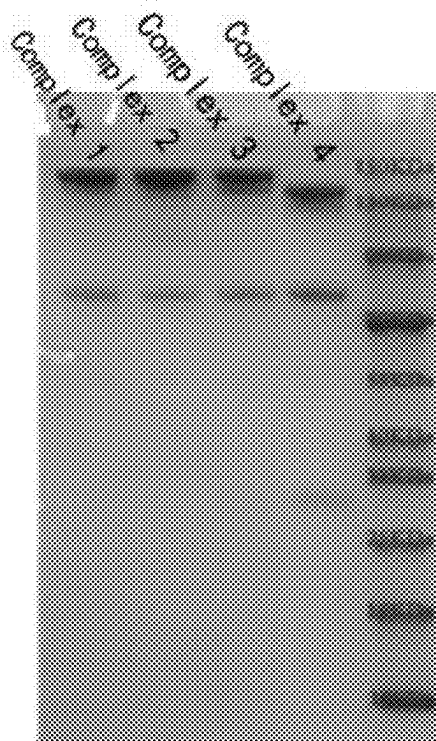


Fig. 32

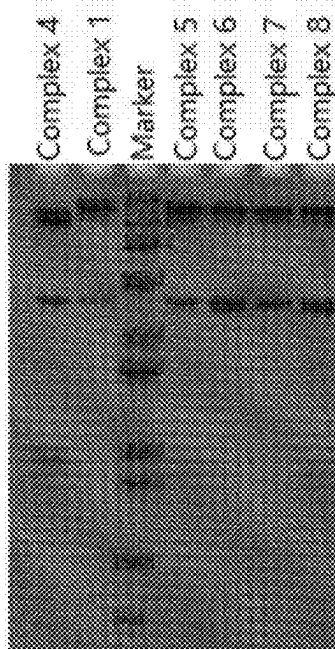
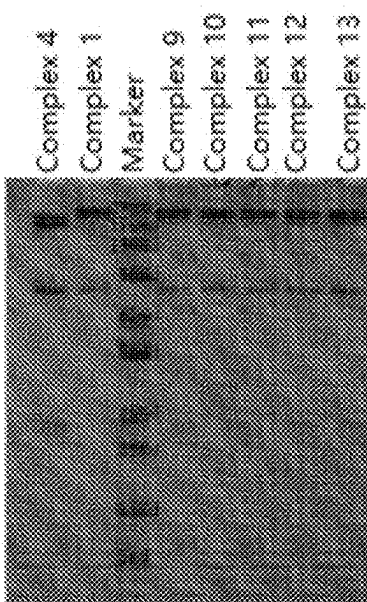


Fig. 33



MULTIFUNCTIONAL BISPECIFIC FUSION POLYPEPTIDE

CROSS-REFERENCE TO RELATED APPLICATION

[0001] The present disclosure claims the priority of a total of 4 Chinese patent applications with Application No. “202110436970.6”, entitled “antibody or functional fragment thereof” filed on Apr. 22, 2021, Application No. “202110871320.4”, entitled “antibody or functional fragment thereof” filed on Jul. 30, 2021, Application No. “202111121937.0”, entitled “multifunctional bispecific fusion polypeptide” filed on Sep. 24, 2021, and Application No. “202210240917.3”, entitled “multifunctional bispecific fusion polypeptide” filed on Mar. 10, 2022 submitted to China Patent Office, and the disclosure of which is incorporated by reference herein in its entirety.

SEQUENCE LISTING

[0002] The instant application contains a Sequence Listing which has submitted electronically in ASCII format and is hereby incorporated by reference in its entirety. Said ASCII copy is named PN214649_SEQ LIST.xml and is 293,063 bytes in size. The sequence listing contains 93 sequences, which is identical in substance to the sequences disclosed in the PCT application, only the priority and the description of the artificial sequence have been added and includes no new matter.

TECHNICAL FIELD

[0003] The present disclosure relates to the technical field of biomedicines, and particularly to a bispecific fusion polypeptide and/or multifunctional fusion polypeptide comprising a ligand and a receptor.

BACKGROUND

[0004] In order to broaden clinical efficacies of protein drugs, global biotechnology science researchers are struggling to develop biomacromolecule drugs with different structures and types. Bispecific antibodies (BsAb) are currently the most clinically popular novel biomacromolecule drug structures. The BsAb are antibodies may bind to two different antigens or different epitopes of an antigen simultaneously, and may exert biological functions that cannot be achieved by monoclonal antibodies through a specific mode of action.

[0005] With the progress of recombinant protein expression and genetic engineering techniques, the forms of the BsAb are increasingly diversified. Until now, more than 20 forms of BsAb have been developed into a technical platform.

[0006] During an assembly of IgG-type bispecific antibodies, 10 possible combinations of two natural heavy chains and two natural light chains may be randomly generated, of which only one is a target bispecific antibody product. The biochemical characteristics of 10 different bispecific antibody products are similar. It is extremely difficult to separate out the target bispecific antibody, such that the yield and the purity of the target bispecific antibody are low, the cost is increased, and the efficacy is affected.

[0007] A core value of a bispecific antibody technical platform is to solve problems of mismatching of heavy chains and mismatching of light chain and heavy chain.

Main technical platforms for solving the mismatching of heavy chains are: Knob-into-Holes (KiH), ART-Ig, a strand-exchange engineered domain (SEED) technology, and XmAb.

[0008] The main technical platforms for solving the mismatching of heavy chain and light chain are:

[0009] 1. Common light chain technology: the technology was developed by Genentech in 2006. Antibodies found from a phage-display screening directed against multiple antigens often share the same VL domain, reflecting a very limited size of a light chain library in a phage library. The use of a common light chain strategy simplifies an antibody engineering and a purification process in an industrial production. Advantages: a problem of LC/HC and HC/HC mismatching is solved. Disadvantages: it is very difficult to find VL commonly shared by antibodies in a phage-display screening, Patent Application WO2008027236.

[0010] 2. CrossMab technology: the technology was developed by Roche in 2007 and solves a light chain problem by exchanging heavy chain and light chain domain sequences of a Fab fragment, which is generally achieved in three ways: exchanging VH and VL, exchanging CH1 and CL or exchanging VH and VL and CH1 and CL. Advantages: a problem of mismatching of light and heavy chain may be solved by combining a KiH structure, and affinities of original antigens are reserved. Disadvantages: the expression level is relatively low, Patent Application WO2009080251.

[0011] The technologies for solving a problem of mismatching of heavy chains are relatively mature. The technologies for solving a problem of mismatching of light chain and heavy chain still need to be improved. Therefore, the present disclosure is specifically provided.

SUMMARY

[0012] The present disclosure aims to solve one technical problem existing in the related art to some extent.

[0013] The inventor provides a concept for developing a novel bispecific antibody. The mismatching of heavy chain and light chain is avoided or reduced by replacing CH1 and CL in the antibody or a functional fragment thereof with a specific affinity for a ligand and a receptor; and further, the replacement may be simultaneously or independently selected from CH2, CH3, and optionally CH4, thereby facilitating formation of a heavy chain heterodimer. In another aspect, the bispecific antibody provided in the present disclosure is a multifunctional fusion protein that may exert a dual-target specificity and also exert a conductive biological activity of a ligand and a receptor.

[0014] In one aspect, the present disclosure provides a bispecific fusion polypeptide, comprising a first antigen-binding moiety which comprises:

[0015] a first polypeptide comprising a first heavy chain variable domain VH1 of a first antibody from the N terminus to the C terminus, which is operably linked to a first conjugate fragment; and

[0016] a second polypeptide comprising a first light chain variable domain VL1 of the first antibody from the N terminus to the C terminus, which is operably linked to a second conjugate fragment, wherein,

[0017] the first conjugate fragment and the second conjugate fragment are capable of specifically binding; and

[0018] the first conjugate fragment is a receptor and the second conjugate fragment is a ligand; or the first

conjugate fragment is a ligand and the second conjugate fragment is a receptor.

[0019] In some embodiments, the bispecific fusion polypeptide further comprises a second antigen-binding moiety, wherein the second antigen-binding moiety is different from the first antigen-binding moiety, and

[0020] the second antigen-binding moiety comprises:

[0021] a third polypeptide comprising a second heavy chain variable domain VH2 of a second antibody from the N terminus to the C terminus, which is operably linked to a third conjugate fragment; and

[0022] a fourth polypeptide comprising a second light chain variable domain VL2 of the second antibody from the N terminus to the C terminus, which is operably linked to a fourth conjugate fragment, wherein,

[0023] the third conjugate fragment and the fourth conjugate fragment are capable of specifically binding; the third conjugate fragment is a receptor and the fourth conjugate fragment is a ligand; or the third conjugate fragment is a ligand and the fourth conjugate fragment is a receptor; and

[0024] the third conjugate fragment and/or the fourth conjugate fragment, and the first conjugate fragment and/or the second conjugate fragment are selected from different receptors and ligands.

[0025] In some embodiments, the bispecific fusion polypeptide further comprises a second antigen-binding moiety, wherein the second antigen-binding moiety is different from the first antigen-binding moiety, and

[0026] the second antigen-binding moiety comprises:

[0027] a third polypeptide comprising a second heavy chain variable domain VH2 of a second antibody from the N terminus to the C terminus, which is operably linked to an antibody heavy chain constant region CH1; and

[0028] a fourth polypeptide comprising a second light chain variable domain VL2 of the second antibody from the N terminus to the C terminus, which is operably linked to an antibody light chain constant region CL.

[0029] In some embodiments, the receptor only comprises an active site that recognizes and binds to a ligand but does not comprise a functionally active site that generates a response reaction.

[0030] In some embodiments, at least one non-natural interchain bond is comprised between the receptor and the ligand, and the non-natural interchain bond is capable of enhancing a specific binding force between the receptor and the ligand. In some embodiments, the non-natural interchain bond is formed between a first mutant residue comprised by the receptor and a second mutant residue comprised by the ligand. In some embodiments, at least one of the first mutant residue and the second mutant residue is a cysteine residue. In some embodiments, the non-natural interchain bond is a disulfide bond.

[0031] In some embodiments, at least one natural glycosylation site is not present in the receptor and/or the ligand.

[0032] In some embodiments, the receptor and the ligand thereof are selected from an interleukin and a receptor thereof.

[0033] In some embodiments, the interleukin and the receptor thereof have a lift-type steric conformation and are selected from IL15/IL15R, IL2/IL2R, IL4/IL-4R α +R γ ,

IL-6/IL-6R, IL-11/IL-11R, IL-13/IL-13R1, IL-20/IL20R α +IL20R β , and/or IL24/IL20R α +IL20R β .

[0034] In some embodiments, the interleukin and the receptor thereof have a pincer-type steric conformation and are selected from IL7/IL7R, IL21/IL21R, and IL23A/IL12B. In some embodiments, the ligand and the receptor are selected from IL15 and IL15R α .

[0035] In some embodiments, the E at position 90 of the IL15 is mutated into C and the P at position 67 of the IL15R α is mutated into C.

[0036] In some embodiments, a non-natural disulfide bond is present between the first heavy chain variable domain VH1 and the first light chain variable domain VL1, and preferably, the first heavy chain variable domain VH1 and the first light chain variable domain VL1 comprise any one of the following mutation combinations:

Combination	VH	VL
1	37C	95C
2	44C	100C
3	44C	101C
4	44C	105C
5	45C	87C
6	45C	98C
7	100C	50C
8	100bC	49C
9	98C	46C
10	101C	46C
11	105C	43C
12	106C	57C
13	108C	43C

[0037] In some embodiments, the D at position 61 of the IL15 is mutated into N, the E at position 64 is mutated into Q, and/or the N at position 65 is mutated into D.

[0038] In some embodiments, at least one N-glycosylation site of the IL15 is not present; preferably, the N-glycosylation site is selected from N71, N79, and/or N112; and preferably, the IL15 comprises the following amino acid mutations: N71Q, N79Q, and/or N112Q.

[0039] In some embodiments, at least one O-glycosylation site of the IL15R α is not present; preferably, the O-glycosylation site is selected from T2, T81, and/or T86; and preferably, the IL15R α comprises the following amino acid mutations: T2A, T81A, and/or T86A.

[0040] In some embodiments, the ligand and the receptor are selected from IL2 and IL2R α .

[0041] In some embodiments, the S at position 75 of the IL2 is mutated into C and the N terminus of the IL2R α is extended by two or three amino acids.

[0042] Preferably, the N terminus of the IL2R α is extended by two amino acids, the second extended amino acid is cysteine, and the first extended amino acid is any one of a non-polar fatty acid amino acid, an aromatic amino acid, an amino acid with an uncharged R group, an amino acid with a positively charged R group or an amino acid with a negatively charged R group.

[0043] Preferably, the N terminus of the IL2R α is extended by three amino acids, the second extended amino acid is cysteine, and the first and third extended amino acids are any one of a non-polar fatty acid amino acid, an aromatic amino acid, an amino acid with an uncharged R group, an amino acid with a positively charged R group or an amino acid with a negatively charged R group.

[0044] In some embodiments, the bispecific fusion polypeptide comprises an antibody Fc constant region. In some embodiments, the antibody Fc constant region is a heterodimer. In some embodiments, the antibody Fc constant region is the heterodimer associated by KiH, a hydrophobic interaction, an electrostatic interaction, a hydrophilic interaction, and/or increased flexibility. In some embodiments, the antibody Fc constant region comprises CH2, CH3, and optionally CH4, and the CH2, the CH3, and/or the optionally CH4 are replaced by the receptor and the ligand thereof.

[0045] In some embodiments, the first antigen-binding moiety and the second antigen-binding moiety bind to different antigens or bind to different epitopes of the same antigen. In some embodiments, the first antigen-binding moiety targets an immune cell and the second antigen-binding moiety targets a tumor cell. In some embodiments, the first antigen-binding moiety and the second antigen-binding moiety both target a tumor cell. In some embodiments, the first antigen-binding moiety and the second antigen-binding moiety both target an immune cell. In some embodiments, the first antigen-binding moiety and the second antigen-binding moiety both target a tumor cell. In some embodiments, the first antigen-binding moiety and the second antigen-binding moiety bind to link a T cell and a tumor antigen. In some embodiments, the first antigen-binding moiety and the second antigen-binding moiety bind to link an NK cell and a tumor antigen. In some embodiments, the first antigen-binding moiety and the second antigen-binding moiety bind to synergistically inhibit a signaling pathway. In some embodiments, the first antigen-binding moiety and the second antigen-binding moiety bind to form a protein complex.

[0046] The present disclosure further relates to an isolated nucleic acid, encoding the bispecific fusion polypeptide.

[0047] The present disclosure further relates a vector containing the nucleic acid.

[0048] The present disclosure further relates to a host cell containing the nucleic acid or the vector.

[0049] The present invention further relates to a method for preparing a bispecific fusion polypeptide, comprising:

[0050] transforming a host cell with the vector;

[0051] culturing the transformed host cell; and

[0052] collecting the bispecific fusion polypeptide expressed in the host cell.

[0053] The present disclosure further relates to a pharmaceutical composition, comprising the bispecific fusion polypeptide, a pharmaceutically acceptable carrier, an excipient, or a stabilizer.

[0054] The present disclosure further relates to use of the bispecific fusion polypeptide in preparing a drug for treating a disease.

BRIEF DESCRIPTION OF THE DRAWINGS

[0055] FIG. 1 shows 4 classical bispecific antibody platforms: FIG. 1A is a KiH heterodimeric Fc modification technology; FIG. 1B is a CrossMab bispecific antibody technology; FIG. 1C is a YBody bispecific antibody technology of Wuhan YZY Med (asymmetric scFv bispecific antibody); and FIG. 1D is a symmetric scFv bispecific antibody;

[0056] FIG. 2 is a novel bispecific antibody FiBody provided by the present disclosure, wherein CH1 and CL of one side Fab are replaced by a ligand and a receptor with a specific affinity;

[0057] FIG. 3 exemplarily shows 4 feasible solutions of FiBody: FIG. 3-1 shows a modified ligand and receptor

having a non-naturally occurring interchain bond between the ligand and the receptor; FIG. 3-2 shows that CH1 and CL of both Fabs are replaced by a receptor and a ligand, and both sides are selected from different ligands and receptors; FIG. 3-3 shows that in addition to that CH1 and CL of a Fab on one side of the antibody are replaced by a ligand and a receptor, a CH3 fragment in an Fc dimer is also replaced by a ligand and a receptor; and FIG. 3-4 shows that in addition to that CH1 and CL of a Fab on one side are replaced by a ligand and a receptor, CH2 in the Fc dimer is also replaced by a ligand and a receptor; and other feasible modification modes are various;

[0058] FIG. 4 shows an example that when the bispecific antibody of the present disclosure is used to treat a tumor, a targeted binding of an antigen-binding moiety of the bispecific antibody comprises 3 exemplary types: FIG. 4-A shows that a first antigen-binding moiety targets a T cell and a second antigen-binding moiety targets a tumor cell; FIG. 4-B shows both a first antigen-binding moiety and a second antigen-binding moiety target a tumor cell; FIG. 4-C shows both a first antigen-binding moiety and a second antigen-binding moiety target a T cell; and FIG. 4-D exemplarily shows that the bispecific antibody of the present disclosure may optionally be a trifunctional fusion protein, is also capable of activating a ligand and receptor pathway and eliciting a biological activity of a ligand and a receptor in addition to exerting binding of different antigens;

[0059] FIG. 5 shows steric conformations of interleukins and receptors thereof, which may be classified into four types: A: lift type, B: bowknot type, C: ballplayer type, and D: pincer type;

[0060] FIG. 6 is an example of four steric conformations of interleukins and receptors thereof, wherein A lift type is IL2/IL2R, B bowknot type is IL22/IL22R, C ballplayer type is IL18/IL18R, and D pincer type is IL21/IL21R;

[0061] FIG. 7 is a FiBody design based on IL15 (ligand) and IL15RA (receptor) in the example of the present disclosure, wherein a second antigen-binding region VH is linked to IL15RA and a second antigen-binding region VL is linked to IL15;

[0062] FIG. 8 is a structural schematic diagram of modification and optimization of a disulfide bond;

[0063] FIG. 9 is a three-dimensional structural schematic diagram of an interaction of IL15/IL15RA with an IL2/15Rβ/γC complex;

[0064] FIG. 10 shows a structural schematic diagram of a mismatched molecule R1042/R1124 in the example of the present disclosure;

[0065] FIG. 11 shows an HPLC-SEC detection result of a sample R0951 in the example of the present disclosure;

[0066] FIG. 12 shows an HPLC-SEC detection result of a sample R1042 in the example of the present disclosure;

[0067] FIG. 13 shows an HPLC-SEC detection result of a sample R0809 in the example of the present disclosure;

[0068] FIG. 14 shows an HPLC-SEC detection result of a sample R1110 in the example of the present disclosure;

[0069] FIG. 15 shows an HPLC-SEC detection result of a sample R1262 in the example of the present disclosure;

[0070] FIG. 16 shows binding activities of TIGIT ends of bispecific antibodies (R0950, R0951, R0952, R0954, R0955, and R0960) and CHO-TIGIT cells detected by an FCM method in the example of the present disclosure;

[0071] FIG. 17 shows binding activities of TIGIT ends of bispecific antibodies (R1123/R1119/R1120/R1124) and

CHO-TIGIT cells detected by an FCM method in the example of the present disclosure;

[0072] FIG. 18 shows binding activities of TIGIT ends of bispecific antibodies (R1042/R1043) and CHO-Tigit cells detected by an FCM method in the example of the present disclosure;

[0073] FIG. 19 shows a binding activity of a TIGIT ends of a bispecific antibody (R0810) and CHO-Tigit cells detected by an FCM method in the example of the present disclosure;

[0074] FIG. 20 shows a binding activity of a TIGIT end of a bispecific antibody (R1262) and CHO-Tigit cells detected by an FCM method in the example of the present disclosure, wherein hIgG1 is a subtype control antibody;

[0075] FIG. 21 shows binding activities of PD-L1 ends of bispecific antibodies (R0950, R0951, R0952, R0954, R0955, and R0960) and CHO-PD-L1 cells detected by an FCM method in the example of the present disclosure;

[0076] FIG. 22 shows binding activities of PD-L1 ends of bispecific antibodies (R1072, R1115-R1120, and R1123-R1124) and CHO-PD-L1 cells detected by an FCM method in the example of the present disclosure;

[0077] FIG. 23 shows binding activities of PD-L1 ends of bispecific antibodies (R0950, R1042, and R1043) and CHO-PD-L1 cells detected by an FCM method in the example of the present disclosure;

[0078] FIG. 24 shows binding activities of PD-L1 ends of bispecific antibodies (R1072 and R1081-R1086) and CHO-PD-L1 cells detected by an FCM method in the example of the present disclosure;

[0079] FIG. 25 shows binding activities of PD-L1 ends of bispecific antibodies (R1072 and R1109-R1111) and CHO-PD-L1 cells detected by an FCM method in the example of the present disclosure;

[0080] FIG. 26 shows a binding activity of a PD-L1 end of a bispecific antibody (R1262) and CHO-PD-L1 cells detected by an FCM method in the example of the present disclosure;

[0081] FIG. 27 shows a detection result of blocking of R1262 on a binding force between a ligand and a target region in the example of the present disclosure;

[0082] FIG. 28 shows binding activities of sample receptor and ligand complexes (IL15/IL15R) (R0950, R0951, R0952, R0954, R0955, and R0960) in the example of the present disclosure;

[0083] FIG. 29 shows binding activities of samples (R1042 and R1043) receptor and ligand complexes (IL15/IL15R) in the example of the present disclosure;

[0084] FIG. 30 shows gel electrophoresis detection results of disulfide bond-modified optimized samples (R1072, R1081, R1082, R0954, and R1084-R1086) in the example of the present disclosure;

[0085] FIG. 31 shows gel electrophoresis detection results of FiBody samples prepared in example 9 of the present disclosure (disulfide bond-modified IL2/IL2R α complexes 1 and 2, and an IL2/IL2R α complex 3 constructed correspondingly in example 8) and R1115 in example 2, wherein the complexes 1 and 2 are disulfide bond-modified IL2/IL2R α complexes, the complex 3 is the prior reported IL2/IL2R α complex, and a complex 4 is a disulfide bond-unmodified IL2/IL2R α complex (sample R1115);

[0086] FIG. 32 shows gel electrophoresis detection results of purified disulfide bond-modified IL2/IL2R α complexes 5, 6, 7, and 8 and a complex 1 prepared in example 14 of the

present disclosure, wherein a disulfide bond-unmodified IL2/IL2R α complex 4 is used as a control; and

[0087] FIG. 33 shows gel electrophoresis detection results of purified disulfide bond-modified IL2/IL2R α complexes 9, 10, 11, 12, and 13 and the complex 1 prepared in example 14 of the present disclosure, wherein the disulfide bond-unmodified IL2/IL2R α complex 4 is used as a control.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0088] A reference will now be made in detail to embodiments of the present disclosure, wherein one or more examples of which are described below. Each example is provided by an explanation, not a limitation, of the present disclosure. Actually, it would have been obvious to a person skilled in the art to make various changes and modifications to the present disclosure without departing from the scope or spirit of the present disclosure. For example, features illustrated or described as a part of one embodiment, may be used on another embodiment to produce a still further embodiment. All references, including publications, patents, and patent applications, cited in the present disclosure are incorporated herein by reference in their entirety.

[0089] Unless otherwise defined, all technical and scientific terms used herein have the same meaning as commonly understood by a person skilled in the technical field of the present disclosure. The terms used in the description of the present disclosure herein are only for the purpose of describing specific examples and are not intended to limit the present disclosure. As used herein, the term “and/or” includes any and all combinations of one or more related items listed.

Terms

[0090] The term “antigen-binding moiety” or “antigen-binding domain” means a portion of an antigen-binding molecule that confers a binding specificity to an antigenic determinant. In some embodiments, the “antigen-binding moiety” is an antibody functional fragment.

[0091] The term “amino acid” means one of 20 naturally occurring amino acids encoded by DNA and RNA.

[0092] Single-letter abbreviations of non-polar fatty acid amino acids are glycine (G), alanine (a), valine (V), leucine (L), isoleucine (I), and methionine (M) respectively; single-letter abbreviations of aromatic amino acids are phenylalanine (F), tryptophan (W), and tyrosine (Y) respectively; single-letter abbreviations of amino acids with an uncharged R group are serine(S), threonine (T), cysteine (C), proline (P), aspartic acid (N), and glutamine (Q) respectively; single-letter abbreviations of amino acids with a positively charged R group are lysine (K), arginine (R), and histidine (H) respectively; and single-letter abbreviations of amino acids with a negatively charged R group are aspartic acid (D) and glutamic acid (E).

[0093] The term “wild-type or WT” means an amino acid sequence or a nucleotide sequence found in nature, including an allelic variation. A WT protein has an amino acid sequence or a nucleotide sequence that has not been intentionally modified.

[0094] The term “antibody” encompasses any immunoglobulin, monoclonal antibody, polyclonal antibody, polyclonal antibody, bispecific (bivalent) antibody, or bispecific fusion polypeptide that may bind to a specific antigen.

A natural and intact antibody comprises two heavy chains and two light chains. Each heavy chain consists of a variable region ("VH") and first, second, and third constant regions (CH1, CH2, and CH3 respectively), and each light chain consists of a variable region ("VL") and a constant region (CL). Mammalian heavy chains may be divided into α , δ , ϵ , γ , and μ , and mammalian light chains into λ or κ .

[0095] The antibody is "Y" shaped, and a backbone consists of the second (CH2), the third (CH3), and optionally the fourth constant regions (CH4) of the two heavy chains, which are bound by disulfide bonds. Each arm of the "Y" structure comprises a variable region (VH) and a first constant region (CH1) of one of the heavy chains, which bind to a variable region (VL) and a constant region (CL) of one of the light chains. The variable regions of the light and heavy chains are responsible for antigen binding. The variable region of each chain contains three hypervariable regions, wherein complementarity determining regions (CDRs) of the light (L) chain include LCDR1, LCDR2, and LCDR3, and the CDRs of the heavy (H) chain include HCDR1, HCDR2, and HCDR3. The three CDRs are separated by flanking continuous portions called framework regions (FRs) that are more highly conserved than the CDRs and form a scaffold-supported hypervariable loop. The HCVR and the LCVR each comprise 4 FRs. The CDRs and the FRs are arranged from an amino terminus to a carboxy terminus in the following order: FR1, CDR1, FR2, CDR2, FR3, CDR3, and FR4.

[0096] The constant regions of the heavy chain and the light chain are not involved in antigen binding, but have multiple effector functions. Antibodies may be divided into several kinds depending on an amino acid sequence of a heavy chain constant region. Depending on whether containing α , δ , ϵ , γ , and μ heavy chains, the antibodies may be divided into five major classes or isoforms respectively: IgA, IgD, IgE, IgG, and IgM. Several main antibody classes may further be divided into subclasses, such as IgG1 (γ 1 heavy chain), IgG2(γ 2 heavy chain), IgG3(γ 3 heavy chain), IgG4(γ 4 heavy chain), IgA1 (α 1 heavy chain), or IgA2 (α 2 heavy chain).

[0097] The hypervariable regions typically comprise about 24-34 (LCDR1; "L" represents the light chain), 50-56 (LCDR2), and 89-97 (LCDR3) amino acid residues in the light chain variable region, and about 31-35 (HCDR1; "H" represents the heavy chain), 50-65 (HCDR2), and 95-102 (HCDR3) amino acid residues in the heavy chain variable region; Kabat et al., SEQUENCES OF PROTEINS OF IMMUNOLOGICAL INTEREST, 5th edition, Public Health Service, National Institutes of Health, Bethesda, Md. (1991) and/or those residues that form hyper-mutated loops (e.g., residues 26-32 (LCDR1), 50-52 (LCDR2), and 91-96 (LCDR3) in the light chain variable region, and 26-32 (HCDR1), 53-55 (HCDR2), and 96-101 (HCDR3) in the heavy chain variable region); and Chothia and Lesk (1987), J. Mol. Biol., 196:901-917).

[0098] In some embodiments, the antibody is a bispecific antibody (BiAb). The term "bispecific" herein refers to two different antigens, or even when both are the same antigen, each of which has a binding specificity directed to different epitopes. The epitopes may be derived from different antigens or the same antigen. The terms "bispecific fusion polypeptide" and "bispecific antibody" herein refer to all prepared products having a full-length antibody or a fragment with an antigen-binding site. The antibody may be a

human antibody, a non-human antibody (e.g., a murine antibody), a humanized antibody, or a chimeric antibody (e.g., a human-mouse chimeric antibody or a chimera of different subtypes of antibodies). In some cases, a variant of the antibody is a conservative modification or a conservative replacement or substitution on a sequence of the antibody provided herein. The "conservative modification" or the "conservative replacement or substitution" refer to the replacement of an amino acid in a protein with other amino acids having similar characteristics (e.g., charge, side chain size, hydrophobicity/hydrophilicity, backbone conformation, rigidity, etc.), such that changes may be made frequently without changing a biological activity of the protein. It is known to a person skilled in the art that, in general, a single amino acid replacement in a non-essential region of a polypeptide does not substantially change a biological activity (see, for example, Watson et al. (1987), Molecular Biology of the Gene, The Benjamin/Cummings Pub. Co., page 224, (the fourth edition)). In addition, replacement of structurally or functionally similar amino acids is unlikely to destroy a biological activity. A person skilled in the art would have been able to determine suitable variants of the antigen-binding molecules set forth herein with well-known technologies. With regard to nucleotide and amino acid sequences, the term "identity" indicates a degree of identity between two nucleic acid or two amino acid sequences when optimally aligned and compared with appropriate insertions or deletions.

[0099] The term "Fab" is a Fab fragment not containing or containing a small portion of a residual Fc fragment in an immunoglobulin, e.g., the Fab fragment comprises variable regions of a heavy chain and a light chain, and a first constant domain of all or a portion. For simplicity, the term "Fab" hereinafter may also refer to a fragment such as F(ab)2.

[0100] The term "Fc" or "Fc region" or "Fc domain" means the constant region of an antibody, in some cases excluding all or a portion of a first constant region immunoglobulin domain (e.g., CH1) or all or a portion of a portion thereof, and in some cases further excluding all or a portion of a hinge. Therefore, Fc may refer to all or a portion of the last two constant region immunoglobulin domains (e.g., CH2 and CH3) of IgA, IgD, and IgG, the last three constant region immunoglobulin domains of IgE and IgM, and optionally a flexible hinge N terminus of these domains. With regard to IgA and IgM, the Fc may comprise a J chain. With regard to IgG, the Fc domain comprises the immunoglobulin domains CH2 and CH3 (C γ 2 and C γ 3) and a lower hinge region located between CH1 (C γ 1) and CH2 (C γ 2). Although a boundary of the Fc region may vary, the Fc region of a human IgG heavy chain is generally defined to include residues E216, C226, or A231 of its carboxy terminus, wherein numbering is performed according to an EU index in Kabat. In some embodiments, as described more comprehensively below, the Fc region is subjected to an amino acid modification, e.g., the Fc is a heterodimer.

[0101] The "modification" herein refers to an amino acid substitution, insertion and/or deletion in a polypeptide sequence or an alteration in a moiety chemically linked to a protein. The term "amino acid modification" herein refers to an amino acid substitution, insertion and/or deletion in a polypeptide sequence. For clarity, unless otherwise indi-

cated, the amino acid modification is always amino acids encoded by DNA, e.g., 20 amino acids with codons in DNA and RNA.

[0102] The “epitope” means herein a determinant that interacts with a specific antigen-binding domain, e.g., a variable region (referred to as a paratope) of an antibody molecule. The epitope is a group of molecules, such as amino acids or sugar side chains, and generally has specific structural characteristics and specific charge characteristics. A single molecule may have more than one epitope. The epitope may comprise amino acid residues directly involved in binding (also referred to as an immunodominant component of the epitope) and other amino acid residues not directly involved in binding, such as amino acid residues effectively blocked by a specific antigen-binding peptide. In other words, the amino acid residues are within a cover area of the specific antigen-binding peptide. The epitope may be either conformational or linear. The conformational epitope is produced by a spatial juxtaposition of amino acids from different segments of a linear polypeptide chain. The linear epitope may be produced by adjacent amino acid residues in a polypeptide chain. The conformational and non-conformational epitopes may be distinguished by a loss of binding to the former, but not the latter, in the presence of a denaturing solvent. The epitope typically comprises at least 3, and more typically at least 5 or 8-10 amino acids in a unique spatial conformation. An antigen-binding molecule that recognizes the same epitope may be verified in a simple immunoassay, demonstrating an ability of one antigen-binding molecule to block binding of another antigen-binding molecule to a target antigen. As outlined below, the present disclosure comprises antigen-binding molecules and antigen-binding domains listed herein, and further comprises antigen-binding molecules and antigen-binding domains that compete for binding with the epitopes to which the listed antigen-binding molecules or antigen-binding domains bind.

[0103] The terms “specific binding”, “selective binding”, “selectively binds to”, and “specifically binds to” refer to a biological binding process that a directed ligand that may be competitively blocked by a corresponding substance interacts with a specific structural site in vitro or in vivo, for example, binding between an antigen and an antibody or between a receptor and a ligand.

[0104] A strength or an affinity of specific binding may be expressed according to a dissociation constant (KD) of an interaction, wherein a smaller KD represents a greater affinity and a greater KD represents a lower affinity, such as the KD is expressed by an antigen-binding force of at least about 10^{-4} M, at least about 10^{-5} M, at least about 10^{-6} M, at least about 10^{-7} M, at least about 10^{-8} M, at least about 10^{-9} M, at least about 10^{-10} M, at least about 10^{-11} M, at least about 10^{-12} M, or greater. A binding characteristic may be determined by methods well-known in the art, such as a biolayer interferometry and a method based on a surface plasmon resonance. In one such method, an association and dissociation rate of an antigen-binding site/an antigen or a receptor/ligand complex needs to be measured, wherein the rate depends on a concentration of a complex partner, an affinity of an interaction, and a geometric parameter that affects the rate equally in both directions. Therefore, the association rate (k_a) and the dissociation rate (KD) may be determined, and a ratio of k_d/k_a is equal to the dissociation

constant KD (Nature, 361:186-187 (1993) and Davies et al. (1990), Annual RevBiochem, 59:439-473).

[0105] The term “immune cell” comprises a cell that is involved in an immune system of protecting the body against both infectious diseases and foreign substances. The immune cells may include, for example, neutrophils, eosinophils, basophils, lymphocytes, such as B cells, T cells, and monocytes. The T cells may include, for example, CD4+, CD8+, T helper cells, cytotoxic T cells, $\gamma\delta$ T cells, regulatory T cells, suppressor T cells, and natural killer cells.

[0106] The term “multifunctional fusion polypeptide” means a non-naturally occurring binding molecule designed to target two or more antigens. The “multifunctional fusion polypeptide” described herein is typically a genetically engineered fusion protein designed to bring two different desired biological functions into a single binding molecule. For example, the multifunctional fusion polypeptide may be a multifunctional binding molecule.

[0107] The term “FiBody” is a bispecific fusion polypeptide or a multifunctional fusion protein obtained by substituting a portion or all of a constant region of a bispecific antibody with a specific affinity of a ligand and a receptor thereof.

[0108] The “YBody” technology mentioned in the present disclosure was developed by Wuhan YZY Med in 2012. The technology is that based on the “Knob-into-Holes” technology, one of a formed heterodimer is a normal heavy chain, the other is an N terminal of an Fc functional region linked to scFv, and an asymmetric bispecific antibody is formed.

[0109] The term “about” or “approximately” refers to quantification, level, value, quantity, frequency, percentage, dimension, size, amount, weight, or length that is 30%, 25%, 20%, 25%, 10%, 9%, 8%, 7%, 6%, 5%, 4%, 3%, 2%, or 1% different from reference quantification, level, value, quantity, frequency, percentage, dimension, size, amount, weight, or length. In particular embodiments, when the term “about” or “approximately” precedes a numerical value, it indicates to a range of the value plus or minus 15%, 10%, 5%, or 1%.

[0110] Unless otherwise specified in the context, the words “include”, “comprise”, and “contain” will be understood to mean comprising the stated step or element or a group of steps or elements but not excluding any other step or element or a group of steps or elements. The “consisting of” means including and being limited to what is followed by the phrase “consisting of”. Therefore, the phrase “consisting of” means that the listed elements are required or necessary and that no other elements may be present. The “essentially consisting of” means including any elements listed thereafter, and being limited to other elements that contribute to or do not interfere with activities or functions of the listed elements as detailed in the present disclosure. Therefore, the phrase “essentially consisting of” means that the listed elements are required or necessary, but other elements are optionally and may be present or not present depending on whether they affect activities or functions of the listed elements.

[0111] “One embodiment”, “embodiment”, “specific embodiment”, “related embodiment”, “certain embodiment”, “another embodiment” or “further embodiment” or a combination thereof mentioned in the whole text of the present disclosure mean that the described specific feature, structure, or characteristic related in the embodiment are included in at least one embodiment of the present disclosure. Therefore, the foregoing phrases appearing in various

places throughout the description are not necessarily all referring to the same embodiment. Furthermore, the specific feature, structure, or characteristics may be combined in any suitable manner in one or more embodiments.

[0112] The term “optionally” is used for a descriptive purpose only and not to be construed as indicating or implying a relative importance. Therefore, a feature defined as “optionally” may explicitly or implicitly include or exclude the feature.

[0113] The terms “first” and “second” in the description and in the claims are used for distinguishing similar elements and not necessarily for describing a sequence or a time order. It should be understood that such used terms are interchangeable under an appropriate circumstance and that the embodiments of the present disclosure described herein are capable of being operated in other sequences different from the description or illustration herein.

Bispecific Fusion Polypeptide

[0114] The present disclosure provides a novel bispecific fusion polypeptide comprising a ligand (or a fragment thereof) and a receptor thereof (or a fragment thereof). The ligand (or the fragment thereof) and the receptor thereof (or the fragment thereof) independently replace CH1 and CL of Fab on one side of an antibody respectively. Specifically, the bispecific fusion polypeptide comprises a first antigen-binding moiety. The first antigen-binding moiety comprises: a first polypeptide comprising a first heavy chain variable domain VH1 of a first antibody from the N terminus to the C terminus, which is operably linked to a first conjugate fragment; and

[0115] a second polypeptide comprising a first light chain variable domain VL1 of the first antibody from the N terminus to the C terminus, which is operably linked to a second conjugate fragment, wherein

[0116] the first conjugate fragment is a receptor and the second conjugate fragment is a ligand; or the first conjugate fragment is a ligand and the second conjugate fragment is a receptor.

[0117] The bispecific fusion polypeptide further comprises a second antigen-binding moiety which is different from the first antigen-binding moiety. A polypeptide fusion method for the second antigen-binding moiety is optionally selected from:

[0118] 1. CH1 and CL of Fab on the other side of the antibody are replaced by another ligand (or a fragment thereof) and a receptor thereof (or a fragment thereof), i.e.,

[0119] the second antigen-binding moiety comprises: a third polypeptide comprising a second heavy chain variable domain VH2 of a second antibody from the N terminus to the C terminus, which is operably linked to a third conjugate fragment; and

[0120] a fourth polypeptide comprising a second light chain variable domain VL2 of the second antibody from the N terminus to the C terminus, which is operably linked to a fourth conjugate fragment, wherein,

[0121] the third conjugate fragment is a receptor and the fourth conjugate fragment is a ligand; or the third conjugate fragment is a ligand and the fourth conjugate fragment is a receptor; and the third conjugate fragment and/or the fourth conjugate fragment, and the first conjugate fragment and/or the second conjugate fragment are selected from different receptors and ligands.

[0122] 2. Original CH1 and CL of Fab on the other side are retained, i.e.,

[0123] the second antigen-binding moiety comprises a third polypeptide comprising a second heavy chain variable domain VH2 of a second antibody from the N terminus to the C terminus, which is operably linked to an antibody heavy chain constant region CH1; and

[0124] a fourth polypeptide comprising a second light chain variable domain VL2 of the second antibody from the N terminus to the C terminus, which is operably linked to an antibody light chain constant region CL.

[0125] The present disclosure uses the unique specific binding force of a ligand and a receptor thereof, and creatively operably links the ligand and the receptor thereof to the antigen-binding region (antibody variable region). The linking comprises linking to one antigen-binding region, and the other antigen-binding region is still linked to CH1 and CL; or both antigen-binding regions are linked to the ligand and the receptor, and only the two antigen-binding regions are linked to different ligands and receptors, thereby avoiding mismatching of different antigen-binding regions.

[0126] In some embodiments, the bispecific fusion polypeptide provided by the present disclosure is a multifunctional fusion polypeptide comprising an antibody Fab. The Fab comprises two different antigen-binding moieties, a first antigen-binding moiety and a second antigen-binding moiety. CH1 and CL of one side of the Fab are independently substituted with a ligand and a receptor thereof, and CH1 and CL of the other side of the Fab are unsubstituted, and the receptor comprises both an active site that recognizes and binds to the ligand and a functionally active site that generates a response reaction; and the light chain of the first antigen-binding moiety is not mismatched to the heavy chain of the second antigen-binding moiety. In some embodiments, CH1 and CL of the other side of the Fab are independently substituted with a second ligand and a receptor thereof, and the first ligand and a receptor thereof are different from the second ligand and a receptor thereof.

[0127] The multifunctional fusion protein may exert a dual-target specificity and also exert a conductive biological activity of a ligand and a receptor. For example, in a specific embodiment, the ligand and the receptor thereof are IL15 and IL15R α . The multifunctional fusion polypeptide has a dual-target targeting effect, and the IL15R α may further present the IL-15 to an IL-2/15R $\beta\gamma$ dimer to form a ternary complex, activate JAK and STAT signaling pathways, promote proliferation and activation of target cells, increase secretion levels of IFN- γ and TNF- α ; JAK/STAT and Ras/MAPK enhance proliferation signals; and up-regulation of Bcl-2 and Bcl-XL (anti-apoptotic proteins), and down-regulation of Bim and Puma (pro-apoptotic proteins) attenuate apoptotic signals.

I. Ligand and Receptor

[0128] The “receptor” is a substance capable of recognizing and binding to a bioactive molecule on a cell membrane or in a cell, and a bioactive substance capable of binding to the receptor is collectively called “ligand”.

[0129] The receptors are classified into two classes, cell surface receptors and intracellular receptors, according to their location in the cells. The receptor itself contains at least two active sites: one is an active site that recognizes and binds to a ligand; and the other is a functionally active site

responsible for generating a response reaction, this site can only generate a response reaction after binding to a ligand to form a binary complex and be allosteric, thereby initiating a series of biochemical reactions that ultimately lead to a biological effect of a target cell.

[0130] The receptors are typically glycoproteins. Binding between the wild-type receptors and the ligands is not mediated by covalent bonds, but rather is primarily through ionic bonds, hydrogen bonds, Van der Waals forces, and hydrophobic interactions. When the receptors bind to the ligands, the receptors have the characteristics of saturation, high affinity, specificity, etc.

[0131] The receptors and ligands, in coordination with each other, have an affinity for relative specific binding, and an optional biological effect. In some embodiments, the receptor only comprises an active site that recognizes and binds to a ligand but does not comprise a functionally active site that generates a response reaction. (e.g., function that activates a biological effect of a downstream signaling pathway). In some embodiments, the receptor and/or ligand is a natural receptor-ligand structure. The receptor comprises both an active site for recognizing a binding ligand and a functionally active site responsible for generating a response and may exert a corresponding biological function. The bispecific fusion protein is a multifunctional fusion protein, has a dual specificity, and also may exert functions of a ligand and a receptor.

[0132] In some embodiments, the receptor and/or ligand is modified from a natural sequence. The modification includes but is not limited to: truncation, insertion and/or mutation. The purposes of these modifications include but are not limited to: increasing or decreasing a binding force of a ligand and a receptor; enhancing, reducing or eliminating biological functions of the ligand and the receptor; increasing, decreasing or eliminating glycosylation sites in receptor and/or ligand proteins; and reducing or eliminating toxicity of a receptor and a ligand.

[0133] In some embodiments, an amino acid sequence of the receptor and/or the ligand each independently consists of 10-1,000 amino acids. In some embodiments, the amino acid sequence of the receptor and/or the ligand each independently consists of 20-800 amino acids. In some embodiments, the amino acid sequence of the receptor and/or the ligand each independently consists of 30-600 amino acids. In some embodiments, the amino acid sequence of the receptor and/or the ligand each independently consists of 40-400 amino acids. In some embodiments, the amino acid sequence of the receptor and/or the ligand each independently consists of 50-300 amino acids. In some embodiments, the amino acid sequence of the receptor and/or the ligand each independently consists of 55-260 amino acids. For example, the amino acid sequence of the receptor and/or the ligand may also be independently selected from 20, 30, 40, 50, 60, 70, 80, 90, 100, 150, 200, 300, 400, 500, 600, 700, 800, and 900 amino acids.

[0134] In some embodiments, the molecular weight of the receptor and/or the ligand is each independently selected from 1 KD-100 KD. In some embodiments, the molecular weight of the receptor and/or the ligand is each independently selected from 2 KD-80 KD. In some embodiments, the molecular weight of the receptor and/or the ligand is each independently selected from 3 KD-70 KD. In some embodiments, the molecular weight of the receptor and/or the ligand is each independently selected from 4 KD-60 KD. In some embodiments, the molecular weight of the receptor and/or the ligand is each independently selected from 4 KD-50 KD. In some embodiments, the molecular weight of the receptor and/or the ligand is each independently selected from 4 KD-40 KD. In some embodiments, the molecular weight of the receptor and/or the ligand is each independently selected from 5 KD-30 KD. For example, the molecular weight of the receptor and/or the ligand may be independently selected from 1 KD, 2 KD, 3 KD, 4 KD, 4.5 KD, 5 KD, 6 KD, 7 KD, 8 KD, 9 KD, 10 KD, 11 KD, 15 KD, 18 KD, 20 KD, 25 KD, 30 KD, 35 KD, 40 KD, 45 KD, 50 KD, 60 KD, 70 KD, 80 KD, 90 KD, and 100 KD.

[0135] A binding mode of the receptor (or a fragment thereof) and a corresponding ligand thereof (or a fragment thereof) may be covalent binding, non-covalent interaction, or a combination thereof. Examples of non-covalent bonds include, but are not limited to, hydrogen bonds, hydrophobic bonds, ionic bonds, and Van der Waals bonds. In some embodiments, the ligand and/or the receptor of an antibody may be modified to increase an affinity when the affinity between inserted or replaced conjugate fragments is lower than expected (e.g., two variable regions in an antigen-binding moiety cannot be brought closer to function to specifically recognize an antigen, or a heavy chain mismatching between 2 or more heavy chain constant regions cannot be prevented, or a mismatching between the antigen-binding moieties cannot be prevented to realize a specific VL-VH portion combination). In some embodiments, at least one non-natural interchain bond is comprised between the receptor and the ligand, and the non-natural interchain bond is capable of enhancing the specific binding force between the receptor and the ligand. In some embodiments, the non-natural interchain bond is formed between a first mutant residue comprised by the receptor and a second mutant residue comprised by the ligand. In some embodiments, at least one of the first mutant residue and the second mutant residue is a cysteine residue. In some embodiments, the non-natural interchain bond is a disulfide bond.

[0136] In some embodiments, at least one natural glycosylation site is not present in the receptor and/or the ligand.

[0137] In some embodiments, the receptor and the ligand thereof are selected from an interleukin and a receptor thereof.

[0138] The inventor studies steric conformations of a large number of interleukins and receptors thereof, and finds that the steric conformations of the interleukins or IFN molecules may be classified into four types: A-lift type, B-bow-knot type, C-ballplayer type, and D-pincer type as shown in Table 3:

TABLE 3

Name of interleukin	type	Ligand	Ligand size	Receptor	Receptor size
Interleukin1 α		IL-1 α	160a.a.	IL-1R	310a.a.
Interleukin1 β	C	IL-1B	155a.a.	IL1R1 + IL1R2	310a.a.

TABLE 3-continued

Name of interleukin	type	Ligand	Ligand size	Receptor	Receptor size
Interleukin 2	A	IL-2	133a.a.	IL-2R α + IL2R β + R γ	220 a.a.
Interleukin 3	C	IL-3	132a.a.	IL-3R	286a.a.
Interleukin 4	A	IL-4	128 a.a.	IL-4R α + R γ	206a.a.
Interleukin 5	D	IL-5	115a.a.	IL-5R	321a.a.
Interleukin 6	A	IL-6	182a.a.	IL-6R	345a.a.
Interleukin 7	D	IL-7	152a.a.	IL-7R	218a.a.
Interleukin 8		IL-8/CXCL8	77a.a.	IL-8RA/IL-8RB	350a.a./360a.a.
Interleukin 9		IL-9	136a.a.	IL-9R	230a.a.
Interleukin 10	B	IL-10	160a.a.	IL-10R1	224a.a.
Interleukin 11	A	IL-11	178a.a.	IL-11R	346a.a.
Interleukin 12		IL-12 α /p35	197a.a.	IL-12 β	306a.a.
Interleukin 12		IL-12 β /p40	306a.a.	IL23A + IL23R + IL12RB1	
Interleukin 13	A	IL-13	122a.a.	IL-13R1	321a.a.
Interleukin 15	A	IL-15	114a.a.	IL-15R α	175a.a.
Interleukin 17		IL-17/CTLA8	132a.a.	IL-17R + IL-17RA	287a.a.
Interleukin 18	C	IL-18	157a.a.	IL18R1	164a.a.
Interleukin 19		IL-19	177 a.a.	IL20R α + IL20R β	
Interleukin 20	A	IL-20	152a.a.	IL20R α + IL20R β	
Interleukin 21	D	IL-21	138a.a.	IL21R	213 a.a.
Interleukin 23	D	IL-23A/p19	170a.a.	IL12B	306a.a.
Interleukin 24	A	IL-24	155a.a.	IL20R α + IL20R β	
Interleukin 27		IL-27A/p28	215a.a.	IL-27B	209 a.a..

[0139] In some embodiments, the ligand and the receptor thereof are selected from type A interleukins and receptors thereof, such as IL15/IL15R, IL2/IL2R, IL4/IL-4R α +R γ , IL-6/IL-6R, IL-11/IL-11R, IL-13/IL-13R1, IL-20/IL20R α +IL20R β , and/or IL24/IL20R α +IL20R β .

[0140] In some embodiments, the ligand and the receptor thereof are selected from type D interleukins and receptors thereof, such as IL7/IL7R, IL21/IL21R, and IL23A/IL12B.

[0141] In some embodiments, the interleukins and receptors thereof have amino acid sequences shown in the following table:

Cytokine or receptor thereof	Sequence
IL2 (human, amino acid sequence in mature form)-133 a.a.	APTSSSTKKTQLQLEHLLLDLQMLNGINNYKNPKLTRMLTFKFPMPKKATELKHLQ CLEEELKPLEEVLNLAQSKNFHLRPDLISNINVIVLELKGSETTFMCEYADETATIVE FLNRWITFCQSIISTLT (SEQ ID NO. 21)
IL2R α (human, amino acid sequence in mature form of IL2R α extracellular domain)-219 a.a.	ELCDDDPPEIPHATFKAMAYKEGTMNLNCECKRGFRRIKSGSLYMLCTGNSSHSSW DNQCQCTSSATRNTTKQVTPQPEEQKERKTTEMQSPMQPVDQASLPGHCREPPP WENEATERIYHFVVGQMVVYQCVCVQGYRALHRGPAESVCKMTHGKTRWTQPQLIC TGEMETSQFPGEKQPASPEGRPESETSLVTTTDFQIQTEMAATMETSIFTTEYQ (SEQ ID NO. 22)
IL2R α (human, amino acid sequence in mature fully truncated form of IL2R α extracellular domain)-166 a.a.	ELCDDDPPEIPHATFKAMAYKEGTMNLNCECKRGFRRIKSGSLYMLCTGNSSHSSW DNQCQCTSSATRNTTKQVTPQPEEQKERKTTEMQSPMQPVDQASLPGHCREPPP WENEATERIYHFVVGQMVVYQCVCVQGYRALHRGPAESVCKMTHGKTRWTQPQLIC TGE (SEQ ID NO. 23)
IL2R α (human, amino acid sequence in mature half truncated form of IL2R α extracellular domain)-191 a.a.	ELCDDDPPEIPHATFKAMAYKEGTMNLNCECKRELCDPPEIPHATFKAMAYKEGT MLNCECKRGFRRIKSGSLYMLCTGNSSHSSWDNQCQCTSSATRNTTKQVTPQPE EQKERKTTEMQSPMQPVDQASLPGHCREPPPWENEATERIYHFVVGQMVVYQCVC QGYRALHRGPAESVCKMTHGKTRWTQPQLICTGEMETSQFPGEKQPASPEGRPE SET (SEQ ID NO. 24)
IL-15 (human, amino acid sequence in mature form)-114 a.a.	NWNVNISDLKKIEDLIQSMHIDATLYTESDVHPSCKVTAMKCFLELQVISLES GDASI HDTVENLIILANNSLSNGNVTESGCKECEELEEKNIKEFLQSFVHVHVMQFINTS (SEQ ID NO. 25)
IL-15R α (human, amino acid sequence in extracellular region)-175 a.a.	ITCPPPMSVEHADIWVKSYSLSRERYICNSGFKRKAGTSSLTECVLNKATNVAHWT TPSLKCI RDPALVHQRPA PPSVTVTAGVTPQPELSPSGKEPAASSPSSNNTAATTA AIVPGSQLMPKSPSTGTTEISSHESHGTPSQTTAKNWE LTASASHQPPGVYPQG HSDTT (SEQ ID NO. 26)
IL-15R α sushi (human, sushi domain amino acid sequence)-77 a.a.	ITCPPPMSVEHADIWVKSYSLSRERYICNSGFKRKAGTSSLTECVLNKATNVAHWT TPSLKCI RDPALVHQRPA PPS (SEQ ID NO. 27)

-continued

Cytokine or receptor thereof	Sequence
IL-15Rasushi (human, sushi domain amino acid sequence)-65 a.a.	ITCPPPMSVEHADIWVKSYSLSRERYICNSGFKRKAGTSSLTECVLNKATNVAHWT TPSLKCIR (SEQ ID NO. 28)
IL-15Rasushi (human, sushi domain amino acid sequence)-73 a.a.	ITCPPPMSVEHADIWVKSYSLSRERYICNSGFKRKAGTSSLTECVLNKATNVAHWT TPSLKCIRDPAIVHQR (SEQ ID NO. 29)
IL-15Rasushi (human, sushi domain amino acid sequence)-86 a.a.	ITCPPPMSVEHADIWVKSYSLSRERYICNSGFKRKAGTSSLTECVLNKATNVAHWT TPSLKCIRDPAIVHQRAPPSTVTTAGVT (SEQ ID NO. 30)
IL-15Rasushi (human, sushi domain amino acid sequence)-102 a.a.	ITCPPPMSVEHADIWVKSYSLSRERYICNSGFKRKAGTSSLTECVLNKATNVAHWT TPSLKCIRDPAIVHQRAPPSTVTTAGVTQPESLSPSGKEPAAS (SEQ ID NO. 31)

[0142] In some embodiments, the receptor-ligand combination is selected from IL15 and IL15R α . In the present disclosure, the “IL15R α ” and “IL15RA” are interchangeable.

[0143] In some embodiments, the IL15 and the IL15R α comprise at least one non-natural interchain bond. In some embodiments, the non-natural interchain bond is a disulfide bond, and the IL15 and/or the IL15R α comprise at least one amino acid mutation to cysteine. In some embodiments, the mutation is located at a contact interface of the IL-15 and the IL15R α . In some embodiments, the cysteine mutation of the IL15 is selected from E90C, and the cysteine mutation of the IL15R α is selected from P67C. A position of the amino acid of the IL15 refers to SEQ ID NO. 26 and a position of the amino acid of the IL15R α refers to SEQ ID NO. 27.

[0144] In some embodiments, at least one natural glycosylation site is not present in the IL15 or the IL15R α . In some embodiments, the glycosylation site is an N-glycosylation site. In some embodiments, the IL15 comprises at least one amino acid mutation: N71Q, N79Q or N112Q. In some embodiments, the glycosylation site is an O-glycosylation site. In some embodiments, the IL15R α comprises at least one amino acid mutation: T2A, T81A, and/or T86A.

[0145] In some embodiments, the receptor-ligand combination is selected from IL2 and IL2R α .

[0146] In some embodiments, the IL2 and the IL2R α comprise at least one non-natural interchain bond. In some embodiments, the non-natural interchain bond is a disulfide bond, and the IL2 and/or the IL2R α comprise at least one amino acid mutation to cysteine. In some embodiments, S at a 75th site of the IL2 is mutated to C, and a N terminus of the IL2R α is extended by two or three amino acids.

[0147] In some embodiments, when the N terminus of the IL2R α is extended by two amino acids, the extended second amino acid is cysteine, and the extended first amino acid is any one of a non-polar fatty acid amino acid, an aromatic amino acid, an amino acid with an uncharged R group, an amino acid with a positively charged R group or an amino acid with a negatively charged R group.

[0148] In some embodiments, when the N terminus of the IL2R α is extended by three amino acids, the extended second amino acid is cysteine, and the extended first and third amino acids are any one of a non-polar fatty acid amino acid, an aromatic amino acid, an amino acid with an uncharged R group, an amino acid with a positively charged R group or an amino acid with a negatively charged R group.

A position of the amino acid of the IL2 refers to SEQ ID NO. 21 and a position of the amino acid of the IL2R α refers to SEQ ID NO. 22 or SEQ ID NO. 23.

[0149] Insertion or replacement positions of the cooperated receptor (or a fragment thereof) and ligand (or a fragment thereof) may be located, for example:

[0150] the receptor or a fragment thereof is inserted into or replaces a CL region, and the ligand or a fragment thereof is inserted into or replaces a CH1 region; or

[0151] the receptor or a fragment thereof is inserted into or replaces a CH1 region, and the ligand or a fragment thereof is inserted into or replaces a CL region.

II. Antigen-Binding Moiety

[0152] The present disclosure provides a bispecific fusion polypeptide comprising a first antigen-binding moiety and a second antigen-binding moiety and having two antigen specificities. The first antigen-binding moiety and the second antigen-binding moiety are different, the first antigen-binding moiety and the second antigen-binding moiety may bind to different antigens, or the first antigen-binding moiety and the second antigen-binding moiety may bind to different epitopes of the same antigen.

[0153] In some embodiments, a target of the bispecific fusion protein is a tumor. In some embodiments, binding targets of the first antigen-binding moiety and the second antigen-binding moiety are both expressed in a tumor cell. In some embodiments, a binding target of the first antigen-binding moiety is in a tumor cell and a binding target of the second antigen-binding moiety is in an immune cell. In some embodiments, binding targets of the first antigen-binding moiety and the second antigen-binding moiety are both in an immune cell.

[0154] T cell redirected killing is a desirable mechanism of action in many therapeutic field. In preclinical and clinical trials, various bispecific antibody forms were involved in T-cell redirection (May C et al., (2012) Biochem Pharmacol, 84 (9): 1105-1112; Frenkel S R, and baeuerle P A, (2013) CURR OPIN Chemical Biology, vol. 17 (3): pages 385-92). All T cell retargeting bispecific antibodies or fragments thereof have been engineered to have at least two antigen-binding sites, one site binds to a surface antigen on a target cell and the other site binds to a T cell surface antigen. Among the T cell surface antigens, an ϵ subunit of human CD3 derived from a TCR protein complex is most often targeted as a target for redirected T cell killing.

[0155] Tumor-associated antigens that may be targeted include but are not limited to: a-fetoprotein (AFP), α -actinin-4, A3, an antigen specific for an A33 antibody, ART-4, B7, Ba 733, BAGE, a BrE3-antigen, CA125, CAMEL, CAP-1, a carbonic anhydrase IX, CASP-8/m, CCCL19, CCCL21, CD1, CD1a, CD2, CD3, CD4, CD5, CD8, CD11A, CD14, CD15, CD16, CD18, CD19, CD20, CD21, CD22, CD23, CD25, CD29, CD30, CD32b, CD33, CD37, CD38, CD40, CD40L, CD44, CD45, CD46, CD52, CD54, CD55, CD59, CD64, CD66a-e, CD67, CD70, CD70L, CD74, CD79a, CD80, CD83, CD95, CD126, CD132, CD133, CD138, CD147, CD154, CDC27, CDK-4/m, CDKN2A, CTLA-4, CXCR4, CXCR7, CXCL12, HIF-1 α , a colon specific antigen p (CSAp), CEA (CEACAM5), CEACAM6, c-Met, DAM, EGFR, EGFRvIII, EGP-1 (TROP-2), EGP-2, ELF2-M, Ep-CAM, a fibroblast growth factor (FGF), Flt-1, Flt-3, a folate receptor, a G250 antigen, Claudin18.2, GAGE, gp100, GRO- β , HLA-DR, HM1.24, human chorionic BCMA gonadotropin (HCG) and subunits thereof, HER2/neu, HMGB-1, a hypoxia-inducible factor (HIF-1), HSP70-2M, HST-2, Ia, IGF-1R, IFN- γ , IFN- α , IFN- β , IFN- λ , IL-4R, IL-6R, IL-13R, IL-15R, IL-17R, IL-18R, IL-2, IL-6, IL-8, IL-12, IL-15, IL-17, IL-18, IL-23, IL-25, insulin-like growth factor-1 (IGF-1), a KC4-antigen, a KS-1 antigen, KSI-4, Le-Y, LDR/FUT, a macrophage migration inhibitory factor (MIF), MAGE, MAGE-3, MART-1, MART-2, NY-ESO-1, TRAG-3, mCRP, MCP-1, MIP-1A, MIP-1B, MIF, MUC1 MUC2, MUC3, MUC4, MUC5ac, MUC13, MUC16, MUM-1/2, MUM-3, NCA66, NCA95, NCA90, a PAM4 antigen, a pancreatic cancer mucin, a PD-1 receptor, a placental growth factor, p53, PLAGL2, a prostatic acid phosphatase, PSA, PRAME, PSMA, PIGF, ILGF, ILGF-1R, IL-6, IL-25, RS5, RANTES, T101, SAGE, S100, survivin, survivin-2B, TAC, TAG-72, tenascin, a TRAIL receptor, TNF- α , a Tn antigen, a Thomson-Friedenreich antigen, tumor necrosis antigen, VEGFR, ED-B fibronectin, WT-1, a 17-1A-antigen, complement factors C3, C3a, C3b, C5a, and C5, angiogenic markers, bcl-2, bcl-6, Kras, oncogene markers, and oncogene products (see, for example, Sensi et al., Clin Cancer Res 2006, 12:5023-32; Pamiani et al., J Immunol 2007, 178:1975-79; Novellino et al., Cancer Immunol Immunother 2005, 54:187-207).

[0156] Although antibodies or other binding molecules specific for effector T cells preferably bind to a CD3 antigen, other antigens expressed on the effector T cells are known and may be targeted by T-cell redirected complexes. Exemplary T-cell antigens include, but are not limited to, CD2, CD3, CD4, CD5, CD6, CD8, CD25, CD28, CD30, CD40, CD40L, CD44, CD45, CD69, and CD90.

[0157] Immune checkpoints are inhibitory pathways in an immune system and are critical to maintaining self-tolerance and regulating duration and magnitude of a physiological immune response in a peripheral tissue to minimize collateral tissue damage. In some embodiments, the targets of the first antigen-binding moiety and the second antigen-binding moiety are immune checkpoints or ligands thereof. The immune checkpoint molecules include but are not limited to: TIGIT, PD-1, TIM-3, LAG3, GTLA4, BTLA, BTN1A1, VISTA, LAIR, CD96, PVRIg, LILRA3, LILRA4, LILRB1, LILRB2, LILRB3, LLRB4, NKG-2A, CD47, CD200R1, CD300, Dectin-1, ICOS, Nkp30, CD28, CD28H, CRTAM, DNAM-1, 4-1-BB, BAFF, CD27, CD30, CD40, DR3,

GITR, HVEM, LIGHT, OX40, TACI, 2B4, CD2, CD48, CD229, SLAM, SLAMF5, GRAAC, TIM1, TIM4, CD7, DPPIV.

[0158] In some embodiments, the target of the first antigen-binding part is PD-1, and the target of the second antigen-binding part is PD-L1. In some embodiments, the target of the first antigen-binding moiety is PD-1, and the target of the second antigen-binding moiety is TIGIT. In some embodiments, the target of the first antigen-binding moiety is PD-1, and the target of the second antigen-binding moiety is GTLA4. In some embodiments, the target of the first antigen-binding moiety is PD-1, and the target of the second antigen-binding moiety is LAG3. In some embodiments, the target of the first antigen-binding moiety is PD-1, and the target of the second antigen-binding moiety is TIM-3. In some embodiments, the target of the first antigen-binding moiety is PD-1, and the target of the second antigen-binding moiety is CD47. In some embodiments, the target of the first antigen-binding moiety is PD-1, and the target of the second antigen-binding moiety is GTLA4. In some embodiments, the target of the first antigen-binding moiety is PD-1, and the target of the second antigen-binding moiety is 4-1-BB. In some embodiments, the target of the first antigen-binding moiety is PD-L1, and the target of the second antigen-binding moiety is 4-1-BB. In some embodiments, the target of the first antigen-binding moiety is PD-L1, and the target of the second antigen-binding moiety is TIGIT.

[0159] In some embodiments, the first antigen-binding moiety targets a tumor-associated antigen, and the second antigen-binding moiety targets an immune checkpoint. In some embodiments, the first antigen-binding moiety targets HER2, and the second antigen-binding moiety targets PD-1. In some embodiments, the first antigen-binding moiety targets VEGF, and the second antigen-binding moiety targets PD-L1. In some embodiments, the first antigen-binding moiety targets Claudin18.2, and the second antigen-binding moiety targets PD-L1. In some embodiments, the first antigen-binding moiety targets HER2, and the second antigen-binding moiety targets CTLA-4. In some embodiments, the first antigen-binding moiety targets CD20, and the second antigen-binding moiety targets CD47. In some embodiments, the first antigen-binding moiety targets HER2, and the second antigen-binding moiety targets CD47.

[0160] In some embodiments, the first antigen-binding moiety and the second antigen-binding moiety target tumor heterogeneity simultaneously. Exemplary common targets for tumors include but are not limited to HGF and VEGF, IGF-1R and VEGF, Her2 and VEGF, CD19 and CD3, CD20 and CD3, Her2 and CD3, CD19 and Fc γ RIIIa, CD20 and Fc γ RIIIa, and Her2 and Fc γ RIIIa. The bispecific fusion polypeptide of the present disclosure may bind to VEGF and phosphatidylserine; VEGF and ErbB3; VEGF and PLGF; VEGF and ROBO4; VEGF and BSG2; VEGF and CDCP1; VEGF and ANPEP; VEGF and c-MET; HER-2 and ERB3; HER-2 and BSG2; HER-2 and CDCP1; HER-2 and ANPEP; EGFR and CD64; EGFR and BSG2; EGFR and CDCP1; EGFR and ANPEP; IGF1R and PDGFR; IGF1R and VEGF; IGF1R and CD20; CD20 and CD74; CD20 and CD30; CD20 and DR4; CD20 and VEGFR2; CD20 and CD52; CD20 and CD4; HGF and c-MET; HGF and NRP1; HGF and phosphatidylserine; ErbB3 and IGF1R; ErbB3 and IGF1,2; c-Met and Her-2; c-Met and NRP1; c-Met and IGF1R; IGF1,2 and PDGFR; IGF1,2 and CD20; IGF1,2 and

IGF1R; IGF2 and EGFR; IGF2 and HER2; IGF2 and CD20; IGF2 and VEGF; IGF2 and IGF1R; IGF1 and IGF2; PDGFRa and VEGFR2; PDGFRa and PLGF; PDGFRa and VEGF; PDGFRa and c-Met; PDGFRa and EGFR; PDGFRb and VEGFR2; PDGFRb and c-Met; PDGFRb and EGFR; RON and c-Met; RON and MTSP1; RON and MSP; RON and CDCP1; VGFR1 and PLGF; VGFR1 and RON; VGFR1 and EGFR; VEGFR2 and PLGF; VEGFR2 and NRP1; VEGFR2 and RON; VEGFR2 and DLL4; VEGFR2 and EGFR; VEGFR2 and ROBO4; VEGFR2 and CD55; LPA and S1P; EPHB2 and RON; CTLA4 and VEGF; CD3 and EPCAM; CD40 and IL6; CD40 and IGF; CD40 and CD56; CD40 and CD70; CD40 and VEGFR1; CD40 and DR5; CD40 and DR4; CD40 and APRIL; CD40 and BCMA; CD40 and RANKL; CD28 and MAPG; CD80 and CD40; CD80 and CD30; CD80 and CD33; CD80 and CD74; CD80 and CD2; CD80 and CD3; CD80 and CD19; CD80 and CD4; CD80 and CD52; CD80 and VEGF; CD80 and DR5; CD80 and VEGFR2; CD22 and CD20; CD22 and CD80; CD22 and CD40; CD22 and CD23; CD22 and CD33; CD22 and CD74; CD22 and CD19; CD22 and DR5; CD22 and DR4; CD22 and VEGF; CD22 and CD52; CD30 and CD20; CD30 and CD22; CD30 and CD23; CD30 and CD40; CD30 and VEGF; CD30 and CD74; CD30 and CD19; CD30 and DR5; CD30 and DR4; CD30 and VEGFR2; CD30 and CD52; CD30 and CD4; CD138 and RANKL; CD33 and FTL3; CD33 and VEGF; CD33 and VEGFR2; CD33 and CD44; CD33 and DR4; CD33 and DR5; DR4 and CD137; DR4 and IGF1,2; DR4 and IGF1R; DR4 and DR5; DR5 and CD40; DR5 and CD137; DR5 and CD20; DR5 and EGFR; DR5 and IGF1,2; DR5 and IGFR; DR5 and HER-2, and EGFR and DLL4. Other target combinations include one or more members of the EGF/erb-2/erb-3 family.

[0161] In addition, exemplary common targets for autoimmune and inflammatory diseases include but are not limited to IL-1 and TNF α , IL-6 and TNF α , IL-6 and IL-1, IgE and IL-13, IL-1 and IL-13, IL-4 and IL-13, IL-5 and IL-13, IL-9 and IL-13, CD19 and Fc γ RIIb, and CD79 and Fc γ RIIb.

[0162] Exemplary targets for treating inflammatory diseases include but are not limited to: TNF and IL-17A; TNF and RANKL; TNF and VEGF; TNF and SOST; TNF and DKK; TNF and α V β 3; TNF and NGF; TNF and IL-23p19; TNF and IL-6; TNF and SOST; TNF and IL-6R; TNF and CD-20; IgE and IL-13; IL-13 and IL23p19; IgE and IL-4; IgE and IL-9; IgE and IL-13; IL-13 and IL-9; IL-13 and IL-4; IL-13 and IL-23p19; IL-6R and VEGF; IL-6R and IL-17A; IL-6R and RANKL; IL-17A and IL-1 β ; IL-1 β and RANKL; IL-1 β and VEGF; RANKL and CD-20; IL-1 α and IL-1 β .

[0163] Targets involved in rheumatoid arthritis (RA) include but are not limited to: TNF and IL-18; TNF and IL-12; TNF and IL-23; TNF and IL-1B; TNF and MIF; TNF and IL-17; and TNF and IL-15.

[0164] Targets for treating systemic lupus erythematosus (SLE) include but are not limited to: CD20, CD22, CD19, CD28, CD4, CD24, CD37, CD38, CD40, CD69, CD72, CD74, CD79A, CD79B, CD80, CD81, CD83, CD86, IL-4, IL-6, IL10, IL2, IL4, IL11, TNFRSF5, TNFRSF6, TNFRSF8, C5, TNFRSF7, TNFRSF5, TNFRSF6, TNFRSF7, BLR1, HDAC4, HDAC5, HDAC7A, HDAC9, ICOSL, IGBP1, MS4A1, RGS1, SLA2, IFNB1, AICDA, BLNK, GALNAC4S-6S T, INHA, INHBA, KLF6, DPP4, FCER2, R2, ILIR2, ITGA2, ITGA3, MS4A1, ST6GALI, CDIC,

CHSTIO, HLA-A, HLA-DRA, NT5E, CTLA4, B7.1, B7.2, BlyS, BAFF, IFN- α , and TNF- α .

[0165] Targets for treating multiple sclerosis (MS) include but are not limited to: IL-12, TWEAK, IL-23, CXCL13, CD40, CD40L, IL-18, VEGF, VLA-4, TNF, CD45RB, CD200, IFN γ , GM-CSF, FGF, C5, CD52, and CCR2.

[0166] Targets for treating sepsis include but are not limited to: TNF, IL-1, MIF, IL-6, IL-8, IL-18, IL-12, IL-10, IL-23, FasL, LPS, a Toll-like receptor, TLR-4, a tissue factor, MIP-2, ADORA2A, IL-1B, CASP1, CASP4, NF κ B1, PROC, TNFRSF1A, CSF3, CCR3, ILIRN, MIF, NF κ B1, PTAFR, TLR2, TLR4, GPR44, HMOX1, midkine, IRAK1, NF κ B2, SERPINA1, SERPINE1, and TREM1.

[0167] To form the bispecific fusion protein of the present disclosure, antibodies may be prepared against any combination of these antigens; that is, each of these antigens may be optionally and independently included or not included by a polyspecific antibody of the present disclosure.

[0168] In some embodiments, the first antigen-binding moiety and the second antigen-binding moiety target different epitopes of the same antigen.

[0169] In some embodiments, at least one of the two antigen-binding fragments may further comprise a secretory signal sequence.

[0170] The secretory signal sequence refers to a sequence that induces secretion of an expressed protein or peptide by being linked to an N terminus of a coding sequence located outside a cell membrane or outside a cell, and may be a peptide sequence consisting of about 18-30 amino acids. All proteins that may be transported to the outside of the cell membrane have different signal sequences, which are cleaved by signal peptidases on the cell membrane. In general, for foreign proteins that are not naturally expressed by the host cell, the secretory signal sequences, or modified sequences that secrete the proteins into cell periplasm or a culture medium may be used.

III. Heterodimeric Fc Fusion Protein

[0171] In some embodiments, a heavy chain constant region Fc is comprised and the constant region Fc is a heterodimer (heterodimeric Fc fusion protein).

[0172] The Fc includes but is not limited to the following combinations:

[0173] CH2;

[0174] CH2+CH3; and

[0175] CH2+CH3+CH4.

[0176] A mutation is introduced into the Fc constant region to avoid mismatching of heavy chains.

[0177] In some embodiments, the mutation introduced into the Fc constant region is based on a KiH technique (Knob-into-Holes), wherein an amino acid mutation is introduced into one heavy chain of the Fc constant region, the volume of the introduced amino acid is greater than the volume of an original amino acid residue, forming a protuberant "Knob"-like structure (Knob), and another mutation is introduced into another chain region of the Fc constant region, wherein the volume of the introduced amino acid is less than the volume of an original amino acid residue, forming a concave, a "Hole"-like structure (Hole), such that the convex heavy chain is more likely to pair with the concave heavy chain, and mismatching of the heavy chains is avoided. This technology was developed by Genentech and described in patent application WO1996027011, which is incorporated herein in its entirety.

[0178] In some embodiments, the mutation introduction of the Fc constant region is based on electrostatic interactions. For example, an ART-Ig technology was developed by Chugai, Roche, specifically alters charges of a Fc constant region domain, facilitates pairing of heterologous heavy chains, and is equivalent to a charged-version KiH technique and described in patent application WO2006106905, which is incorporated herein in its entirety.

[0179] In some embodiments, the mutation introduction of the Fc constant region is based on a SEED technology. A SEED heterodimerization is another design strategy based on a spatial mutation and the strategy utilizes a complementarity of alternating sequences derived from IgG and IgA CH3 domains (also known as AG SEED CH3 and GA SEED CH3). IgG and IgA CH3 derivatives produce complementary sequences. Therefore, when two complementary heavy chain heterodimers are assembled, assembly of homodimers lacking complementarity is excluded. This technology is described in patent application WO2007110205, which is incorporated herein in its entirety.

[0180] In some embodiments, the mutation introduction of the Fc constant region is based on an isoelectric point change, which facilitates improving a heterodimer formation rate and maintaining stability of the Fc region. This technology is described in WO2014145806, which is incorporated herein in its entirety.

[0181] In some embodiments, the Fc constant region is a heterodimer associated based on hydrophilic interactions or increased flexibility.

[0182] In some embodiments, the Fc constant region is a heterodimer associated based on any combination of the above technologies, e.g., in some embodiments, the Fc constant region is mutated based on a combination of the KiH and the electrostatic interactions. For example, an XmAb bispecific platform method may improve thermostability of the bispecific antibody by binding electrostatic interactions, CH3 domain conformations, and hydrogen bonds. In particular, this strategy mutates and exchanges an Fc side chain of natural IgG1 into S364K and K370S heterodimers to form hydrogen bonds between the two heterodimers, followed by L368D/K370S replacement to drive an interaction of a salt bridge to promote heterodimer formation, patent application WO2014145907, which is incorporated herein in its entirety.

[0183] In some embodiments, all or a portion of the CH2, CH3, or CH4 region is inserted into or replaced with a receptor and a ligand thereof.

[0184] In some embodiments, the inserted or replaced regions are independently located in the CH2, CH3, or CH4 region, or any location between adjacent regions (e.g., CH1-CH2 junction, CH2-CH3 junction, and CH3-CH4 junction).

[0185] In some embodiments, when any two constant regions (such as any one of CL-CH1, CH2-CH2, CH3-CH3, or CH4-CH4 regions) mentioned above are inserted or replaced, an affinity between two conjugate fragments that cooperate with each other in the replacement regions is $K_D < 1 \times 10^{-3}$ (M), such as $x \times 10^{-4}$ (M), $x \times 10^{-5}$ (M), $x \times 10^{-6}$ (M), $x \times 10^{-7}$ (M), $x \times 10^{-8}$ (M), $x \times 10^{-9}$ (M), $x \times 10^{-10}$ (M), and $x \times 10^{-11}$ (M); and the value of x may be selected from 1-9, such as 2, 3, 4, 5, 6, 7, and 8.

[0186] In some embodiments, an N terminus and/or a C terminus of the conjugate fragment is linked to the antigen-binding fragment by a linker peptide.

[0187] In some embodiments, the number of amino acids in the linker peptide ranges from 1-30; and may be 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29 or 30, preferably 5-20.

[0188] In some embodiments, the amino acids of the linker peptide are nonsense polypeptides that do not have additional functions (e.g., protein localization, cleavage sites, etc.) other than linking.

[0189] In some embodiments, the linker peptide is a flexible linker peptide.

[0190] In some embodiments, an amino acid sequence of the linker peptide is selected from one or more of Gly, Ser, Pro, Ala, and Glu.

[0191] In some embodiments, the amino acid sequence of the linker peptide is selected from (GGGGS)_n, (GGGS)_n, (GGS)_n, (GS)_n, or (G)_n, wherein n is selected from 1, 2, 3, 4, 5, or 6.

[0192] The linker peptide is generally flexible and may reduce steric hindrance between the fusion protein and a target protein, thereby facilitating proper folding of the protein.

[0193] In further embodiments, the linker peptide is a rigid linker peptide; i.e. a relatively non-flexible peptide linker. The rigid linker peptide does not require a complete lack of flexibility, but is less flexible than the flexible linker peptide such as a glycine-rich peptide linker. Due to its relative lack of flexibility, the rigid linker peptide reduces a movement of two protein domains (in the present case, a stabilizer protein and a thermostable reverse transcriptase) linked together by the rigid linker peptide. A linker peptide providing an ordered chain (e.g., an a helical structure) may provide a rigid linker peptide. For example, arginine, leucine, glutamic acid, glutamine, and methionine all exhibit a relatively high inclination for a helical linker structure. However, a non-helical linker containing many proline residues may also exhibit a significant rigidity. Examples of the rigid linker peptide include polylysine and poly-DL-alanine polylysine. A further description of a rigid peptide linker is provided by Wriggers et al., *Biopolymers*, 80, pages 736-46 (2005). In addition, the rigid linker peptide is described in a linker database described by George et al., *Protein Engineering*, 15, pages 871-79 (2003). Preferably, the rigid linker peptide is also a non-cleavable linker peptide, i.e. a non-cleavable rigid linker peptide.

Isolated Nucleic Acid

[0194] The present disclosure further relates to an isolated nucleic acid encoding a bispecific fusion polypeptide or a multifunctional fusion protein as described above.

[0195] The term "isolated nucleic acid" herein refers to a deoxyribonucleic acid or ribonucleic acid polymer in a single-stranded or double-stranded form. The isolated nucleic acid includes a RNA genomic sequence, DNA (gDNA and cDNA) or a RNA sequence transcribed from DNA. Unless otherwise specified, the polypeptide further includes natural polynucleotide, sugar, or base changed analogs. According to one aspect of the present disclosure, the polynucleotide is a light-chain polynucleotide.

[0196] The isolated nucleic acid includes a nucleotide sequence encoding an amino acid sequence of a protein complex and also includes a nucleotide sequence complementary thereto. The complementary sequence includes a fully complementary sequence and a substantially complementary sequence, which refers to a sequence that hybrid-

izes under a stringent condition known in the art to the nucleotide sequence encoding the amino acid sequence of the protein complex.

[0197] Furthermore, the nucleotide sequence encoding the amino acid sequence of the protein complex may be altered or mutated. Such alterations include additions, deletions, or non-conservative or conservative substitutions. A polynucleotide encoding the amino acid sequence of the protein complex may be interpreted to include a nucleotide sequence that is substantially identical to the isolated nucleic acid. The substantial identity aligns the nucleotide sequence with an additional random sequence to maximize their correspondence. When the aligned sequences are analyzed with an algorithm common in the field, the sequence may show a homology greater than 80%, greater than 90%, or greater than 95%.

Vector

[0198] The present disclosure further relates a vector containing the nucleic acid.

[0199] The term “vector” refers to a nucleic acid delivery vehicle into which a polynucleotide may be inserted. When a vector is capable of expressing a protein encoded by an inserted polynucleotide, the vector is referred to as an expression vector. The vector may be introduced into a host cell by transformation, transduction, or transfection such that a genetic material element it carries is expressed in the host cell. The vectors are well known to a person skilled in the art, and include but are not limited to: plasmid; phagemid; cosmid; artificial chromosomes such as yeast artificial chromosome (YAC), bacterial artificial chromosome (BAC), or P1-derived artificial chromosome (PAC); and phage such as A phage or M13 phage, animal viruses, etc. The animal viruses that may be used as a vector include, but are not limited to, retroviruses (including lentiviruses), adenoviruses, adeno-associated viruses, herpes viruses (e.g., herpes simplex viruses), pox viruses, baculoviruses, papilloma viruses, and papovaviruses (e.g., SV40). The vector may contain a selectable marker (e.g., a tag to facilitate enrichment, such as his tag; or a tag to facilitate detection, such as GFP), and an origin of replication compatible with a cell type prescribed by the cloning vector. The expression vector contains regulatory elements necessary to affect expressions in the prescribed target cells, such as enhancers, promoters, internal ribosome entry sites (IRES), and other expression control elements (e.g., transcription termination signals, or polyadenylation signals and poly-U sequences, etc.). The vector may be a cloning vector or an expression vector. When the antibody or the fragment is expressed or prepared, a prokaryotic expression vector and an eukaryotic expression vector are usually involved, PET series and pGEX series are usually used as the prokaryotic expression vector, and pcDNA3.1, pcDNA3.4, pcDNA4, pEGFP-N1, pSV2, etc. are usually used as the eukaryotic expression vector.

[0200] In the present disclosure, the vector may be a composition, for example a mixture of various plasmids, and different plasmids carrying a portion of an antibody or a fragment thereof.

Host Cell

[0201] The present disclosure further relates to a host cell containing the nucleic acid or the vector.

[0202] Various cultured host cells that may be used include, for example, prokaryotic cells, eukaryotic cells, bacterial cells (such as *Escherichia coli* or *Bacillus stearothermophilus*), fungal cells (such as *Saccharomyces cerevisiae* or *Pichia pastoris*), insect cells (such as Lepidoptera insect cells including *Spodoptera frugiperda* cells), or mammalian cells (such as Chinese Hamster Ovary (CHO) cells, NS0 cells, baby hamster kidney (BHK) cells, monkey kidney cells, Hela cells, human hepatocellular carcinoma cells, or 293 cells, etc.).

Method for Preparing Bispecific Fusion Polypeptide or Multifunctional Fusion Protein

[0203] The bispecific fusion polypeptide or the multifunctional fusion protein of the present disclosure may be prepared according to any well-known methods in the art,

[0204] for example, transforming a host cell with the vector;

[0205] culturing the transformed host cell; and

[0206] collecting the bispecific fusion polypeptide or the multifunctional fusion protein expressed in the host cell.

[0207] In particular, the following methods may be used.

[0208] Early methods for constructing a bispecific antibody include a chemical crosslinking method or a hybrid hybridoma or quadroma method (such as Staerz U D et al., Nature, 314:628-31, 1985; Milstein C et al., Nature, 305: 537-540, 1983; Karpovsky B et al., J.Exp.Md., 160:1686-1701, 1984). The chemical coupling method is to link 2 different monoclonal antibodies together by chemical coupling to prepare a bispecific monoclonal antibody, for example, chemical binding of two different monoclonal antibodies, or for example, chemical binding of two antibody fragments, such as two Fab fragments. The hybrid hybridoma method is to produce a bispecific monoclonal antibody by a cell hybridization method or a triple hybridoma method. These cell hybridomas or ternary hybridomas are fused by established hybridomas or obtained by fusing the established hybridomas with lymphocytes obtained from mice. Although these technologies are used to make BiAb, various produced problems make such complexes difficult to use, such as generation of a mixed population containing different combinations of antigen-binding sites, difficulties in protein expression, need to purify the target BiAb, low yield, high production cost, etc.

[0209] Recent methods utilize genetically engineered constructs that are capable of producing a homogeneous product of a single BiAb without the need for thorough purification to remove unwanted side products. Such constructs include tandem scFv, bispecific antibodies, tandem bispecific antibodies, double variable domain antibodies, and heterodimers using motifs such as Ch1/Ck domain or DNLTm (Chames&Baty, Curr. Opin. Drug. Discov. Devel., 12: 276-83, 2009; Chames&Baty, mAbs, 1:539-47). Related purification technologies are well-known.

[0210] Antibodies may also be produced using a single lymphocyte antibody method by cloning and expressing immunoglobulin variable region cDNA produced by single lymphocytes selected for producing specific antibodies, for example, the methods by Babcook J et al., Proc.Natl.Acad. Sci.USA.93: 7843-7848, 1996; WO 92/02551; WO 2004/051268, and WO 2004/106377.

[0211] Antigenic polypeptides for producing antibodies, e.g., for immunizing a host or for panning a phage display

(yeast cell or bacterial cell surface expression) may be prepared from genetically engineered host cells comprising expression systems by methods well-known in the art, or they may be recovered from natural biological sources. For example, a nucleic acid encoding one or both polypeptide chains of a bispecific antibody may be introduced into a cultured host cell by a variety of known methods (e.g., transformation, transfection, electroporation, bombardment with nucleic acid-coated microparticles, etc.). In some embodiments, the nucleic acid encoding the bispecific antibody may be inserted into a vector suitable for expression in a host cell prior to introduction into the host cell. Typically, the vector may contain a sequence element that enables the inserted nucleic acid to be expressed at RNA and protein levels.

[0212] The bispecific antibody of the present disclosure or a portion thereof may be used to detect any or all of these antigens (e.g., in a biological sample such as serum or plasma) by a conventional immunological assay method, such as enzyme-linked immunosorbent assay (ELISA), radioimmunoassay (RIA), or tissue immunohistochemistry. The present disclosure provides a method of detecting an antigen in a biological sample, comprising: contacting the biological sample with the bispecific antibody, or an antibody portion of the present disclosure, which specifically recognizes the antigen, and detecting the antibody (or antibody portion), or a non-bound antibody (or antibody portion), bound to the antigen, thereby detecting the antigen in the biological sample. The antibody is directly or indirectly labeled with a detectable substance to facilitate detection of a bound or unbound antibody. Suitable detectable substances include various enzymes, repair groups, fluorescent substances, luminescent substances, and radioactive substances. Examples of suitable enzymes include horseradish peroxidase, alkaline phosphatase, B-galactosidase, and acetylcholinesterase; examples of suitable repair group complexes include streptavidin/biotin and avidin/biotin; examples of suitable fluorescent substances include 7-hydroxycoumarin, fluorescein, fluorescein isothiocyanate, rhodamine B, dichlorotriazinylamine fluorescein, dansyl chloride or phycoerythrin; examples of luminescent substances include 3-aminophthalic acid cyclic hydrazine; and examples of suitable radioactive substances include I^{125} , I^{131} , ^{35}S , or 3H .

Pharmaceutical Composition

[0213] The bispecific fusion polypeptide or the multifunctional fusion protein or the nucleic acid encoding same of the present disclosure may be used in preparing a pharmaceutical composition or a sterile composition, e.g., the bispecific fusion polypeptide or the multifunctional fusion protein is mixed with a pharmaceutically acceptable carrier, an excipient, or a stabilizer. The pharmaceutical composition may comprise one or a combination (e.g. two or more different) of the antibody or a functional fragment thereof of the present disclosure. For example, the pharmaceutical composition of the present disclosure may comprise a combination of an antibody or an antibody fragment (or immunoconjugate) with a complementary activity that binds to different epitopes on a target antigen. Preparations of therapeutic and diagnostic agents may be prepared by mixing, for example, freeze-dried powder, slurry, an aqueous solution or a suspension, with a pharmaceutically acceptable carrier, an excipient or a stabilizer.

[0214] The bispecific fusion polypeptide or the multifunctional fusion protein in the pharmaceutical composition may be in a form that is bound to a second activator (functional molecule). The second activator may be a randomly functional molecule capable of preventing or treating a target disease and may include a compound, peptide, polypeptide, nucleic acid, carbohydrate, lipid, or inorganic particle. In the pharmaceutical composition, the bispecific fusion polypeptide or the multifunctional fusion protein may itself have a therapeutic activity; it may function to target the second activator to a specific disease region. The disease region may be those organs, tissues, or cells in which the bispecific antibody that specifically binds to an antigen is aggregated and distributed. A drug targeted to the disease region is present in a high concentration, such that a drug effect is increased compared to the injected amount. Therefore, the pharmaceutical composition may be used to treat drug-resistant tumors and may reduce side effects and adverse drug reactions due to non-specific drug distribution.

[0215] The activator comprising the bispecific fusion polypeptide or the multifunctional fusion protein in the pharmaceutical composition may be contained in microcapsules, or in a drug delivery system of a colloidal nature (such as liposomes, albumin spherules, microemulsions, nanoparticles, and nanocapsules), or in macroemulsions. The microcapsules may be prepared, for example, by a coacervation technology or an interfacial polymerization, examples include hydroxymethyl cellulose or gelatin microcapsules and poly-(methylmethacrylate) microcapsules respectively.

Medical Use and Therapeutic Method

[0216] The present disclosure further relates to use of the bispecific fusion polypeptide or the multifunctional fusion protein in preparing a drug for treating a disease.

[0217] According to one aspect of the present disclosure, the disease may be, for example, cancer, immune disorders, metabolic diseases, and microbial infections.

[0218] The term “cancer” refers to a class of diseases characterized by an uncontrolled growth of abnormal cells in vivo. The “cancer” includes benign and malignant cancers as well as dormant tumors or micrometastases.

[0219] In some embodiments, microorganisms in the microbial infections may be exogenous pathogens or a population of cells bearing the exogenous pathogens, such as viruses. The present disclosure is applicable to the exogenous pathogens such as bacteria, fungi, viruses, mycoplasmas, and parasites. The pathogens that may be treated with the present disclosure may be any infectious organisms known in the art to be pathogenic in animals, including the following organisms such as: gram-negative or gram-positive cocci or bacilli, DNA viruses, and RNA viruses, including, but not limited to, DNA viruses such as papillomaviruses, parvoviruses, adenoviruses, herpesviruses, and vaccinia viruses, and RNA viruses such as arenaviruses, coronaviruses, rhinoviruses, respiratory syncytial viruses, influenza viruses, picornaviruses, paramyxoviruses, reoviruses, retroviruses, and rhabdoviruses. Antibiotic-resistant bacteria of particular interest are such as antibiotic-resistant *Streptococcus* species and *Staphylococcus* species, or bacteria that are sensitive to antibiotics but cause recurrent infections treated with antibiotics so as to finally generate resistant organisms. Such organisms may be treated with the

ligand-immunogen conjugate of the present disclosure in combination with antibiotics at doses lower than those normally administered to patients to avoid generation of these antibiotic-resistant bacterial strains. The present disclosure is also applicable to any fungi, *mycoplasma* species, parasites or other infectious organisms that are pathogenic in animals. Examples of fungi that may be treated by the method of the present disclosure include fungi that grow as molds or yeasts, including, for example, fungi that cause diseases such as: tinea, histoplasmosis, blastomycosis, aspergillosis, cryptococcosis, sporotrichosis, coccidioidomycosis, paracoccidioidomycosis, and candidiasis. The present disclosure may be used to treat parasitic infections, including but not limited to infections caused by the following parasites: tapeworm, schistosome, roundworm, amoeba and *Plasmodium*, *Trypanosoma*, *Leishmania*, and *Toxoplasma* species. Parasites of particular interest are those that express a folate receptor and bind to folate; however, there is a large number of references for ligands exhibiting high affinity for infectious organisms. For example, penicillin and cephalosporins known for their antibiotic activity and specific binding to bacterial cell wall precursors may likewise be used as ligands for preparing the ligand-immunogen conjugates for use according to the present disclosure. The ligand-immunogen conjugate of the present disclosure may also be directed to a cell population bearing an endogenous pathogen, wherein a pathogen-specific antigen is preferentially expressed on a surface of cells bearing the pathogen and serves as a receptor for a ligand that specifically binds to the antigen.

[0220] The present disclosure further relates to a method for preventing and/or treating the disease and administering the therapeutically effective amount of the pharmaceutical composition to prevent and/or treat the disease as described above.

[0221] The method of the present disclosure may be used in human clinical medicine and veterinary medicine. Therefore, host animals bearing a pathogenic organism population and treated with the ligand-immunogen conjugate may be human, or in the case of veterinary medicine application, may be laboratory animals, agricultural animals, domestic animals, or wild animals. The present disclosure may be applicable to the host animals including, but not limited to: human; laboratory animals such as rodents (e.g., mice, rats, hamsters, etc.), rabbits, monkeys, and chimpanzees; domestic animals such as dogs, cats, and rabbits; agricultural animals, such as cattle, horses, pigs, sheep, and goats; and

wild animals raised in captivity, such as bears, pandas, lions, tigers, leopards, elephants, zebras, giraffes, gorillas, dolphins, and whales.

[0222] The pharmaceutical composition may be injected into entities, including rats, mice, domestic animals, and/or human, by a variety of routes. All injection methods can be expected, for example, oral, rectal, intravenous, nasal, abdominal, subcutaneous, or topical injection. The composition may be injected by other methods known in the art.

[0223] The “therapeutically effective amount” means herein an amount sufficient to treat a disease, in view of a reasonable profit/loss ratio. The therapeutically effective amount may vary from patients for a variety of reasons, e.g., type of disease, severity, onset, age, weight, excretion rate, susceptibility to reaction, health status, and/or complications; and/or drug activity, route of injection, injection cycle, and number of injections, and/or pharmaceutical combinations; and may be appropriately selected by a person skilled in the art depending on a treatment purpose. For example, the injection amount may be randomly divided into a plurality of times such that the amount is about 0.001-100 mg/kg of adult body weight.

[0224] The bispecific fusion polypeptide or the multifunctional fusion protein of the present disclosure or the nucleic acid or the polynucleotide encoding the antibody of the present disclosure may also be administered in combination with, for example, standard cancer treatments (e.g., surgery, radiation, and chemotherapy). For example, anti-tumor therapies using these compositions of the present disclosure and/or effector cells equipped with these compositions are used in combination with chemotherapy. Non-limiting examples of antibody combined therapies of the present disclosure include surgery, chemotherapy, radiotherapy, immunotherapy, gene therapy, DNA therapy, RNA therapy, nanotherapy, viral therapy, adjuvant therapy, and a combination thereof.

[0225] The embodiment of the present disclosure will be described in detail below with reference to examples.

Example 1 FiBody Design

[0226] In some examples, FiBody was a bispecific antibody obtained recombinantly with a specific affinity between a ligand and a receptor thereof and replacing CL and CH1 on one side of the bispecific antibody, which may avoid or reduce mismatching between a light chain and a heavy chain of the bispecific antibody.

[0227] In the present example, FiBody was constructed by taking interleukins and receptors thereof as an example, and classified into four types according to steric conformations of the interleukins and receptors thereof shown in FIG. 5:

steric conformation	Classification	
	No.	Interleukin and receptor thereof
Lift type	A	IL15/IL15R, IL2/IL2R, IL4/IL-4Rα + Rγ, IL-6/IL-6R, IL-11/IL-11R, IL-13/IL-13R1, IL-20/IL20Rα + HL20Rβ, and IL24/IL20Rα + HL20Rβ
Bowknot type	B	IL10/IL10R1 and IL22/IL22R
Ballplayer type	C	IL1β/IL1R1, IL3/IL3R, IL5/IL5R, and IL18/IL18R1
Pincer type	D	IL7/IL7R, IL21/IL21R, and IL23A/IL12B

[0228] The bispecific antibodies were respectively constructed based on the 4 types of interleukins and receptors thereof.

Example 2 Construction of FiBody Based on Interleukins and Receptors Thereof

[0229] VH targeting a first antibody was selected, linked to a receptor protein through a Linker, and linked to Fc of an antibody through a Hinge; VL targeting to the first antibody was linked to a ligand protein through a Linker to reduce or avoid mismatching of a light chain and a heavy chain; and the other end was an intact Fab structure targeting a second antibody (anti-TIGIT antibody), and the Fc making up the first antibody and the Fc making up the second antibody had a conventional KiH modification to reduce or avoid mismatching of heavy chains. or

[0230] VH targeting a first antibody was selected, linked to a ligand protein through a Linker, and linked to Fc of an antibody through a Hinge; VL targeting to a second antibody was linked to a receptor protein (IL15) through a Linker to reduce or avoid mismatching of a light chain and a heavy chain; and the other end was an intact Fab structure targeting

a first antibody, and the Fc making up the first antibody and the Fc making up the second antibody had a conventional KiH modification to reduce or avoid mismatching of heavy chains.

Molecule No.	Molecule name	Replacement mode of antibody structure		Classification of steric conformation
		Replacement of CH1 of heavy chain	Replacement of CL of light chain	
Molecule 1	R0950	IL15RA	IL15	A
Molecule 2	R0951	IL15RA	IL15	A
Molecule 3	R0952	IL15	IL15RA	A
Molecule 4	R1115	IL2RA	IL2	A
Molecule 5	R1116	IL2	IL2RA	A
Molecule 6	R1117	IL22RA_D1D2	IL22	B
Molecule 7	R1118	IL22RA_D1	IL22	B
Molecule 8	R1119	IL18R1_D1D2	IL18	C
Molecule 9	R1120	IL18R1_D1D2D3	IL18	C
Molecule 10	R1123	IL21R	IL21	D

[0231] Some specific implementations were shown in the following table:

Molecule 1	R0950: PD-L1_VH_IL15RA/PD-L1_VL_IL15/TIGIT-Fab (FIG. 7)			
1st polypeptide	@PD-L1_VH_IL15RA_Fc-Knob (SEQ ID NO. 1) EVQLLESGGGLVQPQGLRLSCAASGFTSSYIMMWVRQAPGKLEWVSSIYPSSGGITFYADTVKGRFTISRDNSKNTLYLQMNSLRAEDTAVYYCARIKLGTVTTVDYWGQGLTVTVSSGGGGSGGGSGGGSGGGSGITCPPPMSEHADIWVKSYSLSYRERYICNSGFKRKAGTSSLTECVLNKATNVAHWTPSLKLCIRDPAHVHQRPAAPPSTVTAGVTPQPESLSPSGKEPAASSPSSNNTAATTAIIVPGSQLMPKSPSTGTTEISSHESHGTPSQTTAKNWELTASASHQPPGVYPQGHSDTTEPRGPTIKPCPPCKCPAPNLLGGPSVFIIPPPIKDVLMLISLSPIVTCWVVDVSEDDPDVQISWVFNNEVEVHTAQTHREDYNSTLRVVSALPIQHQQDWMGKEFKCKVNNKDLPAPIERTISKPKGSVRAPQVYVLPPEEEMTKKQVTLWCMTDFMPEDIYVEWTNNGKTELNYDNTEPVLDSGSGYFMYSDLRVEKKNWVERNSYSCSVVHEGLHNHHTTKSFSRTPGK			
second polypeptide	@PD-L1_VL_IL15 (SEQ ID NO. 2) QSALTQPAASVSGSPGQSITISCTGTSSDVGGYNYVSWYQQHPGKAPKLMYDVSNNRPSGVSNRFGSGSKSGNTASLTISGLQAEDEADYYCYSYSSSTRVFGTGTQVTVLGGGGSGGGSGGGSGGGSGGNWVNVISDLKKIEDLIQSMHIDATLYTESDVHPSCKVTAMKCFLELQVISLESGLASIHDTVENLIILANNSLSSNGNVTESGCKECELEEKNIKEFLQSFVHVQMFINTS			
third polypeptide	@TIGIT_VH_CH1_Fc-Hole (SEQ ID NO. 3) QVQLESEGGFLFKPTDTLTCTVSGSSLSSSYMSWVRQAPGKLEWIGIIGSNGNTYYANWAKGRFTISKSTSTVELKITSPPTEDTATYFCARGGYRTSGMDPWGPGTLTVTVSSAKTTAPSVYPLAPVCGDITGSSVTLGCLVKGYFPEPVLTWNISGSLSSGVHTFPAVLQSDLYTLSSSVTVTSTWPSQITCNVAHPASSTKVDKKIEPRGPTIKPCPPCKCPAPNLLGGPSVFIIPPPIKDVLMLISLSPIVTCVVDVSEDDPDVQISWVFNNEVEVHTAQTHREDYNSTLRVVSALPIQHQQDWMGKEFKCKVNNKDLPAPIERTISKPKGSVRAPQVYVLPPEEEMTKKQVTLWCMTDFMPEDIYVEWTNNGKTELNYDNTEPVLDSGSGYFMYSDLRVEKKNWVERNSYSCSVVHEGLHNHHTTKSFSRTPGK			
fourth polypeptide	@TIGIT_VL_CL (SEQ ID NO. 4) DIVMTQTPASVEVAVGGTVTIKQASQSISSYLNWYQQKPGQPPKLLIYDALKLASGVPSRFGSGSGTEYTLTISGVESSADAATYYCQEHSSVGNVDNVFGGGTEVVVKRTDAAPTVISIFPPSSEQLTSGGASVVCPLNNFYPKIDINVWKIDGSEKQNGVLNWDQDSKSDSTYSMSSTLTLTKEDEYERHNSYTCBATHKTSTSPIVKSFNRNEC			
Molecule 2	R0951: TIGIT_VH_IL15RA/TIGIT_VL_IL15/PD-L1-Fab			
first polypeptide	@TIGIT_VH_IL15RA_Fc-Knob (SEQ ID NO. 5) QVQLESEGGFLFKPTDTLTCTVSGSSLSSSYMSWVRQAPGKLEWIGIIGSNGNTYYANWAKGRFTISKSTSTVELKITSPPTEDTATYFCARGGYRTSGMDPWGPGTLTVTVSSGGGGSGGGSGGGSGGGSGGNWVNVISDLKKIEDLIQSMHIVKSYSLSYRERYICNSGFKRKAGTSSLTECVLNKATNVAHWTPSLKLCIRDPAHVHQRPAAPPSTVTAGVTPQPESLSPSGKEPAASSPSSNNTAATTAIIVPGSQLMPKSPSTGTTEISSHESHGTPSQTTAKNWELTASASHQPPGVYPQGHSDTTEPRGPTIKPCPPCKCPAPNLLGGPSVFIIPPPIKDVLMLISLSPIVTCVVDVSEDDPDVQISWVFNNEVEVHTAQTHREDYNSTLRVVSALPIQHQQDWMGKEFKCKVNNKDLPAPIERTISKPKGSVRAPQVYVLPPEEEMTKKQVTLWCMTDFMPEDIYVEWTNNGKTELNYDNTEPVLDSGSGYFMYSDLRVEKKNWVERNSYSCSVVHEGLHNHHTTKSFSRTPGK			
second polypeptide	@TIGIT_VL_IL15 (SEQ ID NO. 6) DIVMTQTPASVEVAVGGTVTIKQASQSISSYLNWYQQKPGQPPKLLIYDALKLASGVPSRFGSGSGTEYTLTISGVESSADAATYYCQEHSSVGNVDNVFGGGTEVVVKGGGGSGGGSGGGSGGGSGGNWVNVISDLKKIEDLIQSMHIDATLYTESDVHPSCKVTAMKCFLELQVISLESGLASIHDTVENLIILANNSLSSNGNVTESGCKECELEEKNIKEFLQSFVHVQMFINTS			

-continued

third polypeptide	@PD-L1_VH_CH1_Fc-Hole (SEQ ID NO. 7) EVQLLESGGGLVQPGGSLRLSCAASGFTFSYIMMWVRQAPGKGLEWVSSIYPSSGGITFYADTVKGRFTISRDNSTLTLVQMNLSRAEDTAVYYCARIKLGTVTTVDYWGQGLTVTVSSAKTTAPSVYPLAPVCGDITGSSVTLGCLVKGYFPEPVTLTWNSSGLSSGVHTFPAVLQSDLYTLSSSVTVTSSTWPSQSTICNVHPASSTKVDKKIEPRGPTIKPCPPCKCPAPNLLGGPSVFIFPPKIKDVLMISSLPIVTCVVVDVSEDDPDVQISWVFNNEVHTAQQTTHREDYNSTLRVVSALPIQHODWMSGKEFKCKVNNKDLPAPIERTISKPKGSVRAPQVYVLPPEEEMTKKQVTLTCRVTDMPEDIYVEWTNNGKTELNYKNTPEVLKSDGSYFMAKSLRVEKKNWVERNSYSCSVVHEGLHNHHTTKSFSRTPGK
fourth polypeptide	@PD-L1_VL_CL (SEQ ID NO. 8) QSALTQPASVSGSPGQSITISCTGTSSDVGGINVSWYQQHPGKAPKLMYDVSNRPSGVSNRFGSGSKGNTASLTI SGLQAEDEADYYCSSYSSSTRVFGTGKTVTLRTDAAPTVISIFPPSSEQLTSGGASVVCFLNNFYPKDINVKKIDGSRQNGVLNSWTDQSDKSDSTYSMSSTLTLTCKDEYERHNSYTCEATHKTSSTPIVKSFNRENC
Molecule 3	R0952: TIGIT_VH_IL15/TIGIT_VL_IL15RA/PD-L1-Fab (FIG. 7)
first polypeptide	@TIGIT_VH_IL15_Fc-Knob (SEQ ID NO. 9) QVQLESEGGGLFKPTDTLTLCTVSGSSLSSSYMSWVRQAPGKGLEWIGIIGSNGNTYYANWAKGRFTISKSTTTVELKITSPPTEDTATYFCARGGYRTSGMDPWGPGLTVTVSSGGGSGGGGSGGGGSGGGGSGNNVNVISDLKKIEDLIQSMHIDATLYTESDVHPSCVKVTAMKCFLELQVISLESGLASIHDTVENLILANNLSLSSNGNVTESGCKECELEEKNIKEFLQSFVHIVQMFINTSEPRGPTIKPCPPCKCPAPNLLGGPSVFIFPPKIKDVLMISSLPIVTCVVVDVSEDDPDVQISWVFNNEVHTAQQTTHREDYNSTLRVVSALPIQHODWMSGKEFKCKVNNKDLPAPIERTISKPKGSVRAPQVYVLPPEEEMTKKQVTLWCMVTDMPEDIYVEWTNNGKTELNYDNTPEVLSDGSYFMYSDLRVEKKNWVERNSYSCSVVHEGLHNHHTTKSFSRTPGK
second polypeptide	@TIGIT_VL_IL15RA (SEQ ID NO. 10) DIVMTQTPASVEVAVGGTVTIKCQASQSISSYLNWYQQKPGQPPKLLIYDALKLASGVPSRFGSGSGTEYTLTISGVESSADAATYYCQEHVSVGNVDNVFSGGTEVVVKGSGGGGSGGGGSGGGGSGGITCAPPMSVEHADIWVKSYSLYSRERYICNSGFKRKAGTSSSLTECVLNKATNVAHWTTPSLKCIKRDPAVHQRPAPPSTVTTAGVTPQPESLSPSGKEPAASSPSSNNTAATTAIVPGSQLMPKSPSTGTTEISSHESHGTPSQTTAKNWELETASASHQPPGVYPQGHSDTT
third polypeptide	@PD-L1_VH_CH1_Fc-Hole (SEQ ID NO. 7)
fourth polypeptide	@PD-L1_VL_CL (SEQ ID NO. 8)
Molecule 4	R1115: TIGIT_VH_IL2RA/TIGIT_VL_IL2/PD-L1-Fab
first polypeptide	@TIGIT_VH_IL2RA_Fc-Hole (SEQ ID NO. 32) QVQLVQSGAEVKKPGSSVKVSKASGYSTFSYWMNWVRQAPGQGLEWIGMIRPSDSETRLNQMFKDRVITITVDKSTSTAYMELSSLRSEDATAVYYCAGIHDYGHGAYWGQGLTVTVSSGGGSGGGGSGGGGSGGGSGELCDDDPPEIPHATFKAMAYKEGTMNLNCECKRGFRRIKSGSLYMLCTGNSSHSWSDNQCCQCTSSATRNITTKQVTPQPEEQKERKTEMQSPMQPVDQASLPGHCREPPPWENEATERIYHFVVGQMVVYQCVQGYRALHHRGPAESVCKMTHGKTRWTQ PQLICTGEPKSSDKTHTCCPPAPELGGPSVFLFPPKPKDTLMIISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCTRVKGFYPSDIAVEWESNGQPENNYKTTTPPVLKSDGSFFLASKLTVDKSRWQQGNVFCSCVMHEALHNNHTQKSLSLSPGK
second polypeptide	@TIGIT_VL_IL2 (SEQ ID NO. 33) DIQMTQSPSSLSASVGRVTITCRASENIYSNLAWYQQKPGKSPKLLVYAASHLPDGVPSRFGSGSGTDYSLTISLLQPEDFATYYCQHFWGTPRTFTGQGTKLEIKGGGSGGGGSGGGGSGGGGSGAPTSSSTKKTQLQLEHLLLDLQMILNGINNYKNPKLTRMLTFKFYMPKKATELKHLCLEELKPLEEVLNLAQSKNPHLRPRDLISINIVILELKGSETTFMCEYADETATIFEFLNRWITFCQSIISTLT
third polypeptide	@PD-L1_VH_CH1_Fc-Knob (SEQ ID NO. 34) EVQLQESGPGGLVKPSETLSLTCAVYGDSITSGYWNWIRKPPGKGLEWIMGYISYTGSTYQNPSLKSRIITFSRDTSKNQYYLKLSSVTAADTATYYCARSAWIRTYFDYWGQGLTVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVWTVPSSSLGTQTYICNVNHKPSNTKVDKRVPEPKSCDKTHTCPPECAPPELLGGPSVFLFPPKPKDTLMIISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCTRVKGFYPSDIAVEWESNGQPENNYDTPPVLDSDGSFFLYSDLTVDKSRWQQGNVFCSCVMHEALHNNHTQKSLSLSPGK
fourth polypeptide	@PD-L1_VL_CL (SEQ ID NO. 35) EIVLTQSPDFQSVTPKEKVTITCSVSSSISSNLHWYQQKPDQSPKLLIYGTSNLSAGVPSRFGSGSGTDFTLTINSL EADEAATYYCQWSSYPLTFGQGTKLEIKRTVAAPSVFIFPPSDEQLKSGTASVVCFLNNFYPREAKVQWKVDNALQSGNSQESVTEQDSKSDSTYSLSSLTLSLKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC
Molecule 5	R1116: TIGIT_VH_IL2/TIGIT_VL_IL2RA/PD-L1-Fab
first polypeptide	@TIGIT_VH_IL2_Fc-Hole (SEQ ID NO. 39) QVQLVQSGAEVKKPGSSVKVSKASGYSTFSYWMNWVRQAPGQGLEWIGMIRPSDSETRLNQMFKDRVITITVDKSTSTAYMELSSLRSEDATAVYYCAGIHDYGHGAYWGQGLTVTVSSGGGSGGGGSGGGGSGGGSGAPTSSSTKKTQLQLEHLLLDLQMILNGINNYKNPKLTRMLTFKFYMPKKATELKHLCLEELKPLEEVLNLAQSKNPHLRPRDLISINIVILELKGSETTFMCEYADETATIFEFLNRWITFCQSIISTLTPEPKSSDKTHTCCPPAPELGGPSVFLFPPKPKDTLMIISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCTRVKGFYPSDIAVEWESNGQPENNYKTTTPPVLKSDGSFFLASKLTVDKSRWQQGNVFCSCVMHEALHNNHTQKSLSLSPGK

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second polypeptide	@TIGIT_VL_IL2RA (SEQ ID NO. 40) DIQMTQSPSSLSASVGDRTITCRASENIYSNLAWYQQKPKGKSPKLLVYAASHLPDGVPSRFSGSGSGTDYSLTISSLQPEDFATYYCQHFHWGTPRTFGQGTKLEIKGGGGSGGGSGGGSGGGSGGGSGELCDDDPPEIPHATFKAMAYKEGTMLNCECKRGFRRIKSGSLYMLCTGNSSHSWDNQCCQTSSATRNTTKQVTPQPPEEQKERKTTEMQSPMQPVDQASLPGHCREPPPWENEATERIYHFVVGQMVVYQCVQGYRALHRGPAESVCKMTHGKTRWTQPQLICTG
third polypeptide	@PD-L1_VH_CH1_Fc-Knob (SEQ ID NO. 34)
fourth polypeptide	@PD-L1_VL_CL (SEQ ID NO. 35)
Molecule 6	R1117: TIGIT_VH_IL22RA_D1D2/TIGIT_VL_IL22/PD-L1-Fab
first polypeptide	@TIGIT_VH_IL22RA_D1D2_Fc-Hole (SEQ ID NO. 41) QVQLVQSGAEVKKPGSSVKVSKASGYSFTSYWMNWVRQAPGGGLEWIGMIRPSDSETRLNQMFKDRVITITVDKSTSTAYMELSSLRSEDTAVYYCAGIHDYGHGAYWGQGTLLTVSSGGGGSGGGSGGGSGGGSGGTQSTHESLKPQRVQFQSRNPFHNILQWQPGRALTGNSSVYFVQYKIMFSCSMKSSHQKPSGCWQHISCNFPGCRTLAKYGRQWKNKEDCWGTQELSCDLTSETSDIQEPPYGRVRAASAGSYSEWSMTPRFTPWETKIDEPKSSDKTHTCPCPAPELLGGPSVFLFPPPKPDTLMI SRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCTRVKGFYPSPDIAVEWESNGQPENNYKTTTPPVLKSDGSFFLASKLTVDKSRWQQGNVFCSCVMHEALHNHYTQKSLSLSPGK
second polypeptide	@TIGIT_VL_IL22 (SEQ ID NO. 42) DIQMTQSPSSLSASVGDRTITCRASENIYSNLAWYQQKPKGKSPKLLVYAASHLPDGVPSRFSGSGSGTDYSLTISSLQPEDFATYYCQHFHWGTPRTFGQGTKLEIKGGGGSGGGSGGGSGGGSGGGSGAPISSSHCRDLKSNFQQPYITNRTFMLAKEASLADNNTDVRLLIGEKLPHGVSMSERCYLMKQVLNFTLEEVLPQSDRFPQYMQEVVPFLARLSNRLSTCHIEGDDLHIQRNVQKLKDTVKKLGESGEIKAGELDLLFMSLRNACI
third polypeptide	@PD-L1_VH_CH1_Fc-Knob (SEQ ID NO. 34)
fourth polypeptide	@PD-L1_VL_CL (SEQ ID NO. 35)
Molecule 7	R1118: TIGIT_VH_IL22RA_D1/TIGIT_VL_IL22/PD-L1-Fab
first polypeptide	@TIGIT_VH_IL22RA_D1_Fc-Hole (SEQ ID NO. 43) QVQLVQSGAEVKKPGSSVKVSKASGYSFTSYWMNWVRQAPGGGLEWIGMIRPSDSETRLNQMFKDRVITITVDKSTSTAYMELSSLRSEDTAVYYCAGIHDYGHGAYWGQGTLLTVSSGGGGSGGGSGGGSGGGSGGTQSTHESLKPQRVQFQSRNPFHNILQWQPGRALTGNSSVYFVQYKIMFEPKSSDKTHTCPCPAPELLGGPSVFLFPPPKPDTLMI SRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCTRVKGFYPSPDIAVEWESNGQPENNYKTTTPPVLKSDGSFFLASKLTVDKSRWQQGNVFCSCVMHEALHNHYTQKSLSLSPGK
second polypeptide	@TIGIT_VL_IL22 (SEQ ID NO. 42)
third polypeptide	@PD-L1_VH_CH1_Fc-Knob (SEQ ID NO. 34)
fourth polypeptide	@PD-L1_VL_CL (SEQ ID NO. 35)
Molecule 8	R1119: TIGIT_VH_IL18R1_D1D2/TIGIT_VL_IL18/PD-L1-Fab
first polypeptide	@TIGIT_VH_IL18R1_D1D2_Fc-Hole: (SEQ ID NO. 44) QVQLVQSGAEVKKPGSSVKVSKASGYSFTSYWMNWVRQAPGGGLEWIGMIRPSDSETRLNQMFKDRVITITVDKSTSTAYMELSSLRSEDTAVYYCAGIHDYGHGAYWGQGTLLTVSSGGGGSGGGSGGGSGGGSGGTQSTHESLKPQRVQFQSRNPFHNILQWQPGRALTGNSSVYFVQYKIMFEPKSSDKTHTCPCPAPELLGGPSVFLFPPPKPDTLMI SRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCTRVKGFYPSPDIAVEWESNGQPENNYKTTTPPVLKSDGSFFLASKLTVDKSRWQQGNVFCSCVMHEALHNHYTQKSLSLSPGK
second polypeptide	@TIGIT_VL_IL18 (SEQ ID NO. 45) DIQMTQSPSSLSASVGDRTITCRASENIYSNLAWYQQKPKGKSPKLLVYAASHLPDGVPSRFSGSGSGTDYSLTISSLQPEDFATYYCQHFHWGTPRTFGQGTKLEIKGGGGSGGGSGGGSGGGSGGGSGYFGKLESKLSVIRNLNDQVLFIDQGNRPLFEDMTDSDCRDNAPRTIFIIISMYKDSQPRGMAVTISVKCEKISTLSKENKIIISFKEMNPPDNIKDTSDIIFQRSVPGHDKNMQFESSSYEGYFLACEKERDLFKLILKKEDELGDRSIMFTVQNE
third polypeptide	@PD-L1_VH_CH1_Fc-Knob (SEQ ID NO. 34)

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fourth polypeptide	@PD-L1_VL_CL (SEQ ID NO. 35)
Molecule 9	R1120; TIGIT_VH_IL18R1_D1D2D3/TIGIT_VL_IL18/PD-L1-Fab
first polypeptide	@TIGIT_VH_IL18R1_D1D2D3_Fc-Hole (SEQ ID NO. 46) QVQLVQSGAEVKKPGSSVKVSKASGYSTSYWMNWVRQAPGQGLEWIGMIRPSDSETRLNQMFKDRVITITVDKS TSTAYMELSSLRSEDTAVYYCAGIHGYGHGAYWGQGLTVTVSSGGGGGGGGGGGGGGGGGGGCTSRPHITVV EGEPFYLKHCSCSLAHEIETTTKSWYKSSGSQEHVELNPRSSRIALHDCVLEFWPVELNDTGSYFFQMKNYTQKW KLNVIIRNKHSCPTERQVTSKIVEVKKFFQITCENSYYQTLVNSTSLYKNCKKLLLENNKNPTIKKNAEFEDQGYYSKV HFLHNHGKLFNITKTNITIVEDRSNIVPVLLGPKLNHVAVELGKNVRLNCSALLNEEDVIYWMFGEENGSDPNIEEEK EMRIMTPEGKWHASKVLRIENIGESNLNVLYNCTVASTGGTDTKSFIEPKSSDKTHTCPPCPAPELLGGPSVFLFPPK PKDTLMISRTPEVTCVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKC KVSNAKLPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCRVKGFYPSDIAVEWESNGQPENNYKTTTPVL KSDGSFFLASKLTVDKSRWQQGNVFCSCVMHEALHNHYTQKSLSLSPGK
second polypeptide	@TIGIT_VL_IL18 (SEQ ID NO. 45)
third polypeptide	@PD-L1_VH_CH1_Fc-Knob (SEQ ID NO. 34)
fourth polypeptide	@PD-L1_VL_CL (SEQ ID NO. 35)
Molecule 10	R1123; TIGIT_VH_IL21R/TIGIT_VL_IL21/PD-L1-Fab
first polypeptide	@TIGIT_VH_IL21R_Fc-Hole (SEQ ID NO. 47) QVQLVQSGAEVKKPGSSVKVSKASGYSTSYWMNWVRQAPGQGLEWIGMIRPSDSETRLNQMFKDRVITITVDKS TSTAYMELSSLRSEDTAVYYCAGIHGYGHGAYWGQGLTVTVSSGGGGGGGGGGGGGGGGGGGSGCPDLVCYTDY LQTVICILEMWNLHPSTLTLTWQDQYEELKDEATCSLHRSAHNATHATYTCCHMDVFHFMADDIFSVQITDQSGQYS QECGSLAASIKPAPPFDVTVTFSGQYQISWRSYEDPAFYMLKGLQYELQYRNRGDPWAVSPRRKLISVDSRSV SLLPLEFRKDSSEYELQVRAGPMPGSSYQGTWSEWSDPVIQTQSEELKEPKSSDKTHTCPPCPAPELLGGPSVFL FPPKPKDTLMISRTPEVTCVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKE YKCKVSNALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCRVKGFYPSDIAVEWESNGQPENNYKTTTP VVLKSDGSFFLASKLTVDKSRWQQGNVFCSCVMHEALHNHYTQKSLSLSPGK
second polypeptide	@TIGIT_VL_IL21 (SEQ ID NO. 48) DIQMTQSPSSLSASVGRVTITCRASENIYSNLAWYQQKPKGKSPKLLVYAASHLPDGVPSRFGSGSGTDYSLTISSL QPEDFATYYCQHFWGTPRTFGQTKLEIKGGGGSGGGGGGGGGGGGGGHKSSSQGDRHIMRMRLIDIVD QLKNYVNDLVPEFLPAPEDVETNCEWSAFSCFQKAQLKSANTGNNERIINVSIKKLKRKPPSTNAGRQKHRLTCPS CDSYEKKPKPEFLERFKSLQKMIHQHLSRTHGSEDS
third polypeptide	@PD-L1_VH_CH1_Fc-Knob (SEQ ID NO. 34)
fourth polypeptide	@PD-L1_VL_CL (SEQ ID NO. 35)

Example 3 Bispecific Antibodies with Disulfide-Bond Modification

[0232] 3.1 To further improve stability and extend a half-life period of bispecific antibodies, the bispecific antibodies were subjected to a disulfide-bond modification (see FIG. 8).

[0233] Disulfide-bond modification of ligand and receptor: VH targeting a second antibody (anti-TIGIT) was selected, linked to a receptor protein (IL15RA) through a Linker, and linked to Fc of an antibody through a Hinge; VL

targeting to the second antibody was linked to a ligand protein (IL15) through a Linker; and the other end was an intact Fab structure targeting a first antibody, and the Fc making up the first antibody and the Fc making up the second antibody had a conventional KiH modification to avoid mismatching of heavy chains. Meanwhile, the receptor and ligand proteins were mutated to form an intermolecular disulfide bond, so as to further improve stability of molecules, specifically, for example:

Molecule No.	R0954: TIGIT_VH_IL15RA (D96 + C97)/TIGIT_VL_IL15 (E87C)/PD-L1-Fab
first polypeptide	@TIGIT_VH_IL15RA (D96 + C97)_Fc-Knob (SEQ ID NO. 13) QVQLESEGGFLKPTDTLTLTCTVSGSSLSSSYMSWVRQAPGKLEWIGIIGSNGNTYYANWAKGR FTISKTSITTVELKI TSPTTDDATYFCARGGYRTSGMDPWGPGLTVTVSSGGGGSGGGGGGGG SGGGGGGITCPPPMSVEHADIWVKSYSLSYRERYICNSGFKRKAGTSSLTECVLNKATNVAHWTT PSLKCIRDPAIVHQRAPPSTVTTAGVTTPQESLSPSGKEPAASSPSSNNTAATTAIVPGSQLMP SKSPSTGTTTEISSHESHTPSQTTAKNWELTASASHQPPGVYPQGHSDTTEPRGPTIKPCPPCK CPAPNLLGGPSVFI FPPKIKDVLMI SLSPIVTCVVDVSEDDPDVQISWFWNNVEVHTAQTQTHRED YNSTLRVVSALPIQHQQDWSGKEFKCKVNNKDLPAPIERTISKPKGSVRAPQVYVLPPEEEMTK KQVTLWCMVTDMPEDIYVEWTNNGKTELNYDNTEPVLDSGGSYFMSYDLRVEKKNWVERNSY SCSVVHEGLHNHHTTKSFSRTPGK

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second polypeptide	@TIGIT_VL_IL15 (E87C) (SEQ ID NO. 14) DIVMTQTPASVEVAVGGTVTIKCQASQSISSYLNWYQQKPGQPPKLLIYDALKLASGVPSRFRSGSG SGTEYTLTISGVESADAATYYCQQEHSVGNVDNVFGGGTEVVVKGGGSGGGGSGGGGSGGG SGSNWNVNVIDLKKIEDLIQSMHIDATLYTESDVHPSCVKVTAMKCFLLLELQVISLES GDASIHDTVEN LIILANNSLSNGNVTESGCKCEELEEKNIKEFLQSFVHIVQMFINTS
third polypeptide	@PD-L1_VH_CH1_Fc-Hole (SEQ ID NO. 7)
fourth polypeptide	@PD-L1_VL_CL (SEQ ID NO. 8)
Molecule No.	R1085: TIGIT_VH_IL15RA (P67C)/TIGIT_VL_IL15 (E90C)/PD-L1-Fab
first polypeptide	@TIGIT_VH_IL15RA (P67C)_Fc-Knob (SEQ ID NO. 49) QVQLVQSGAEVKKPGSSVKVSCKASGYSFTSYWMNWVRQAPGGGLEWIGMIRPSDSETRLNQ MFKDRVITITVDKSTSTAYMELSSLRSEDTAVYYCAGIHDYGHGAYWGQGLTVTVSSGGGGSGGG SGGGGGSGGGSGGITCPPPMSVEHADIWVKSYSLYSRERYI CNSGFKRKAGTSSLTECVLNKAT NVAHWTTPSLKCIRDALVHQRPAAPPSTVTTAGVTPQPESLSPSGKEPAASSPSSNNTAATTAAIV PGSQLMPKSPSTGTTEISSHESHGTPSQTTAKNWELTASASHQPPGVYPQGHSDTTEPKSSD KTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHN AKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPP PSREEMTKNQVSLTCTRVKGFYPSPDIAVEWESNGQPENNYKTTPPVLKSDGSFFLASKLTVDKSR WQQGNVFPSCSVMHEALHNHYTQKSLSLSPGK
second polypeptide	@TIGIT_VL_IL15 (E90C) (SEQ ID NO. 50) DIQMTQSPSSLSASVGRVTITCRASENIYSNLAWYQQKPGKSPKLLVYAASHLPDGVPSRFRSGS SGGTDYSLTISLQPEDFATYYCQHPFGTPTRTFGQGTKLEIKGGGGSGGGSGGGSGGGSGG NWNVNISDLKKIEDLIQSMHIDATLYTESDVHPSCVKVTAMKCFLLLELQVISLES GDASIHNTVQDLIIL ANNSLSNGNVTESGCKECCLEEKNIKEFLQSFVHIVQMFINTS
third polypeptide	@PD-L1_VH_CH1_Fc-Knob (SEQ ID NO. 51) QVQLQESGPGLVKPSSETLSLTCAVYGDSITSGYWNWIRKPPGKGLLEYIGIYSYTGSTYQNPSLKS ITMSRDTSKNQYYLKLSSVTAADTAVYYCARSAWIRTYFDYWGQGLTVTVSSASTKGPSVFPLAP SSKTSGGTAALGCLVKDYFPEPTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGT QTYICNVNHKPSNTKVDKRVPEPKSCDKTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTC VVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV SNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLWCLVKGFYPSPDIAVEWESNGQPEN NYDTTPPVLDSDGSFFLYSDDLTVDKSRWQQGNVFPSCSVMHEALHNHYTQKSLSLSPGK
fourth polypeptide	@PD-L1_VL_CL (SEQ ID NO. 52) DIQMTQSPSSLSASVGRVTITCSVSSSISSNLHWYQQKPGKAPKAPWIIYGTSNLASGVPSRFRSG SGSGTDYSLTISLQPEDFATYYCQWSSYPLTFGQGTKLEIKRTVAAPSVFIFPPSDEQLKSGTAS VVCLLNNFYPREAKVQWKVDNALQSGNSQESVTEQDSKSTYLSLSSTLTLSKADYEEKHKVYACE VTHQGLSSPVTKSFNRGEC
Molecule No.	R1086: TIGIT_VH_IL15RA (R35C)/TIGIT_VL_IL15 (E93C)/PD-L1-Fab
first polypeptide	@TIGIT_VH_IL15RA (R35C)_Fc-Knob (SEQ ID NO. 65) QVQLVQSGAEVKKPGSSVKVSCKASGYSFTSYWMNWVRQAPGGGLEWIGMIRPSDSETRLNQ MFKDRVITITVDKSTSTAYMELSSLRSEDTAVYYCAGIHDYGHGAYWGQGLTVTVSSGGGGSGGG SGGGGGSGGGSGGITCPPPMSVEHADIWVKSYSLYSRERYI CNSGFKRKAGTSSLTECVLNKAT NVAHWTTPSLKCIRDALVHQRPAAPPSTVTTAGVTPQPESLSPSGKEPAASSPSSNNTAATTAAIVP GSQMLPKSPSTGTTEISSHESHGTPSQTTAKNWELTASASHQPPGVYPQGHSDTTEPKSSDK THTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNA KTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPP SREEMTKNQVSLTCTRVKGFYPSPDIAVEWESNGQPENNYKTTPPVLKSDGSFFLASKLTVDKSRW QQGNVFPSCSVMHEALHNHYTQKSLSLSPGK
second polypeptide	@TIGIT_VL_IL15 (E93C) (SEQ ID NO. 12) DIQMTQSPSSLSASVGRVTITCRASENIYSNLAWYQQKPGKSPKLLVYAASHLPDGVPSRFRSGS SGGTDYSLTISLQPEDFATYYCQHPFGTPTRTFGQGTKLEIKGGGGSGGGSGGGSGGGSGG NWNVNISDLKKIEDLIQSMHIDATLYTESDVHPSCVKVTAMKCFLLLELQVISLES GDASIHNTVQDLIIL ANNSLSNGNVTESGCKECCLEEKNIKEFLQSFVHIVQMFINTS
third polypeptide	@PD-L1_VH_CH1_Fc-Knob (SEQ ID NO. 51)
fourth polypeptide	@PD-L1_VL_CL (SEQ ID NO. 52)

[0234] 3.2 The disulfide-bond modification was designed between VH and VL to form dsFv to facilitate a covalent disulfide-bond linkage between non-covalently designed heavy and light chains of a receptor and a ligand. Disulfide-bond modification sites include, but are not limited to, the following mutation sites:

Combination	VH	VL
1	37C	95C
2	44C	100C
3	44C	101C
4	44C	105C

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Combination	VH	VL
5	45C	87C
6	45C	98C
7	100C	50C
8	100bC	49C
9	98C	46C
10	101C	46C
11	105C	43C
12	106C	57C
13	108C	43C

[0235] Specifically, for example:

Molecule No.	R1081: TIGIT_VH (G44C)_IL15RA/TIGIT_VL(Q100C)_IL15/PD-L1-Fab
first polypeptide	@TIGIT_VH (G44C)_IL15RA_Fc-Knob (SEQ ID NO. 53) QVQLVQSGAEVKKPGSSVKVSCKASGYSTSYWMNWVRQAPGQCLEWIGMIRPSDSETRLNQ MFKDRVTITVDKSTSTAYMELSSLRSEDTAVYYCAGIHDYGHGAYWGQGTLLVTVSSGGGGSGG GGSGGGGGGGGGSGITCPPPMSVEHADIWVKSYSLYSRERYICNSGFKRKAGTSSLTECVLNK ATNVAHWTTPSLKCIRDPAIVHQRPAAPPSTVTTAGVTPQPESLSPSGKEPAASSPSSNNTAATTA AIVPGSQLMPKSPSTGTTEISSHESHGTPSQTTAKNWELTASASHQPPGVYPQGHSDTTEPK SSDKTHTCPPCPAPELLGGPSVFLFPPKPKDTLMI SRTPEVTCVVVDVSHEDPEVKFNWYVDGV EVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREP QVYTLPPSREEMTKNQVSLTCTRVKGFYPSPDIAVEWESNGQPENNYKTTTPVLKSDGSFFLASKL TVDKSRWQQGNVFCSCVMHEALHNHYTQKSLSLSPGK
second polypeptide	@TIGIT_VL (Q100C)_IL15 (SEQ ID NO. 54) DIQMTQSPSSLSASVGRVTITCRASENIYSNLAWYQQKPKGKSPKLLVYAASHLPDGVPSRFSG SGSGTDYSLTISSLQPEDFATYYCQHFQWGTPTFTFGCTKLEIKGGGGSGGGSGGGSGGGG SGNWNVISDLKKIEDLIQSMHIDATLYTESDVHPSCKVTAMKCFLELQVISLESGLDASIHNTVQ DLIILANNSLSSNGNVTESGCKECELEEKNIKEFLQSFVHIVQMFINTS
third polypeptide	@PD-L1_VH_CH1_Fc-Knob (SEQ ID NO. 51)
fourth polypeptide	@PD-L1_VL_CL (SEQ ID NO. 52)
Molecule No.	R1082: TIGIT_VH (W108C)_IL15RA/TIGIT_VL(S43C)_IL15/PD-L1-Fab
first polypeptide	@TIGIT_VH (W108C)_IL15RA_Fc-Knob (SEQ ID NO. 55) QVQLVQSGAEVKKPGSSVKVSCKASGYSTSYWMNWVRQAPGQCLEWIGMIRPSDSETRLNQ MFKDRVTITVDKSTSTAYMELSSLRSEDTAVYYCAGIHDYGHGAYWGQGTLLVTVSSGGGGSGG GGSGGGGGGGGGSGITCPPPMSVEHADIWVKSYSLYSRERYICNSGFKRKAGTSSLTECVLNK ATNVAHWTTPSLKCIRDPAIVHQRPAAPPSTVTTAGVTPQPESLSPSGKEPAASSPSSNNTAATTA AIVPGSQLMPKSPSTGTTEISSHESHGTPSQTTAKNWELTASASHQPPGVYPQGHSDTTEPK SSDKTHTCPPCPAPELLGGPSVFLFPPKPKDTLMI SRTPEVTCVVVDVSHEDPEVKFNWYVDGV EVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREP QVYTLPPSREEMTKNQVSLTCTRVKGFYPSPDIAVEWESNGQPENNYKTTTPVLKSDGSFFLASKL TVDKSRWQQGNVFCSCVMHEALHNHYTQKSLSLSPGK
second polypeptide	@TIGIT_VL (S43C)_IL15 (SEQ ID NO. 56) DIQMTQSPSSLSASVGRVTITCRASENIYSNLAWYQQKPKGKCPKLLVYAASHLPDGVPSRFSG SGSGTDYSLTISSLQPEDFATYYCQHFQWGTPTFTFGQGTKLEIKGGGGSGGGSGGGSGGGG SGNWNVISDLKKIEDLIQSMHIDATLYTESDVHPSCKVTAMKCFLELQVISLESGLDASIHNTVQ DLIILANNSLSSNGNVTESGCKECELEEKNIKEFLQSFVHIVQMFINTS
third polypeptide	@PD-L1_VH_CH1_Fc-Knob (SEQ ID NO. 51)
fourth polypeptide	@PD-L1_VL_CL (SEQ ID NO. 52)
Molecule No.	R1083: TIGIT_VH (L45C)_IL15RA/TIGIT_VL(F98C)_IL15/PD-L1-Fab
first polypeptide	@TIGIT_VH (L45C)_IL15RA_Fc-Knob (SEQ ID NO. 57) QVQLVQSGAEVKKPGSSVKVSCKASGYSTSYWMNWVRQAPGQGCLEWIGMIRPSDSETRLN QMFKDRVTITVDKSTSTAYMELSSLRSEDTAVYYCAGIHDYGHGAYWGQGTLLVTVSSGGGGSG GGSGGGGGGGGGSGITCPPPMSVEHADIWVKSYSLYSRERYICNSGFKRKAGTSSLTECVLN KATNVAHWTTPSLKCIRDPAIVHQRPAAPPSTVTTAGVTPQPESLSPSGKEPAASSPSSNNTAATT

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	AAIVPGSQLMPKSPSTGTTEISSHESHSHTPSQTTAKNWELTASASHQPPGVYPQGHSDTTEP KSSDKTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDG VEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPRE PQVYITLPPSREEMTKNQVSLTCRVKGFYPSDIAVEWESNGQPENNYKTTTPVLKSDGSFFLASK LTVDKSRWQQGNVFCSCVMHEALHNHYTQKSLSLSPGK
second polypeptide	@TIGIT_VL (F98C)_IL15 (SEQ ID NO. 58) DIQMTQSPSSLSASVGDRTITCRASENIYSNLAWYQQKPKGKSPKLLVYAASHLPDGVPSRFSG SGSGTDYSLTISSLQPEDFATYYCQHFEGTPRTCGQGTLEIKGGGSGGGSGGGSGGGSGGG SGNWNVNVIDLKKIEDLIQSMHIDATLYTESDVHPSCVKVTAMKCFLELQVISLES GDASIHNTVQ DLIILANNSLSSNGNVTESGCKECEELEEKNIKEFLQSFVHIVQMFINTS
third polypeptide	@PD-L1_VH_CH1_Fc-Knob (SEQ ID NO. 51)
fourth polypeptide	@PD-L1_VL_CL (SEQ ID NO. 51)
Molecule No.	R1084: PD-L1_VH (G44C)_IL15RA/PD-L1_VL (Q101C)_IL15/TIGIT-Fab
first polypeptide	@PD-L1_VH (G44C)_IL15RA_Fc-Knob (SEQ ID NO. 59) QVQLQESGPGLVKPSSETLSLTCAVYGDSITSGYWNWIRKPPGKCLEYIGYISYTGSTYQNPSLKS RITMSRDTSKNQYYLKLSSVTAADTAVYYCARSAWIRTYFDYWGQGLTVTVSSGGGSGGGG SGGGSGGGSGGITCPPPMSVEHADIWVKSYSLSRERYICNSGFKRKAGTSSLTCEVLNKAT NVAHWTPSLKCIKDPALVHQRPAAPPSTVTTAGVTPQPELSLSPGKEPAASSPSSNNTAATTA VPGSQLMPKSPSTGTTEISSHESHSHTPSQTTAKNWELTASASHQPPGVYPQGHSDTTEPKS SDKLTGTPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVE VHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQ VYITLPPSREEMTKNQVSLTCRVKGFYPSDIAVEWESNGQPENNYKTTTPVLKSDGSFFLASKLT VDKSRWQQGNVFCSCVMHEALHNHYTQKSLSLSPGK
second polypeptide	@PD-L1_VL (Q101C)_IL15 (SEQ ID NO. 60) DIQMTQSPSSLSASVGDRTITCSVSSSISSNLHWYQQKPKAPKPYGTSNLASGVPSRFSG GSGSGTDYTLTISSLQPEDFATYYCQWSSYPLTFGCGTKLEIKGGGSGGGSGGGSGGGSGGG SGNWNVNVIDLKKIEDLIQSMHIDATLYTESDVHPSCVKVTAMKCFLELQVISLES GDASIHNTV QDLIILANNSLSSNGNVTESGCKECEELEEKNIKEFLQSFVHIVQMFINTS
third polypeptide	@TIGIT_VH_CH1_Fc-Knob (SEQ ID NO. 61) QVQLVQSGAEVKKPGSSVKVCKASGYSTSYWMNWVRQAPGQGLEWIGMIRPSDSETRLNQ MFKDRVTITVDKSTSTAYMELSSLRSEDTAVYYCAGIHDYGHGAYWGQGLTVTVSSASTKGPSV FPLAPSSKSTSGGTAAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTP SSSLGTQTYICNVNHKPSNTKVDKRVPEKSCDKTHTCPPCPAPELLGGPSVFLFPPKPKDTLMIS RTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNG KEYKCKVSNKALPAPIEKTISKAKGQPREPQVYITLPPSREEMTKNQVSLWCLVKGFYPSDIAVEW ESNGQPENNYDTTPVLDSDGSFFLYSDLTVDKSRWQQGNVFCSCVMHEALHNHYTQKSLSLSPGK
fourth polypeptide	@TIGIT_VL_CL (SEQ ID NO. 62) DIQMTQSPSSLSASVGDRTITCRASENIYSNLAWYQQKPKGKSPKLLVYAASHLPDGVPSRFSG SGSGTDYSLTISSLQPEDFATYYCQHFEGTPRTFGQGTLEIKRTVAAPSVFIFPPSDEQLKSGT ASVCLLNNFYPREAKVQWKVDNALQSGNSQESVTEQDSKSTSYLSSTLTLSKADYEKHKVY ACEVTHQGLSPVTKSFNRGEC

Example 4 Mutation Modification to Eliminate Glycosylation

[0236] Construction method of Fab-IL15/IL15RA_Fc: VH targeting a second antibody (anti-TIGIT) was specifically selected, linked to a receptor protein (IL15RA) through a Linker, and linked to Fc of an antibody through a Hinge; VL targeting to the second antibody was linked to a ligand protein (IL15) through a Linker; and the other end was an

intact Fab structure targeting a first antibody, and the Fc making up the first antibody and the Fc making up the second antibody had a conventional KiH modification to avoid mismatching of heavy chains. Meanwhile, the receptor and ligand proteins were mutated to form an intermolecular disulfide bond, so as to further improve stability of molecules. Further, glycosylation sites on the receptor and ligand proteins were modified to eliminate molecular heterogeneity, specifically, for example:

Molecule No. R1072: TIGIT_VH_IL15RA/TIGIT_VL_IL15/PD-L1-Fab

first polypeptide	@TIGIT_VH_IL15RA_Fc-Knob (SEQ ID NO. 63) QVQLVQSGAEVKKPGSSVKVCKASGYSTSYWMNWVRQAPGQGLEWIGMIRPSDSETRLNQMFKDRVTITVDKSTSTAYMELSSLRSEDTAVYYCAGIHDYGHGAYWGQGLTVTVSSGGGSGGGSGGGSGGGSGGGSGGITCPPPMSVEHADIWVKSYSLSRERYICNSGFKRKAGTSSLTCEVLNKATNVAHWTPSLKCIKDPALVHQRPAAPPSTVTTAGVTPQPELSLSPGKEPAASSPSSNNTAATTAIVPGSQLMPKSPSTGTTEISSHESHSHTPSQTTAKNWELTASASHQPPGVYPQGHSDTTEPKSSD
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	KHTCTPCPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRV VSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCTRVKGFYPSDIAVEWES NGQPENNYKTTPPVLKSDGSFFLASKLTVDKSRWQQGNVFCSCVMHEALHNHYTQKSLSLSPGK
second polypeptide	@TIGIT_VL_IL15 (SEQ ID NO. 64) DIQMTQSPSSLSASVGDRVTITCRASENIYSNLAWYQQKPKGSKPLLVAASHLPDGVPSRFSGSGSGTDYSLTISSQLQPE DFATYYCQHFPGWTPRTFGQGTKLEIKGGGSGGGGSGGGSGGGSGNWNVISDLKKIEDLIQSMHIDATLYTESDVHPS CKVTAMKCFLEELQVISLESGLDASIHNTVQDLIILANNSLSNGNVTESGCKECEELEEKNIKEFLQSFVHIVQMFINTS
third polypeptide	@PD-L1_VH_CH1_Fc-Knob (SEQ ID NO. 51)
fourth polypeptide	@PD-L1_VL_CL (SEQ ID NO. 52)
Molecule No. R0955: TIGIT_VH_IL15RA/TIGIT_VL_IL15 (N71Q, N79Q, N112Q)/PD-L1-Fab	
first polypeptide	@TIGIT_VH_IL15RA_Fc-Knob (SEQ ID NO.11) QVQLSEEGGLFPKPTDRLTLTCTVSGSSLSSSYMSWVRQAPGKGLEWIGIIGSNGNTYYANWAKGRFTISKSTTVELKITS PTTEDTATYFCARGGYRTSGMDPWGPGTLVTVSSGGGSGGGGSGGGGSGGGSGITCPPPMSVEHADIWVKSYSLYSRER YICNSGFKRKAGTCSLTCVLNKAHVAHWTPSLKCIKDPALVHQRPAAPPSTVTAGVTPQPELSPSGKEPAASSPSSN NTAATTAIAIVPGSQLMPKSPSTGTTEISSHESHGTPSQTTAKNWELTASASHQPPGVYPQGHSDTTEPRGPTIKPCPPC KCPAPNLLGGPSVFIFPPKIKDVLMISSLPIVTCVVVDVSEDDPDVQISWVFNNEVHTAQTQTHREDYNSTLRVVSALPI QHQQDWSGKEFKCKVNNKDLPAIERTISKPKGSVRAPQVYVLPPEEEMTKKQVTLWCMVTDPMFEDIYVEWTNNGKTEL NYDNTPEVLVDSGYSFMYSDLRVEKKNWVERNSYSCSVHEGLHNHHTTKSFRTPGK
second polypeptide	@TIGIT_VL_IL15 (N71Q, N79Q, N112Q) (SEQ ID NO. 15) DIVMTQTPASVEVAVGGTVTIKCQASQSISSYLNWVQKPGQPKLLIYDALKLASGVPSRFSGSGSGTEYTLTISGVESA DAATYYCQQEHSVGNVDNVFPGGTEVVVKGSGGGGSGGGGSGGGSGGNWNVISDLKKIEDLIQSMHIDATLYTESDVH PSCKVTAMKCFLEELQVISCESGLDASIHDTVENLIILAQNLSLSNGQVTEGCKECEELEEKNIKEFLQSFVHIVQMFIQT S
third polypeptide	@PD-L1_VH_CH1_Fc-Hole (SEQ ID NO. 7)
fourth polypeptide	@PD-L1_VL_CL (SEQ ID NO. 8)
Molecule No. R1109: TIGIT_VH_IL15RA (sushi65)/TIGIT_VL_IL15/PD-L1-Fab	
first polypeptide	@TIGIT_VH_IL15RA (sushi65)_Fc-Knob (SEQ ID NO. 66) QVQLVQSGAEVKKPGSSVKVSCKASGYSTSYWMNWVRQAPGQGLEWIGMIRPSDSETRLNQMFKDRVITITVDKSTSTA YMESSLRSEDATVYYCAGIHDYGHGAYWGQGTTLVTVSSGGGSGGGGSGGGGSGGGSGITCPPPMSVEHADIWVKSY SLYSRERYICNSGFKRKAGTSSLTECVLNKAHVAHWTPSLKCIKREPSSDKTHTCPPCPAPELLGGPSVFLFPPKPKDT LMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAP IEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCTRVKGFYPSDIAVEWESNGQPENNYKTTPPVLKSDGSFFLASKLTV DKSRWQQGNVFCSCVMHEALHNHYTQKSLSLSPGK
second polypeptide	@TIGIT_VL_IL15 (SEQ ID NO. 64)
third polypeptide	@PD-L1_VH_CH1_Fc-Knob (SEQ ID NO. 51)
fourth polypeptide	@PD-L1_VL_CL (SEQ ID NO. 52)
Molecule No. R1110: TIGIT_VH_IL15RA (sushi65 T2A)/TIGIT_VL_IL15/PD-L1-Fab	
first polypeptide	@TIGIT_VH_IL15RA (sushi65 T2A)_Fc-Knob (SEQ ID NO. 67) QVQLVQSGAEVKKPGSSVKVSCKASGYSTSYWMNWVRQAPGQGLEWIGMIRPSDSETRLNQMFKDRVITITVDKSTSTA YMESSLRSEDATVYYCAGIHDYGHGAYWGQGTTLVTVSSGGGSGGGGSGGGGSGGGSGIACPPPMSVEHADIWVKSY SLYSRERYICNSGFKRKAGTSSLTECVLNKAHVAHWTPSLKCIKREPSSDKTHTCPPCPAPELLGGPSVFLFPPKPKDT LMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAP IEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCTRVKGFYPSDIAVEWESNGQPENNYKTTPPVLKSDGSFFLASKLTV DKSRWQQGNVFCSCVMHEALHNHYTQKSLSLSPGK
second polypeptide	@TIGIT_VL_IL15 (SEQ ID NO. 64)
third polypeptide	@PD-L1_VH_CH1_Fc-Knob (SEQ ID NO. 51)
fourth polypeptide	@PD-L1_VL_CL (SEQ ID NO. 52)

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Molecule No. R1111: TIGIT_VH_IL15RA (sushi65 T2A, T81A, T86A)/TIGIT_VL_IL15/PD-L1-Fab	
first	@TIGIT_VH_IL15RA (sushi65 T2A, T81A, T86A)_Fc-Knob (SEQ ID NO. 68)
polypeptide	QVQLVQSGAEVKKPGSSVKVSKASGYSTSYWMNWVRQAPGQGLEWIGMIRPSDSETRLNQMFKDRVTITVDKSTSTA YMEISSLRSEDTAIVYYCAGIHGYGHGAYWGQGLTVTVSSGGGGSGGGGGSGGGSGGIACPPPMSEVHADIVVKS SLYSRERYICNSGFKRKAGTSSLTECVLNKATNVAHWTTPSLKCIKIRDPALVHQRPAAPPSTVATAGVAPQPELSPSGKEPA ASSPSSNNTAATTAEPKSSDKHTCPPELPGGSPVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGV EVHNAKTKPREQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQV SLTCTRVKGFYPDSIAVEWESNGQPENNYKTTPPVLKSDGSFPLASKLTVDKSRWQQGNVFCSCVMHEALHNHYTQKSLSLS PGK
second	@TIGIT_VL_IL15 (SEQ ID NO. 64)
polypeptide	
third	@PD-L1_VH_CH1_Fc-Knob (SEQ ID NO. 51)
polypeptide	
fourth	@PD-L1_VL_CL (SEQ ID NO. 52)
polypeptide	

Example 5 Modified IL15/IL15RA and Bispecific
Antibody Composed Thereof with Reduced Affinity
of IL15/IL15RA with IL2/15Rβ/γC Complex

[0237] In certain applications, in order to avoid interaction of IL15/IL15RA and a bispecific antibody composed thereof with an IL2/15Rβ/γC complex, causing unwanted non-specific binding, the IL15/IL15RA and the bispecific antibody composed thereof were modified to reduce or completely lose an affinity of the IL15/IL15RA with the IL2/15Rβ/γC complex. By examining a crystal structure of an interaction interface of the IL15/IL15RA and the IL2/15Rβ/γC complex, and performing modeling using a Molecular Operating Environment (MOE; Chemical Computing Group, Montreal, Quebec, Canada) software, it was predicted that an amino acid mutation modification could be

performed at an IL15/IL15RA interface so as to reduce or completely lose the affinity of the IL15/IL15RA with the IL2/15Rβ/γC complex, as depicted in FIG. 9.

[0238] Construction method of Fab-IL15/IL15RA_Fc: VH targeting a second antibody (anti-TIGIT) was specifically selected, linked to a receptor protein (IL15RA) through a Linker, and linked to Fc of an antibody through a Hinge; VL targeting to the second antibody was linked to a ligand protein (IL15) through a Linker; and the other end was an intact Fab structure targeting a first antibody, and the Fc making up the first antibody and the Fc making up the second antibody had a conventional KiH modification to avoid mismatching of heavy chains. At the same time, the ligand protein was mutated to reduce or inactivate biological functions of the receptor-ligand complex, specifically, for example:

Molecule	
No. R0960: TIGIT_VH_IL15RA/TIGIT_VL_IL15 (D61N, E64Q, N65D)/PD-L1-Fab	
first	@TIGIT_VH_IL15RA_Fc-Knob (SEQ ID NO. 5)
polypeptide	QVQLESEGGLFKPTDTLTLTCTVSGSSLSSSYMSWVRQAPGKGLEWIGIIGSNNGNTYYANWAKGRFTISKSTSTVELKITSP TTEDTATYFCARGGYRTSGMDPWGPGTLTVTVSSGGGGSGGGGGSGGGSGGITCPPPMSEVHADIVVKSYSLSRERYIC NSGFKRKAGTSSLTECVLNKATNVAHWTTPSLKCIKIRDPALVHQRPAAPPSTVTTAGVTPQPELSPSGKEPAASSPSSNNTAA TTAAIVPGSQLMPKSPSTGTTEISSHESHGTPSQTTAKNWELTASASHQPPGVYPQGHSDTTEPRGPTIKPCPPCKCPAP NLLGGPSVFIFPPKIKDVLMSISPIVTCVVVDVSEDDPDVQISWVFNVEVHTAQTHREDYNTSLRVVSALPIQHQQDWM SGKEFKCKVNNKDLPAPIERTISKPKGSVRAPQVYVLPPEEEMTKKQVTLWCMVTDMPEDIYVEWTNNGKTELNYDNTEP VLSDSGSYFMYSDLRVEKKNNWVERNSYSCSVVHEGLHNHHTTKSFSRTPGK
second	@TIGIT_VL_IL15 (D61N, E64Q, N65D) (SEQ ID NO. 16)
polypeptide	DIVMTQTPASVEVAVGGTVTIKCQASQSISSYLNWYQQKPGQPPLLIYDALKLASGVPSRFGSGSGTEYTLTISGVESAD AATYYCQQEHSVGNVDNVFGGTEVVVKGGGGSGGGGGSGGGSGGNWVNVISDLKKIEDLIQSMHIDATLYTESDVHP SCKVTAMKCFLELQVISLESQDASIHNTYQDLIILANNSLSSNGNVTESGCKECELEEKNKEPLQSPFVHVQMFINTS
third	@PD-L1_VH_CH1_Fc-Hole (SEQ ID NO. 7)
polypeptide	
fourth	@PD-L1_VL_CL (SEQ ID NO. 8)
polypeptide	

Example 6 Construction of Bispecific Antibodies
Based on scFv and CrossMab Structures as
Experimental Control

[0239] As described above, both scFv and CrossMab were commonly used technological means for constructing bispecific antibodies, which were used as designed controls to compare with our molecules:
[0240] Construction method of bispecific antibody based on scFv structure: VH targeting a second antibody (anti-

TIGIT) was specifically selected, linked to VL of the second antibody through a Linker to form an scFv structure, and linked to Fc of an antibody through a Hinge; and the other end was an intact Fab structure (this structure was similar to Y-Body of Wuhan YZY Med), and the Fc making up the first antibody and the Fc making up the second antibody had a conventional KiH modification to avoid mismatching of heavy chains. Specifically, for example:

Molecule No.	R0809: Y-Body TIGIT_scFv/PD-L1-Fab (FIG. 1-C)
scFv arm	@TIGIT_scFv_Fc-Knob (SEQ ID NO. 17) QVQLESEGGLFKPTDTLTCTVSGSSLSSSYMSWVRQAPGKGLEWIGIIGSNGNTYYANWAKGRFTISKSTTTVELKITSP TTEDTATYFCARGGYRTSGMDPWGPGTLVTVSSGGGGSGGGSGGGSDIVMTQTPASVEVAVGGTVTIKQASQSISSYLN WYQQKPGQPPKLLIYDALKLASGVPSRFSGSGSTEYTLTISGVESADAATYYCQEHSGVGNVDNVFPGGTEVVVGAAAE RGPTIKPCPPCKCPAPNLLGGPSVFIPPPFIKIDVLMISLSPIVTCVVVDVSEDDPDVQISWFWNNVEVHTAQQTQTHREDYNS TLRVVSALPIQHQQDWMGKEFKCKVNNKDLPAPIERTISKPGSVRAPQVYVLPPEEEMTKKQVTLWCMVTDMPEDIYVE WTNNGKTELNYDNTEPVLDSGYSFMYSDLRVEKKNWVERNSYSCSVVHEGLHNHHTTKSFSTRTPGK
Fab arm-VH	@PD-L1_VH_CH1_Fc-Hole (SEQ ID NO. 7)
Fab arm-VL	@PD-L1_VL_CL (SEQ ID NO. 8)

[0241] Construction method of bispecific antibody based on scFv structure: VH targeting a second antibody (anti-TIGIT) was specifically selected, linked to VL of the second antibody through a Linker to form an scFv structure, and linked to a C terminus of an intact Fc targeting a first antibody through a Linker to form a symmetric structure. Specifically, for example:

Molecule No.	R0810: Symmetric scFv PD-L1_VH_CH1_Fc_@TIGIT_scFv/PD-L1_VL_CL (FIG. 1-D)
Heavy chain	@PD-L1_VH_CH1_Fc_@TIGIT_scFv (SFQ ID NO. 20) EVQLLESGGGLVQPGGSLRLSCAASGFTFSYIMMWVRQAPGKGLEWVSSIYPSGGITFYADTVKG RFTISRDNKNTLYLQMNLSRAEDTAVYYCARIKLGTVTTVDYWGQGLTVTVSSAKTTAPSVYPLAPV CGDITGSSVTLGCLVKGYFPEPVTLTWNSGSLSSGVHTFPAVLQSDLYTLSSSVTVTSSTWPSQSI CNVAHPASSTKVDKKIEPRGPTIKPCPPCKCPAPNLLGGPSVFIPPPFIKIDVLMISLSPIVTCVVVDVSE DDPDVQISWFWNNVEVHTAQQTQTHREDYNSTLRVVSALPIQHQQDWMGKEFKCKVNNKDLPAPIER TISKPGSVRAPQVYVLPPEEEMTKKQVTLTCMVTDMPEDIYVEWTNNGKTELNYKNTPEVLDS GSYFMYSKLRVEKKNWVERNSYSCSVVHEGLHNHHTTKSFSTRTPGAGGGGGGGSGGGSGG GGSQVQLESEGGLFKPTDTLTCTVSGSSLSSSYMSWVRQAPGKGLEWIGIIGSNGNTYYANWAK GRFTISKSTTTVELKITSPTTEDTATYFCARGGYRTSGMDPWGPGTLVTVSSGGGGSGGGSGG GSDIVMTQTPASVEVAVGGTVTIKQASQSISSYLNWYQQKPGQPPKLLIYDALKLASGVPSRFSGS SGSTEYTLTISGVESADAATYYCQEHSGVGNVDNVFPGGTEVVVK
Light chain	@PD-L1_VL_CL (SEQ ID NO. 8)

[0242] Construction method of bispecific antibody based on CrossMab structure: VH targeting a second antibody (anti-TIGIT) was specifically selected, linked to a CL domain, and linked to Fc of an antibody through a Hinge; VL targeting to the second antibody (anti-TIGIT) was linked to a CH1 domain to form a light chain; and the other end was an intact Fab structure targeting a first antibody, and the Fc making up the first antibody and the Fc making up the second antibody had a conventional KiH modification to avoid mismatching of heavy chains. Specifically, for example:

Molecule No.	R0959: CrossMab TIGIT_VH_CL/TIGIT_VL_CH1/PD-L1_Fab (FIG. 1-B)
First polypeptide	@TIGIT_VH_CL_Fc-Knob (SEQ ID NO. 18) QVQLESEGGLFKPTDTLTCTVSGSSLSSSYMSWVRQAPGKGLEWIGIIGSNGNTYYANWAK RFTISKSTTTVELKITSPTTEDTATYFCARGGYRTSGMDPWGPGTLVTVSSLRDAAPTVSIFPPS

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Molecule No.	R0959: CrossMab TIGIT_VH_CL/TIGIT_VL_CH1/PD-L1_Fab (FIG. 1-B)
	SEQLTSGGASVVCFLNNFYPKDINVWKIDGSEKQNGVLNSWTDQDSKDYMSSTLTTLTKDE YERHNSYTCEATHKTSTSPIVKSFNNECEPRGPTIKPCPPCKCPAPNLLGGPSVFIFPPKIKDVL MISLSPIVTCVVVDVSEDDPDVQISWVFNNEVHTAQTQTHREDYNSTLRVVSALPIQHQQDWMS GKEFKCKVNNKDLPAPIERTISKPKGSVRAPQVYVLPPEEEMTKKQVTLWCMVTDMPEDIYVE WTNNGKTELNYDNTPEVLDSDGSYFMYSDLRVEKKNWVERNSYSCSVVHEGLHNHHTTKSFS RTPGK
Second polypeptide	@TIGIT_VL_CH1 (SEQ ID NO. 19) DIVMTQTTPASVEVAVGGTVTIKCQASQSISSYLNWYQQKPGQPPKLLIYDALKLASGVPSRFGSGS GSGTEYTLTISGVESADAATYYCQEHSESVGNVDNVFGGGTEVVKAKTTAPSVYPLAPVCGDIT GSSVTLGCLVKGYFPEPVTLTWNSGSLSSGVHTFPAVLQSDLYTLSSSVTVTSSTWPSQSIITCNV AHPASSTKVDKKI
Third polypeptide	@PD-L1_VH_CH1_Fc-Hole (SEQ ID NO. 7)
Fourth polypeptide	@PD-L1_VL_CL (SEQ ID NO. 8)

Example 7 Antibody Heavy-Light Chain Mismatching Assay

[0243] Light-heavy chain mismatching was a difficult problem faced by a bispecific-antibody platform. In order to verify an anti-mismatching performance of the platform, a receptor and a ligand were specially designed to be distributed on Fab of two sides of an antibody, a mismatched light-chain structure was intentionally designed, and an expression verification was performed.

Construction Method of Fab-IL15/IL15RA_Fc Mismatching:

[0244] R1042: VH targeting a first antibody (anti-PD-L1 antibody) was specifically selected, linked to a receptor protein (IL15RA) through a Linker, and linked to Fc of an antibody through a Hinge; VL targeting to a second antibody (anti-TIGIT antibody) was linked to a ligand protein (IL15) through a Linker; the other end was VH targeting the second antibody through a conventional sequence to be linked to CH1, and linked to Fc of an antibody through a Hinge; and VL targeting the first antibody was linked to CL through a conventional sequence, and both Fc had a conventional KiH modification to avoid mismatching of heavy chains. Specifically, for example:

Molecule No.	R1042: PD-L1_VH_IL15RA/TIGIT_VL_IL15/TIGIT_VH/PD-L1-VL (FIG. 10 left)
First polypeptide	@PD-L1_VH_IL15RA_Fc-Knob (SEQ ID NO. 1)
Second polypeptide	@TIGIT_VL_IL15 (SEQ ID NO. 6)
Third polypeptide	@TIGIT_VH_CH1_Fc-Hole (SEQ ID NO. 3)
Fourth polypeptide	@PD-L1_VL_CL (SEQ ID NO. 8)

[0245] R1043: VH targeting a second antibody (anti-TIGIT antibody) was specifically selected, linked to a receptor protein (IL15RA) through a Linker, and linked to Fc of an antibody through a Hinge; VL targeting to a first antibody (anti-PD-L1 antibody) was linked to a ligand protein (IL15) through a Linker; the other end was VH targeting the first antibody through a conventional sequence to be linked to CH1, and linked to Fc of an antibody through a Hinge; and VL targeting the second antibody was linked to CL through a conventional sequence, and both Fc had a conventional KiH modification to avoid mismatching of heavy chains.

Molecule No.	R1043: PD-L1_VH_IL15RA/TIGIT_VL_IL15/TIGIT_VH/PD-L1-VL
First polypeptide	@TIGIT_VH_IL15RA_Fc-Knob (SEQ ID NO. 5)
Second polypeptide	@PD-L1_VL_IL15 (SEQ ID NO. 2)
Third polypeptide	@PD-L1_VH_CH1_Fc-Hole (SEQ ID NO. 7)
Fourth polypeptide	@TIGIT_VL_CL (SEQ ID NO. 4)

Construction Method of Fab-IL21/IL21R_Fc Mismatching:

[0246] VH targeting a second antibody (anti-PD-L1 antibody) was selected, linked to a receptor protein (IL21R) through a Linker, and linked to Fc of an antibody through a Hinge; VL targeting to a first antibody (anti-TIGIT antibody) was linked to a ligand protein (IL21) through a Linker; the other end was VL targeting the second antibody (anti-PD-L1 antibody) linked to CL; and VH targeting the first antibody (anti-TIGIT antibody) was linked to CH1, and linked to Fc of an antibody through a Hinge, and Fc on both ends had a conventional KiH modification. Specifically, for example:

Molecule No.	R1124: PD-L1_VH_IL21R/TIGIT_VL_IL21/TIGIT_VH/PD-L1-VL (FIG. 10 right)
First polypeptide	@PD-L1_VH_IL21R_Fc-Knob (SEQ ID NO: 69) EVQLQESGPGGLVKPSETLSLTCAVYGDSTSGYWNWIRKPPGKGLGYMGIISYTGSTYQNPSLKSRI TFSRDTSKNQYLLKLSVTAADATYYCARSAWIRTYFDYWGQGLVTVSSGGGGGGGGSGGG GSGGGGSGCPDLVCYTDYLTQVTCILEMWNLHPSTLTLTWQDQYEELKDEATSCSLHRSAHNATHAT YTCHMDVFHFMAIDIFSQVITDQSGQYSQECGSLLAESIKPAPPFDVTVTFSGQYQISWRSDYEDP AFYMLKGLQYELQYRNRPDPAVSPRRKLI SVDSRSVSLPLEFRKIDSSYELQVRAGMPGSSYQ GTWSEWSDPVIQTQSEELKEPKSSDKTHTCPPCPAPPELLGGPSVFLFPPPKDITLMISRTPEVTC

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Molecule No.	R1124: PD-L1_VH_IL21R/TIGIT_VL_IL21/TIGIT_VH/PD-L1-VL (FIG. 10 right)
	VVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCRVKGFYPSDIAVEWESNGQPENNYKTTPPVLKSDGSFFLASKLTVDKSRWQQGNVFSQSVMHREALHNYHTQKSLSLSPG
Second polypeptide	@TIGIT_VL_IL21 (SEQ ID NO: 48)
Third polypeptide	@TIGIT_VH_CH1_Fc-Hole (SEQ ID NO. 70) QVQLVQSGAEVKKPGSSVKVSCASGYSTSYWMNWVRQAPGQGLEWIGMIRPSDSETRLNQMFKDRVITITVDKSTSTAYMELSSLRSEDTAVYYCAGIHDYGHGAYWGQGLTVTVSSASTKGPSVFPLAPSSKSTSGGTAAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSNTKVDKRVEPKCDKTHCTCPPELLEGPSVFLFPPKPKDITLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLWCLVKGFYPSDIAVEWESNGQPENNYDTTPVLDSGDSFFLYSLTVDKSRWQQGNVFSQSVMHREALHNYHTQKSLSLSPGK
Fourth polypeptide	@PD-L1_VL_CL (SEQ ID NO. 35)

Example 8 Bispecific Antibodies with Disulfide-Bond Modification

[0247] A disulfide-bond modification was designed between IL2 and IL2Rα (the amino acid position 75 in the amino acid sequence of wild-type interleukin 2 (SEQ ID NO. 21) was mutated to cysteine; the N-terminal of the amino acid sequence of interleukin-2 receptor α (SEQ ID NO. 22 or SEQ ID NO. 23) is extended by two or three amino acids, in which the second amino acid is cysteine) to help form a covalent disulfide bond between non-covalent linked IL2 and IL2Rα.

[0248] A construction method of a bispecific antibody with a disulfide-bond modification (disulfide bond-modified IL2/IL2Rα complex) was exemplarily constructed: a heavy chain variable region (VH) of an antibody (such as TIGIT antibody) targeting a surface antigen of a tumor cell or an immune cell was linked to an IL2Rα mutant through a Linker, and linked to Fc of a hIgG1 antibody through a Hinge region; a light chain variable region (VL) of the antibody (TIGIT antibody) targeting the same antigen was linked to an IL2 mutant through a Linker; VH of an antibody (such as a PD-L1 antibody) targeting a surface antigen of an immune cell or a tumor cell was directly linked to a constant

region (hIgG1) of the hIgG1 antibody; and VL of the antibody (PD-L1 antibody) targeting the same antigen was directly linked to a light-chain kappa antibody (K-Ig LC) of human immunoglobulin, so as to prepare a targeted disulfide bond-modified IL2/IL2Rα complex.

[0249] According to the above construction method, the inventor designed 2 disulfide bond-modified IL2/IL2Rα complexes, namely disulfide bond-modified IL2/IL2Rα complex 1 and disulfide bond-modified IL2/IL2Rα complex 2. IL2/IL2Rα complex 3 was the complex reported in the prior art and complex 4 was a disulfide bond-unmodified IL2/IL2Rα complex. Specific compositions and amino acid sequences of complexes 1, 2, 3, and 4 were as follows:

[0250] The sequence structure of the disulfide bond-modified IL2/IL2Rα complex 1 (sample R1262) was: TIGIT VH-IL2Rα mutant-Fc, TIGIT VL-IL2 mutant, PD-L1 VH-hIgG1, and PD-L1 VL-κ-Ig LC, wherein the amino acid sequence of the TIGIT VH-IL2Rα mutant-Fc was shown in SEQ ID NO. 77, an amino acid sequence of the TIGIT VL-IL2 mutant was shown in SEQ ID NO. 78, the amino acid sequence of the PD-L1 VH-hIgG1 was shown in SEQ ID NO. 79, and the amino acid sequence of the PD-L1 VL-κ-Ig LC was shown in SEQ ID NO. 80.

Molecule No.	R1262: TIGIT VH-IL2Rα mutant-Fc/TIGIT VL-IL2 mutant/PD-L1_VH-hIgG1/PD-L1 VL-κ-Ig LC
First polypeptide	@TIGIT VH-IL2Rα mutant-Fc (SEQ ID NO. 77) QVQLVQSGAEVKKPGSSVKVSCASGYSTSYWMNWVRQAPGQGLEWIGMIRPSDSETRLNQMFKDRVITITVDKSTSTAYMELSSLRSEDTAVYYCAGIHDYGHGAYWGQGLTVTVSSGGGGSGGGSGGGSGGGGCGGELCDDDDPPEIPHATFKAMAYKEGTMLNCECKRGFRRIKSGSLYMLCTGNSSSHSSWDNQCQCTSSATRNNTTKQVTPQPEEKERKTTMQSPMQPVDQASLPGHCREPPPWENEATERIYHFVVGQMVYYQCQVQGYRALHRGPAESVCKMTHGKTRWTPQQLICTGEPKSSDKTHTCPPELLEGPSVFLFPPKPKDITLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCRVKGFYPSDIAVEWESNGQPENNYKTTPPVLKSDGSFFLASKLTVDKSRWQQGNVFSQSVMHREALHNYHTQKSLSLSPGK
Second polypeptide	@TIGIT VL-IL2 mutant (SEQ ID NO. 78) DIQMTQSPSSLSASVGDRVTITCRASENIYSNLAHYQQKPGKSPKLLVYAASHLPDGVPSRFSGSGSGTDYSLTISLQPEDFATYYCQHFQGTPTFTFGQGTLEIKGGGGSGGGSGGGSGGGSGGAPTSSSTKKTQLQLEHLLLDLQMLNGINNYKNPKLTRLMTFKFYMPKKATELKHLCLEELKPLEEVLNLAQCKNPFHLRPRDLISININVIVLELKGSETTFMCEYADETATIEFLNRWITFCQSIISTLT
Third polypeptide	@PD-L1 VH-hIgG1 (SEQ ID NO. 79) EVQLQESGPGLVKPSSETLSLTCAVYGDSTISGYWNWIRKPPGKGLGYMGIYISYTGSTYQNPGLSKRITFSRDTSKNQYLLKLSVTAADTATYYCARSAWIRTYFDYWGQGLTVTVSSASTKGPSVFPLAPSSKSTSGGTA

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Molecule No.	R1262: TIGIT VH-IL2R α mutant-Fc/TIGIT VL-IL2 mutant/PD-L1_VH-hIgG1/PD-L1 VL- κ -Ig LC
	ALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKV DKRVEPKSCDKTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVDVSHEDPEVKFNWYVDG VEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLF PSREEMTKNQVSLWCLVKGFYPSDIAVEWESNGQPENNYDTTPPVLDSDGSFFLYSDLTVDKSRWQQGN VFSCSVMHEALHNHYTQKSLSLSPGK
Fourth polypeptide	@PD-L1_VL- κ -Ig LC (SEQ ID NO. 80) EIVLTQSPDFQSVTPKEKVTITCSVSSSISSSNLHWYQQKPDQSPKLLIYGTSNLSAGVPSRFRSGSGSGTDF TLTINSLEAEDAATYYCQQWSSYPLTFGQGTKEIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPRE AKVQWKVDNALQSGNSQESVTEQDSKSTYLSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC

[0251] The sequence structure of the disulfide bond-modified IL2/IL2R α complex 2 was: TIGIT VH-IL2 mutant-Fc, TIGIT VL-IL2R α mutant, PD-L1 VH-hIgG1, and PD-L1 VL- κ -Ig LC, wherein an amino acid sequence of the TIGIT VH-IL2 mutant-Fc was shown in SEQ ID NO. 81, an amino acid sequence of the TIGIT VL-IL2R α mutant was shown in SEQ ID NO. 82, an amino acid sequence of the PD-L1 VH-hIgG1 was shown in SEQ ID NO. 79, and an amino acid sequence of the PD-L1 VL- κ -Ig LC was shown in SEQ ID NO. 80.

Molecule No.	Disulfide bond-modified IL2/IL2R α complex 2: TIGIT VH-IL2R α mutant-Fc/TIGIT VL-IL2 mutant/PD-L1 VH-hIgG1/PD-L1 VL- κ -Ig LC
First polypeptide	@TIGIT VH-IL2 mutant-Fc (SEQ ID NO. 81) QVQLVQSGAEVKKPGSSVKVCKASGYSFTSYWMNWVRQAPGQGLEWIGMIRPSDSETRLNQ MPKDRVTITVDKSTSTAYMELSSLRSEDTAVYYCAGIHDYGHGAYWGQGTTLVTVSSGGGGSGGG GSGGGSGGGSGGAPTSSSTKKTQLQLEHLLDLQMLNGINNYKNPKLTRMLTFKFYMPKKAT ELKHLQCLEEELKPLEEVLNLAQCKNFHLRPRDLISNINIVLELKGSETTFMCEYADETATIVEFLN RWITFCQSIISTLTPEKSSDKTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVDVSH EDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPI EKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCRVKGFYPSDIAVEWESNGQPENNYKTTPP VLKSDGSFFLASKLTVDKSRWQQGNVVFSCSVMHEALHNHYTQKSLSLSPGK
Second polypeptide	@TIGIT VL-IL2R α mutant (SEQ ID NO. 82) DIQMTQSPSSLSASVGDRVTITCRASENIYSNLAWYQQKPGKSPKLLVYAASHLPDGVPSRFRSGS GSGTDYSLTISSLQPEDFATYYCQHFWGTPTRTFGQGTKEIKGGGSGGGSGGGSGGGGCG GELCDDDPPEIPHATFKAMAYKEGTMNLNCECKRGFRRIKSGSLYMLCTGNSHSSWDNQCCQT SSATRNTTKQVTPQPEEQKERKTTQMSPMQPVDQASLPGHCREPPPWENEATERIYHFVVGQ MYYYQCQVQGYRALHRGPAESVCKMTHGKTRWTPQLICTG
Third polypeptide	@PD-L1_VH-hIgG1 (SEQ ID NO. 79)
Fourth polypeptide	@PD-L1_VL- κ -Ig LC (SEQ ID NO. 80)

[0252] The sequence structure of IL2/IL2R α complex 3 was: TIGIT VH-IL2R α (L42C)-Fc, TIGIT VL-IL2 (F42C), PD-L1 VH-hIgG1, and PD-L1 VL- κ -Ig LC, wherein the amino acid sequence of the TIGIT VH-IL2R α (L42C)-Fc was shown in SEQ ID NO. 83, the amino acid sequence of the TIGIT VL-IL2 (F42C) was shown in SEQ ID NO. 84, the amino acid sequence of the PD-L1 VH-hIgG1 was

shown in SEQ ID NO. 79, and the amino acid sequence of the PD-L1 VL- κ -Ig LC was shown in SEQ ID NO. 80. IL2R α (L42C): the position 42 of leucine in the amino acid sequence of IL2R α was mutated to cysteine, and IL2 (F42C): the position 42 of phenylalanine in the amino acid sequence of IL2 was mutated to cysteine.

Molecule No.	Disulfide bond-modified IL2/IL2R α complex 3: TIGIT VH-IL2R α (L42C)-Fc/TIGIT VL-IL2 (F42C)/PD-L1 VH-hIgG1/PD-L1 VL- κ -Ig LC
First polypeptide	@TIGIT VH-IL2R α (L42C)-Fc (SEQ ID NO. 83) QVQLVQSGAEVKKPGSSVKVCKASGYSFTSYWMNWVRQAPGQGLEWIGMIRPSDSETRLNQ MPKDRVTITVDKSTSTAYMELSSLRSEDTAVYYCAGIHDYGHGAYWGQGTTLVTVSSGGGGSGGG GSGGGSGGGSGGELCDDDPPEIPHATFKAMAYKEGTMNLNCECKRGFRRIKSGSCYMLCTGN

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Molecule No.	Disulfide bond-modified IL2/IL2R α complex 3: TIGIT VH-IL2Ra (L42C)-Fc/TIGIT VL-IL2 (F42C)/PD-L1 VH-hIgG1/PD-L1 VL-K-Ig LC
	SSHSSWDNQCQCTSSATRNTTKQVTPQPEEQKERKTTEMQSPMQPVDQASLPGHCREPPPW ENEATERIYHFVVGQMVYYQCVQGYRALHRGPAESVCKMTHGKTRWTPQLICTGEPKSSDKT HTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNA KTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTIISKAKGQPREPQVYTLTP PSREEMTKNQVSLTCRVKGFYPSDIAVEWESNGQPENNYKTTTPVLKSDGSFFLASKLTVDKSR WQQGNVFSQCSVMHEALHNHYTQKSLSLSPGK
Second polypeptide	STIGIT VL-IL2 (F42C) (SEQ ID NO. 84) DIQMTQSPSSLSASVGDRVTITCRASENIYSNLAWYQQKPGKSPKLLVYAASHLPDGVPSRFSGS GSGTDYSLTISSLQPEDFATYYCQHFWGTPRTFGQGTKLEIKGGGSGGGSGGGSGGGSGGGG GAPTSSSTKKTQLQLEHLLLDLQMLNGINNYKNPKLTRMLTCKFYMPKKATELKHLCLEELKP LLEEVNLAQSKNFHLRPRDLISNINVIVLELKGSETTFMCEYADETATIVEFLNRWITFCQSIISTLT
Third polypeptide	@PD-L1 VH-hIgG1 (SEQ ID NO. 79)
Fourth polypeptide	@PD-L1 VL-K-Ig LC (SEQ ID NO. 80)

Example 9 Preparation of FiBody Sample

Transient Expression of Proteins:

[0253] After a plasmid containing a target gene formed a cationic complex with a transfection reagent PEI, the cationic complex was introduced into a host cell Expi293, and an exogenous gene on the plasmid was transcribed and translated in the cell during the plasmid was in the cell, so as to obtain a target protein.

[0254] The Expi293 cells were cultured at 37° C. in 8% carbon dioxide at 130 rpm and counted before transfection, and 2E6 cells were inoculated into a 1-L shake flask at a culture system of about 300 mL. Preparation of transfection complex for transfection: firstly, 750 μ g target plasmid was added into a 50-mL centrifuge tube containing 15 mL Opti-MEM reagent, and the mixture was gently mixed uniformly and marked as tube A; 1.5 mg transfection reagent PEI was added into a 50-mL centrifuge tube containing 15 mL Opti-MEM reagent, and the mixture was gently mixed uniformly, incubated at room temperature for 5 min, and marked as tube B; and the PEI diluent of the B tube was dropwise added the into the DNA diluent of the tube A, the mixture was gently mixed uniformly, and incubated at room temperature for 15 min, the PEI-target plasmid complex was added into the Expi293 cells after the incubation was finished, and the cells were placed in a shaking table at 37° C. for a continuous culture. The sample was collected until D7-D10.

Protein Purification:

[0255] The supernatant of the transient cell expression solution was collected after 9000 rpm/20 min centrifugation, and then filtered by 0.22 μ m filter membrane. A purification was performed with ProA affinity chromatography. The process was as follows: an AKTA avant 150 chromatography device was used, a chromatography column (e.g., Mab-SelectSuRe LX, GE) was equilibrated with at least 5 CV of an equilibration buffer (10 mM PBS), the sample was loaded onto a chromatographic column, such that the target protein was adsorbed onto the chromatographic column, and other impurities passed through to be separated. After the sample was loaded, the chromatographic column was washed again

with at least 5 CV of an equilibration buffer (10 mM PBS), then the target protein was eluted with an elution buffer (20 mM NaAc, pH=3.4), and the neutralizing buffer (1 M Tris, pH 8.0) was pre-added into a collection tube, wherein the volume of the added neutralizing buffer was determined based on the estimated content of the eluted sample, generally 10% of the elution volume.

Example 10 Physical and Chemical Detection of FiBody

[0256] The samples were subjected to a one-step purification, then the purities were detected by HPLC-SEC (analytical column TOSOH, TSKgel G2000), and the expression level and purity of each sample were shown in the following table.

Type of bispecific antibody	Number of samples	Expression level mg/L	Purity (%)
FiBody-A	R0950	25.1	89.2
	R0951	16.1	90.8
	R0952	14.6	84.7
	R1072	122.1	91.7
	R1109	70.2	98.4
	R1115	83.9	
FiBody-C	R1116	89.3	94.0
	R1119	22.9	95.3
	R1120	18.0	50.7
FiBody-D	R1123	49.6	92.9
FiBody-disulfide bond-modified	R0954	17.1	91.0
	R1085	99.6	90.4
	R1086	66.13	56.1
FiBody-glycosylation modified	R0955	7.6	90.9
	R1110	98.4	98.6
	R1111	107.6	92.4
FiBody-affinity modified	R0960	17.0	90.4
ScFv-asymmetric (Y-Body)	R0809	1.5	36.8
ScFv-symmetric	R0810	10.0	68.9
CrossMab	R0959	1.4	41.4
FiBody-mismatching	R1042	12.9	41.9
	R1043	19.5	86.8
	R1124	14.5	65.6

[0257] An HPLC-SEC detection result of the sample R0951 was shown in FIG. 11, an HPLC-SEC detection result of the sample R1042 was shown in FIG. 12, an

HPLC-SEC detection result of the sample R0809 was shown in FIG. 13, and an HPLC-SEC detection result of the sample R1110 was shown in FIG. 14.

[0258] The results showed that bispecific antibodies (including various modified optimized antibodies) prepared by the FiBody platform had higher expression level and/or higher purity than those with asymmetric scFv (Y-Body and R0809), symmetric scFv (R0810), and CrossMab (R0959) structures.

[0259] Surprisingly, the bispecific antibodies with a mismatching pattern (samples R1042, R1043, and R1124) were also expressed and had the expression level similar to that of the normal molecule, but had a significantly lower purity.

[0260] The HPLC-SEC detection result of R1262 was shown in FIG. 15 and indicated that free light chains were successfully removed, and the purity of the disulfide bond-modified IL2/IL2Ra compound 1 prepared by the present disclosure was high.

Example 11 Detection of Antigen Affinity of FiBody

Analysis of Binding Activity of TIGIT End

[0261] A binding activity of a bispecific antibody molecule (TIGIT end) to CHO-TIGIT cells was detected by an FCM experimental method. Preparation of 3% BSA buffer: 4.5 g BSA was weighed into 150 mL 1×PBS and mixed uniformly, and the mixture was placed on ice for later use; antibody dilution: the tested antibody and the positive control were diluted to an initial concentration of 800 nM with 3% BSA, the subtype control was diluted to an initial concentration of 20 µg/mL at a volume of 300 µL, and 10 points were obtained by 3 times of gradient dilution (100+200); detection of binding activity: counting and plate-spreading of cells: after the R0254-3 cells were counted, the cells were distributed into 96-well V-shaped plates according to 100 µL 2E+05/well; 50 µL antibody with different concentrations was firstly added into the cells, the cells were incubated at 2-8° C. for 0.5 h, then 50 µL ligand was added, and the cells were incubated at 2-8° C. for 0.5 h; the cells were centrifuged at 350×g for 5 min, the supernatant was removed, and 200 µL/well of 3% BSA was added; the cells were centrifuged at 350×g for 5 min, the supernatant was removed, fluorescent antibodies PE Goat anti-human IgG Fc and PE Goat anti-mouse IgG Fc (diluted at 1:500×) were prepared with 3% BSA, the antibodies were added into a corresponding 96-well plate according to 100 µL/well, and the cells were incubated at 2-8° C. for 30 min; the cells were centrifuged at 350 g for 5 min, a supernatant was removed, and the cells were washed once with 3% BSA; after the cells were centrifuged at 350×g for 5 min, the supernatant was removed, and 1×PBS was added to resuspend the cells at 100 µL/well; and the resuspended cells were detected according to a standard operation procedure of a CytoFLEX flow cytometer.

[0262] Results of R0950, R0951, R0952, R0954, R0955, R0960, R1123/R1119/R1120/R1124, R1042/R1043, and R0810 were shown in FIGS. 16-19. Surprisingly, the binding activity of R0950 (@TIGIT at Fab terminal) was lower than that of R0951-R0960 (@TIGIT at IL15/IL15R terminal); and the binding activity of R0951-R0960 was similar to that of the positive control R0226 (Tigit monoclonal antibody, OMP-313R12, WO2016191643). After CH1 and CL of A-type and D-type interleukins and receptors thereof were

replaced, a binding force of the target region was not affected and the equivalent affinity was shown compared to the positive control (R0226/R0774 (the VH sequence was shown in SEQ ID NO: 73, and the VL sequence was shown in SEQ ID NO: 74) and Tigit monoclonal antibody). After CH1 and CL of C-type molecules were replaced, the target region was affected, and the binding force of the target region was obviously lower than that of the positive control (R0226/R0774 and Tigit monoclonal antibody).

[0263] Modified molecules of disulfide bond-modified samples R1081 and R1085 and the glycosylated sample had an affinity result of the tigit terminal equivalent to that of the molecules before the modification.

[0264] R1042, R1043, and R1124 were mismatched test molecules, and had an obviously reduced TIGIT binding activity. R0810 was an ScFv structural molecule and had the binding activity weaker than that of the control molecule R0226.

[0265] The result of R1262 was shown in FIG. 20 and showed that the disulfide bond-modified IL2/IL2Ra complex 1 (R1262) had an affinity equivalent to that of the disulfide bond-unmodified IL2/IL2Ra complex (R1115), indicating that the disulfide-bond modification did not affect the affinity of the target region.

Analysis of Binding Activity of PD-L1 Terminal

[0266] A binding activity of a double-antibody molecule (PD-L1 end) and CHO-PD-L1 cells was detected by an FCM experiment method. Preparation of 3% BSA buffer: 4.5 g BSA was weighed into 150 mL 1×PBS and mixed uniformly, and the mixture was placed on ice for later use; antibody dilution: the tested antibody and the positive control were diluted to an initial concentration of 800 nM with 3% BSA, the subtype control was diluted to an initial concentration of 20 µg/mL at a volume of 300 µL, and 10 points were obtained by 3 times of gradient dilution (100+200); detection of binding activity: cells were counted and spread on plates: after the R0254-3 cells were counted, the cells were distributed into 96-well V-shaped plates according to 100 µL 2E+05/well; 50 µL antibody with different concentrations was firstly added into the cells, the cells were incubated at 2-8° C. for 0.5 h, then 50 µL ligand was added, and the cells were incubated at 2-8° C. for 0.5 h; the cells were centrifuged at 350×g for 5 min, the supernatant was removed, and 200 µL/well of 3% BSA was added; the cells were centrifuged at 350×g for 5 min, the supernatant was removed, fluorescent antibodies PE Goat anti-human IgG Fc and PE Goat anti-mouse IgG Fc (diluted at 1:500×) were prepared with 3% BSA, the antibodies were added into a corresponding 96-well plate according to 100 µL/well, and the cells were incubated at 2-8° C. for 30 min; the cells were centrifuged at 350 g for 5 min, the supernatant was removed, and the cells were washed once with 3% BSA; after the cells were centrifuged at 350×g for 5 min, the supernatant was removed, and 1×PBS was added to resuspend the cells at 100 µL/well; and the resuspended cells were detected according to a standard operation procedure of a CytoFLEX flow cytometer.

[0267] Results of R0950, R0951, R0952, R0954, R0955, R0960, R1072, R1115-R1120 and R1123-R1124; R0950, R1042 and R1043; R1072 and R1081-R1086; R1072 and R1109-R1111 were shown in FIG. 21-FIG. 25. Surprisingly, the activity of @PD-L1 at an IL15 end was better than that at a Fab end. R1042, R1043, and R1124 were corresponding

mismatched molecules with a significantly weaker activity, and other FiBody all showed an equivalent affinity to that of a positive antibody (PD-L1 monoclonal antibody). R0802 was a PD-L1 monoclonal antibody, 176F9. The VH sequence was shown in SEQ ID NO: 36, the VL sequence was shown in SEQ ID NO: 37, R0514 was a PD-L1 monoclonal antibody, Avelumab, R0919 was a PD-L1 monoclonal antibody, the VH sequence was shown in SEQ ID NO: 75, the VL sequence was shown in SEQ ID NO: 76, and R0968 was a PD-L1 monoclonal antibody, the VH sequence was shown in SEQ ID NO: 71, and the VL sequence was shown in SEQ ID NO: 72.

[0268] The result of R1262 was shown in FIG. 26 and showed that the disulfide bond-modified IL2/IL2R α complex 1 (R1262) had an affinity equivalent to that of the antibody of the disulfide bond-unmodified IL2/IL2R α complex 4 (R1115), indicating that the disulfide-bond modification did not affect the affinity of the target region.

Analysis of Binding Blockade of TIGIT End

[0269] A binding activity of a bispecific-antibody molecule (TIGIT end) blockade ligand and CHO-TIGIT cells was detected by an FCM experiment method. Preparation of 3% BSA buffer: 4.5 g BSA was weighed into 150 mL 1 $\times$ PBS and mixed uniformly, and the mixture was placed on ice for later use; antibody dilution: the tested antibody (R1262) and the positive control antibody (R1115) were diluted to an initial concentration of 800 nM with 3% BSA, and the subtype control antibody (hIgG1 antibody) was diluted to an initial concentration of 20 μ g/mL at a volume of 300 μ L with 3% BSA. 10 concentrations were obtained by 3 times of gradient dilution (100 μ L+200 μ L); counting and plate-spreading of cells: after the R0254-3 cells were counted, the cells were distributed into 96-well V-shaped plates according to 100 μ L 2E+05/well; 50 μ L of an antibody with different concentrations was firstly added into the cells, the cells were incubated at 2-8 $^{\circ}$ C. for 0.5 h, then 50 μ L of a ligand was added, and the cells were incubated at 2-8 $^{\circ}$ C. for 0.5 h; the cells were centrifuged at 350 $\times$ g for 5 min, the supernatant was removed, and 200 μ L/well of 3% BSA was added; the cells were centrifuged at 350 $\times$ g for 5 min, the supernatant was removed, fluorescent antibodies PE Goat anti-human IgG Fc and PE Goat anti-mouse IgG Fc (diluted at 1:500 $\times$) were prepared with 3% BSA, the antibodies were added into a corresponding 96-well plate according to 100 μ L/well, and the cells were incubated at 2-8 $^{\circ}$ C. for 30 min; the cells were centrifuged at 350 g for 5 min, the supernatant was removed, and the cells were washed once with 3% BSA; after the cells were centrifuged at 350 $\times$ g for 5 min, a supernatant was removed, and 1 $\times$ PBS was added to resuspend the cells at 100 μ L/well; and the resuspended cells were detected according to a standard operation procedure of a CytoFLEX flow cytometer.

[0270] The detection result of R1262 was shown in FIG. 27 and showed that compared with the disulfide bond-unmodified IL2/IL2R α complex 4 (R1115), the disulfide bond-modified IL2/IL2R α complex 1 (R1262) had an equivalent blocking effect on a binding force of the ligand and the target region.

Example 12 Binding Activity of FiBody Receptor and Ligand Complex (IL15/IL15R)

[0271] Antibody dilution: all molecules were diluted with an FACS buffer to an initial concentration 400 nM at a

volume of 180 μ L; 10 concentrations were obtained by 3 times of gradient dilution (60+120); counting and plate-spreading of cells: R0255-2 (CHO-mTigit)/293T-IL15R-28 cells were centrifuged at 250 g for 5 min, a supernatant was removed, a cell density was adjusted to 2E+06 with an FACS buffer, and the cells were uniformly distributed into 96-well V-shaped plates at a 100 μ L/tube; the diluted antibody was added into the cells at 100 μ L/well and the cells were incubated at 2-8 $^{\circ}$ C. for 0.5 h; the 96-well plate was taken out and centrifuged at 250 g for 5 min, a supernatant was carefully removed, the FACS buffer was added at 200 μ L/well, and the cells were centrifuged at 250 g for 5 min again, and a supernatant was carefully removed; a PE fluorescent secondary antibody (diluted at 1:500) was prepared with the FACS buffer and added into a corresponding 96-well plate at 100 μ L/well, and the cells were resuspended and incubated at 2-8 $^{\circ}$ C. for 30 min; the 96-well plate was taken out, the cells were centrifuged at 250 g for 5 min, the supernatant was carefully removed, the FACS buffer at 200 μ L/well, the cells were centrifuged at 250 g again for 5 min, and the supernatant was carefully removed; and the cells were resuspended with 1 $\times$ PBS at 100 μ L/well and detected by FACS.

[0272] Results were shown in FIG. 28 and FIG. 29: R0952 (IL15 and IL15RA rearranged) and R0960 (deactivated) molecules had a very low IL15 receptor complex binding activity; the IL15 activity of R0955 (deglycosylated) was decreased; and the activity of the R0953 and R0954 after covalent linking was similar to that of R0951, indicating that there was little effect on the structure.

[0273] The mismatched molecules R1042 and R1043 had an activity equivalent to R0951, indicating that no mismatching was generated and even a wrong Fv may also be expressed, wherein R0655 was IL15/IL15RFc fusion protein (see SEQ ID NO: 38).

Example 13 Electrophoresis Detection of Disulfide Bond-Modified Molecules

[0274] The disulfide bond-modified optimized samples (R1072, R1081, R1082, R0954, and R1084-R1086) in the example of the present disclosure were subjected to an SDS-PAGE electrophoresis detection.

[0275] The result was shown in FIG. 30. The R1072 molecule without disulfide-bond modification had a band with a molecular weight of 25 KD-35 KD, indicating that a free light chain exists; the ligand and receptor disulfide bond-modified molecules were R0954, R1085, and R1086, wherein the electrophoresis result of R0954 and R1086 showed that the non-covalent light chain still existed (a band existed between 25 KD and 35 KD), and R1085 was free of non-covalent light chain (a band did not exist between 25 KD and 35 KD), indicating that the disulfide-bond modification of R1085 was successful.

[0276] The light and heavy chain disulfide bond-modified molecules were R1081, R1082, and R1084. The electrophoresis result showed that no non-covalent light chain existed (no band existed between 25 KD and 35 KD), indicating that the disulfide-bond modification of R1081, R1082, and R1084 was successful.

[0277] The FiBody samples prepared in example 9 (disulfide bond-modified IL2/IL2R α complexes 1 and 2, and IL2/IL2R α complex 3 constructed in example 8) and R1115 in example 2 were subjected to an SDS-PAGE electrophoresis detection.

[0278] The detection result was shown in FIG. 31, wherein R1115 was a disulfide bond-unmodified molecule, and a band existed between a molecular weight of 25 KD and 35 KD, indicating that a free light chain existed. The disulfide bond-modified IL2/IL2Rα complexes 1 and 2 had no bands with a molecular weight of 25 KD-35 KD, which was the same as the IL2/IL2Rα complex 3, indicating that the free light chain was successfully removed, and the disulfide-bond modification was successful.

Example 14 Effect of Types of First and Third Amino Acids Extended from N Terminus of Amino Acid Sequence of IL2Rα Mutant on Formation of Disulfide Bond Introduced after Modification of IL2/IL2Rα Complex

[0279] When the second antigen-binding moiety was indirectly linked to the N terminus of the IL2Rα mutant through a linker, the N terminus of the amino acid sequence of the IL2Rα mutant extended three amino acids. To further clarify an effect of types of first and third amino acids extended from the N terminus of the amino acid sequence of the IL2Rα mutant on formation of a disulfide bond introduced after modification of an IL2/IL2Rα complex, the following experiments were performed:

1. Effect of Type of Extended Third Amino Acid on Formation of Disulfide Bond Introduced after Modification of IL2/IL2Rα Complex

[0280] The extended third amino acid was selected from any one of an aromatic amino acid (for example, phenyl-

alanine), an amino acid with an uncharged R group (for example, serine), an amino acid with a positively charged R group (for example, lysine) or an amino acid with a negatively charged R group (for example, aspartic acid) to construct and prepare disulfide bond-modified IL2/IL2Rα complexes 5, 6, 7, and 8 (abbreviated to R1493, R1494, R1495, and R1496 successively), the complexes, together with a disulfide bond-modified IL2/IL2Rα complex 1 (the third amino acid extended from the N terminus of the amino acid sequence of the IL2Rα mutant in the IL2/IL2Rα complex 1 was glycine (non-polar fatty acid amino acid)) were used as experimental groups, and a disulfide bond-unmodified IL2/IL2Rα complex 4 was used as a control group.

[0281] The disulfide bond-modified IL2/IL2Rα complexes 5, 6, 7, and 8 had the same preparation process as the disulfide bond-modified IL2/IL2Rα complex 1 in example 8.

[0282] Sequence structures of the disulfide bond-modified IL2/IL2Rα complexes 5, 6, 7, and 8 were all as follows: TIGIT VH-IL2Rα mutant-Fc, TIGIT VL-IL2 mutant (SEQ ID NO: 78), PDL1 VH-hIgG1 (SEQ ID NO: 79), and PDL1 VL-κ-Ig LC (SEQ ID NO: 80), wherein amino acid sequences of TIGIVH-IL2Rα mutant-Fc of the disulfide bond-modified IL2/IL2Rα complexes 5, 6, 7, and 8 were shown in SEQ ID NO: 85, SEQ ID NO: 86, SEQ ID NO: 87, and SEQ ID NO: 88 successively (wherein the three amino acids extended from the N terminus of the amino acid sequence of the IL2Rα mutant were shown in bold, and the extended third amino acid was underlined).

Molecule No.	R1493: TIGIT VH-IL2Rα mutant-Fc/TIGIT VL4L2 mutant/PD-L1 VH-hIgG1/PD-L1 VL-κ-Ig LC
First polypeptide	@TIGIT VH-IL2Rα mutant-Fc (SEQ ID NO. 85) QVQLVQSGAEVKKPGSSVKVSCASGYSTSYWMNWVRQAPGGLEWIGMIRPSDSETRLNQM FKDRVITITVDKSTSTAYMELSSLRSEDTAVYYCAGIHDYGHGAYWGQGLTVTVSSGGGGSGGGGS GGGGSGGGG FC GEELCDDDPPEIPHATFKAMAYKEGTMLNCECKRGFRRIKSGSLYMLCTGNSSHS SWDNQCQCTSSATRNNTTKQVTPQPEEQKERKTTMQSPMQPVDQASLPGHCREPPPWENEATE RIYHFVVGQMYYQCVQGYRALHRGPAESVCKMTHGKTRWTQPQLICTGEPKSSDKTHTCPCPP APELLGGPSVFLFPPPKDITLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQ YNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKN QVSLTCTRVKGFYPSDIAVEWESNGQPENNYKTTTPVLKSDGSFFLASKLTVDKSRWQQGNVFSCS VMHEALHNHYTQKSLSLSPGK
Second polypeptide	@TIGIT VL-IL2 mutant (SEQ ID NO. 78)
Third polypeptide	@PD-L1 VH-hIgG1 (SEQ ID NO. 79)
Fourth polypeptide	@PD-L1 VL-κ-Ig LC (SEQ ID NO. 80)
Molecule No.	R1494: TIGIT VH-IL2Rα mutant-Fc/TIGIT VL-IL2 mutant/PD-L1 VH-hIgG1/PD-L1 VL-κ-Ig LC
First polypeptide	@TIGIT VH-IL2Rα mutant-Fc (SEQ ID NO. 86) QVQLVQSGAEVKKPGSSVKVSCASGYSTSYWMNWVRQAPGGLEWIGMIRPSDSETRLNQM FKDRVITITVDKSTSTAYMELSSLRSEDTAVYYCAGIHDYGHGAYWGQGLTVTVSSGGGGSGGGGS GGGGSGGGG CG ELCDDDPPEIPHATFKAMAYKEGTMLNCECKRGFRRIKSGSLYMLCTGNSSHS SWDNQCQCTSSATRNNTTKQVTPQPEEQKERKTTMQSPMQPVDQASLPGHCREPPPWENEATE RIYHFVVGQMYYQCVQGYRALHRGPAESVCKMTHGKTRWTQPQLICTGEPKSSDKTHTCPCPP APELLGGPSVFLFPPPKDITLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQ YNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKN QVSLTCTRVKGFYPSDIAVEWESNGQPENNYKTTTPVLKSDGSFFLASKLTVDKSRWQQGNVFSCS VMHEALHNHYTQKSLSLSPGK
Second polypeptide	@TIGIT VL-IL2 mutant (SEQ ID NO. 78)
Third polypeptide	@PD-L1 VH-hIgG1 (SEQ ID NO. 79)

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Fourth polypeptide	@PD-L1 VL-κ-Ig LC (SEQ ID NO. 80)
Molecule No.	R1495: TIGIT VH-IL2Rα mutant-Fc/TIGIT VL-IL2 mutant/PD-L1 VH-hIgG1/PD-L1 VL-κ-Ig LC
First polypeptide	@TIGIT VIH-IL2Rα mutant-Fc (SEQ ID NO. 87) QVQLVQSGAEVKKPGSSVKVCKASGYSFTSYWMNWRQAPGQGLEWIGMIRPSDSETRLNQMFKDRVTITVDKSTSTAYMELSSLRSEDTAVYYCAGIHDYGHGAYWGQGLTVTVSSGGGGSGGGSGGGSGGGGKCGELCDDDPPEIPHATFKAMAYKEGTMNLNCECKRGFRRIKSGSLYMLCTGNSSHSWDNQCCQTSSATRNNTTKQVTPQPEEQKERKTTEMQSPMQPVDQASLPGHCREPPPWENEATERIYHFVVGQMVYYQCVQGYRALHRGPAESVCKMTHGKTRWTPQLICTGEPKSSDKTHTCPPCPAPPELLGGPSVFLFPPPKPDTLMI SRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCTRVKGFYPSDIAVEWESNGQPENNYKTTTPVLKSDGSFFLASKLTVDKSRWQQGNVFSCLVMHEALHNHYTQKSLSLSPGK
Second polypeptide	@TIGIT VL-IL2 mutant (SEQ ID NO. 78)
Third polypeptide	@PD-L1 VH-hIgG1 (SEQ ID NO. 79)
Fourth polypeptide	@PD-L1 VL-κ-Ig LC (SEQ ID NO. 80)
Molecule No.	R1496: TIGIT VH-IL2Rα mutant-Fc/TIGIT VL-IL2 mutant/PD-L1 VH-hIgG1/PD-L1 VL-κ-Ig LC
First polypeptide	@TIGIT VH-IL2Rα mutant-Fc (SEQ ID NO. 88) QVQLVQSGAEVKKPGSSVKVCKASGYSFTSYWMNWRQAPGQGLEWIGMIRPSDSETRLNQMFKDRVTITVDKSTSTAYMELSSLRSEDTAVYYCAGIHDYGHGAYWGQGLTVTVSSGGGGSGGGSGGGSGGGGDCGELCDDDPPEIPHATFKAMAYKEGTMNLNCECKRGFRRIKSGSLYMLCTGNSSHSWDNQCCQTSSATRNNTTKQVTPQPEEQKERKTTEMQSPMQPVDQASLPGHCREPPPWENEATERIYHFVVGQMVYYQCVQGYRALHRGPAESVCKMTHGKTRWTPQLICTGEPKSSDKTHTCPPCPAPPELLGGPSVFLFPPPKPDTLMI SRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCTRVKGFYPSDIAVEWESNGQPENNYKTTTPVLKSDGSFFLASKLTVDKSRWQQGNVFSCLVMHEALHNHYTQKSLSLSPGK
Second polypeptide	@TIGIT VL-IL2 mutant (SEQ ID NO. 78)
Third polypeptide	@PD-L1 VH-hIgG1 (SEQ ID NO. 79)
Fourth polypeptide	@PD-L1 VL-κ-Ig LC (SEQ ID NO. 80)

[0283] The obtained disulfide bond-modified IL2/IL2Rα complexes 5, 6, 7, and 8 and the complex 1 were subjected to an SDS-PAGE electrophoresis detection, and the disulfide bond-unmodified IL2/IL2Rα complex 4 was used as a control group to investigate the effect of the type of the third amino acid extended from the N terminus of the amino acid sequence of the IL2Rα mutant on formation of the disulfide bond introduced after the modification of the IL2/IL2Rα complex. The experimental result was shown in FIG. 32. It may be seen that compared with the control group, the free light chains were successfully removed from the disulfide bond-modified IL2/IL2Rα complexes 5, 6, 7, and 8, and the disulfide-bond modification was successful, indicating that the extended third amino acid selected from the non-polar fatty acid amino acid, the aromatic amino acid, the amino acid with an uncharged R group, the amino acid with a positively charged R group or the amino acid with a negatively charged R group all did not affect the formation of the disulfide bond introduced after the modification of the IL2/IL2Rα complex.

2. Effect of Combined Types of Extended First and Third Amino Acids on Formation of Disulfide Bond after Modification of IL2/IL2Rα Complex

[0284] The extended first and third amino acids were selected from any combination of a non-polar fatty acid amino acid (for example, glycine) an aromatic amino acid (for example, phenylalanine), an amino acid with an uncharged R group (for example, serine), an amino acid with a positively charged R group (for example, lysine) or an amino acid with a negatively charged R group (for example, aspartic acid) to construct and prepare disulfide bond-modified IL2/IL2Rα complexes 9, 10, 11, 12, and 13 (abbreviated to R1662, R1663, R1664, R1665, and R1666 successively), the complexes, together with the disulfide bond-modified IL2/IL2Rα complex 1 were used as experimental groups, and the disulfide bond-unmodified IL2/IL2Rα complex 4 was used as a control group.

[0285] The disulfide bond-modified IL2/IL2Rα complexes 9, 10, 11, 12, and 13 had the same preparation process as the disulfide bond-modified IL2/IL2Rα complex 1 in example 8.

[0286] Sequence structures of the disulfide bond-modified IL2/IL2R α complexes 9, 10, 11, 12, and 13 were all as follows: TIGIT VH-IL2R α mutant-Fc, TIGIT VL-IL2 mutant (SEQ ID NO: 78), PDL1 VH-hIgG1 (SEQ ID NO: 79), and PDL1 VL-k-Ig LC (SEQ ID NO: 80), wherein amino acid sequences of TIGIT VH-IL2R α mutant-Fc of the disulfide bond-modified IL2/IL2R α complexes 9, 10, 11, 12,

and 13 were shown in SEQ ID NO: 89, SEQ ID NO: 90, SEQ ID NO: 91, SEQ ID NO: 92, and SEQ ID NO: 93 successively (wherein the three amino acids extending from the N terminus of the amino acid sequence of the IL2R α mutant were shown in bold, and the extended first and third amino acids were underlined).

Molecule No.	R1662: TIGIT VH-IL2R α mutant-Fc/TIGIT VL-IL2 mutant/PD-L1 VH-hIgG1/PD-L1 VL-k-Ig LC
First polypeptide	@TIGIT VH-IL2R α mutant-Fc (SEQ ID NO. 89) QVQLVQSGAEVKKPGSSVKVSCASGYSFTSYWMNWVRQAPGQGLEWIGMIRPSDSETRL NQMPKDRVTITVDKSTSTAYMELSSLRSED TAVYYCAGIHDYGHGAYWGQGT LVTVSSGGG GSGGGGSGGGGSGGGFCSELCDDDPPEIPHATFKAMAYKEGTMNLNCECKRGFRRIKSGSL YMLCTGNSHSSWDNQCCQTSSATRN TTKQVTPQPEEQKERKTT EMQSPMPVDQASLP GHCREPPPWENEATERIYHFVVGQMVYYQCVQGYRALHRGPAESVCKMTHGKTRWTQPQ LICTEGPKSSDKTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEV KFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEK TISKAKGQPREPQVYTLPPSREEMTKNQVSLTCTRVKGFYPSDIAVEWESNGQPENNYKTTTP VLKSDGSFPLASKLTVDKSRWQQGNVFCSCVMHEALHNHYTQKSLSLSPGK
Second polypeptide	@TIGIT VL-IL2 mutant (SEQ ID NO. 78)
Third polypeptide	@PD-L1 VH-hIgG1 (SEQ ID NO. 79)
Fourth polypeptide	@PD-L1 VL-k-Ig LC (SEQ ID NO. 80)
Molecule No.	R1663: TIGIT VH-IL2R α mutant-Fc/TIGIT VL-IL2 mutant/PD-L1 VH-hIgG1/PD-L1 VL-k-Ig LC
First polypeptide	@TIGIT VH-IL2R α mutant-Fc (SEQ ID NO. 90) QVQLVQSGAEVKKPGSSVKVSCASGYSFTSYWMNWVRQAPGQGLEWIGMIRPSDSETRL NQMPKDRVTITVDKSTSTAYMELSSLRSED TAVYYCAGIHDYGHGAYWGQGT LVTVSSGGG GSGGGGSGGGGSGGGKDELCD DDPPEIPHATFKAMAYKEGTMNLNCECKRGFRRIKSGSL YMLCTGNSHSSWDNQCCQTSSATRN TTKQVTPQPEEQKERKTT EMQSPMPVDQASLP GHCREPPPWENEATERIYHFVVGQMVYYQCVQGYRALHRGPAESVCKMTHGKTRWTQPQ LICTEGPKSSDKTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEV KFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEK TISKAKGQPREPQVYTLPPSREEMTKNQVSLTCTRVKGFYPSDIAVEWESNGQPENNYKTTTP VLKSDGSFPLASKLTVDKSRWQQGNVFCSCVMHEALHNHYTQKSLSLSPGK
Second polypeptide	@TIGIT VL-IL2 mutant (SEQ ID NO. 78)
Third polypeptide	@PD-L1 VH-hIgG1 (SEQ ID NO. 79)
Fourth polypeptide	@PD-L1 VL-k-Ig LC (SEQ ID NO. 80)
Molecule No.	R1664: TIGIT VH-IL2R α mutant-Fc/TIGIT VL-IL2 mutant/PD-L1 VH-hIgG1/PD-L1 VL-k-Ig LC
First polypeptide	@TIGIT VH-IL2R α mutant (SEQ ID NO. 91) QVQLVQSGAEVKKPGSSVKVSCASGYSFTSYWMNWVRQAPGQGLEWIGMIRPSDSETRL NQMPKDRVTITVDKSTSTAYMELSSLRSED TAVYYCAGIHDYGHGAYWGQGT LVTVSSGGG GSGGGGSGGGGSGGGKDELCD DDPPEIPHATFKAMAYKEGTMNLNCECKRGFRRIKSGSL YMLCTGNSHSSWDNQCCQTSSATRN TTKQVTPQPEEQKERKTT EMQSPMPVDQASLP GHCREPPPWENEATERIYHFVVGQMVYYQCVQGYRALHRGPAESVCKMTHGKTRWTQPQ LICTEGPKSSDKTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEV KENWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEK TISKAKGQPREPQVYTLPPSREEMTKNQVSLTCTRVKGFYPSDIAVEWESNGQPENNYKTTTP VLKSDGSFPLASKLTVDKSRWQQGNVFCSCVMHEALHNHYTQKSLSLSPGK
Second polypeptide	@TIGIT VL-IL2 mutant (SEQ ID NO. 78)
Third polypeptide	@PD-L1 VH-hIgG1 (SEQ ID NO. 79)
Fourth polypeptide	@PD-L1 VL-k-Ig LC (SEQ ID NO. 80)

-continued

Molecule No.	R1665: TIGIT VH-IL2Rα mutant-Fc/TIGIT VL-IL2 mutant/PD-L1 VH-hIgG1/PD-L1 VL-κ-Ig LC
First polypeptide	@TIGIT VH-IL2Rα mutant (SEQ ID NO. 92) QVQLVQSGAEVKKPGSSVKVSKASGYSTSYWMNWVRQAPGGGLEWIGMIRPSDSETRL NQMFKDRVTITVDKSTSTAYMELSSLRSEDAVYYCAGIHDYGHGAYWGQGLTVTVSSGGG SGGGGGSGGGGGSGGDCFELCDDDPPEIPHATFKAMAYKEGTMNCECKRGFRRIKSGSL YMLCTGNSSSHSSWDNQCTSSATRNTTKQVTPQPEEQKERKTTEMQSPMQPVDQASLP GHCREPPPWENEATERIYHFFVVGQMVYYQCVQGYRALHRGPAESVCKMTHGKTRWTQPQ LICTGPEKSSDKTHTCPPCPAPELLGGPSVFLFPPPKPDTLMISRTPEVTCVVVDVSHEDPEV KFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYCKVSNKALPAPIEK TISKAKGQPREPQVYTLPPSREEMTKNQVSLTCRVKGFYPSDIAVEWESNGQPENNYKTTTP VLKSDGSFFLASKLTVDKSRWQQGNVFSQVMHEALHNHYTQKSLSLSPGK
Second polypeptide	@TIGIT VL-IL2 mutant (SEQ ID NO. 78)
Third polypeptide	@PD-L1 VH-hIgG1 (SEQ ID NO. 79)
Fourth polypeptide	@PD-L1 VL-κ-Ig LC (SEQ ID NO. 80)
Molecule No.	R1666: TIGIT VH-IL2Rα mutant-Fc/TIGIT VL-IL2 mutant/PD-L1 VH-hIgG1/PD-L1 VL-κ-Ig LC
First polypeptide	@TIGIT VH-IL2Rα mutant-Fc (SEQ ID NO. 93) QVQLVQSGAEVKKPGSSVKVSKASGYSTSYWMNWVRQAPGGGLEWIGMIRPSDSETRL NQMFKDRVTITVDKSTSTAYMELSSLRSEDAVYYCAGIHDYGHGAYWGQGLTVTVSSGGG SGGGGGSGGGGGSGGCKELCDDDPPEIPHATFKAMAYKEGTMNCECKRGFRRIKSGSL YMLCTGNSSSHSSWDNQCTSSATRNTTKQVTPQPEEQKERKTTEMQSPMQPVDQASLP GHCREPPPWENEATERIYHFFVVGQMVYYQCVQGYRALHRGPAESVCKMTHGKTRWTQPQ LICTGPEKSSDKTHTCPPCPAPELLGGPSVFLFPPPKPDTLMISRTPEVTCVVVDVSHEDPEV KFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYCKVSNKALPAPIEK TISKAKGQPREPQVYTLPPSREEMTKNQVSLTCRVKGFYPSDIAVEWESNGQPENNYKTTTP VLKSDGSFFLASKLTVDKSRWQQGNVFSQVMHEALHNHYTQKSLSLSPGK
Second polypeptide	@TIGIT VL-IL2 mutant (SEQ ID NO. 78)
Third polypeptide	@PD-L1 VH-hIgG1 (SEQ ID NO. 79)
Fourth polypeptide	@PD-L1 VL-κ-Ig LC (SEQ ID NO. 80)

[0287] The obtained disulfide bond-modified IL2/IL2Rα complexes 9, 10, 11, 12, and 13 and the complex 1 were subjected to an SDS-PAGE electrophoresis detection, and the disulfide bond-unmodified IL2/IL2Rα complex 4 was used as a control group to investigate the effect of the combined types of the first and third amino acids extended from the N terminus of the amino acid sequence of the IL2Rα mutant on formation of the disulfide bond introduced after the modification of the IL2/IL2Rα complex. The experimental result was shown in FIG. 33. It may be seen that compared with the control group, the free light chains were successfully removed from the disulfide bond-modified IL2/IL2Rα complexes 9, 10, 11, 12, and 13, and the disulfide-bond modification was successful, indicating that the extended first and third amino acids selected from any combination of the non-polar fatty acid amino acid, the aromatic amino acid, the amino acid with an uncharged R group, the amino acid with a positively charged R group or the amino acid with a negatively charged R group all did not

affect the formation of the disulfide bond introduced after the modification of the IL2/IL2Rα complex.

[0288] The technical characteristics of the above examples may be combined arbitrarily. For a concise description, all possible combinations of all technical characteristics of the examples may not be described; however, these combinations of the technical characteristics should be construed as disclosed in the description as long as no contradiction occurs.

[0289] The above embodiments only express several embodiments of the present disclosure and the description is relatively specific and in detail, but they should not therefore be construed as limiting the scope of the present disclosure. It should be noted that a person of ordinary skill in the art may further make several variations and improvements without departing from the conception of the present disclosure. These variations and improvements all fall within the protection scope of the present disclosure. Therefore, the scope of protection of the patent of the present disclosure shall be in accordance with the appended claims.

Sequence 36: @PD-L1: VH (SEQ ID NO: 36)
EVQLLESQGGGLVQPGGSLRLSCAASGFTFSSYIMMWVRQAPGKLEWVSSIYPSSGGITFYAD
TVKGRFTISRDNKNTLYLQMNSLRAEDTAVYYCARIKLGTVTTVDYWGQGLTVTVSS.

Sequence 37: @PD-L1: VL (SEQ ID NO: 37)
QSALTQPASVSGSPGQSITISCTGTSSDVGGYNYVSWYQQHPGKAPKLMYDVSNRPSGVSN
RFGSKSGNTASLTISGLQAEDEADYYCSSYTSSSTRVFGTGTKVTVL.

Amino acid sequence of R0655 (SEQ ID NO: 38)
EPRGPTIKPCPPCKCAPNLLGGPSVFIFPPKIKDVLMISSLPIVTCVVVDVSEDDPDVQISWV
NNVEVHTAQQTQTHREDYNSTLRVVSALPIQHQDWMGKEFKCKVNNKDLPAPIERTISKPKGSVR
APQVYVLPPEEEMTKQVTLTCMVTDFMPEDIYVEWTNNGKTELNYKNTEPVLDSGGSYFMY
KLRVEKKNWVERNSYSCSVVHEGLHNHHTTKSFRTPGAGGGSGGGSGGGSGGGSGGIT
CPPPMSVEHADIVVKSYSLSRERYICNSGFKRKAGTSSLTECVLNKATNVAHWTTPSLKCIRDPA
LVHQRPAAPPSTVTTAGVTPQESLSPSGKEPAASSPSSNNTAATTAAIVPGSQLMPKSPSTGTTEI
SSHESSHGTPSQTTAKNWEHTASASHQPPGVYPQGHSDTTSGSGGGSGGGSGGGSLQNW
VNVISDLKKIEDLIQSMHIDATLYTESDVHPSCKVTAMKCFLELQVISLESGDASIHDTVENLIILAND
SLSSNGNVTESGCKECEELEEKNIKEFLQSFVHIVQMFINTS.

Amino acid sequence of R0968PD-L1 VH (SEQ ID NO: 71)
QVQLQESGPGLVKPSSETLSLTCAVYGDSITSGYWNWIRKPPGKCLEYIGYISYTGSTYQNPSLK
SRITMSRDTSKNQYLLKSSVTAADTAVYYCARSAWIRTYFDYWGQGLTVTVSS.

Amino acid sequence of R0968PD-L1 VL (SEQ ID NO: 72)
DIQMTQSPSSLSASVGDRVTITCSVSSISSNLHWYQQKPKAPKPIYGTSLASGVPSRF
SGSGSGTDYTLTISSLQPEDFATYYCQQWSSYPLTFGCGTKLEIK.

Amino acid sequence of R0774TIGIT VH (SEQ ID NO: 73)
QVQLVQSGAEVKKPGSSVKVSKASGYSFTSYWMNWRQAPGGLEWIGMIRPSDSETRLN
QMFKDRVTITVDKSTSTAYMELSSLRSEDTAVYYCAGIHDYGHGAYWGQGLTVTVSS.

Amino acid sequence of R0774TIGIT VL (SEQ ID NO: 74)
DIQMTQSPSSLSASVGDRVTITCRASENIYSLAWYQQKPKSLVYAASHLPDGVPSRFS
GSGSGTDYSLTISSLQPEDFATYYCQHFVGTPTFTFGQGTKLEIK.

Amino acid sequence of R0919PD-L1 VH (SEQ ID NO: 75)
EVQLQESGPGLVKPSSETLSLTCAVYGDSITSGYWNWIRKPPGKLEWIMGYISYTGSTYQNPSL
KSRITFSRDTSKNQYLLKSSVTAADTATYYCARSAWIRTYFDYWGQGLTVTVSS.

Amino acid sequence of R0919PD-L1 VL (SEQ ID NO: 76)
EIVLTQSPDFQSVTPKEKVTITCSVSSISSNLHWYQQKPKLLIYGTSLASGVPSRFS
GSGSGTDFTLTINSLEAEDAATYYCQQWSSYPLTFGQGTKLEIK.

SEQUENCE LISTING

<160> NUMBER OF SEQ ID NOS: 93

<210> SEQ ID NO 1

<211> LENGTH: 549

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<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 1

Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
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Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
20          25          30
Ile Met Met Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35          40          45
Ser Ser Ile Tyr Pro Ser Gly Gly Ile Thr Phe Tyr Ala Asp Thr Val
50          55          60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
65          70          75          80
Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
85          90          95
Ala Arg Ile Lys Leu Gly Thr Val Thr Thr Val Asp Tyr Trp Gly Gln
100         105         110
Gly Thr Leu Val Thr Val Ser Ser Gly Gly Gly Gly Ser Gly Gly Gly
115         120         125
Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Ile Thr Cys
130         135         140
Pro Pro Pro Met Ser Val Glu His Ala Asp Ile Trp Val Lys Ser Tyr
145         150         155         160
Ser Leu Tyr Ser Arg Glu Arg Tyr Ile Cys Asn Ser Gly Phe Lys Arg
165         170         175
Lys Ala Gly Thr Ser Ser Leu Thr Glu Cys Val Leu Asn Lys Ala Thr
180         185         190
Asn Val Ala His Trp Thr Thr Pro Ser Leu Lys Cys Ile Arg Asp Pro
195         200         205
Ala Leu Val His Gln Arg Pro Ala Pro Pro Ser Thr Val Thr Thr Ala
210         215         220
Gly Val Thr Pro Gln Pro Glu Ser Leu Ser Pro Ser Gly Lys Glu Pro
225         230         235         240
Ala Ala Ser Ser Pro Ser Ser Asn Asn Thr Ala Ala Thr Thr Ala Ala
245         250         255
Ile Val Pro Gly Ser Gln Leu Met Pro Ser Lys Ser Pro Ser Thr Gly
260         265         270
Thr Thr Glu Ile Ser Ser His Glu Ser Ser His Gly Thr Pro Ser Gln
275         280         285
Thr Thr Ala Lys Asn Trp Glu Leu Thr Ala Ser Ala Ser His Gln Pro
290         295         300
Pro Gly Val Tyr Pro Gln Gly His Ser Asp Thr Thr Glu Pro Arg Gly
305         310         315         320
Pro Thr Ile Lys Pro Cys Pro Pro Cys Lys Cys Pro Ala Pro Asn Leu
325         330         335
Leu Gly Gly Pro Ser Val Phe Ile Phe Pro Pro Lys Ile Lys Asp Val
340         345         350
Leu Met Ile Ser Leu Ser Pro Ile Val Thr Cys Val Val Val Asp Val
355         360         365

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Ser Glu Asp Asp Pro Asp Val Gln Ile Ser Trp Phe Val Asn Asn Val
 370 375 380
 Glu Val His Thr Ala Gln Thr Gln Thr His Arg Glu Asp Tyr Asn Ser
 385 390 395 400
 Thr Leu Arg Val Val Ser Ala Leu Pro Ile Gln His Gln Asp Trp Met
 405 410 415
 Ser Gly Lys Glu Phe Lys Cys Lys Val Asn Asn Lys Asp Leu Pro Ala
 420 425 430
 Pro Ile Glu Arg Thr Ile Ser Lys Pro Lys Gly Ser Val Arg Ala Pro
 435 440 445
 Gln Val Tyr Val Leu Pro Pro Pro Glu Glu Glu Met Thr Lys Lys Gln
 450 455 460
 Val Thr Leu Trp Cys Met Val Thr Asp Phe Met Pro Glu Asp Ile Tyr
 465 470 475 480
 Val Glu Trp Thr Asn Asn Gly Lys Thr Glu Leu Asn Tyr Asp Asn Thr
 485 490 495
 Glu Pro Val Leu Asp Ser Asp Gly Ser Tyr Phe Met Tyr Ser Asp Leu
 500 505 510
 Arg Val Glu Lys Lys Asn Trp Val Glu Arg Asn Ser Tyr Ser Cys Ser
 515 520 525
 Val Val His Glu Gly Leu His Asn His His Thr Thr Lys Ser Phe Ser
 530 535 540
 Arg Thr Pro Gly Lys
 545

<210> SEQ ID NO 2
 <211> LENGTH: 245
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 2

Gln Ser Ala Leu Thr Gln Pro Ala Ser Val Ser Gly Ser Pro Gly Gln
 1 5 10 15
 Ser Ile Thr Ile Ser Cys Thr Gly Thr Ser Ser Asp Val Gly Gly Tyr
 20 25 30
 Asn Tyr Val Ser Trp Tyr Gln Gln His Pro Gly Lys Ala Pro Lys Leu
 35 40 45
 Met Ile Tyr Asp Val Ser Asn Arg Pro Ser Gly Val Ser Asn Arg Phe
 50 55 60
 Ser Gly Ser Lys Ser Gly Asn Thr Ala Ser Leu Thr Ile Ser Gly Leu
 65 70 75 80
 Gln Ala Glu Asp Glu Ala Asp Tyr Tyr Cys Ser Ser Tyr Thr Ser Ser
 85 90 95
 Ser Thr Arg Val Phe Gly Thr Gly Thr Lys Val Thr Val Leu Gly Gly
 100 105 110
 Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly
 115 120 125
 Gly Ser Gly Asn Trp Val Asn Val Ile Ser Asp Leu Lys Lys Ile Glu
 130 135 140
 Asp Leu Ile Gln Ser Met His Ile Asp Ala Thr Leu Tyr Thr Glu Ser
 145 150 155 160

-continued

Asp	Val	His	Pro	Ser	Cys	Lys	Val	Thr	Ala	Met	Lys	Cys	Phe	Leu	Leu
				165					170					175	
Glu	Leu	Gln	Val	Ile	Ser	Leu	Glu	Ser	Gly	Asp	Ala	Ser	Ile	His	Asp
			180					185					190		
Thr	Val	Glu	Asn	Leu	Ile	Ile	Leu	Ala	Asn	Asn	Ser	Leu	Ser	Ser	Asn
		195					200					205			
Gly	Asn	Val	Thr	Glu	Ser	Gly	Cys	Lys	Glu	Cys	Glu	Glu	Leu	Glu	Glu
	210					215					220				
Lys	Asn	Ile	Lys	Glu	Phe	Leu	Gln	Ser	Phe	Val	His	Ile	Val	Gln	Met
225					230					235					240
Phe	Ile	Asn	Thr	Ser											
				245											

<210> SEQ ID NO 3
 <211> LENGTH: 445
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 3

Gln	Val	Gln	Leu	Glu	Ser	Glu	Gly	Gly	Leu	Phe	Lys	Pro	Thr	Asp	Thr
1				5					10					15	
Leu	Thr	Leu	Thr	Cys	Thr	Val	Ser	Gly	Ser	Ser	Leu	Ser	Ser	Ser	Tyr
			20					25					30		
Met	Ser	Trp	Val	Arg	Gln	Ala	Pro	Gly	Lys	Gly	Leu	Glu	Trp	Ile	Gly
		35					40					45			
Ile	Ile	Gly	Ser	Asn	Gly	Asn	Thr	Tyr	Tyr	Ala	Asn	Trp	Ala	Lys	Gly
	50					55					60				
Arg	Phe	Thr	Ile	Ser	Lys	Thr	Ser	Thr	Thr	Val	Glu	Leu	Lys	Ile	Thr
65					70					75					80
Ser	Pro	Thr	Thr	Glu	Asp	Thr	Ala	Thr	Tyr	Phe	Cys	Ala	Arg	Gly	Gly
				85					90					95	
Tyr	Arg	Thr	Ser	Gly	Met	Asp	Pro	Trp	Gly	Pro	Gly	Thr	Leu	Val	Thr
			100					105					110		
Val	Ser	Ser	Ala	Lys	Thr	Thr	Ala	Pro	Ser	Val	Tyr	Pro	Leu	Ala	Pro
			115					120					125		
Val	Cys	Gly	Asp	Thr	Thr	Gly	Ser	Ser	Val	Thr	Leu	Gly	Cys	Leu	Val
	130					135					140				
Lys	Gly	Tyr	Phe	Pro	Glu	Pro	Val	Thr	Leu	Thr	Trp	Asn	Ser	Gly	Ser
145					150					155					160
Leu	Ser	Ser	Gly	Val	His	Thr	Phe	Pro	Ala	Val	Leu	Gln	Ser	Asp	Leu
			165						170						175
Tyr	Thr	Leu	Ser	Ser	Ser	Val	Thr	Val	Thr	Ser	Ser	Thr	Trp	Pro	Ser
			180					185						190	
Gln	Ser	Ile	Thr	Cys	Asn	Val	Ala	His	Pro	Ala	Ser	Ser	Thr	Lys	Val
		195						200					205		
Asp	Lys	Lys	Ile	Glu	Pro	Arg	Gly	Pro	Thr	Ile	Lys	Pro	Cys	Pro	Pro
	210					215					220				
Cys	Lys	Cys	Pro	Ala	Pro	Asn	Leu	Leu	Gly	Gly	Pro	Ser	Val	Phe	Ile
225					230					235					240
Phe	Pro	Pro	Lys	Ile	Lys	Asp	Val	Leu	Met	Ile	Ser	Leu	Ser	Pro	Ile
				245					250						255

-continued

Val	Thr	Cys	Val	Val	Val	Asp	Val	Ser	Glu	Asp	Asp	Pro	Asp	Val	Gln
			260					265					270		
Ile	Ser	Trp	Phe	Val	Asn	Asn	Val	Glu	Val	His	Thr	Ala	Gln	Thr	Gln
		275					280					285			
Thr	His	Arg	Glu	Asp	Tyr	Asn	Ser	Thr	Leu	Arg	Val	Val	Ser	Ala	Leu
		290				295					300				
Pro	Ile	Gln	His	Gln	Asp	Trp	Met	Ser	Gly	Lys	Glu	Phe	Lys	Cys	Lys
305					310					315					320
Val	Asn	Asn	Lys	Asp	Leu	Pro	Ala	Pro	Ile	Glu	Arg	Thr	Ile	Ser	Lys
				325						330				335	
Pro	Lys	Gly	Ser	Val	Arg	Ala	Pro	Gln	Val	Tyr	Val	Leu	Pro	Pro	Pro
				340				345					350		
Glu	Glu	Glu	Met	Thr	Lys	Lys	Gln	Val	Thr	Leu	Thr	Cys	Arg	Val	Thr
		355					360					365			
Asp	Phe	Met	Pro	Glu	Asp	Ile	Tyr	Val	Glu	Trp	Thr	Asn	Asn	Gly	Lys
	370					375					380				
Thr	Glu	Leu	Asn	Tyr	Lys	Asn	Thr	Glu	Pro	Val	Leu	Lys	Ser	Asp	Gly
385					390					395					400
Ser	Tyr	Phe	Met	Ala	Ser	Lys	Leu	Arg	Val	Glu	Lys	Lys	Asn	Trp	Val
				405					410					415	
Glu	Arg	Asn	Ser	Tyr	Ser	Cys	Ser	Val	Val	His	Glu	Gly	Leu	His	Asn
				420				425					430		
His	His	Thr	Thr	Lys	Ser	Phe	Ser	Arg	Thr	Pro	Gly	Lys			
		435					440					445			

<210> SEQ ID NO 4
 <211> LENGTH: 217
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 4

Asp	Ile	Val	Met	Thr	Gln	Thr	Pro	Ala	Ser	Val	Glu	Val	Ala	Val	Gly
1				5					10					15	
Gly	Thr	Val	Thr	Ile	Lys	Cys	Gln	Ala	Ser	Gln	Ser	Ile	Ser	Ser	Tyr
		20					25						30		
Leu	Asn	Trp	Tyr	Gln	Gln	Lys	Pro	Gly	Gln	Pro	Pro	Lys	Leu	Leu	Ile
		35				40						45			
Tyr	Asp	Ala	Leu	Lys	Leu	Ala	Ser	Gly	Val	Pro	Ser	Arg	Phe	Ser	Gly
	50				55					60					
Ser	Gly	Ser	Gly	Thr	Glu	Tyr	Thr	Leu	Thr	Ile	Ser	Gly	Val	Glu	Ser
65				70					75					80	
Ala	Asp	Ala	Ala	Thr	Tyr	Tyr	Cys	Gln	Gln	Glu	His	Ser	Val	Gly	Asn
				85				90						95	
Val	Asp	Asn	Val	Phe	Gly	Gly	Gly	Thr	Glu	Val	Val	Val	Lys	Arg	Thr
			100					105					110		
Asp	Ala	Ala	Pro	Thr	Val	Ser	Ile	Phe	Pro	Pro	Ser	Ser	Glu	Gln	Leu
		115					120					125			
Thr	Ser	Gly	Gly	Ala	Ser	Val	Val	Cys	Phe	Leu	Asn	Asn	Phe	Tyr	Pro
	130				135						140				
Lys	Asp	Ile	Asn	Val	Lys	Trp	Lys	Ile	Asp	Gly	Ser	Glu	Arg	Gln	Asn
145				150						155					160

-continued

Gly	Val	Leu	Asn	Ser	Trp	Thr	Asp	Gln	Asp	Ser	Lys	Asp	Ser	Thr	Tyr
			165						170					175	
Ser	Met	Ser	Ser	Thr	Leu	Thr	Leu	Thr	Lys	Asp	Glu	Tyr	Glu	Arg	His
			180					185					190		
Asn	Ser	Tyr	Thr	Cys	Glu	Ala	Thr	His	Lys	Thr	Ser	Thr	Ser	Pro	Ile
		195					200					205			
Val	Lys	Ser	Phe	Asn	Arg	Asn	Glu	Cys							
	210					215									

<210> SEQ ID NO 5
 <211> LENGTH: 544
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 5

Gln	Val	Gln	Leu	Glu	Ser	Glu	Gly	Gly	Leu	Phe	Lys	Pro	Thr	Asp	Thr
1			5						10					15	
Leu	Thr	Leu	Thr	Cys	Thr	Val	Ser	Gly	Ser	Ser	Leu	Ser	Ser	Ser	Tyr
		20						25					30		
Met	Ser	Trp	Val	Arg	Gln	Ala	Pro	Gly	Lys	Gly	Leu	Glu	Trp	Ile	Gly
		35				40						45			
Ile	Ile	Gly	Ser	Asn	Gly	Asn	Thr	Tyr	Tyr	Ala	Asn	Trp	Ala	Lys	Gly
	50					55					60				
Arg	Phe	Thr	Ile	Ser	Lys	Thr	Ser	Thr	Thr	Val	Glu	Leu	Lys	Ile	Thr
65					70					75				80	
Ser	Pro	Thr	Thr	Glu	Asp	Thr	Ala	Thr	Tyr	Phe	Cys	Ala	Arg	Gly	Gly
			85					90					95		
Tyr	Arg	Thr	Ser	Gly	Met	Asp	Pro	Trp	Gly	Pro	Gly	Thr	Leu	Val	Thr
		100					105						110		
Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly
		115					120					125			
Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Ile	Thr	Cys	Pro	Pro	Pro	Met	Ser
		130				135					140				
Val	Glu	His	Ala	Asp	Ile	Trp	Val	Lys	Ser	Tyr	Ser	Leu	Tyr	Ser	Arg
145					150					155					160
Glu	Arg	Tyr	Ile	Cys	Asn	Ser	Gly	Phe	Lys	Arg	Lys	Ala	Gly	Thr	Ser
			165					170						175	
Ser	Leu	Thr	Glu	Cys	Val	Leu	Asn	Lys	Ala	Thr	Asn	Val	Ala	His	Trp
		180						185					190		
Thr	Thr	Pro	Ser	Leu	Lys	Cys	Ile	Arg	Asp	Pro	Ala	Leu	Val	His	Gln
		195				200						205			
Arg	Pro	Ala	Pro	Pro	Ser	Thr	Val	Thr	Thr	Ala	Gly	Val	Thr	Pro	Gln
	210					215					220				
Pro	Glu	Ser	Leu	Ser	Pro	Ser	Gly	Lys	Glu	Pro	Ala	Ala	Ser	Ser	Pro
225					230					235					240
Ser	Ser	Asn	Asn	Thr	Ala	Ala	Thr	Thr	Ala	Ala	Ile	Val	Pro	Gly	Ser
			245					250						255	
Gln	Leu	Met	Pro	Ser	Lys	Ser	Pro	Ser	Thr	Gly	Thr	Thr	Glu	Ile	Ser
		260						265					270		
Ser	His	Glu	Ser	Ser	His	Gly	Thr	Pro	Ser	Gln	Thr	Thr	Ala	Lys	Asn
	275						280						285		

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Trp	Glu	Leu	Thr	Ala	Ser	Ala	Ser	His	Gln	Pro	Pro	Gly	Val	Tyr	Pro
	290					295					300				
Gln	Gly	His	Ser	Asp	Thr	Thr	Glu	Pro	Arg	Gly	Pro	Thr	Ile	Lys	Pro
305					310					315					320
Cys	Pro	Pro	Cys	Lys	Cys	Pro	Ala	Pro	Asn	Leu	Leu	Gly	Gly	Pro	Ser
			325						330					335	
Val	Phe	Ile	Phe	Pro	Pro	Lys	Ile	Lys	Asp	Val	Leu	Met	Ile	Ser	Leu
			340					345					350		
Ser	Pro	Ile	Val	Thr	Cys	Val	Val	Val	Asp	Val	Ser	Glu	Asp	Asp	Pro
		355					360					365			
Asp	Val	Gln	Ile	Ser	Trp	Phe	Val	Asn	Asn	Val	Glu	Val	His	Thr	Ala
	370					375					380				
Gln	Thr	Gln	Thr	His	Arg	Glu	Asp	Tyr	Asn	Ser	Thr	Leu	Arg	Val	Val
385					390					395					400
Ser	Ala	Leu	Pro	Ile	Gln	His	Gln	Asp	Trp	Met	Ser	Gly	Lys	Glu	Phe
			405						410					415	
Lys	Cys	Lys	Val	Asn	Asn	Lys	Asp	Leu	Pro	Ala	Pro	Ile	Glu	Arg	Thr
			420					425					430		
Ile	Ser	Lys	Pro	Lys	Gly	Ser	Val	Arg	Ala	Pro	Gln	Val	Tyr	Val	Leu
		435					440					445			
Pro	Pro	Pro	Glu	Glu	Glu	Met	Thr	Lys	Lys	Gln	Val	Thr	Leu	Trp	Cys
		450				455					460				
Met	Val	Thr	Asp	Phe	Met	Pro	Glu	Asp	Ile	Tyr	Val	Glu	Trp	Thr	Asn
465					470					475					480
Asn	Gly	Lys	Thr	Glu	Leu	Asn	Tyr	Asp	Asn	Thr	Glu	Pro	Val	Leu	Asp
			485						490					495	
Ser	Asp	Gly	Ser	Tyr	Phe	Met	Tyr	Ser	Asp	Leu	Arg	Val	Glu	Lys	Lys
			500					505					510		
Asn	Trp	Val	Glu	Arg	Asn	Ser	Tyr	Ser	Cys	Ser	Val	Val	His	Glu	Gly
		515					520					525			
Leu	His	Asn	His	His	Thr	Thr	Lys	Ser	Phe	Ser	Arg	Thr	Pro	Gly	Lys
	530					535					540				

<210> SEQ ID NO 6

<211> LENGTH: 245

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 6

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1				5					10					15	
Gly	Thr	Val	Thr	Ile	Lys	Cys	Gln	Ala	Ser	Gln	Ser	Ile	Ser	Ser	Tyr
			20					25					30		
Leu	Asn	Trp	Tyr	Gln	Gln	Lys	Pro	Gly	Gln	Pro	Pro	Lys	Leu	Leu	Ile
		35					40					45			
Tyr	Asp	Ala	Leu	Lys	Leu	Ala	Ser	Gly	Val	Pro	Ser	Arg	Phe	Ser	Gly
	50					55				60					
Ser	Gly	Ser	Gly	Thr	Glu	Tyr	Thr	Leu	Thr	Ile	Ser	Gly	Val	Glu	Ser
65				70						75				80	
Ala	Asp	Ala	Ala	Thr	Tyr	Tyr	Cys	Gln	Gln	Glu	His	Ser	Val	Gly	Asn
				85				90						95	

-continued

Val	Asp	Asn	Val	Phe	Gly	Gly	Gly	Thr	Glu	Val	Val	Val	Lys	Gly	Gly		
			100					105					110				
Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly		
		115					120					125					
Gly	Ser	Gly	Asn	Trp	Val	Asn	Val	Ile	Ser	Asp	Leu	Lys	Lys	Ile	Glu		
	130					135					140						
Asp	Leu	Ile	Gln	Ser	Met	His	Ile	Asp	Ala	Thr	Leu	Tyr	Thr	Glu	Ser		
145					150					155					160		
Asp	Val	His	Pro	Ser	Cys	Lys	Val	Thr	Ala	Met	Lys	Cys	Phe	Leu	Leu		
			165						170					175			
Glu	Leu	Gln	Val	Ile	Ser	Leu	Glu	Ser	Gly	Asp	Ala	Ser	Ile	His	Asp		
		180						185					190				
Thr	Val	Glu	Asn	Leu	Ile	Ile	Leu	Ala	Asn	Asn	Ser	Leu	Ser	Ser	Asn		
	195						200					205					
Gly	Asn	Val	Thr	Glu	Ser	Gly	Cys	Lys	Glu	Cys	Glu	Glu	Leu	Glu	Glu		
	210					215					220						
Lys	Asn	Ile	Lys	Glu	Phe	Leu	Gln	Ser	Phe	Val	His	Ile	Val	Gln	Met		
225					230					235					240		
Phe	Ile	Asn	Thr	Ser													
				245													

<210> SEQ ID NO 7
 <211> LENGTH: 450
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 7

Glu	Val	Gln	Leu	Leu	Glu	Ser	Gly	Gly	Gly	Leu	Val	Gln	Pro	Gly	Gly		
1				5					10					15			
Ser	Leu	Arg	Leu	Ser	Cys	Ala	Ala	Ser	Gly	Phe	Thr	Phe	Ser	Ser	Tyr		
		20						25					30				
Ile	Met	Met	Trp	Val	Arg	Gln	Ala	Pro	Gly	Lys	Gly	Leu	Glu	Trp	Val		
		35				40						45					
Ser	Ser	Ile	Tyr	Pro	Ser	Gly	Gly	Ile	Thr	Phe	Tyr	Ala	Asp	Thr	Val		
	50					55					60						
Lys	Gly	Arg	Phe	Thr	Ile	Ser	Arg	Asp	Asn	Ser	Lys	Asn	Thr	Leu	Tyr		
65					70				75						80		
Leu	Gln	Met	Asn	Ser	Leu	Arg	Ala	Glu	Asp	Thr	Ala	Val	Tyr	Tyr	Cys		
			85						90				95				
Ala	Arg	Ile	Lys	Leu	Gly	Thr	Val	Thr	Thr	Val	Asp	Tyr	Trp	Gly	Gln		
		100						105					110				
Gly	Thr	Leu	Val	Thr	Val	Ser	Ser	Ala	Lys	Thr	Thr	Ala	Pro	Ser	Val		
	115						120					125					
Tyr	Pro	Leu	Ala	Pro	Val	Cys	Gly	Asp	Thr	Thr	Gly	Ser	Ser	Val	Thr		
	130					135					140						
Leu	Gly	Cys	Leu	Val	Lys	Gly	Tyr	Phe	Pro	Glu	Pro	Val	Thr	Leu	Thr		
145					150					155					160		
Trp	Asn	Ser	Gly	Ser	Leu	Ser	Ser	Gly	Val	His	Thr	Phe	Pro	Ala	Val		
			165					170						175			
Leu	Gln	Ser	Asp	Leu	Tyr	Thr	Leu	Ser	Ser	Ser	Val	Thr	Val	Thr	Ser		
			180					185					190				

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Ser	Thr	Trp	Pro	Ser	Gln	Ser	Ile	Thr	Cys	Asn	Val	Ala	His	Pro	Ala
		195					200					205			
Ser	Ser	Thr	Lys	Val	Asp	Lys	Lys	Ile	Glu	Pro	Arg	Gly	Pro	Thr	Ile
	210					215					220				
Lys	Pro	Cys	Pro	Pro	Cys	Lys	Cys	Pro	Ala	Pro	Asn	Leu	Leu	Gly	Gly
225					230					235				240	
Pro	Ser	Val	Phe	Ile	Phe	Pro	Pro	Lys	Ile	Lys	Asp	Val	Leu	Met	Ile
			245						250					255	
Ser	Leu	Ser	Pro	Ile	Val	Thr	Cys	Val	Val	Val	Asp	Val	Ser	Glu	Asp
			260					265					270		
Asp	Pro	Asp	Val	Gln	Ile	Ser	Trp	Phe	Val	Asn	Asn	Val	Glu	Val	His
		275					280					285			
Thr	Ala	Gln	Thr	Gln	Thr	His	Arg	Glu	Asp	Tyr	Asn	Ser	Thr	Leu	Arg
	290					295					300				
Val	Val	Ser	Ala	Leu	Pro	Ile	Gln	His	Gln	Asp	Trp	Met	Ser	Gly	Lys
305					310					315				320	
Glu	Phe	Lys	Cys	Lys	Val	Asn	Asn	Lys	Asp	Leu	Pro	Ala	Pro	Ile	Glu
			325						330					335	
Arg	Thr	Ile	Ser	Lys	Pro	Lys	Gly	Ser	Val	Arg	Ala	Pro	Gln	Val	Tyr
			340					345					350		
Val	Leu	Pro	Pro	Pro	Glu	Glu	Glu	Met	Thr	Lys	Lys	Gln	Val	Thr	Leu
		355					360					365			
Thr	Cys	Arg	Val	Thr	Asp	Phe	Met	Pro	Glu	Asp	Ile	Tyr	Val	Glu	Trp
	370					375					380				
Thr	Asn	Asn	Gly	Lys	Thr	Glu	Leu	Asn	Tyr	Lys	Asn	Thr	Glu	Pro	Val
385					390					395				400	
Leu	Lys	Ser	Asp	Gly	Ser	Tyr	Phe	Met	Ala	Ser	Lys	Leu	Arg	Val	Glu
			405						410				415		
Lys	Lys	Asn	Trp	Val	Glu	Arg	Asn	Ser	Tyr	Ser	Cys	Ser	Val	Val	His
			420					425				430			
Glu	Gly	Leu	His	Asn	His	His	Thr	Thr	Lys	Ser	Phe	Ser	Arg	Thr	Pro
		435					440					445			
Gly	Lys														
	450														

<210> SEQ ID NO 8

<211> LENGTH: 217

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 8

Gln	Ser	Ala	Leu	Thr	Gln	Pro	Ala	Ser	Val	Ser	Gly	Ser	Pro	Gly	Gln
1				5					10				15		
Ser	Ile	Thr	Ile	Ser	Cys	Thr	Gly	Thr	Ser	Ser	Asp	Val	Gly	Gly	Tyr
		20						25					30		
Asn	Tyr	Val	Ser	Trp	Tyr	Gln	Gln	His	Pro	Gly	Lys	Ala	Pro	Lys	Leu
		35					40					45			
Met	Ile	Tyr	Asp	Val	Ser	Asn	Arg	Pro	Ser	Gly	Val	Ser	Asn	Arg	Phe
	50					55					60				
Ser	Gly	Ser	Lys	Ser	Gly	Asn	Thr	Ala	Ser	Leu	Thr	Ile	Ser	Gly	Leu
65					70					75				80	

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Gln Ala Glu Asp Glu Ala Asp Tyr Tyr Cys Ser Ser Tyr Thr Ser Ser
85 90 95

Ser Thr Arg Val Phe Gly Thr Gly Thr Lys Val Thr Val Leu Arg Thr
100 105 110

Asp Ala Ala Pro Thr Val Ser Ile Phe Pro Pro Ser Ser Glu Gln Leu
115 120 125

Thr Ser Gly Gly Ala Ser Val Val Cys Phe Leu Asn Asn Phe Tyr Pro
130 135 140

Lys Asp Ile Asn Val Lys Trp Lys Ile Asp Gly Ser Glu Arg Gln Asn
145 150 155 160

Gly Val Leu Asn Ser Trp Thr Asp Gln Asp Ser Lys Asp Ser Thr Tyr
165 170 175

Ser Met Ser Ser Thr Leu Thr Leu Thr Lys Asp Glu Tyr Glu Arg His
180 185 190

Asn Ser Tyr Thr Cys Glu Ala Thr His Lys Thr Ser Thr Ser Pro Ile
195 200 205

Val Lys Ser Phe Asn Arg Asn Glu Cys
210 215

<210> SEQ ID NO 9
 <211> LENGTH: 483
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 9

Gln Val Gln Leu Glu Ser Glu Gly Gly Leu Phe Lys Pro Thr Asp Thr
1 5 10 15

Leu Thr Leu Thr Cys Thr Val Ser Gly Ser Ser Leu Ser Ser Ser Tyr
20 25 30

Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Ile Gly
35 40 45

Ile Ile Gly Ser Asn Gly Asn Thr Tyr Tyr Ala Asn Trp Ala Lys Gly
50 55 60

Arg Phe Thr Ile Ser Lys Thr Ser Thr Thr Val Glu Leu Lys Ile Thr
65 70 75 80

Ser Pro Thr Thr Glu Asp Thr Ala Thr Tyr Phe Cys Ala Arg Gly Gly
85 90 95

Tyr Arg Thr Ser Gly Met Asp Pro Trp Gly Pro Gly Thr Leu Val Thr
100 105 110

Val Ser Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly
115 120 125

Gly Ser Gly Gly Gly Gly Ser Gly Asn Trp Val Asn Val Ile Ser Asp
130 135 140

Leu Lys Lys Ile Glu Asp Leu Ile Gln Ser Met His Ile Asp Ala Thr
145 150 155 160

Leu Tyr Thr Glu Ser Asp Val His Pro Ser Cys Lys Val Thr Ala Met
165 170 175

Lys Cys Phe Leu Leu Glu Leu Gln Val Ile Ser Leu Glu Ser Gly Asp
180 185 190

Ala Ser Ile His Asp Thr Val Glu Asn Leu Ile Ile Leu Ala Asn Asn
195 200 205

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Ser	Leu	Ser	Ser	Asn	Gly	Asn	Val	Thr	Glu	Ser	Gly	Cys	Lys	Glu	Cys
210						215					220				
Glu	Glu	Leu	Glu	Glu	Lys	Asn	Ile	Lys	Glu	Phe	Leu	Gln	Ser	Phe	Val
225					230					235					240
His	Ile	Val	Gln	Met	Phe	Ile	Asn	Thr	Ser	Glu	Pro	Arg	Gly	Pro	Thr
			245						250					255	
Ile	Lys	Pro	Cys	Pro	Pro	Cys	Lys	Cys	Pro	Ala	Pro	Asn	Leu	Leu	Gly
			260					265					270		
Gly	Pro	Ser	Val	Phe	Ile	Phe	Pro	Pro	Lys	Ile	Lys	Asp	Val	Leu	Met
	275						280					285			
Ile	Ser	Leu	Ser	Pro	Ile	Val	Thr	Cys	Val	Val	Val	Asp	Val	Ser	Glu
	290					295					300				
Asp	Asp	Pro	Asp	Val	Gln	Ile	Ser	Trp	Phe	Val	Asn	Asn	Val	Glu	Val
305					310					315					320
His	Thr	Ala	Gln	Thr	Gln	Thr	His	Arg	Glu	Asp	Tyr	Asn	Ser	Thr	Leu
			325						330					335	
Arg	Val	Val	Ser	Ala	Leu	Pro	Ile	Gln	His	Gln	Asp	Trp	Met	Ser	Gly
			340					345					350		
Lys	Glu	Phe	Lys	Cys	Lys	Val	Asn	Asn	Lys	Asp	Leu	Pro	Ala	Pro	Ile
		355					360					365			
Glu	Arg	Thr	Ile	Ser	Lys	Pro	Lys	Gly	Ser	Val	Arg	Ala	Pro	Gln	Val
	370					375					380				
Tyr	Val	Leu	Pro	Pro	Pro	Glu	Glu	Glu	Met	Thr	Lys	Lys	Gln	Val	Thr
385					390					395					400
Leu	Trp	Cys	Met	Val	Thr	Asp	Phe	Met	Pro	Glu	Asp	Ile	Tyr	Val	Glu
			405					410					415		
Trp	Thr	Asn	Asn	Gly	Lys	Thr	Glu	Leu	Asn	Tyr	Asp	Asn	Thr	Glu	Pro
		420					425						430		
Val	Leu	Asp	Ser	Asp	Gly	Ser	Tyr	Phe	Met	Tyr	Ser	Asp	Leu	Arg	Val
		435					440					445			
Glu	Lys	Lys	Asn	Trp	Val	Glu	Arg	Asn	Ser	Tyr	Ser	Cys	Ser	Val	Val
	450					455					460				
His	Glu	Gly	Leu	His	Asn	His	His	Thr	Thr	Lys	Ser	Phe	Ser	Arg	Thr
465				470						475					480

Pro Gly Lys

<210> SEQ ID NO 10
 <211> LENGTH: 306
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 10

Asp	Ile	Val	Met	Thr	Gln	Thr	Pro	Ala	Ser	Val	Glu	Val	Ala	Val	Gly
1				5				10						15	
Gly	Thr	Val	Thr	Ile	Lys	Cys	Gln	Ala	Ser	Gln	Ser	Ile	Ser	Ser	Tyr
		20					25					30			
Leu	Asn	Trp	Tyr	Gln	Gln	Lys	Pro	Gly	Gln	Pro	Pro	Lys	Leu	Leu	Ile
	35					40						45			
Tyr	Asp	Ala	Leu	Lys	Leu	Ala	Ser	Gly	Val	Pro	Ser	Arg	Phe	Ser	Gly
	50				55					60					

Ser Gly Ser Gly Thr Glu Tyr Thr Leu Thr Ile Ser Gly Val Glu Ser

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65	70	75	80
Ala Asp Ala Ala Thr Tyr Tyr Cys Gln Gln Glu His Ser Val Gly Asn	85	90	95
Val Asp Asn Val Phe Gly Gly Gly Thr Glu Val Val Val Lys Gly Gly	100	105	110
Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly	115	120	125
Gly Ser Gly Ile Thr Cys Pro Pro Pro Met Ser Val Glu His Ala Asp	130	135	140
Ile Trp Val Lys Ser Tyr Ser Leu Tyr Ser Arg Glu Arg Tyr Ile Cys	145	150	155
Asn Ser Gly Phe Lys Arg Lys Ala Gly Thr Ser Ser Leu Thr Glu Cys	165	170	175
Val Leu Asn Lys Ala Thr Asn Val Ala His Trp Thr Thr Pro Ser Leu	180	185	190
Lys Cys Ile Arg Asp Pro Ala Leu Val His Gln Arg Pro Ala Pro Pro	195	200	205
Ser Thr Val Thr Thr Ala Gly Val Thr Pro Gln Pro Glu Ser Leu Ser	210	215	220
Pro Ser Gly Lys Glu Pro Ala Ala Ser Ser Pro Ser Ser Asn Asn Thr	225	230	235
Ala Ala Thr Thr Ala Ala Ile Val Pro Gly Ser Gln Leu Met Pro Ser	245	250	255
Lys Ser Pro Ser Thr Gly Thr Thr Glu Ile Ser Ser His Glu Ser Ser	260	265	270
His Gly Thr Pro Ser Gln Thr Thr Ala Lys Asn Trp Glu Leu Thr Ala	275	280	285
Ser Ala Ser His Gln Pro Pro Gly Val Tyr Pro Gln Gly His Ser Asp	290	295	300
Thr Thr			
305			

<210> SEQ ID NO 11
 <211> LENGTH: 544
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 11

Gln Val Gln Leu Glu Ser Glu Gly Gly Leu Phe Lys Pro Thr Asp Thr	1	5	10	15
Leu Thr Leu Thr Cys Thr Val Ser Gly Ser Ser Leu Ser Ser Ser Tyr	20	25	30	
Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Ile Gly	35	40	45	
Ile Ile Gly Ser Asn Gly Asn Thr Tyr Tyr Ala Asn Trp Ala Lys Gly	50	55	60	
Arg Phe Thr Ile Ser Lys Thr Ser Thr Thr Val Glu Leu Lys Ile Thr	65	70	75	80
Ser Pro Thr Thr Glu Asp Thr Ala Thr Tyr Phe Cys Ala Arg Gly Gly	85	90	95	
Tyr Arg Thr Ser Gly Met Asp Pro Trp Gly Pro Gly Thr Leu Val Thr				

[illegible]

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Asn Trp Val Glu Arg Asn Ser Tyr Ser Cys Ser Val Val His Glu Gly
515 520 525

Leu His Asn His His Thr Thr Lys Ser Phe Ser Arg Thr Pro Gly Lys
530 535 540

<210> SEQ ID NO 12
<211> LENGTH: 242
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 12

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1 5 10 15

Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Glu Asn Ile Tyr Ser Asn
20 25 30

Leu Ala Trp Tyr Gln Gln Lys Pro Gly Lys Ser Pro Lys Leu Leu Val
35 40 45

Tyr Ala Ala Ser His Leu Pro Asp Gly Val Pro Ser Arg Phe Ser Gly
50 55 60

Ser Gly Ser Gly Thr Asp Tyr Ser Leu Thr Ile Ser Ser Leu Gln Pro
65 70 75 80

Glu Asp Phe Ala Thr Tyr Tyr Cys Gln His Phe Trp Gly Thr Pro Arg
85 90 95

Thr Phe Gly Gln Gly Thr Lys Leu Glu Ile Lys Gly Gly Gly Ser
100 105 110

Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Ser Gly
115 120 125

Asn Trp Val Asn Val Ile Ser Asp Leu Lys Lys Ile Glu Asp Leu Ile
130 135 140

Gln Ser Met His Ile Asp Ala Thr Leu Tyr Thr Glu Ser Asp Val His
145 150 155 160

Pro Ser Cys Lys Val Thr Ala Met Lys Cys Phe Leu Leu Glu Leu Gln
165 170 175

Val Ile Ser Leu Glu Ser Gly Asp Ala Ser Ile His Asn Thr Val Gln
180 185 190

Asp Leu Ile Ile Leu Ala Asn Asn Ser Leu Ser Ser Asn Gly Asn Val
195 200 205

Thr Glu Ser Gly Cys Lys Glu Cys Glu Glu Leu Glu Cys Lys Asn Ile
210 215 220

Lys Glu Phe Leu Gln Ser Phe Val His Ile Val Gln Met Phe Ile Asn
225 230 235 240

Thr Ser

<210> SEQ ID NO 13
<211> LENGTH: 544
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 13

Gln Val Gln Leu Glu Ser Glu Gly Gly Leu Phe Lys Pro Thr Asp Thr
1 5 10 15

Leu	Thr	Leu	Thr	Cys	Thr	Val	Ser	Gly	Ser	Ser	Leu	Ser	Ser	Thr	
			20				25				30				
Met	Ser	Trp	Val	Arg	Gln	Ala	Pro	Gly	Lys	Gly	Leu	Glu	Trp	Ile	Gly
			35				40				45				
Ile	Ile	Gly	Ser	Asn	Gly	Asn	Thr	Tyr	Tyr	Ala	Asn	Trp	Ala	Lys	Gly
			50				55				60				
Arg	Phe	Thr	Ile	Ser	Lys	Thr	Ser	Thr	Thr	Val	Glu	Leu	Lys	Ile	Thr
			65				70				75				
Ser	Pro	Thr	Thr	Glu	Asp	Thr	Ala	Thr	Tyr	Phe	Cys	Ala	Arg	Gly	Gly
			85				90				95				
Tyr	Arg	Thr	Ser	Gly	Met	Asp	Pro	Trp	Gly	Pro	Gly	Thr	Leu	Val	Thr
			100				105				110				
Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly
			115				120				125				
Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Ile	Thr	Cys	Pro	Pro	Pro	Met	Ser
			130				135				140				
Val	Glu	His	Ala	Asp	Ile	Trp	Val	Lys	Ser	Tyr	Ser	Leu	Tyr	Ser	Arg
			145				150				155				
Glu	Arg	Tyr	Ile	Cys	Asn	Ser	Gly	Phe	Lys	Arg	Lys	Ala	Gly	Thr	Ser
			165				170				175				
Ser	Leu	Thr	Glu	Cys	Val	Leu	Asn	Lys	Ala	Thr	Asn	Val	Ala	His	Trp
			180				185				190				
Thr	Thr	Pro	Ser	Leu	Lys	Cys	Ile	Arg	Asp	Pro	Ala	Leu	Val	His	Gln
			195				200				205				
Arg	Pro	Ala	Pro	Pro	Ser	Thr	Val	Thr	Thr	Ala	Gly	Val	Thr	Pro	Gln
			210				215				220				
Pro	Glu	Ser	Leu	Ser	Pro	Ser	Gly	Lys	Glu	Pro	Ala	Ala	Ser	Ser	Pro
			225				230				235				
Ser	Ser	Asn	Asn	Thr	Ala	Ala	Thr	Thr	Ala	Ala	Ile	Val	Pro	Gly	Ser
			245				250				255				
Gln	Leu	Met	Pro	Ser	Lys	Ser	Pro	Ser	Thr	Gly	Thr	Thr	Glu	Ile	Ser
			260				265				270				
Ser	His	Glu	Ser	Ser	His	Gly	Thr	Pro	Ser	Gln	Thr	Thr	Ala	Lys	Asn
			275				280				285				
Trp	Glu	Leu	Thr	Ala	Ser	Ala	Ser	His	Gln	Pro	Pro	Gly	Val	Tyr	Pro
			290				295				300				
Gln	Gly	His	Ser	Asp	Thr	Thr	Glu	Pro	Arg	Gly	Pro	Thr	Ile	Lys	Pro
			305				310				315				
Cys	Pro	Pro	Cys	Lys	Cys	Pro	Ala	Pro	Asn	Leu	Leu	Gly	Gly	Pro	Ser
			325				330				335				
Val	Phe	Ile	Phe	Pro	Pro	Lys	Ile	Lys	Asp	Val	Leu	Met	Ile	Ser	Leu
			340				345				350				
Ser	Pro	Ile	Val	Thr	Cys	Val	Val	Val	Asp	Val	Ser	Glu	Asp	Asp	Pro
			355				360				365				
Asp	Val	Gln	Ile	Ser	Trp	Phe	Val	Asn	Asn	Val	Glu	Val	His	Thr	Ala
			370				375				380				
Gln	Thr	Gln	Thr	His	Arg	Glu	Asp	Tyr	Asn	Ser	Thr	Leu	Arg	Val	Val
			385				390				395				
Ser	Ala	Leu	Pro	Ile	Gln	His	Gln	Asp	Trp	Met	Ser	Gly	Lys	Glu	Phe
			405				410				415				

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Lys	Cys	Lys	Val	Asn	Asn	Lys	Asp	Leu	Pro	Ala	Pro	Ile	Glu	Arg	Thr
			420					425					430		
Ile	Ser	Lys	Pro	Lys	Gly	Ser	Val	Arg	Ala	Pro	Gln	Val	Tyr	Val	Leu
		435					440					445			
Pro	Pro	Pro	Glu	Glu	Glu	Met	Thr	Lys	Lys	Gln	Val	Thr	Leu	Trp	Cys
		450				455					460				
Met	Val	Thr	Asp	Phe	Met	Pro	Glu	Asp	Ile	Tyr	Val	Glu	Trp	Thr	Asn
465					470					475					480
Asn	Gly	Lys	Thr	Glu	Leu	Asn	Tyr	Asp	Asn	Thr	Glu	Pro	Val	Leu	Asp
			485						490					495	
Ser	Asp	Gly	Ser	Tyr	Phe	Met	Tyr	Ser	Asp	Leu	Arg	Val	Glu	Lys	Lys
			500					505					510		
Asn	Trp	Val	Glu	Arg	Asn	Ser	Tyr	Ser	Cys	Ser	Val	Val	His	Glu	Gly
		515					520					525			
Leu	His	Asn	His	His	Thr	Thr	Lys	Ser	Phe	Ser	Arg	Thr	Pro	Gly	Lys
	530					535					540				

<210> SEQ ID NO 14
 <211> LENGTH: 245
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 14

Asp	Ile	Val	Met	Thr	Gln	Thr	Pro	Ala	Ser	Val	Glu	Val	Ala	Val	Gly
1				5					10					15	
Gly	Thr	Val	Thr	Ile	Lys	Cys	Gln	Ala	Ser	Gln	Ser	Ile	Ser	Ser	Tyr
			20					25					30		
Leu	Asn	Trp	Tyr	Gln	Gln	Lys	Pro	Gly	Gln	Pro	Pro	Lys	Leu	Leu	Ile
		35					40					45			
Tyr	Asp	Ala	Leu	Lys	Leu	Ala	Ser	Gly	Val	Pro	Ser	Arg	Phe	Ser	Gly
	50					55					60				
Ser	Gly	Ser	Gly	Thr	Glu	Tyr	Thr	Leu	Thr	Ile	Ser	Gly	Val	Glu	Ser
65				70						75				80	
Ala	Asp	Ala	Ala	Thr	Tyr	Tyr	Cys	Gln	Gln	Glu	His	Ser	Val	Gly	Asn
			85					90						95	
Val	Asp	Asn	Val	Phe	Gly	Gly	Gly	Thr	Glu	Val	Val	Val	Lys	Gly	Gly
		100						105					110		
Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly
		115					120					125			
Gly	Ser	Gly	Asn	Trp	Val	Asn	Val	Ile	Ser	Asp	Leu	Lys	Lys	Ile	Glu
	130					135					140				
Asp	Leu	Ile	Gln	Ser	Met	His	Ile	Asp	Ala	Thr	Leu	Tyr	Thr	Glu	Ser
145					150					155					160
Asp	Val	His	Pro	Ser	Cys	Lys	Val	Thr	Ala	Met	Lys	Cys	Phe	Leu	Leu
			165						170					175	
Glu	Leu	Gln	Val	Ile	Ser	Leu	Glu	Ser	Gly	Asp	Ala	Ser	Ile	His	Asp
		180					185						190		
Thr	Val	Glu	Asn	Leu	Ile	Ile	Leu	Ala	Asn	Asn	Ser	Leu	Ser	Ser	Asn
	195						200					205			
Gly	Asn	Val	Thr	Glu	Ser	Gly	Cys	Lys	Cys	Cys	Glu	Glu	Leu	Glu	Glu
	210					215					220				

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Lys Asn Ile Lys Glu Phe Leu Gln Ser Phe Val His Ile Val Gln Met
225 230 235 240

Phe Ile Asn Thr Ser
245

<210> SEQ ID NO 15
<211> LENGTH: 245
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 15

Asp Ile Val Met Thr Gln Thr Pro Ala Ser Val Glu Val Ala Val Gly
1 5 10 15

Gly Thr Val Thr Ile Lys Cys Gln Ala Ser Gln Ser Ile Ser Ser Tyr
20 25 30

Leu Asn Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu Leu Ile
35 40 45

Tyr Asp Ala Leu Lys Leu Ala Ser Gly Val Pro Ser Arg Phe Ser Gly
50 55 60

Ser Gly Ser Gly Thr Glu Tyr Thr Leu Thr Ile Ser Gly Val Glu Ser
65 70 75 80

Ala Asp Ala Ala Thr Tyr Tyr Cys Gln Gln Glu His Ser Val Gly Asn
85 90 95

Val Asp Asn Val Phe Gly Gly Gly Thr Glu Val Val Val Lys Gly Gly
100 105 110

Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly
115 120 125

Gly Ser Gly Asn Trp Val Asn Val Ile Ser Asp Leu Lys Lys Ile Glu
130 135 140

Asp Leu Ile Gln Ser Met His Ile Asp Ala Thr Leu Tyr Thr Glu Ser
145 150 155 160

Asp Val His Pro Ser Cys Lys Val Thr Ala Met Lys Cys Phe Leu Leu
165 170 175

Glu Leu Gln Val Ile Ser Cys Glu Ser Gly Asp Ala Ser Ile His Asp
180 185 190

Thr Val Glu Asn Leu Ile Ile Leu Ala Gln Asn Ser Leu Ser Ser Asn
195 200 205

Gly Gln Val Thr Glu Ser Gly Cys Lys Glu Cys Glu Glu Leu Glu Glu
210 215 220

Lys Asn Ile Lys Glu Phe Leu Gln Ser Phe Val His Ile Val Gln Met
225 230 235 240

Phe Ile Gln Thr Ser
245

<210> SEQ ID NO 16
<211> LENGTH: 245
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 16

Asp Ile Val Met Thr Gln Thr Pro Ala Ser Val Glu Val Ala Val Gly
1 5 10 15

-continued

Gly Thr Val Thr Ile Lys Cys Gln Ala Ser Gln Ser Ile Ser Ser Tyr
 20 25 30
 Leu Asn Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu Leu Ile
 35 40 45
 Tyr Asp Ala Leu Lys Leu Ala Ser Gly Val Pro Ser Arg Phe Ser Gly
 50 55 60
 Ser Gly Ser Gly Thr Glu Tyr Thr Leu Thr Ile Ser Gly Val Glu Ser
 65 70 75 80
 Ala Asp Ala Ala Thr Tyr Tyr Cys Gln Gln Glu His Ser Val Gly Asn
 85 90 95
 Val Asp Asn Val Phe Gly Gly Gly Thr Glu Val Val Val Lys Gly Gly
 100 105 110
 Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly
 115 120 125
 Gly Ser Gly Asn Trp Val Asn Val Ile Ser Asp Leu Lys Lys Ile Glu
 130 135 140
 Asp Leu Ile Gln Ser Met His Ile Asp Ala Thr Leu Tyr Thr Glu Ser
 145 150 155 160
 Asp Val His Pro Ser Cys Lys Val Thr Ala Met Lys Cys Phe Leu Leu
 165 170 175
 Glu Leu Gln Val Ile Ser Leu Glu Ser Gly Asp Ala Ser Ile His Asn
 180 185 190
 Thr Val Gln Asp Leu Ile Ile Leu Ala Asn Asn Ser Leu Ser Ser Asn
 195 200 205
 Gly Asn Val Thr Glu Ser Gly Cys Lys Glu Cys Glu Glu Leu Glu Glu
 210 215 220
 Lys Asn Ile Lys Glu Phe Leu Gln Ser Phe Val His Ile Val Gln Met
 225 230 235 240
 Phe Ile Asn Thr Ser
 245

<210> SEQ ID NO 17
 <211> LENGTH: 477
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 17

Gln Val Gln Leu Glu Ser Glu Gly Gly Leu Phe Lys Pro Thr Asp Thr
 1 5 10 15
 Leu Thr Leu Thr Cys Thr Val Ser Gly Ser Ser Leu Ser Ser Tyr
 20 25 30
 Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Ile Gly
 35 40 45
 Ile Ile Gly Ser Asn Gly Asn Thr Tyr Tyr Ala Asn Trp Ala Lys Gly
 50 55 60
 Arg Phe Thr Ile Ser Lys Thr Ser Thr Thr Val Glu Leu Lys Ile Thr
 65 70 75 80
 Ser Pro Thr Thr Glu Asp Thr Ala Thr Tyr Phe Cys Ala Arg Gly Gly
 85 90 95
 Tyr Arg Thr Ser Gly Met Asp Pro Trp Gly Pro Gly Thr Leu Val Thr
 100 105 110


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<210> SEQ ID NO 18
<211> LENGTH: 456
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
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-continued

<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 18

Gln Val Gln Leu Glu Ser Glu Gly Gly Leu Phe Lys Pro Thr Asp Thr
 1 5 10 15
 Leu Thr Leu Thr Cys Thr Val Ser Gly Ser Ser Leu Ser Ser Ser Tyr
 20 25 30
 Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Ile Gly
 35 40 45
 Ile Ile Gly Ser Asn Gly Asn Thr Tyr Tyr Ala Asn Trp Ala Lys Gly
 50 55 60
 Arg Phe Thr Ile Ser Lys Thr Ser Thr Thr Val Glu Leu Lys Ile Thr
 65 70 75 80
 Ser Pro Thr Thr Glu Asp Thr Ala Thr Tyr Phe Cys Ala Arg Gly Gly
 85 90 95
 Tyr Arg Thr Ser Gly Met Asp Pro Trp Gly Pro Gly Thr Leu Val Thr
 100 105 110
 Val Ser Ser Leu Arg Thr Asp Ala Ala Pro Thr Val Ser Ile Phe Pro
 115 120 125
 Pro Ser Ser Glu Gln Leu Thr Ser Gly Gly Ala Ser Val Val Cys Phe
 130 135 140
 Leu Asn Asn Phe Tyr Pro Lys Asp Ile Asn Val Lys Trp Lys Ile Asp
 145 150 155 160
 Gly Ser Glu Arg Gln Asn Gly Val Leu Asn Ser Trp Thr Asp Gln Asp
 165 170 175
 Ser Lys Asp Ser Thr Tyr Ser Met Ser Ser Thr Leu Thr Leu Thr Lys
 180 185 190
 Asp Glu Tyr Glu Arg His Asn Ser Tyr Thr Cys Glu Ala Thr His Lys
 195 200 205
 Thr Ser Thr Ser Pro Ile Val Lys Ser Phe Asn Arg Asn Glu Cys Glu
 210 215 220
 Pro Arg Gly Pro Thr Ile Lys Pro Cys Pro Pro Cys Lys Cys Pro Ala
 225 230 235 240
 Pro Asn Leu Leu Gly Gly Pro Ser Val Phe Ile Phe Pro Pro Lys Ile
 245 250 255
 Lys Asp Val Leu Met Ile Ser Leu Ser Pro Ile Val Thr Cys Val Val
 260 265 270
 Val Asp Val Ser Glu Asp Asp Pro Asp Val Gln Ile Ser Trp Phe Val
 275 280 285
 Asn Asn Val Glu Val His Thr Ala Gln Thr Gln Thr His Arg Glu Asp
 290 295 300
 Tyr Asn Ser Thr Leu Arg Val Val Ser Ala Leu Pro Ile Gln His Gln
 305 310 315 320
 Asp Trp Met Ser Gly Lys Glu Phe Lys Cys Lys Val Asn Asn Lys Asp
 325 330 335
 Leu Pro Ala Pro Ile Glu Arg Thr Ile Ser Lys Pro Lys Gly Ser Val
 340 345 350
 Arg Ala Pro Gln Val Tyr Val Leu Pro Pro Pro Glu Glu Glu Met Thr
 355 360 365
 Lys Lys Gln Val Thr Leu Trp Cys Met Val Thr Asp Phe Met Pro Glu
 370 375 380

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Asp	Ile	Tyr	Val	Glu	Trp	Thr	Asn	Asn	Gly	Lys	Thr	Glu	Leu	Asn	Tyr
385					390					395					400

Asp	Asn	Thr	Glu	Pro	Val	Leu	Asp	Ser	Asp	Gly	Ser	Tyr	Phe	Met	Tyr
			405						410					415	

Ser	Asp	Leu	Arg	Val	Glu	Lys	Lys	Asn	Trp	Val	Glu	Arg	Asn	Ser	Tyr
		420					425						430		

Ser	Cys	Ser	Val	Val	His	Glu	Gly	Leu	His	Asn	His	His	Thr	Thr	Lys
	435						440					445			

Ser	Phe	Ser	Arg	Thr	Pro	Gly	Lys
450						455	

<210> SEQ ID NO 19
 <211> LENGTH: 207
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 19

Asp	Ile	Val	Met	Thr	Gln	Thr	Pro	Ala	Ser	Val	Glu	Val	Ala	Val	Gly
1				5					10					15	

Gly	Thr	Val	Thr	Ile	Lys	Cys	Gln	Ala	Ser	Gln	Ser	Ile	Ser	Ser	Tyr
		20					25						30		

Leu	Asn	Trp	Tyr	Gln	Gln	Lys	Pro	Gly	Gln	Pro	Pro	Lys	Leu	Leu	Ile
	35					40						45			

Tyr	Asp	Ala	Leu	Lys	Leu	Ala	Ser	Gly	Val	Pro	Ser	Arg	Phe	Ser	Gly
	50				55					60					

Ser	Gly	Ser	Gly	Thr	Glu	Tyr	Thr	Leu	Thr	Ile	Ser	Gly	Val	Glu	Ser
65				70					75					80	

Ala	Asp	Ala	Ala	Thr	Tyr	Tyr	Cys	Gln	Gln	Glu	His	Ser	Val	Gly	Asn
		85						90						95	

Val	Asp	Asn	Val	Phe	Gly	Gly	Gly	Thr	Glu	Val	Val	Val	Lys	Ala	Lys
		100					105						110		

Thr	Thr	Ala	Pro	Ser	Val	Tyr	Pro	Leu	Ala	Pro	Val	Cys	Gly	Asp	Thr
		115				120						125			

Thr	Gly	Ser	Ser	Val	Thr	Leu	Gly	Cys	Leu	Val	Lys	Gly	Tyr	Phe	Pro
	130				135						140				

Glu	Pro	Val	Thr	Leu	Thr	Trp	Asn	Ser	Gly	Ser	Leu	Ser	Ser	Gly	Val
145				150					155					160	

His	Thr	Phe	Pro	Ala	Val	Leu	Gln	Ser	Asp	Leu	Tyr	Thr	Leu	Ser	Ser
		165						170					175		

Ser	Val	Thr	Val	Thr	Ser	Ser	Thr	Trp	Pro	Ser	Gln	Ser	Ile	Thr	Cys
		180					185						190		

Asn	Val	Ala	His	Pro	Ala	Ser	Ser	Thr	Lys	Val	Asp	Lys	Lys	Ile
	195					200					205			

<210> SEQ ID NO 20
 <211> LENGTH: 710
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 20

Glu	Val	Gln	Leu	Leu	Glu	Ser	Gly	Gly	Gly	Leu	Val	Gln	Pro	Gly	Gly
1			5					10					15		

-continued

Ser	Leu	Arg	Leu	Ser	Cys	Ala	Ala	Ser	Gly	Phe	Thr	Phe	Ser	Ser	Tyr
			20					25					30		
Ile	Met	Met	Trp	Val	Arg	Gln	Ala	Pro	Gly	Lys	Gly	Leu	Glu	Trp	Val
		35					40					45			
Ser	Ser	Ile	Tyr	Pro	Ser	Gly	Gly	Ile	Thr	Phe	Tyr	Ala	Asp	Thr	Val
	50					55					60				
Lys	Gly	Arg	Phe	Thr	Ile	Ser	Arg	Asp	Asn	Ser	Lys	Asn	Thr	Leu	Tyr
65					70					75					80
Leu	Gln	Met	Asn	Ser	Leu	Arg	Ala	Glu	Asp	Thr	Ala	Val	Tyr	Tyr	Cys
			85						90				95		
Ala	Arg	Ile	Lys	Leu	Gly	Thr	Val	Thr	Thr	Val	Asp	Tyr	Trp	Gly	Gln
			100					105					110		
Gly	Thr	Leu	Val	Thr	Val	Ser	Ser	Ala	Lys	Thr	Thr	Ala	Pro	Ser	Val
		115						120				125			
Tyr	Pro	Leu	Ala	Pro	Val	Cys	Gly	Asp	Thr	Thr	Gly	Ser	Ser	Val	Thr
	130					135					140				
Leu	Gly	Cys	Leu	Val	Lys	Gly	Tyr	Phe	Pro	Glu	Pro	Val	Thr	Leu	Thr
145					150					155					160
Trp	Asn	Ser	Gly	Ser	Leu	Ser	Ser	Gly	Val	His	Thr	Phe	Pro	Ala	Val
			165						170					175	
Leu	Gln	Ser	Asp	Leu	Tyr	Thr	Leu	Ser	Ser	Ser	Val	Thr	Val	Thr	Ser
			180					185					190		
Ser	Thr	Trp	Pro	Ser	Gln	Ser	Ile	Thr	Cys	Asn	Val	Ala	His	Pro	Ala
		195					200					205			
Ser	Ser	Thr	Lys	Val	Asp	Lys	Lys	Ile	Glu	Pro	Arg	Gly	Pro	Thr	Ile
	210					215					220				
Lys	Pro	Cys	Pro	Pro	Cys	Lys	Cys	Pro	Ala	Pro	Asn	Leu	Leu	Gly	Gly
225					230					235					240
Pro	Ser	Val	Phe	Ile	Phe	Pro	Pro	Lys	Ile	Lys	Asp	Val	Leu	Met	Ile
			245						250					255	
Ser	Leu	Ser	Pro	Ile	Val	Thr	Cys	Val	Val	Val	Asp	Val	Ser	Glu	Asp
			260					265					270		
Asp	Pro	Asp	Val	Gln	Ile	Ser	Trp	Phe	Val	Asn	Asn	Val	Glu	Val	His
		275					280					285			
Thr	Ala	Gln	Thr	Gln	Thr	His	Arg	Glu	Asp	Tyr	Asn	Ser	Thr	Leu	Arg
	290					295					300				
Val	Val	Ser	Ala	Leu	Pro	Ile	Gln	His	Gln	Asp	Trp	Met	Ser	Gly	Lys
305					310					315					320
Glu	Phe	Lys	Cys	Lys	Val	Asn	Asn	Lys	Asp	Leu	Pro	Ala	Pro	Ile	Glu
			325						330					335	
Arg	Thr	Ile	Ser	Lys	Pro	Lys	Gly	Ser	Val	Arg	Ala	Pro	Gln	Val	Tyr
			340					345					350		
Val	Leu	Pro	Pro	Pro	Glu	Glu	Glu	Met	Thr	Lys	Lys	Gln	Val	Thr	Leu
		355					360					365			
Thr	Cys	Met	Val	Thr	Asp	Phe	Met	Pro	Glu	Asp	Ile	Tyr	Val	Glu	Trp
	370					375					380				
Thr	Asn	Asn	Gly	Lys	Thr	Glu	Leu	Asn	Tyr	Lys	Asn	Thr	Glu	Pro	Val
385					390					395					400
Leu	Asp	Ser	Asp	Gly	Ser	Tyr	Phe	Met	Tyr	Ser	Lys	Leu	Arg	Val	Glu
				405					410					415	

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Lys	Lys	Asn	Trp	Val	Glu	Arg	Asn	Ser	Tyr	Ser	Cys	Ser	Val	Val	His
		420						425					430		
Glu	Gly	Leu	His	Asn	His	His	Thr	Thr	Lys	Ser	Phe	Ser	Arg	Thr	Pro
		435					440					445			
Gly	Ala	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly
	450					455					460				
Ser	Gly	Gly	Gly	Gly	Ser	Gln	Val	Gln	Leu	Glu	Ser	Glu	Gly	Gly	Leu
465					470					475					480
Phe	Lys	Pro	Thr	Asp	Thr	Leu	Thr	Leu	Thr	Cys	Thr	Val	Ser	Gly	Ser
			485						490					495	
Ser	Leu	Ser	Ser	Ser	Tyr	Met	Ser	Trp	Val	Arg	Gln	Ala	Pro	Gly	Lys
			500					505					510		
Gly	Leu	Glu	Trp	Ile	Gly	Ile	Ile	Gly	Ser	Asn	Gly	Asn	Thr	Tyr	Tyr
		515					520					525			
Ala	Asn	Trp	Ala	Lys	Gly	Arg	Phe	Thr	Ile	Ser	Lys	Thr	Ser	Thr	Thr
	530					535					540				
Val	Glu	Leu	Lys	Ile	Thr	Ser	Pro	Thr	Thr	Glu	Asp	Thr	Ala	Thr	Tyr
545					550					555					560
Phe	Cys	Ala	Arg	Gly	Gly	Tyr	Arg	Thr	Ser	Gly	Met	Asp	Pro	Trp	Gly
			565					570						575	
Pro	Gly	Thr	Leu	Val	Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly
			580					585					590		
Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Asp	Ile	Val	Met	Thr	Gln	Thr	Pro
		595					600					605			
Ala	Ser	Val	Glu	Val	Ala	Val	Gly	Gly	Thr	Val	Thr	Ile	Lys	Cys	Gln
	610					615					620				
Ala	Ser	Gln	Ser	Ile	Ser	Ser	Tyr	Leu	Asn	Trp	Tyr	Gln	Gln	Lys	Pro
625					630					635					640
Gly	Gln	Pro	Pro	Lys	Leu	Leu	Ile	Tyr	Asp	Ala	Leu	Lys	Leu	Ala	Ser
			645						650					655	
Gly	Val	Pro	Ser	Arg	Phe	Ser	Gly	Ser	Gly	Ser	Gly	Thr	Glu	Tyr	Thr
			660					665					670		
Leu	Thr	Ile	Ser	Gly	Val	Glu	Ser	Ala	Asp	Ala	Ala	Thr	Tyr	Tyr	Cys
		675					680					685			
Gln	Gln	Glu	His	Ser	Val	Gly	Asn	Val	Asp	Asn	Val	Phe	Gly	Gly	Gly
	690					695					700				
Thr	Glu	Val	Val	Val	Lys										
705					710										

<210> SEQ ID NO 21
 <211> LENGTH: 133
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 21

Ala	Pro	Thr	Ser	Ser	Ser	Thr	Lys	Lys	Thr	Gln	Leu	Gln	Leu	Glu	His
1				5					10					15	
Leu	Leu	Leu	Asp	Leu	Gln	Met	Ile	Leu	Asn	Gly	Ile	Asn	Asn	Tyr	Lys
			20					25					30		
Asn	Pro	Lys	Leu	Thr	Arg	Met	Leu	Thr	Phe	Lys	Phe	Tyr	Met	Pro	Lys
		35					40					45			

-continued

Lys Ala Thr Glu Leu Lys His Leu Gln Cys Leu Glu Glu Glu Leu Lys
 50 55 60
 Pro Leu Glu Glu Val Leu Asn Leu Ala Gln Ser Lys Asn Phe His Leu
 65 70 75 80
 Arg Pro Arg Asp Leu Ile Ser Asn Ile Asn Val Ile Val Leu Glu Leu
 85 90 95
 Lys Gly Ser Glu Thr Thr Phe Met Cys Glu Tyr Ala Asp Glu Thr Ala
 100 105 110
 Thr Ile Val Glu Phe Leu Asn Arg Trp Ile Thr Phe Cys Gln Ser Ile
 115 120 125
 Ile Ser Thr Leu Thr
 130

<210> SEQ ID NO 22
 <211> LENGTH: 219
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 22

Glu Leu Cys Asp Asp Asp Pro Pro Glu Ile Pro His Ala Thr Phe Lys
 1 5 10 15
 Ala Met Ala Tyr Lys Glu Gly Thr Met Leu Asn Cys Glu Cys Lys Arg
 20 25 30
 Gly Phe Arg Arg Ile Lys Ser Gly Ser Leu Tyr Met Leu Cys Thr Gly
 35 40 45
 Asn Ser Ser His Ser Ser Trp Asp Asn Gln Cys Gln Cys Thr Ser Ser
 50 55 60
 Ala Thr Arg Asn Thr Thr Lys Gln Val Thr Pro Gln Pro Glu Glu Gln
 65 70 75 80
 Lys Glu Arg Lys Thr Thr Glu Met Gln Ser Pro Met Gln Pro Val Asp
 85 90 95
 Gln Ala Ser Leu Pro Gly His Cys Arg Glu Pro Pro Pro Trp Glu Asn
 100 105 110
 Glu Ala Thr Glu Arg Ile Tyr His Phe Val Val Gly Gln Met Val Tyr
 115 120 125
 Tyr Gln Cys Val Gln Gly Tyr Arg Ala Leu His Arg Gly Pro Ala Glu
 130 135 140
 Ser Val Cys Lys Met Thr His Gly Lys Thr Arg Trp Thr Gln Pro Gln
 145 150 155 160
 Leu Ile Cys Thr Gly Glu Met Glu Thr Ser Gln Phe Pro Gly Glu Glu
 165 170 175
 Lys Pro Gln Ala Ser Pro Glu Gly Arg Pro Glu Ser Glu Thr Ser Cys
 180 185 190
 Leu Val Thr Thr Thr Asp Phe Gln Ile Gln Thr Glu Met Ala Ala Thr
 195 200 205
 Met Glu Thr Ser Ile Phe Thr Thr Glu Tyr Gln
 210 215

<210> SEQ ID NO 23
 <211> LENGTH: 166
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:

-continued

<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 23

Glu Leu Cys Asp Asp Asp Pro Pro Glu Ile Pro His Ala Thr Phe Lys
 1 5 10 15
 Ala Met Ala Tyr Lys Glu Gly Thr Met Leu Asn Cys Glu Cys Lys Arg
 20 25 30
 Gly Phe Arg Arg Ile Lys Ser Gly Ser Leu Tyr Met Leu Cys Thr Gly
 35 40 45
 Asn Ser Ser His Ser Ser Trp Asp Asn Gln Cys Gln Cys Thr Ser Ser
 50 55 60
 Ala Thr Arg Asn Thr Thr Lys Gln Val Thr Pro Gln Pro Glu Glu Gln
 65 70 75 80
 Lys Glu Arg Lys Thr Thr Glu Met Gln Ser Pro Met Gln Pro Val Asp
 85 90 95
 Gln Ala Ser Leu Pro Gly His Cys Arg Glu Pro Pro Pro Trp Glu Asn
 100 105 110
 Glu Ala Thr Glu Arg Ile Tyr His Phe Val Val Gly Gln Met Val Tyr
 115 120 125
 Tyr Gln Cys Val Gln Gly Tyr Arg Ala Leu His Arg Gly Pro Ala Glu
 130 135 140
 Ser Val Cys Lys Met Thr His Gly Lys Thr Arg Trp Thr Gln Pro Gln
 145 150 155 160
 Leu Ile Cys Thr Gly Glu
 165

<210> SEQ ID NO 24

<211> LENGTH: 223

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 24

Glu Leu Cys Asp Asp Asp Pro Pro Glu Ile Pro His Ala Thr Phe Lys
 1 5 10 15
 Ala Met Ala Tyr Lys Glu Gly Thr Met Leu Asn Cys Glu Cys Lys Arg
 20 25 30
 Glu Leu Cys Asp Asp Asp Pro Pro Glu Ile Pro His Ala Thr Phe Lys
 35 40 45
 Ala Met Ala Tyr Lys Glu Gly Thr Met Leu Asn Cys Glu Cys Lys Arg
 50 55 60
 Gly Phe Arg Arg Ile Lys Ser Gly Ser Leu Tyr Met Leu Cys Thr Gly
 65 70 75 80
 Asn Ser Ser His Ser Ser Trp Asp Asn Gln Cys Gln Cys Thr Ser Ser
 85 90 95
 Ala Thr Arg Asn Thr Thr Lys Gln Val Thr Pro Gln Pro Glu Glu Gln
 100 105 110
 Lys Glu Arg Lys Thr Thr Glu Met Gln Ser Pro Met Gln Pro Val Asp
 115 120 125
 Gln Ala Ser Leu Pro Gly His Cys Arg Glu Pro Pro Pro Trp Glu Asn
 130 135 140
 Glu Ala Thr Glu Arg Ile Tyr His Phe Val Val Gly Gln Met Val Tyr
 145 150 155 160

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Tyr Gln Cys Val Gln Gly Tyr Arg Ala Leu His Arg Gly Pro Ala Glu
 165 170 175

Ser Val Cys Lys Met Thr His Gly Lys Thr Arg Trp Thr Gln Pro Gln
 180 185 190

Leu Ile Cys Thr Gly Glu Met Glu Thr Ser Gln Phe Pro Gly Glu Glu
 195 200 205

Lys Pro Gln Ala Ser Pro Glu Gly Arg Pro Glu Ser Glu Thr Ser
 210 215 220

<210> SEQ ID NO 25
 <211> LENGTH: 114
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 25

Asn Trp Val Asn Val Ile Ser Asp Leu Lys Lys Ile Glu Asp Leu Ile
 1 5 10 15

Gln Ser Met His Ile Asp Ala Thr Leu Tyr Thr Glu Ser Asp Val His
 20 25 30

Pro Ser Cys Lys Val Thr Ala Met Lys Cys Phe Leu Leu Glu Leu Gln
 35 40 45

Val Ile Ser Leu Glu Ser Gly Asp Ala Ser Ile His Asp Thr Val Glu
 50 55 60

Asn Leu Ile Ile Leu Ala Asn Asn Ser Leu Ser Ser Asn Gly Asn Val
 65 70 75 80

Thr Glu Ser Gly Cys Lys Glu Cys Glu Glu Leu Glu Glu Lys Asn Ile
 85 90 95

Lys Glu Phe Leu Gln Ser Phe Val His Ile Val Gln Met Phe Ile Asn
 100 105 110

Thr Ser

<210> SEQ ID NO 26
 <211> LENGTH: 175
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 26

Ile Thr Cys Pro Pro Pro Met Ser Val Glu His Ala Asp Ile Trp Val
 1 5 10 15

Lys Ser Tyr Ser Leu Tyr Ser Arg Glu Arg Tyr Ile Cys Asn Ser Gly
 20 25 30

Phe Lys Arg Lys Ala Gly Thr Ser Ser Leu Thr Glu Cys Val Leu Asn
 35 40 45

Lys Ala Thr Asn Val Ala His Trp Thr Thr Pro Ser Leu Lys Cys Ile
 50 55 60

Arg Asp Pro Ala Leu Val His Gln Arg Pro Ala Pro Pro Ser Thr Val
 65 70 75 80

Thr Thr Ala Gly Val Thr Pro Gln Pro Glu Ser Leu Ser Pro Ser Gly
 85 90 95

Lys Glu Pro Ala Ala Ser Ser Pro Ser Ser Asn Asn Thr Ala Ala Thr
 100 105 110

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Thr Ala Ala Ile Val Pro Gly Ser Gln Leu Met Pro Ser Lys Ser Pro
    115                      120                      125

Ser Thr Gly Thr Thr Glu Ile Ser Ser His Glu Ser Ser His Gly Thr
    130                      135                      140

Pro Ser Gln Thr Thr Ala Lys Asn Trp Glu Leu Thr Ala Ser Ala Ser
    145                      150                      155                      160

His Gln Pro Pro Gly Val Tyr Pro Gln Gly His Ser Asp Thr Thr
    165                      170                      175

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<210> SEQ ID NO 27
<211> LENGTH: 77
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthesized

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<400> SEQUENCE: 27

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Ile Thr Cys Pro Pro Pro Met Ser Val Glu His Ala Asp Ile Trp Val
1      5      10      15

Lys Ser Tyr Ser Leu Tyr Ser Arg Glu Arg Tyr Ile Cys Asn Ser Gly
    20      25      30

Phe Lys Arg Lys Ala Gly Thr Ser Ser Leu Thr Glu Cys Val Leu Asn
    35      40      45

Lys Ala Thr Asn Val Ala His Trp Thr Thr Pro Ser Leu Lys Cys Ile
    50      55      60

Arg Asp Pro Ala Leu Val His Gln Arg Pro Ala Pro Pro
65      70      75

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<210> SEQ ID NO 28
<211> LENGTH: 65
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthesized

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<400> SEQUENCE: 28

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Ile Thr Cys Pro Pro Pro Met Ser Val Glu His Ala Asp Ile Trp Val
1      5      10      15

Lys Ser Tyr Ser Leu Tyr Ser Arg Glu Arg Tyr Ile Cys Asn Ser Gly
    20      25      30

Phe Lys Arg Lys Ala Gly Thr Ser Ser Leu Thr Glu Cys Val Leu Asn
    35      40      45

Lys Ala Thr Asn Val Ala His Trp Thr Thr Pro Ser Leu Lys Cys Ile
    50      55      60

Arg
65

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<210> SEQ ID NO 29
<211> LENGTH: 73
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthesized

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<400> SEQUENCE: 29

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Ile Thr Cys Pro Pro Pro Met Ser Val Glu His Ala Asp Ile Trp Val
1      5      10      15

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Lys Ser Tyr Ser Leu Tyr Ser Arg Glu Arg Tyr Ile Cys Asn Ser Gly
      20                25                30

Phe Lys Arg Lys Ala Gly Thr Ser Ser Leu Thr Glu Cys Val Leu Asn
      35                40                45

Lys Ala Thr Asn Val Ala His Trp Thr Thr Pro Ser Leu Lys Cys Ile
      50                55                60

Arg Asp Pro Ala Leu Val His Gln Arg
      65                70

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<210> SEQ ID NO 30
<211> LENGTH: 86
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthesized

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<400> SEQUENCE: 30

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Ile Thr Cys Pro Pro Pro Met Ser Val Glu His Ala Asp Ile Trp Val
1          5          10          15

Lys Ser Tyr Ser Leu Tyr Ser Arg Glu Arg Tyr Ile Cys Asn Ser Gly
      20                25                30

Phe Lys Arg Lys Ala Gly Thr Ser Ser Leu Thr Glu Cys Val Leu Asn
      35                40                45

Lys Ala Thr Asn Val Ala His Trp Thr Thr Pro Ser Leu Lys Cys Ile
      50                55                60

Arg Asp Pro Ala Leu Val His Gln Arg Pro Ala Pro Pro Ser Thr Val
      65                70                75                80

Thr Thr Ala Gly Val Thr
      85

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<210> SEQ ID NO 31
<211> LENGTH: 102
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthesized

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<400> SEQUENCE: 31

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Ile Thr Cys Pro Pro Pro Met Ser Val Glu His Ala Asp Ile Trp Val
1          5          10          15

Lys Ser Tyr Ser Leu Tyr Ser Arg Glu Arg Tyr Ile Cys Asn Ser Gly
      20                25                30

Phe Lys Arg Lys Ala Gly Thr Ser Ser Leu Thr Glu Cys Val Leu Asn
      35                40                45

Lys Ala Thr Asn Val Ala His Trp Thr Thr Pro Ser Leu Lys Cys Ile
      50                55                60

Arg Asp Pro Ala Leu Val His Gln Arg Pro Ala Pro Pro Ser Thr Val
      65                70                75                80

Thr Thr Ala Gly Val Thr Pro Gln Pro Glu Ser Leu Ser Pro Ser Gly
      85                90                95

Lys Glu Pro Ala Ala Ser
      100

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<210> SEQ ID NO 32
<211> LENGTH: 536
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence

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-continued

<220> FEATURE:

<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 32

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Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ser
1          5          10          15

Ser Val Lys Val Ser Cys Lys Ala Ser Gly Tyr Ser Phe Thr Ser Tyr
          20          25          30

Trp Met Asn Trp Val Arg Gln Ala Pro Gly Gln Gly Leu Glu Trp Ile
          35          40          45

Gly Met Ile Arg Pro Ser Asp Ser Glu Thr Arg Leu Asn Gln Met Phe
          50          55          60

Lys Asp Arg Val Thr Ile Thr Val Asp Lys Ser Thr Ser Thr Ala Tyr
          65          70          75          80

Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Val Tyr Tyr Cys
          85          90          95

Ala Gly Ile His Asp Tyr Gly His Gly Ala Tyr Trp Gly Gln Gly Thr
          100          105          110

Leu Val Thr Val Ser Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
          115          120          125

Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Glu Leu Cys Asp Asp
          130          135          140

Asp Pro Pro Glu Ile Pro His Ala Thr Phe Lys Ala Met Ala Tyr Lys
          145          150          155          160

Glu Gly Thr Met Leu Asn Cys Glu Cys Lys Arg Gly Phe Arg Arg Ile
          165          170          175

Lys Ser Gly Ser Leu Tyr Met Leu Cys Thr Gly Asn Ser Ser His Ser
          180          185          190

Ser Trp Asp Asn Gln Cys Gln Cys Thr Ser Ser Ala Thr Arg Asn Thr
          195          200          205

Thr Lys Gln Val Thr Pro Gln Pro Glu Glu Gln Lys Glu Arg Lys Thr
          210          215          220

Thr Glu Met Gln Ser Pro Met Gln Pro Val Asp Gln Ala Ser Leu Pro
          225          230          235          240

Gly His Cys Arg Glu Pro Pro Pro Trp Glu Asn Glu Ala Thr Glu Arg
          245          250          255

Ile Tyr His Phe Val Val Gly Gln Met Val Tyr Tyr Gln Cys Val Gln
          260          265          270

Gly Tyr Arg Ala Leu His Arg Gly Pro Ala Glu Ser Val Cys Lys Met
          275          280          285

Thr His Gly Lys Thr Arg Trp Thr Gln Pro Gln Leu Ile Cys Thr Gly
          290          295          300

Glu Pro Lys Ser Ser Asp Lys Thr His Thr Cys Pro Pro Cys Pro Ala
          305          310          315          320

Pro Glu Leu Leu Gly Gly Pro Ser Val Phe Leu Phe Pro Pro Lys Pro
          325          330          335

Lys Asp Thr Leu Met Ile Ser Arg Thr Pro Glu Val Thr Cys Val Val
          340          345          350

Val Asp Val Ser His Glu Asp Pro Glu Val Lys Phe Asn Trp Tyr Val
          355          360          365

Asp Gly Val Glu Val His Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln
          370          375          380

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Tyr Asn Ser Thr Tyr Arg Val Val Ser Val Leu Thr Val Leu His Gln
 385 390 395 400
 Asp Trp Leu Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala
 405 410 415
 Leu Pro Ala Pro Ile Glu Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro
 420 425 430
 Arg Glu Pro Gln Val Tyr Thr Leu Pro Pro Ser Arg Glu Glu Met Thr
 435 440 445
 Lys Asn Gln Val Ser Leu Thr Cys Arg Val Lys Gly Phe Tyr Pro Ser
 450 455 460
 Asp Ile Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr
 465 470 475 480
 Lys Thr Thr Pro Pro Val Leu Lys Ser Asp Gly Ser Phe Phe Leu Ala
 485 490 495
 Ser Lys Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe
 500 505 510
 Ser Cys Ser Val Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys
 515 520 525
 Ser Leu Ser Leu Ser Pro Gly Lys
 530 535

<210> SEQ ID NO 33
 <211> LENGTH: 261
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 33

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
 1 5 10 15
 Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Glu Asn Ile Tyr Ser Asn
 20 25 30
 Leu Ala Trp Tyr Gln Gln Lys Pro Gly Lys Ser Pro Lys Leu Leu Val
 35 40 45
 Tyr Ala Ala Ser His Leu Pro Asp Gly Val Pro Ser Arg Phe Ser Gly
 50 55 60
 Ser Gly Ser Gly Thr Asp Tyr Ser Leu Thr Ile Ser Ser Leu Gln Pro
 65 70 75 80
 Glu Asp Phe Ala Thr Tyr Tyr Cys Gln His Phe Trp Gly Thr Pro Arg
 85 90 95
 Thr Phe Gly Gln Gly Thr Lys Leu Glu Ile Lys Gly Gly Gly Ser
 100 105 110
 Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Ser Gly
 115 120 125
 Ala Pro Thr Ser Ser Ser Thr Lys Lys Thr Gln Leu Gln Leu Glu His
 130 135 140
 Leu Leu Leu Asp Leu Gln Met Ile Leu Asn Gly Ile Asn Asn Tyr Lys
 145 150 155 160
 Asn Pro Lys Leu Thr Arg Met Leu Thr Phe Lys Phe Tyr Met Pro Lys
 165 170 175
 Lys Ala Thr Glu Leu Lys His Leu Gln Cys Leu Glu Glu Glu Leu Lys
 180 185 190

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Pro Leu Glu Glu Val Leu Asn Leu Ala Gln Ser Lys Asn Phe His Leu
    195                200                205

Arg Pro Arg Asp Leu Ile Ser Asn Ile Asn Val Ile Val Leu Glu Leu
    210                215                220

Lys Gly Ser Glu Thr Thr Phe Met Cys Glu Tyr Ala Asp Glu Thr Ala
    225                230                235                240

Thr Ile Val Glu Phe Leu Asn Arg Trp Ile Thr Phe Cys Gln Ser Ile
                245                250                255

Ile Ser Thr Leu Thr
                260

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<210> SEQ ID NO 34
<211> LENGTH: 449
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthesized

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<400> SEQUENCE: 34

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Glu Val Gln Leu Gln Glu Ser Gly Pro Gly Leu Val Lys Pro Ser Glu
1          5          10          15

Thr Leu Ser Leu Thr Cys Ala Val Tyr Gly Asp Ser Ile Thr Ser Gly
    20          25          30

Tyr Trp Asn Trp Ile Arg Lys Pro Pro Gly Lys Gly Leu Glu Tyr Met
    35          40          45

Gly Tyr Ile Ser Tyr Thr Gly Ser Thr Tyr Gln Asn Pro Ser Leu Lys
    50          55          60

Ser Arg Ile Thr Phe Ser Arg Asp Thr Ser Lys Asn Gln Tyr Tyr Leu
    65          70          75          80

Lys Leu Ser Ser Val Thr Ala Ala Asp Thr Ala Thr Tyr Tyr Cys Ala
    85          90          95

Arg Ser Arg Ala Trp Ile Arg Thr Tyr Phe Asp Tyr Trp Gly Gln Gly
    100         105         110

Thr Leu Val Thr Val Ser Ser Ala Ser Thr Lys Gly Pro Ser Val Phe
    115         120         125

Pro Leu Ala Pro Ser Ser Lys Ser Thr Ser Gly Gly Thr Ala Ala Leu
    130         135         140

Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val Thr Val Ser Trp
    145         150         155         160

Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro Ala Val Leu
    165         170         175

Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val Thr Val Pro Ser
    180         185         190

Ser Ser Leu Gly Thr Gln Thr Tyr Ile Cys Asn Val Asn His Lys Pro
    195         200         205

Ser Asn Thr Lys Val Asp Lys Arg Val Glu Pro Lys Ser Cys Asp Lys
    210         215         220

Thr His Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu Leu Gly Gly Pro
    225         230         235         240

Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile Ser
    245         250         255

Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser His Glu Asp
    260         265         270

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Pro Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val Glu Val His Asn
275 280 285

Ala Lys Thr Lys Pro Arg Glu Glu Gln Tyr Asn Ser Thr Tyr Arg Val
290 295 300

Val Ser Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly Lys Glu
305 310 315 320

Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu Pro Ala Pro Ile Glu Lys
325 330 335

Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr Thr
340 345 350

Leu Pro Pro Ser Arg Glu Glu Met Thr Lys Asn Gln Val Ser Leu Trp
355 360 365

Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu
370 375 380

Ser Asn Gly Gln Pro Glu Asn Asn Tyr Asp Thr Thr Pro Pro Val Leu
385 390 395 400

Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Asp Leu Thr Val Asp Lys
405 410 415

Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met His Glu
420 425 430

Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro Gly
435 440 445

Lys

<210> SEQ ID NO 35
<211> LENGTH: 215
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 35

Glu Ile Val Leu Thr Gln Ser Pro Asp Phe Gln Ser Val Thr Pro Lys
1 5 10 15

Glu Lys Val Thr Ile Thr Cys Ser Val Ser Ser Ser Ile Ser Ser Ser
20 25 30

Asn Leu His Trp Tyr Gln Gln Lys Pro Asp Gln Ser Pro Lys Leu Leu
35 40 45

Ile Tyr Gly Thr Ser Asn Leu Ala Ser Gly Val Pro Ser Arg Phe Ser
50 55 60

Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Asn Ser Leu Glu
65 70 75 80

Ala Glu Asp Ala Ala Thr Tyr Tyr Cys Gln Gln Trp Ser Ser Tyr Pro
85 90 95

Leu Thr Phe Gly Gln Gly Thr Lys Leu Glu Ile Lys Arg Thr Val Ala
100 105 110

Ala Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu Gln Leu Lys Ser
115 120 125

Gly Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe Tyr Pro Arg Glu
130 135 140

Ala Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln Ser Gly Asn Ser
145 150 155 160

<400> SEQUENCE: 36

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<210> SEQ ID NO 37
<211> LENGTH: 110
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthesized
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<400> SEQUENCE: 37

Gln	Ser	Ala	Leu	Thr	Gln	Pro	Ala	Ser	Val	Ser	Gly	Ser	Pro	Gly	Gln
1				5					10					15	
Ser	Ile	Thr	Ile	Ser	Cys	Thr	Gly	Thr	Ser	Ser	Asp	Val	Gly	Gly	Tyr
			20					25					30		
Asn	Tyr	Val	Ser	Trp	Tyr	Gln	Gln	His	Pro	Gly	Lys	Ala	Pro	Lys	Leu
		35					40					45			
Met	Ile	Tyr	Asp	Val	Ser	Asn	Arg	Pro	Ser	Gly	Val	Ser	Asn	Arg	Phe
	50					55					60				
Ser	Gly	Ser	Lys	Ser	Gly	Asn	Thr	Ala	Ser	Leu	Thr	Ile	Ser	Gly	Leu
65					70					75					80
Gln	Ala	Glu	Asp	Glu	Ala	Asp	Tyr	Tyr	Cys	Ser	Ser	Tyr	Thr	Ser	Ser
				85					90					95	
Ser	Thr	Arg	Val	Phe	Gly	Thr	Gly	Thr	Lys	Val	Thr	Val	Leu		
			100					105					110		

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<210> SEQ ID NO 38
<211> LENGTH: 563
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 38

Glu Pro Arg Gly Pro Thr Ile Lys Pro Cys Pro Pro Cys Lys Cys Pro
1      5      10      15
Ala Pro Asn Leu Leu Gly Gly Pro Ser Val Phe Ile Phe Pro Pro Lys
20     25     30
Ile Lys Asp Val Leu Met Ile Ser Leu Ser Pro Ile Val Thr Cys Val
35     40     45
Val Val Asp Val Ser Glu Asp Asp Pro Asp Val Gln Ile Ser Trp Phe
50     55     60
Val Asn Asn Val Glu Val His Thr Ala Gln Thr Gln Thr His Arg Glu
65     70     75     80
Asp Tyr Asn Ser Thr Leu Arg Val Val Ser Ala Leu Pro Ile Gln His
85     90     95
Gln Asp Trp Met Ser Gly Lys Glu Phe Lys Cys Lys Val Asn Asn Lys
100    105    110
Asp Leu Pro Ala Pro Ile Glu Arg Thr Ile Ser Lys Pro Lys Gly Ser
115    120    125
Val Arg Ala Pro Gln Val Tyr Val Leu Pro Pro Pro Glu Glu Glu Met
130    135    140
Thr Lys Lys Gln Val Thr Leu Thr Cys Met Val Thr Asp Phe Met Pro
145    150    155    160
Glu Asp Ile Tyr Val Glu Trp Thr Asn Asn Gly Lys Thr Glu Leu Asn
165    170    175
Tyr Lys Asn Thr Glu Pro Val Leu Asp Ser Asp Gly Ser Tyr Phe Met
180    185    190
Tyr Ser Lys Leu Arg Val Glu Lys Lys Asn Trp Val Glu Arg Asn Ser
195    200    205
Tyr Ser Cys Ser Val Val His Glu Gly Leu His Asn His His Thr Thr
210    215    220
Lys Ser Phe Ser Arg Thr Pro Gly Ala Gly Gly Gly Ser Gly Gly
225    230    235    240
Gly Gly Ser Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Ile Thr
245    250    255
Cys Pro Pro Pro Met Ser Val Glu His Ala Asp Ile Trp Val Lys Ser
260    265    270
Tyr Ser Leu Tyr Ser Arg Glu Arg Tyr Ile Cys Asn Ser Gly Phe Lys
275    280    285
Arg Lys Ala Gly Thr Ser Ser Leu Thr Glu Cys Val Leu Asn Lys Ala
290    295    300
Thr Asn Val Ala His Trp Thr Thr Pro Ser Leu Lys Cys Ile Arg Asp
305    310    315    320
Pro Ala Leu Val His Gln Arg Pro Ala Pro Pro Ser Thr Val Thr Thr
325    330    335
Ala Gly Val Thr Pro Gln Pro Glu Ser Leu Ser Pro Ser Gly Lys Glu
340    345    350

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Pro Ala Ala Ser Ser Pro Ser Ser Asn Asn Thr Ala Ala Thr Thr Ala
 355 360 365
 Ala Ile Val Pro Gly Ser Gln Leu Met Pro Ser Lys Ser Pro Ser Thr
 370 375 380
 Gly Thr Thr Glu Ile Ser Ser His Glu Ser Ser His Gly Thr Pro Ser
 385 390 395 400
 Gln Thr Thr Ala Lys Asn Trp Glu Leu Thr Ala Ser Ala Ser His Gln
 405 410 415
 Pro Pro Gly Val Tyr Pro Gln Gly His Ser Asp Thr Thr Ser Gly Gly
 420 425 430
 Ser Gly Gly Gly Gly Ser Gly Gly Gly Ser Gly Gly Gly Gly Ser Leu
 435 440 445
 Gln Asn Trp Val Asn Val Ile Ser Asp Leu Lys Lys Ile Glu Asp Leu
 450 455 460
 Ile Gln Ser Met His Ile Asp Ala Thr Leu Tyr Thr Glu Ser Asp Val
 465 470 475 480
 His Pro Ser Cys Lys Val Thr Ala Met Lys Cys Phe Leu Leu Glu Leu
 485 490 495
 Gln Val Ile Ser Leu Glu Ser Gly Asp Ala Ser Ile His Asp Thr Val
 500 505 510
 Glu Asn Leu Ile Ile Leu Ala Asn Asp Ser Leu Ser Ser Asn Gly Asn
 515 520 525
 Val Thr Glu Ser Gly Cys Lys Glu Cys Glu Glu Leu Glu Glu Lys Asn
 530 535 540
 Ile Lys Glu Phe Leu Gln Ser Phe Val His Ile Val Gln Met Phe Ile
 545 550 555 560
 Asn Thr Ser

<210> SEQ ID NO 39
 <211> LENGTH: 504
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 39

Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ser
 1 5 10 15
 Ser Val Lys Val Ser Cys Lys Ala Ser Gly Tyr Ser Phe Thr Ser Tyr
 20 25 30
 Trp Met Asn Trp Val Arg Gln Ala Pro Gly Gln Gly Leu Glu Trp Ile
 35 40 45
 Gly Met Ile Arg Pro Ser Asp Ser Glu Thr Arg Leu Asn Gln Met Phe
 50 55 60
 Lys Asp Arg Val Thr Ile Thr Val Asp Lys Ser Thr Ser Thr Ala Tyr
 65 70 75 80
 Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95
 Ala Gly Ile His Asp Tyr Gly His Gly Ala Tyr Trp Gly Gln Gly Thr
 100 105 110
 Leu Val Thr Val Ser Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
 115 120 125

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Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Ala	Pro	Thr	Ser	Ser
130						135					140				
Ser	Thr	Lys	Lys	Thr	Gln	Leu	Gln	Leu	Glu	His	Leu	Leu	Leu	Asp	Leu
145					150					155					160
Gln	Met	Ile	Leu	Asn	Gly	Ile	Asn	Asn	Tyr	Lys	Asn	Pro	Lys	Leu	Thr
				165					170					175	
Arg	Met	Leu	Thr	Phe	Lys	Phe	Tyr	Met	Pro	Lys	Lys	Ala	Thr	Glu	Leu
			180					185					190		
Lys	His	Leu	Gln	Cys	Leu	Glu	Glu	Glu	Leu	Lys	Pro	Leu	Glu	Glu	Val
		195					200					205			
Leu	Asn	Leu	Ala	Gln	Ser	Lys	Asn	Phe	His	Leu	Arg	Pro	Arg	Asp	Leu
	210					215					220				
Ile	Ser	Asn	Ile	Asn	Val	Ile	Val	Leu	Glu	Leu	Lys	Gly	Ser	Glu	Thr
225					230					235					240
Thr	Phe	Met	Cys	Glu	Tyr	Ala	Asp	Glu	Thr	Ala	Thr	Ile	Val	Glu	Phe
				245					250					255	
Leu	Asn	Arg	Trp	Ile	Thr	Phe	Cys	Gln	Ser	Ile	Ile	Ser	Thr	Leu	Thr
			260					265					270		
Glu	Pro	Lys	Ser	Ser	Asp	Lys	Thr	His	Thr	Cys	Pro	Pro	Cys	Pro	Ala
		275					280					285			
Pro	Glu	Leu	Leu	Gly	Gly	Pro	Ser	Val	Phe	Leu	Phe	Pro	Pro	Lys	Pro
	290					295					300				
Lys	Asp	Thr	Leu	Met	Ile	Ser	Arg	Thr	Pro	Glu	Val	Thr	Cys	Val	Val
305					310					315					320
Val	Asp	Val	Ser	His	Glu	Asp	Pro	Glu	Val	Lys	Phe	Asn	Trp	Tyr	Val
				325					330					335	
Asp	Gly	Val	Glu	Val	His	Asn	Ala	Lys	Thr	Lys	Pro	Arg	Glu	Glu	Gln
			340					345					350		
Tyr	Asn	Ser	Thr	Tyr	Arg	Val	Val	Ser	Val	Leu	Thr	Val	Leu	His	Gln
		355					360					365			
Asp	Trp	Leu	Asn	Gly	Lys	Glu	Tyr	Lys	Cys	Lys	Val	Ser	Asn	Lys	Ala
	370					375					380				
Leu	Pro	Ala	Pro	Ile	Glu	Lys	Thr	Ile	Ser	Lys	Ala	Lys	Gly	Gln	Pro
385					390					395					400
Arg	Glu	Pro	Gln	Val	Tyr	Thr	Leu	Pro	Pro	Ser	Arg	Glu	Glu	Met	Thr
				405					410					415	
Lys	Asn	Gln	Val	Ser	Leu	Thr	Cys	Arg	Val	Lys	Gly	Phe	Tyr	Pro	Ser
			420					425					430		
Asp	Ile	Ala	Val	Glu	Trp	Glu	Ser	Asn	Gly	Gln	Pro	Glu	Asn	Asn	Tyr
		435						440				445			
Lys	Thr	Thr	Pro	Pro	Val	Leu	Lys	Ser	Asp	Gly	Ser	Phe	Phe	Leu	Ala
	450					455					460				
Ser	Lys	Leu	Thr	Val	Asp	Lys	Ser	Arg	Trp	Gln	Gln	Gly	Asn	Val	Phe
465					470					475					480
Ser	Cys	Ser	Val	Met	His	Glu	Ala	Leu	His	Asn	His	Tyr	Thr	Gln	Lys
				485					490					495	
Ser	Leu	Ser	Leu	Ser	Pro	Gly	Lys								
				500											

<210> SEQ ID NO 40

<211> LENGTH: 293

<212> TYPE: PRT

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<213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 40

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Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1           5           10           15
Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Glu Asn Ile Tyr Ser Asn
20          25          30
Leu Ala Trp Tyr Gln Gln Lys Pro Gly Lys Ser Pro Lys Leu Leu Val
35          40          45
Tyr Ala Ala Ser His Leu Pro Asp Gly Val Pro Ser Arg Phe Ser Gly
50          55          60
Ser Gly Ser Gly Thr Asp Tyr Ser Leu Thr Ile Ser Ser Leu Gln Pro
65          70          75          80
Glu Asp Phe Ala Thr Tyr Tyr Cys Gln His Phe Trp Gly Thr Pro Arg
85          90          95
Thr Phe Gly Gln Gly Thr Lys Leu Glu Ile Lys Gly Gly Gly Gly Ser
100         105         110
Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly
115         120         125
Glu Leu Cys Asp Asp Asp Pro Pro Glu Ile Pro His Ala Thr Phe Lys
130         135         140
Ala Met Ala Tyr Lys Glu Gly Thr Met Leu Asn Cys Glu Cys Lys Arg
145         150         155         160
Gly Phe Arg Arg Ile Lys Ser Gly Ser Leu Tyr Met Leu Cys Thr Gly
165         170         175
Asn Ser Ser His Ser Ser Trp Asp Asn Gln Cys Gln Cys Thr Ser Ser
180         185         190
Ala Thr Arg Asn Thr Thr Lys Gln Val Thr Pro Gln Pro Glu Glu Gln
195         200         205
Lys Glu Arg Lys Thr Thr Glu Met Gln Ser Pro Met Gln Pro Val Asp
210         215         220
Gln Ala Ser Leu Pro Gly His Cys Arg Glu Pro Pro Pro Trp Glu Asn
225         230         235         240
Glu Ala Thr Glu Arg Ile Tyr His Phe Val Val Gly Gln Met Val Tyr
245         250         255
Tyr Gln Cys Val Gln Gly Tyr Arg Ala Leu His Arg Gly Pro Ala Glu
260         265         270
Ser Val Cys Lys Met Thr His Gly Lys Thr Arg Trp Thr Gln Pro Gln
275         280         285
Leu Ile Cys Thr Gly
290

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<210> SEQ ID NO 41
 <211> LENGTH: 511
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 41

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Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ser
1           5           10           15

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Ser	Val	Lys	Val	Ser	Cys	Lys	Ala	Ser	Gly	Tyr	Ser	Phe	Thr	Ser	Tyr
			20					25					30		
Trp	Met	Asn	Trp	Val	Arg	Gln	Ala	Pro	Gly	Gln	Gly	Leu	Glu	Trp	Ile
		35					40					45			
Gly	Met	Ile	Arg	Pro	Ser	Asp	Ser	Glu	Thr	Arg	Leu	Asn	Gln	Met	Phe
	50					55					60				
Lys	Asp	Arg	Val	Thr	Ile	Thr	Val	Asp	Lys	Ser	Thr	Ser	Thr	Ala	Tyr
	65				70					75					80
Met	Glu	Leu	Ser	Ser	Leu	Arg	Ser	Glu	Asp	Thr	Ala	Val	Tyr	Tyr	Cys
			85						90					95	
Ala	Gly	Ile	His	Asp	Tyr	Gly	His	Gly	Ala	Tyr	Trp	Gly	Gln	Gly	Thr
			100					105					110		
Leu	Val	Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser
		115					120					125			
Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Thr	Gln	Ser	Thr	His
	130					135					140				
Glu	Ser	Leu	Lys	Pro	Gln	Arg	Val	Gln	Phe	Gln	Ser	Arg	Asn	Phe	His
	145				150					155					160
Asn	Ile	Leu	Gln	Trp	Gln	Pro	Gly	Arg	Ala	Leu	Thr	Gly	Asn	Ser	Ser
			165						170					175	
Val	Tyr	Phe	Val	Gln	Tyr	Lys	Ile	Met	Phe	Ser	Cys	Ser	Met	Lys	Ser
			180					185					190		
Ser	His	Gln	Lys	Pro	Ser	Gly	Cys	Trp	Gln	His	Ile	Ser	Cys	Asn	Phe
		195					200					205			
Pro	Gly	Cys	Arg	Thr	Leu	Ala	Lys	Tyr	Gly	Gln	Arg	Gln	Trp	Lys	Asn
	210					215					220				
Lys	Glu	Asp	Cys	Trp	Gly	Thr	Gln	Glu	Leu	Ser	Cys	Asp	Leu	Thr	Ser
	225				230					235					240
Glu	Thr	Ser	Asp	Ile	Gln	Glu	Pro	Tyr	Tyr	Gly	Arg	Val	Arg	Ala	Ala
			245						250					255	
Ser	Ala	Gly	Ser	Tyr	Ser	Glu	Trp	Ser	Met	Thr	Pro	Arg	Phe	Thr	Pro
		260						265					270		
Trp	Trp	Glu	Thr	Lys	Ile	Asp	Glu	Pro	Lys	Ser	Ser	Asp	Lys	Thr	His
		275					280					285			
Thr	Cys	Pro	Pro	Cys	Pro	Ala	Pro	Glu	Leu	Leu	Gly	Gly	Pro	Ser	Val
	290					295					300				
Phe	Leu	Phe	Pro	Pro	Lys	Pro	Lys	Asp	Thr	Leu	Met	Ile	Ser	Arg	Thr
	305				310					315					320
Pro	Glu	Val	Thr	Cys	Val	Val	Val	Asp	Val	Ser	His	Glu	Asp	Pro	Glu
			325						330					335	
Val	Lys	Phe	Asn	Trp	Tyr	Val	Asp	Gly	Val	Glu	Val	His	Asn	Ala	Lys
			340					345					350		
Thr	Lys	Pro	Arg	Glu	Glu	Gln	Tyr	Asn	Ser	Thr	Tyr	Arg	Val	Val	Ser
		355					360					365			
Val	Leu	Thr	Val	Leu	His	Gln	Asp	Trp	Leu	Asn	Gly	Lys	Glu	Tyr	Lys
	370					375					380				
Cys	Lys	Val	Ser	Asn	Lys	Ala	Leu	Pro	Ala	Pro	Ile	Glu	Lys	Thr	Ile
	385				390					395					400
Ser	Lys	Ala	Lys	Gly	Gln	Pro	Arg	Glu	Pro	Gln	Val	Tyr	Thr	Leu	Pro
			405						410					415	
Pro	Ser	Arg	Glu	Glu	Met	Thr	Lys	Asn	Gln	Val	Ser	Leu	Thr	Cys	Arg

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420	425	430
Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu Ser Asn		
435	440	445
Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val Leu Lys Ser		
450	455	460
Asp Gly Ser Phe Phe Leu Ala Ser Lys Leu Thr Val Asp Lys Ser Arg		
465	470	475
Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met His Glu Ala Leu		
485	490	495
His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro Gly Lys		
500	505	510

<210> SEQ ID NO 42
 <211> LENGTH: 274
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 42

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1 5 10 15
Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Glu Asn Ile Tyr Ser Asn
20 25 30
Leu Ala Trp Tyr Gln Gln Lys Pro Gly Lys Ser Pro Lys Leu Leu Val
35 40 45
Tyr Ala Ala Ser His Leu Pro Asp Gly Val Pro Ser Arg Phe Ser Gly
50 55 60
Ser Gly Ser Gly Thr Asp Tyr Ser Leu Thr Ile Ser Ser Leu Gln Pro
65 70 75 80
Glu Asp Phe Ala Thr Tyr Tyr Cys Gln His Phe Trp Gly Thr Pro Arg
85 90 95
Thr Phe Gly Gln Gly Thr Lys Leu Glu Ile Lys Gly Gly Gly Gly Ser
100 105 110
Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly
115 120 125
Ala Pro Ile Ser Ser His Cys Arg Leu Asp Lys Ser Asn Phe Gln Gln
130 135 140
Pro Tyr Ile Thr Asn Arg Thr Phe Met Leu Ala Lys Glu Ala Ser Leu
145 150 155 160
Ala Asp Asn Asn Thr Asp Val Arg Leu Ile Gly Glu Lys Leu Phe His
165 170 175
Gly Val Ser Met Ser Glu Arg Cys Tyr Leu Met Lys Gln Val Leu Asn
180 185 190
Phe Thr Leu Glu Glu Val Leu Phe Pro Gln Ser Asp Arg Phe Gln Pro
195 200 205
Tyr Met Gln Glu Val Val Pro Phe Leu Ala Arg Leu Ser Asn Arg Leu
210 215 220
Ser Thr Cys His Ile Glu Gly Asp Asp Leu His Ile Gln Arg Asn Val
225 230 235 240
Gln Lys Leu Lys Asp Thr Val Lys Lys Leu Gly Glu Ser Gly Glu Ile
245 250 255
Lys Ala Ile Gly Glu Leu Asp Leu Leu Phe Met Ser Leu Arg Asn Ala

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260	265	270
Cys Ile		
<210> SEQ ID NO 43		
<211> LENGTH: 418		
<212> TYPE: PRT		
<213> ORGANISM: Artificial Sequence		
<220> FEATURE:		
<223> OTHER INFORMATION: Synthesized		
<400> SEQUENCE: 43		
Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ser		
1 5 10 15		
Ser Val Lys Val Ser Cys Lys Ala Ser Gly Tyr Ser Phe Thr Ser Tyr		
20 25 30		
Trp Met Asn Trp Val Arg Gln Ala Pro Gly Gln Gly Leu Glu Trp Ile		
35 40 45		
Gly Met Ile Arg Pro Ser Asp Ser Glu Thr Arg Leu Asn Gln Met Phe		
50 55 60		
Lys Asp Arg Val Thr Ile Thr Val Asp Lys Ser Thr Ser Thr Ala Tyr		
65 70 75 80		
Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Val Tyr Tyr Cys		
85 90 95		
Ala Gly Ile His Asp Tyr Gly His Gly Ala Tyr Trp Gly Gln Gly Thr		
100 105 110		
Leu Val Thr Val Ser Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser		
115 120 125		
Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Thr Gln Ser Thr His		
130 135 140		
Glu Ser Leu Lys Pro Gln Arg Val Gln Phe Gln Ser Arg Asn Phe His		
145 150 155 160		
Asn Ile Leu Gln Trp Gln Pro Gly Arg Ala Leu Thr Gly Asn Ser Ser		
165 170 175		
Val Tyr Phe Val Gln Tyr Lys Ile Met Phe Glu Pro Lys Ser Ser Asp		
180 185 190		
Lys Thr His Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu Leu Gly Gly		
195 200 205		
Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile		
210 215 220		
Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser His Glu		
225 230 235 240		
Asp Pro Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val Glu Val His		
245 250 255		
Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Tyr Asn Ser Thr Tyr Arg		
260 265 270		
Val Val Ser Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly Lys		
275 280 285		
Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu Pro Ala Pro Ile Glu		
290 295 300		
Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr		
305 310 315 320		
Thr Leu Pro Pro Ser Arg Glu Glu Met Thr Lys Asn Gln Val Ser Leu		
325 330 335		

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Thr Cys Arg Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp
 340 345 350

Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val
 355 360 365

Leu Lys Ser Asp Gly Ser Phe Phe Leu Ala Ser Lys Leu Thr Val Asp
 370 375 380

Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met His
 385 390 395 400

Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro
 405 410 415

Gly Lys

<210> SEQ ID NO 44
 <211> LENGTH: 564
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 44

Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ser
 1 5 10 15

Ser Val Lys Val Ser Cys Lys Ala Ser Gly Tyr Ser Phe Thr Ser Tyr
 20 25 30

Trp Met Asn Trp Val Arg Gln Ala Pro Gly Gln Gly Leu Glu Trp Ile
 35 40 45

Gly Met Ile Arg Pro Ser Asp Ser Glu Thr Arg Leu Asn Gln Met Phe
 50 55 60

Lys Asp Arg Val Thr Ile Thr Val Asp Lys Ser Thr Ser Thr Ala Tyr
 65 70 75 80

Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95

Ala Gly Ile His Asp Tyr Gly His Gly Ala Tyr Trp Gly Gln Gly Thr
 100 105 110

Leu Val Thr Val Ser Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
 115 120 125

Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Cys Thr Ser Arg Pro
 130 135 140

His Ile Thr Val Val Glu Gly Glu Pro Phe Tyr Leu Lys His Cys Ser
 145 150 155 160

Cys Ser Leu Ala His Glu Ile Glu Thr Thr Thr Lys Ser Trp Tyr Lys
 165 170 175

Ser Ser Gly Ser Gln Glu His Val Glu Leu Asn Pro Arg Ser Ser Ser
 180 185 190

Arg Ile Ala Leu His Asp Cys Val Leu Glu Phe Trp Pro Val Glu Leu
 195 200 205

Asn Asp Thr Gly Ser Tyr Phe Phe Gln Met Lys Asn Tyr Thr Gln Lys
 210 215 220

Trp Lys Leu Asn Val Ile Arg Arg Asn Lys His Ser Cys Phe Thr Glu
 225 230 235 240

Arg Gln Val Thr Ser Lys Ile Val Glu Val Lys Lys Phe Phe Gln Ile
 245 250 255

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Thr	Cys	Glu	Asn	Ser	Tyr	Tyr	Gln	Thr	Leu	Val	Asn	Ser	Thr	Ser	Leu
			260					265					270		
Tyr	Lys	Asn	Cys	Lys	Lys	Leu	Leu	Leu	Glu	Asn	Asn	Lys	Asn	Pro	Thr
		275					280					285			
Ile	Lys	Lys	Asn	Ala	Glu	Phe	Glu	Asp	Gln	Gly	Tyr	Tyr	Ser	Cys	Val
	290					295					300				
His	Phe	Leu	His	His	Asn	Gly	Lys	Leu	Phe	Asn	Ile	Thr	Lys	Thr	Phe
305					310					315					320
Asn	Ile	Thr	Ile	Val	Glu	Asp	Arg	Ser	Asn	Ile	Val	Glu	Pro	Lys	Ser
			325					330						335	
Ser	Asp	Lys	Thr	His	Thr	Cys	Pro	Pro	Cys	Pro	Ala	Pro	Glu	Leu	Leu
			340					345					350		
Gly	Gly	Pro	Ser	Val	Phe	Leu	Phe	Pro	Pro	Lys	Pro	Lys	Asp	Thr	Leu
		355					360					365			
Met	Ile	Ser	Arg	Thr	Pro	Glu	Val	Thr	Cys	Val	Val	Val	Asp	Val	Ser
	370					375					380				
His	Glu	Asp	Pro	Glu	Val	Lys	Phe	Asn	Trp	Tyr	Val	Asp	Gly	Val	Glu
385					390					395					400
Val	His	Asn	Ala	Lys	Thr	Lys	Pro	Arg	Glu	Glu	Gln	Tyr	Asn	Ser	Thr
			405						410					415	
Tyr	Arg	Val	Val	Ser	Val	Leu	Thr	Val	Leu	His	Gln	Asp	Trp	Leu	Asn
			420					425					430		
Gly	Lys	Glu	Tyr	Lys	Cys	Lys	Val	Ser	Asn	Lys	Ala	Leu	Pro	Ala	Pro
		435					440					445			
Ile	Glu	Lys	Thr	Ile	Ser	Lys	Ala	Lys	Gly	Gln	Pro	Arg	Glu	Pro	Gln
	450					455					460				
Val	Tyr	Thr	Leu	Pro	Pro	Ser	Arg	Glu	Glu	Met	Thr	Lys	Asn	Gln	Val
465					470					475					480
Ser	Leu	Thr	Cys	Arg	Val	Lys	Gly	Phe	Tyr	Pro	Ser	Asp	Ile	Ala	Val
			485					490						495	
Glu	Trp	Glu	Ser	Asn	Gly	Gln	Pro	Glu	Asn	Asn	Tyr	Lys	Thr	Thr	Pro
			500					505					510		
Pro	Val	Leu	Lys	Ser	Asp	Gly	Ser	Phe	Phe	Leu	Ala	Ser	Lys	Leu	Thr
		515					520					525			
Val	Asp	Lys	Ser	Arg	Trp	Gln	Gln	Gly	Asn	Val	Phe	Ser	Cys	Ser	Val
	530					535					540				
Met	His	Glu	Ala	Leu	His	Asn	His	Tyr	Thr	Gln	Lys	Ser	Leu	Ser	Leu
545					550					555					560

Ser Pro Gly Lys

<210> SEQ ID NO 45
 <211> LENGTH: 285
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 45

Asp	Ile	Gln	Met	Thr	Gln	Ser	Pro	Ser	Ser	Leu	Ser	Ala	Ser	Val	Gly
1				5					10					15	
Asp	Arg	Val	Thr	Ile	Thr	Cys	Arg	Ala	Ser	Glu	Asn	Ile	Tyr	Ser	Asn
		20					25						30		
Leu	Ala	Trp	Tyr	Gln	Gln	Lys	Pro	Gly	Lys	Ser	Pro	Lys	Leu	Leu	Val

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35					40					45					
Tyr	Ala	Ala	Ser	His	Leu	Pro	Asp	Gly	Val	Pro	Ser	Arg	Phe	Ser	Gly
50						55					60				
Ser	Gly	Ser	Gly	Thr	Asp	Tyr	Ser	Leu	Thr	Ile	Ser	Ser	Leu	Gln	Pro
65					70					75				80	
Glu	Asp	Phe	Ala	Thr	Tyr	Tyr	Cys	Gln	His	Phe	Trp	Gly	Thr	Pro	Arg
				85					90					95	
Thr	Phe	Gly	Gln	Gly	Thr	Lys	Leu	Glu	Ile	Lys	Gly	Gly	Gly	Gly	Ser
			100					105					110		
Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly
		115				120					125				
Tyr	Phe	Gly	Lys	Leu	Glu	Ser	Lys	Leu	Ser	Val	Ile	Arg	Asn	Leu	Asn
	130					135					140				
Asp	Gln	Val	Leu	Phe	Ile	Asp	Gln	Gly	Asn	Arg	Pro	Leu	Phe	Glu	Asp
145					150					155				160	
Met	Thr	Asp	Ser	Asp	Cys	Arg	Asp	Asn	Ala	Pro	Arg	Thr	Ile	Phe	Ile
			165					170						175	
Ile	Ser	Met	Tyr	Lys	Asp	Ser	Gln	Pro	Arg	Gly	Met	Ala	Val	Thr	Ile
		180						185					190		
Ser	Val	Lys	Cys	Glu	Lys	Ile	Ser	Thr	Leu	Ser	Cys	Glu	Asn	Lys	Ile
		195					200					205			
Ile	Ser	Phe	Lys	Glu	Met	Asn	Pro	Pro	Asp	Asn	Ile	Lys	Asp	Thr	Lys
	210					215					220				
Ser	Asp	Ile	Ile	Phe	Phe	Gln	Arg	Ser	Val	Pro	Gly	His	Asp	Asn	Lys
225					230					235				240	
Met	Gln	Phe	Glu	Ser	Ser	Ser	Tyr	Glu	Gly	Tyr	Phe	Leu	Ala	Cys	Glu
			245						250					255	
Lys	Glu	Arg	Asp	Leu	Phe	Lys	Leu	Ile	Leu	Lys	Lys	Glu	Asp	Glu	Leu
		260					265					270			
Gly	Asp	Arg	Ser	Ile	Met	Phe	Thr	Val	Gln	Asn	Glu	Asp			
	275						280					285			

<210> SEQ ID NO 46

<211> LENGTH: 662

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 46

Gln	Val	Gln	Leu	Val	Gln	Ser	Gly	Ala	Glu	Val	Lys	Lys	Pro	Gly	Ser
1			5					10					15		
Ser	Val	Lys	Val	Ser	Cys	Lys	Ala	Ser	Gly	Tyr	Ser	Phe	Thr	Ser	Tyr
		20					25					30			
Trp	Met	Asn	Trp	Val	Arg	Gln	Ala	Pro	Gly	Gln	Gly	Leu	Glu	Trp	Ile
	35					40					45				
Gly	Met	Ile	Arg	Pro	Ser	Asp	Ser	Glu	Thr	Arg	Leu	Asn	Gln	Met	Phe
	50				55					60					
Lys	Asp	Arg	Val	Thr	Ile	Thr	Val	Asp	Lys	Ser	Thr	Ser	Thr	Ala	Tyr
65				70					75					80	
Met	Glu	Leu	Ser	Ser	Leu	Arg	Ser	Glu	Asp	Thr	Ala	Val	Tyr	Tyr	Cys
			85					90					95		
Ala	Gly	Ile	His	Asp	Tyr	Gly	His	Gly	Ala	Tyr	Trp	Gly	Gln	Gly	Thr

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100					105					110				
Leu	Val	Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Ser
	115						120					125		
Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Cys	Thr	Ser	Arg
	130					135					140			Pro
His	Ile	Thr	Val	Val	Glu	Gly	Glu	Pro	Phe	Tyr	Leu	Lys	His	Cys
145					150					155				160
Cys	Ser	Leu	Ala	His	Glu	Ile	Glu	Thr	Thr	Thr	Lys	Ser	Trp	Tyr
				165					170					175
Ser	Ser	Gly	Ser	Gln	Glu	His	Val	Glu	Leu	Asn	Pro	Arg	Ser	Ser
			180						185				190	
Arg	Ile	Ala	Leu	His	Asp	Cys	Val	Leu	Glu	Phe	Trp	Pro	Val	Glu
	195						200					205		Leu
Asn	Asp	Thr	Gly	Ser	Tyr	Phe	Phe	Gln	Met	Lys	Asn	Tyr	Thr	Gln
	210					215					220			Lys
Trp	Lys	Leu	Asn	Val	Ile	Arg	Arg	Asn	Lys	His	Ser	Cys	Phe	Thr
225					230					235				240
Arg	Gln	Val	Thr	Ser	Lys	Ile	Val	Glu	Val	Lys	Lys	Phe	Phe	Gln
				245					250					255
Thr	Cys	Glu	Asn	Ser	Tyr	Tyr	Gln	Thr	Leu	Val	Asn	Ser	Thr	Ser
		260						265					270	Leu
Tyr	Lys	Asn	Cys	Lys	Lys	Leu	Leu	Leu	Glu	Asn	Asn	Lys	Asn	Pro
	275						280					285		Thr
Ile	Lys	Lys	Asn	Ala	Glu	Phe	Glu	Asp	Gln	Gly	Tyr	Tyr	Ser	Cys
	290					295					300			Val
His	Phe	Leu	His	His	Asn	Gly	Lys	Leu	Phe	Asn	Ile	Thr	Lys	Thr
305					310					315				320
Asn	Ile	Thr	Ile	Val	Glu	Asp	Arg	Ser	Asn	Ile	Val	Pro	Val	Leu
				325					330					335
Gly	Pro	Lys	Leu	Asn	His	Val	Ala	Val	Glu	Leu	Gly	Lys	Asn	Val
		340						345					350	Arg
Leu	Asn	Cys	Ser	Ala	Leu	Leu	Asn	Glu	Glu	Asp	Val	Ile	Tyr	Trp
	355						360					365		Met
Phe	Gly	Glu	Glu	Asn	Gly	Ser	Asp	Pro	Asn	Ile	His	Glu	Glu	Lys
	370				375						380			Glu
Met	Arg	Ile	Met	Thr	Pro	Glu	Gly	Lys	Trp	His	Ala	Ser	Lys	Val
385					390					395				400
Arg	Ile	Glu	Asn	Ile	Gly	Glu	Ser	Asn	Leu	Asn	Val	Leu	Tyr	Asn
				405					410					415
Thr	Val	Ala	Ser	Thr	Gly	Gly	Thr	Asp	Thr	Lys	Ser	Phe	Ile	Glu
		420						425					430	Pro
Lys	Ser	Ser	Asp	Lys	Thr	His	Thr	Cys	Pro	Pro	Cys	Pro	Ala	Pro
		435					440					445		Glu
Leu	Leu	Gly	Gly	Pro	Ser	Val	Phe	Leu	Phe	Pro	Pro	Lys	Pro	Lys
	450					455						460		Asp
Thr	Leu	Met	Ile	Ser	Arg	Thr	Pro	Glu	Val	Thr	Cys	Val	Val	Val
465					470					475				480
Val	Ser	His	Glu	Asp	Pro	Glu	Val	Lys	Phe	Asn	Trp	Tyr	Val	Asp
				485					490					495
Val	Glu	Val	His	Asn	Ala	Lys	Thr	Lys	Pro	Arg	Glu	Glu	Gln	Tyr
			500					505					510	Asn

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Ser Thr Tyr Arg Val Val Ser Val Leu Thr Val Leu His Gln Asp Trp
   515                               520                               525

Leu Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu Pro
   530                               535                               540

Ala Pro Ile Glu Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu
   545                               550                               555                               560

Pro Gln Val Tyr Thr Leu Pro Pro Ser Arg Glu Glu Met Thr Lys Asn
   565                               570                               575

Gln Val Ser Leu Thr Cys Arg Val Lys Gly Phe Tyr Pro Ser Asp Ile
   580                               585                               590

Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr
   595                               600                               605

Thr Pro Pro Val Leu Lys Ser Asp Gly Ser Phe Phe Leu Ala Ser Lys
   610                               615                               620

Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys
   625                               630                               635                               640

Ser Val Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu
   645                               650                               655

Ser Leu Ser Pro Gly Lys
   660

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<210> SEQ ID NO 47
<211> LENGTH: 584
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthesized

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<400> SEQUENCE: 47

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Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ser
 1           5           10           15

Ser Val Lys Val Ser Cys Lys Ala Ser Gly Tyr Ser Phe Thr Ser Tyr
 20           25           30

Trp Met Asn Trp Val Arg Gln Ala Pro Gly Gln Gly Leu Glu Trp Ile
 35           40           45

Gly Met Ile Arg Pro Ser Asp Ser Glu Thr Arg Leu Asn Gln Met Phe
 50           55           60

Lys Asp Arg Val Thr Ile Thr Val Asp Lys Ser Thr Ser Thr Ala Tyr
 65           70           75           80

Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Val Tyr Tyr Cys
 85           90           95

Ala Gly Ile His Asp Tyr Gly His Gly Ala Tyr Trp Gly Gln Gly Thr
100           105           110

Leu Val Thr Val Ser Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
115           120           125

Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Cys Pro Asp Leu Val
130           135           140

Cys Tyr Thr Asp Tyr Leu Gln Thr Val Ile Cys Ile Leu Glu Met Trp
145           150           155           160

Asn Leu His Pro Ser Thr Leu Thr Leu Thr Trp Gln Asp Gln Tyr Glu
165           170           175

Glu Leu Lys Asp Glu Ala Thr Ser Cys Ser Leu His Arg Ser Ala His
180           185           190

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Asn	Ala	Thr	His	Ala	Thr	Tyr	Thr	Cys	His	Met	Asp	Val	Phe	His	Phe
	195						200					205			
Met	Ala	Asp	Asp	Ile	Phe	Ser	Val	Gln	Ile	Thr	Asp	Gln	Ser	Gly	Gln
	210					215					220				
Tyr	Ser	Gln	Glu	Cys	Gly	Ser	Phe	Leu	Leu	Ala	Glu	Ser	Ile	Lys	Pro
225					230					235					240
Ala	Pro	Pro	Phe	Asp	Val	Thr	Val	Thr	Phe	Ser	Gly	Gln	Tyr	Gln	Ile
				245					250					255	
Ser	Trp	Arg	Ser	Asp	Tyr	Glu	Asp	Pro	Ala	Phe	Tyr	Met	Leu	Lys	Gly
			260					265					270		
Lys	Leu	Gln	Tyr	Glu	Leu	Gln	Tyr	Arg	Asn	Arg	Gly	Asp	Pro	Trp	Ala
	275						280					285			
Val	Ser	Pro	Arg	Arg	Lys	Leu	Ile	Ser	Val	Asp	Ser	Arg	Ser	Val	Ser
	290					295					300				
Leu	Leu	Pro	Leu	Glu	Phe	Arg	Lys	Asp	Ser	Ser	Tyr	Glu	Leu	Gln	Val
305					310					315					320
Arg	Ala	Gly	Pro	Met	Pro	Gly	Ser	Ser	Tyr	Gln	Gly	Thr	Trp	Ser	Glu
				325					330					335	
Trp	Ser	Asp	Pro	Val	Ile	Phe	Gln	Thr	Gln	Ser	Glu	Glu	Leu	Lys	Glu
			340					345					350		
Glu	Pro	Lys	Ser	Ser	Asp	Lys	Thr	His	Thr	Cys	Pro	Pro	Cys	Pro	Ala
		355					360					365			
Pro	Glu	Leu	Leu	Gly	Gly	Pro	Ser	Val	Phe	Leu	Phe	Pro	Pro	Lys	Pro
	370					375					380				
Lys	Asp	Thr	Leu	Met	Ile	Ser	Arg	Thr	Pro	Glu	Val	Thr	Cys	Val	Val
385					390					395					400
Val	Asp	Val	Ser	His	Glu	Asp	Pro	Glu	Val	Lys	Phe	Asn	Trp	Tyr	Val
				405					410					415	
Asp	Gly	Val	Glu	Val	His	Asn	Ala	Lys	Thr	Lys	Pro	Arg	Glu	Glu	Gln
			420					425					430		
Tyr	Asn	Ser	Thr	Tyr	Arg	Val	Val	Ser	Val	Leu	Thr	Val	Leu	His	Gln
	435						440					445			
Asp	Trp	Leu	Asn	Gly	Lys	Glu	Tyr	Lys	Cys	Lys	Val	Ser	Asn	Lys	Ala
	450					455					460				
Leu	Pro	Ala	Pro	Ile	Glu	Lys	Thr	Ile	Ser	Lys	Ala	Lys	Gly	Gln	Pro
465					470					475					480
Arg	Glu	Pro	Gln	Val	Tyr	Thr	Leu	Pro	Pro	Ser	Arg	Glu	Glu	Met	Thr
				485					490					495	
Lys	Asn	Gln	Val	Ser	Leu	Thr	Cys	Arg	Val	Lys	Gly	Phe	Tyr	Pro	Ser
			500					505					510		
Asp	Ile	Ala	Val	Glu	Trp	Glu	Ser	Asn	Gly	Gln	Pro	Glu	Asn	Asn	Tyr
	515						520					525			
Lys	Thr	Thr	Pro	Pro	Val	Leu	Lys	Ser	Asp	Gly	Ser	Phe	Phe	Leu	Ala
	530					535					540				
Ser	Lys	Leu	Thr	Val	Asp	Lys									

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<210> SEQ ID NO 48
<211> LENGTH: 266
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 48

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1 5 10 15
Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Glu Asn Ile Tyr Ser Asn
20 25 30
Leu Ala Trp Tyr Gln Gln Lys Pro Gly Lys Ser Pro Lys Leu Leu Val
35 40 45
Tyr Ala Ala Ser His Leu Pro Asp Gly Val Pro Ser Arg Phe Ser Gly
50 55 60
Ser Gly Ser Gly Thr Asp Tyr Ser Leu Thr Ile Ser Ser Leu Gln Pro
65 70 75 80
Glu Asp Phe Ala Thr Tyr Tyr Cys Gln His Phe Trp Gly Thr Pro Arg
85 90 95
Thr Phe Gly Gln Gly Thr Lys Leu Glu Ile Lys Gly Gly Gly Gly Ser
100 105 110
Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly
115 120 125
His Lys Ser Ser Ser Gln Gly Gln Asp Arg His Met Ile Arg Met Arg
130 135 140
Gln Leu Ile Asp Ile Val Asp Gln Leu Lys Asn Tyr Val Asn Asp Leu
145 150 155 160
Val Pro Glu Phe Leu Pro Ala Pro Glu Asp Val Glu Thr Asn Cys Glu
165 170 175
Trp Ser Ala Phe Ser Cys Phe Gln Lys Ala Gln Leu Lys Ser Ala Asn
180 185 190
Thr Gly Asn Asn Glu Arg Ile Ile Asn Val Ser Ile Lys Lys Leu Lys
195 200 205
Arg Lys Pro Pro Ser Thr Asn Ala Gly Arg Arg Gln Lys His Arg Leu
210 215 220
Thr Cys Pro Ser Cys Asp Ser Tyr Glu Lys Lys Pro Pro Lys Glu Phe
225 230 235 240
Leu Glu Arg Phe Lys Ser Leu Leu Gln Lys Met Ile His Gln His Leu
245 250 255
Ser Ser Arg Thr His Gly Ser Glu Asp Ser
260 265

<210> SEQ ID NO 49
<211> LENGTH: 546
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 49

Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ser
1 5 10 15
Ser Val Lys Val Ser Cys Lys Ala Ser Gly Tyr Ser Phe Thr Ser Tyr
20 25 30

Trp	Met	Asn	Trp	Val	Arg	Gln	Ala	Pro	Gly	Gln	Gly	Leu	Glu	Trp	Ile			
35																		
Gly	Met	Ile	Arg	Pro	Ser	Asp	Ser	Glu	Thr	Arg	Leu	Asn	Gln	Met	Phe			
50																		
Lys	Asp	Arg	Val	Thr	Ile	Thr	Val	Asp	Lys	Ser	Thr	Ser	Thr	Ala	Tyr			
65		70					75					80						
Met	Glu	Leu	Ser	Ser	Leu	Arg	Ser	Glu	Asp	Thr	Ala	Val	Tyr	Tyr	Cys			
				85												95		
Ala	Gly	Ile	His	Asp	Tyr	Gly	His	Gly	Ala	Tyr	Trp	Gly	Gln	Gly	Thr			
			100		105					110								
Leu	Val	Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser			
115																		
Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Ile	Thr	Cys	Pro	Pro			
130																		
Pro	Met	Ser	Val	Glu	His	Ala	Asp	Ile	Trp	Val	Lys	Ser	Tyr	Ser	Leu			
145		150					155					160						
Tyr	Ser	Arg	Glu	Arg	Tyr	Ile	Cys	Asn	Ser	Gly	Phe	Lys	Arg	Lys	Ala			
			165		170					175								
Gly	Thr	Ser	Ser	Leu	Thr	Glu	Cys	Val	Leu	Asn	Lys	Ala	Thr	Asn	Val			
			180		185					190								
Ala	His	Trp	Thr	Thr	Pro	Ser	Leu	Lys	Cys	Ile	Arg	Asp	Cys	Ala	Leu			
195		200					205											
Val	His	Gln	Arg	Pro	Ala	Pro	Pro	Ser	Thr	Val	Thr	Thr	Ala	Gly	Val			
210		215					220											
Thr	Pro	Gln	Pro	Glu	Ser	Leu	Ser	Pro	Ser	Gly	Lys	Glu	Pro	Ala	Ala			
225		230					235					240						
Ser	Ser	Pro	Ser	Ser	Asn	Asn	Thr	Ala	Ala	Thr	Thr	Ala	Ala	Ile	Val			
				245		250					255							
Pro	Gly	Ser	Gln	Leu	Met	Pro	Ser	Lys	Ser	Pro	Ser	Thr	Gly	Thr	Thr			
			260		265					270								
Glu	Ile	Ser	Ser	His	Glu	Ser	Ser	His	Gly	Thr	Pro	Ser	Gln	Thr	Thr			
275		280					285											
Ala	Lys	Asn	Trp	Glu	Leu	Thr	Ala	Ser	Ala	Ser	His	Gln	Pro	Pro	Gly			
290		295					300											
Val	Tyr	Pro	Gln	Gly	His	Ser	Asp	Thr	Thr	Glu	Pro	Lys	Ser	Ser	Asp			
305		310					315					320						
Lys	Thr	His	Thr	Cys	Pro	Pro	Cys	Pro	Ala	Pro	Glu	Leu	Leu	Gly	Gly			
			325		330					335								
Pro	Ser	Val	Phe	Leu	Phe	Pro	Pro	Lys	Pro	Lys	Asp	Thr	Leu	Met	Ile			
			340		345					350								
Ser	Arg	Thr	Pro	Glu	Val	Thr	Cys	Val	Val	Val	Asp	Val	Ser	His	Glu			
355				360					365									
Asp	Pro	Glu	Val	Lys	Phe	Asn	Trp	Tyr	Val	Asp	Gly	Val	Glu	Val	His			
370		375					380											
Asn	Ala	Lys	Thr	Lys	Pro	Arg	Glu	Glu	Gln	Tyr	Asn	Ser	Thr	Tyr	Arg			
385		390					395					400						
Val	Val	Ser	Val	Leu	Thr	Val	Leu	His	Gln	Asp	Trp	Leu	Asn	Gly	Lys			
				405		410					415							
Glu	Tyr	Lys	Cys	Lys	Val	Ser	Asn	Lys	Ala	Leu	Pro	Ala	Pro	Ile	Glu			
420		425					430											

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Lys	Thr	Ile	Ser	Lys	Ala	Lys	Gly	Gln	Pro	Arg	Glu	Pro	Gln	Val	Tyr
		435					440					445			
Thr	Leu	Pro	Pro	Ser	Arg	Glu	Glu	Met	Thr	Lys	Asn	Gln	Val	Ser	Leu
	450					455					460				
Thr	Cys	Arg	Val	Lys	Gly	Phe	Tyr	Pro	Ser	Asp	Ile	Ala	Val	Glu	Trp
465					470					475					480
Glu	Ser	Asn	Gly	Gln	Pro	Glu	Asn	Asn	Tyr	Lys	Thr	Thr	Pro	Pro	Val
				485					490					495	
Leu	Lys	Ser	Asp	Gly	Ser	Phe	Phe	Leu	Ala	Ser	Lys	Leu	Thr	Val	Asp
			500					505					510		
Lys	Ser	Arg	Trp	Gln	Gln	Gly	Asn	Val	Phe	Ser	Cys	Ser	Val	Met	His
		515					520					525			
Glu	Ala	Leu	His	Asn	His	Tyr	Thr	Gln	Lys	Ser	Leu	Ser	Leu	Ser	Pro
	530					535					540				
Gly	Lys														
545															

<210> SEQ ID NO 50
 <211> LENGTH: 242
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 50

Asp	Ile	Gln	Met	Thr	Gln	Ser	Pro	Ser	Ser	Leu	Ser	Ala	Ser	Val	Gly
1				5					10					15	
Asp	Arg	Val	Thr	Ile	Thr	Cys	Arg	Ala	Ser	Glu	Asn	Ile	Tyr	Ser	Asn
		20						25					30		
Leu	Ala	Trp	Tyr	Gln	Gln	Lys	Pro	Gly	Lys	Ser	Pro	Lys	Leu	Leu	Val
		35					40					45			
Tyr	Ala	Ala	Ser	His	Leu	Pro	Asp	Gly	Val	Pro	Ser	Arg	Phe	Ser	Gly
	50					55				60					
Ser	Gly	Ser	Gly	Thr	Asp	Tyr	Ser	Leu	Thr	Ile	Ser	Ser	Leu	Gln	Pro
65					70					75				80	
Glu	Asp	Phe	Ala	Thr	Tyr	Tyr	Cys	Gln	His	Phe	Trp	Gly	Thr	Pro	Arg
			85					90					95		
Thr	Phe	Gly	Gln	Gly	Thr	Lys	Leu	Glu	Ile	Lys	Gly	Gly	Gly	Gly	Ser
		100						105					110		
Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly
		115				120					125				
Asn	Trp	Val	Asn	Val	Ile	Ser	Asp	Leu	Lys	Lys	Ile	Glu	Asp	Leu	Ile
	130					135					140				
Gln	Ser	Met	His	Ile	Asp	Ala	Thr	Leu	Tyr	Thr	Glu	Ser	Asp	Val	His
145					150					155					160
Pro	Ser	Cys	Lys	Val	Thr	Ala	Met	Lys	Cys	Phe	Leu	Leu	Glu	Leu	Gln
			165					170						175	
Val	Ile	Ser	Leu	Glu	Ser	Gly	Asp	Ala	Ser	Ile	His	Asn	Thr	Val	Gln
		180						185					190		
Asp	Leu	Ile	Ile	Leu	Ala	Asn	Asn	Ser	Leu	Ser	Ser	Asn	Gly	Asn	Val
	195					200						205			
Thr	Glu	Ser	Gly	Cys	Lys	Glu	Cys	Glu	Cys	Leu	Glu	Glu	Lys	Asn	Ile
	210					215					220				

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Lys Glu Phe Leu Gln Ser Phe Val His Ile Val Gln Met Phe Ile Asn
225 230 235 240

Thr Ser

<210> SEQ ID NO 51
 <211> LENGTH: 449
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 51

Gln Val Gln Leu Gln Glu Ser Gly Pro Gly Leu Val Lys Pro Ser Glu
1 5 10 15

Thr Leu Ser Leu Thr Cys Ala Val Tyr Gly Asp Ser Ile Thr Ser Gly
20 25 30

Tyr Trp Asn Trp Ile Arg Lys Pro Pro Gly Lys Gly Leu Glu Tyr Ile
35 40 45

Gly Tyr Ile Ser Tyr Thr Gly Ser Thr Tyr Gln Asn Pro Ser Leu Lys
50 55 60

Ser Arg Ile Thr Met Ser Arg Asp Thr Ser Lys Asn Gln Tyr Tyr Leu
65 70 75 80

Lys Leu Ser Ser Val Thr Ala Ala Asp Thr Ala Val Tyr Tyr Cys Ala
85 90 95

Arg Ser Arg Ala Trp Ile Arg Thr Tyr Phe Asp Tyr Trp Gly Gln Gly
100 105 110

Thr Leu Val Thr Val Ser Ser Ala Ser Thr Lys Gly Pro Ser Val Phe
115 120 125

Pro Leu Ala Pro Ser Ser Lys Ser Thr Ser Gly Gly Thr Ala Ala Leu
130 135 140

Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val Thr Val Ser Trp
145 150 155 160

Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro Ala Val Leu
165 170 175

Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val Thr Val Pro Ser
180 185 190

Ser Ser Leu Gly Thr Gln Thr Tyr Ile Cys Asn Val Asn His Lys Pro
195 200 205

Ser Asn Thr Lys Val Asp Lys Arg Val Glu Pro Lys Ser Cys Asp Lys
210 215 220

Thr His Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu Leu Gly Gly Pro
225 230 235 240

Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile Ser
245 250 255

Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser His Glu Asp
260 265 270

Pro Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val Glu Val His Asn
275 280 285

Ala Lys Thr Lys Pro Arg Glu Glu Gln Tyr Asn Ser Thr Tyr Arg Val
290 295 300

Val Ser Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly Lys Glu
305 310 315 320

Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu Pro Ala Pro Ile Glu Lys

325								330							335				
Thr	Ile	Ser	Lys	Ala	Lys	Gly	Gln	Pro	Arg	Glu	Pro	Gln	Val	Tyr	Thr				
			340						345						350				
Leu	Pro	Pro	Ser	Arg	Glu	Glu	Met	Thr	Lys	Asn	Gln	Val	Ser	Leu	Trp				
			355						360						365				
Cys	Leu	Val	Lys	Gly	Phe	Tyr	Pro	Ser	Asp	Ile	Ala	Val	Glu	Trp	Glu				
			370						375						380				
Ser	Asn	Gly	Gln	Pro	Glu	Asn	Asn	Tyr	Asp	Thr	Thr	Pro	Pro	Val	Leu				
			385						390						395				
Asp	Ser	Asp	Gly	Ser	Phe	Phe	Leu	Tyr	Ser	Asp	Leu	Thr	Val	Asp	Lys				
			405						410						415				
Ser	Arg	Trp	Gln	Gln	Gly	Asn	Val	Phe	Ser	Cys	Ser	Val	Met	His	Glu				
			420						425						430				
Ala	Leu	His	Asn	His	Tyr	Thr	Gln	Lys	Ser	Leu	Ser	Leu	Ser	Pro	Gly				
			435						440						445				
Lys																			
<210> SEQ ID NO 52																			
<211> LENGTH: 215																			
<212> TYPE: PRT																			
<213> ORGANISM: Artificial Sequence																			
<220> FEATURE:																			
<223> OTHER INFORMATION: Synthesized																			
<400> SEQUENCE: 52																			
Asp	Ile	Gln	Met	Thr	Gln	Ser	Pro	Ser	Ser	Leu	Ser	Ala	Ser	Val	Gly				
1				5						10						15			
Asp	Arg	Val	Thr	Ile	Thr	Cys	Ser	Val	Ser	Ser	Ser	Ile	Ser	Ser	Ser				
			20						25						30				
Asn	Leu	His	Trp	Tyr	Gln	Gln	Lys	Pro	Gly	Lys	Ala	Pro	Lys	Pro	Trp				
			35						40						45				
Ile	Tyr	Gly	Thr	Ser	Asn	Leu	Ala	Ser	Gly	Val	Pro	Ser	Arg	Phe	Ser				
			50						55						60				
Gly	Ser	Gly	Ser	Gly	Thr	Asp	Tyr	Thr	Leu	Thr	Ile	Ser	Ser	Leu	Gln				
			65						70						75				
Pro	Glu	Asp	Phe	Ala	Thr	Tyr	Tyr	Cys	Gln	Gln	Trp	Ser	Ser	Tyr	Pro				
			85						90						95				
Leu	Thr	Phe	Gly	Gln	Gly	Thr	Lys	Leu	Glu	Ile	Lys	Arg	Thr	Val	Ala				
			100						105						110				
Ala	Pro	Ser	Val	Phe	Ile	Phe	Pro	Pro	Ser	Asp	Glu	Gln	Leu	Lys	Ser				
			115						120						125				
Gly	Thr	Ala	Ser	Val	Val	Cys	Leu	Leu	Asn	Asn	Phe	Tyr	Pro	Arg	Glu				
			130						135						140				
Ala	Lys	Val	Gln	Trp	Lys	Val	Asp	Asn	Ala	Leu	Gln	Ser	Gly	Asn	Ser				
			145						150						155				
Gln	Glu	Ser	Val	Thr	Glu	Gln	Asp	Ser	Lys	Asp	Ser	Thr	Tyr	Ser	Leu				
			165						170						175				
Ser	Ser	Thr	Leu	Thr	Leu	Ser	Lys	Ala	Asp	Tyr	Glu	Lys	His	Lys	Val				
			180						185						190				
Tyr	Ala	Cys	Glu	Val	Thr	His	Gln	Gly	Leu	Ser	Ser	Pro	Val	Thr	Lys				
			195						200						205				
Ser	Phe	Asn	Arg	Gly	Glu	Cys													
			210						215										

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<210> SEQ ID NO 53
<211> LENGTH: 546
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 53

Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ser
1          5              10              15

Ser Val Lys Val Ser Cys Lys Ala Ser Gly Tyr Ser Phe Thr Ser Tyr
20          25              30

Trp Met Asn Trp Val Arg Gln Ala Pro Gly Gln Cys Leu Glu Trp Ile
35          40              45

Gly Met Ile Arg Pro Ser Asp Ser Glu Thr Arg Leu Asn Gln Met Phe
50          55              60

Lys Asp Arg Val Thr Ile Thr Val Asp Lys Ser Thr Ser Thr Ala Tyr
65          70              75              80

Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Val Tyr Tyr Cys
85          90              95

Ala Gly Ile His Asp Tyr Gly His Gly Ala Tyr Trp Gly Gln Gly Thr
100         105              110

Leu Val Thr Val Ser Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
115         120              125

Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Ile Thr Cys Pro Pro
130         135              140

Pro Met Ser Val Glu His Ala Asp Ile Trp Val Lys Ser Tyr Ser Leu
145         150              155              160

Tyr Ser Arg Glu Arg Tyr Ile Cys Asn Ser Gly Phe Lys Arg Lys Ala
165         170              175

Gly Thr Ser Ser Leu Thr Glu Cys Val Leu Asn Lys Ala Thr Asn Val
180         185              190

Ala His Trp Thr Thr Pro Ser Leu Lys Cys Ile Arg Asp Pro Ala Leu
195         200              205

Val His Gln Arg Pro Ala Pro Pro Ser Thr Val Thr Thr Ala Gly Val
210         215              220

Thr Pro Gln Pro Glu Ser Leu Ser Pro Ser Gly Lys Glu Pro Ala Ala
225         230              235              240

Ser Ser Pro Ser Ser Asn Asn Thr Ala Ala Thr Thr Ala Ala Ile Val
245         250              255

Pro Gly Ser Gln Leu Met Pro Ser Lys Ser Pro Ser Thr Gly Thr Thr
260         265              270

Glu Ile Ser Ser His Glu Ser Ser His Gly Thr Pro Ser Gln Thr Thr
275         280              285

Ala Lys Asn Trp Glu Leu Thr Ala Ser Ala Ser His Gln Pro Pro Gly
290         295              300

Val Tyr Pro Gln Gly His Ser Asp Thr Thr Glu Pro Lys Ser Ser Asp
305         310              315              320

Lys Thr His Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu Leu Gly Gly
325         330              335

Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile
340         345              350

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Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser His Glu
 355 360 365
 Asp Pro Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val Glu Val His
 370 375 380
 Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Tyr Asn Ser Thr Tyr Arg
 385 390 395 400
 Val Val Ser Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly Lys
 405 410 415
 Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu Pro Ala Pro Ile Glu
 420 425 430
 Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr
 435 440 445
 Thr Leu Pro Pro Ser Arg Glu Glu Met Thr Lys Asn Gln Val Ser Leu
 450 455 460
 Thr Cys Arg Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp
 465 470 475 480
 Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val
 485 490 495
 Leu Lys Ser Asp Gly Ser Phe Phe Leu Ala Ser Lys Leu Thr Val Asp
 500 505 510
 Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met His
 515 520 525
 Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro
 530 535 540
 Gly Lys
 545

<210> SEQ ID NO 54
 <211> LENGTH: 242
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 54

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
 1 5 10 15
 Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Glu Asn Ile Tyr Ser Asn
 20 25 30
 Leu Ala Trp Tyr Gln Gln Lys Pro Gly Lys Ser Pro Lys Leu Leu Val
 35 40 45
 Tyr Ala Ala Ser His Leu Pro Asp Gly Val Pro Ser Arg Phe Ser Gly
 50 55 60
 Ser Gly Ser Gly Thr Asp Tyr Ser Leu Thr Ile Ser Ser Leu Gln Pro
 65 70 75 80
 Glu Asp Phe Ala Thr Tyr Tyr Cys Gln His Phe Trp Gly Thr Pro Arg
 85 90 95
 Thr Phe Gly Cys Gly Thr Lys Leu Glu Ile Lys Gly Gly Gly Gly Ser
 100 105 110
 Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly
 115 120 125
 Asn Trp Val Asn Val Ile Ser Asp Leu Lys Lys Ile Glu Asp Leu Ile
 130 135 140

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Gln Ser Met His Ile Asp Ala Thr Leu Tyr Thr Glu Ser Asp Val His
 145 150 155 160

Pro Ser Cys Lys Val Thr Ala Met Lys Cys Phe Leu Leu Glu Leu Gln
 165 170 175

Val Ile Ser Leu Glu Ser Gly Asp Ala Ser Ile His Asn Thr Val Gln
 180 185 190

Asp Leu Ile Ile Leu Ala Asn Asn Ser Leu Ser Ser Asn Gly Asn Val
 195 200 205

Thr Glu Ser Gly Cys Lys Glu Cys Glu Glu Leu Glu Glu Lys Asn Ile
 210 215 220

Lys Glu Phe Leu Gln Ser Phe Val His Ile Val Gln Met Phe Ile Asn
 225 230 235 240

Thr Ser

<210> SEQ ID NO 55
 <211> LENGTH: 546
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 55

Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ser
 1 5 10 15

Ser Val Lys Val Ser Cys Lys Ala Ser Gly Tyr Ser Phe Thr Ser Tyr
 20 25 30

Trp Met Asn Trp Val Arg Gln Ala Pro Gly Gln Gly Leu Glu Trp Ile
 35 40 45

Gly Met Ile Arg Pro Ser Asp Ser Glu Thr Arg Leu Asn Gln Met Phe
 50 55 60

Lys Asp Arg Val Thr Ile Thr Val Asp Lys Ser Thr Ser Thr Ala Tyr
 65 70 75 80

Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95

Ala Gly Ile His Asp Tyr Gly His Gly Ala Tyr Cys Gly Gln Gly Thr
 100 105 110

Leu Val Thr Val Ser Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
 115 120 125

Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Ile Thr Cys Pro Pro
 130 135 140

Pro Met Ser Val Glu His Ala Asp Ile Trp Val Lys Ser Tyr Ser Leu
 145 150 155 160

Tyr Ser Arg Glu Arg Tyr Ile Cys Asn Ser Gly Phe Lys Arg Lys Ala
 165 170 175

Gly Thr Ser Ser Leu Thr Glu Cys Val Leu Asn Lys Ala Thr Asn Val
 180 185 190

Ala His Trp Thr Thr Pro Ser Leu Lys Cys Ile Arg Asp Pro Ala Leu
 195 200 205

Val His Gln Arg Pro Ala Pro Pro Ser Thr Val Thr Thr Ala Gly Val
 210 215 220

Thr Pro Gln Pro Glu Ser Leu Ser Pro Ser Gly Lys Glu Pro Ala Ala
 225 230 235 240

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Ser	Ser	Pro	Ser	Ser	Asn	Asn	Thr	Ala	Ala	Thr	Thr	Ala	Ala	Ile	Val
					245				250					255	
Pro	Gly	Ser	Gln	Leu	Met	Pro	Ser	Lys	Ser	Pro	Ser	Thr	Gly	Thr	Thr
			260					265					270		
Glu	Ile	Ser	Ser	His	Glu	Ser	Ser	His	Gly	Thr	Pro	Ser	Gln	Thr	Thr
		275						280				285			
Ala	Lys	Asn	Trp	Glu	Leu	Thr	Ala	Ser	Ala	Ser	His	Gln	Pro	Pro	Gly
	290					295					300				
Val	Tyr	Pro	Gln	Gly	His	Ser	Asp	Thr	Thr	Glu	Pro	Lys	Ser	Ser	Asp
305					310					315					320
Lys	Thr	His	Thr	Cys	Pro	Pro	Cys	Pro	Ala	Pro	Glu	Leu	Leu	Gly	Gly
				325					330					335	
Pro	Ser	Val	Phe	Leu	Phe	Pro	Pro	Lys	Pro	Lys	Asp	Thr	Leu	Met	Ile
			340					345					350		
Ser	Arg	Thr	Pro	Glu	Val	Thr	Cys	Val	Val	Val	Asp	Val	Ser	His	Glu
		355					360					365			
Asp	Pro	Glu	Val	Lys	Phe	Asn	Trp	Tyr	Val	Asp	Gly	Val	Glu	Val	His
	370					375					380				
Asn	Ala	Lys	Thr	Lys	Pro	Arg	Glu	Glu	Gln	Tyr	Asn	Ser	Thr	Tyr	Arg
385					390					395					400
Val	Val	Ser	Val	Leu	Thr	Val	Leu	His	Gln	Asp	Trp	Leu	Asn	Gly	Lys
				405					410					415	
Glu	Tyr	Lys	Cys	Lys	Val	Ser	Asn	Lys	Ala	Leu	Pro	Ala	Pro	Ile	Glu
			420					425					430		
Lys	Thr	Ile	Ser	Lys	Ala	Lys	Gly	Gln	Pro	Arg	Glu	Pro	Gln	Val	Tyr
		435					440					445			
Thr	Leu	Pro	Pro	Ser	Arg	Glu	Glu	Met	Thr	Lys	Asn	Gln	Val	Ser	Leu
	450					455					460				
Thr	Cys	Arg	Val	Lys	Gly	Phe	Tyr	Pro	Ser	Asp	Ile	Ala	Val	Glu	Trp
465					470					475					480
Glu	Ser	Asn	Gly	Gln	Pro	Glu	Asn	Asn	Tyr	Lys	Thr	Thr	Pro	Pro	Val
				485					490					495	
Leu	Lys	Ser	Asp	Gly	Ser	Phe	Phe	Leu	Ala	Ser	Lys	Leu	Thr	Val	Asp
			500					505					510		
Lys	Ser	Arg	Trp	Gln	Gln	Gly	Asn	Val	Phe	Ser	Cys	Ser	Val	Met	His
		515					520					525			
Glu	Ala	Leu	His	Asn	His	Tyr	Thr	Gln	Lys	Ser	Leu	Ser	Leu	Ser	Pro
	530					535					540				
Gly	Lys														
545															

<210> SEQ ID NO 56
 <211> LENGTH: 242
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 56

Asp	Ile	Gln	Met	Thr	Gln	Ser	Pro	Ser	Ser	Leu	Ser	Ala	Ser	Val	Gly
1				5					10					15	
Asp	Arg	Val	Thr	Ile	Thr	Cys	Arg	Ala	Ser	Glu	Asn	Ile	Tyr	Ser	Asn
		20					25					30			

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Leu Ala Trp Tyr Gln Gln Lys Pro Gly Lys Cys Pro Lys Leu Leu Val
 35 40 45
 Tyr Ala Ala Ser His Leu Pro Asp Gly Val Pro Ser Arg Phe Ser Gly
 50 55 60
 Ser Gly Ser Gly Thr Asp Tyr Ser Leu Thr Ile Ser Ser Leu Gln Pro
 65 70 75 80
 Glu Asp Phe Ala Thr Tyr Tyr Cys Gln His Phe Trp Gly Thr Pro Arg
 85 90 95
 Thr Phe Gly Gln Gly Thr Lys Leu Glu Ile Lys Gly Gly Gly Gly Ser
 100 105 110
 Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly
 115 120 125
 Asn Trp Val Asn Val Ile Ser Asp Leu Lys Lys Ile Glu Asp Leu Ile
 130 135 140
 Gln Ser Met His Ile Asp Ala Thr Leu Tyr Thr Glu Ser Asp Val His
 145 150 155 160
 Pro Ser Cys Lys Val Thr Ala Met Lys Cys Phe Leu Leu Glu Leu Gln
 165 170 175
 Val Ile Ser Leu Glu Ser Gly Asp Ala Ser Ile His Asn Thr Val Gln
 180 185 190
 Asp Leu Ile Ile Leu Ala Asn Asn Ser Leu Ser Ser Asn Gly Asn Val
 195 200 205
 Thr Glu Ser Gly Cys Lys Glu Cys Glu Glu Leu Glu Glu Lys Asn Ile
 210 215 220
 Lys Glu Phe Leu Gln Ser Phe Val His Ile Val Gln Met Phe Ile Asn
 225 230 235 240
 Thr Ser

<210> SEQ ID NO 57
 <211> LENGTH: 546
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 57

Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ser
 1 5 10 15
 Ser Val Lys Val Ser Cys Lys Ala Ser Gly Tyr Ser Phe Thr Ser Tyr
 20 25 30
 Trp Met Asn Trp Val Arg Gln Ala Pro Gly Gln Gly Cys Glu Trp Ile
 35 40 45
 Gly Met Ile Arg Pro Ser Asp Ser Glu Thr Arg Leu Asn Gln Met Phe
 50 55 60
 Lys Asp Arg Val Thr Ile Thr Val Asp Lys Ser Thr Ser Thr Ala Tyr
 65 70 75 80
 Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95
 Ala Gly Ile His Asp Tyr Gly His Gly Ala Tyr Trp Gly Gln Gly Thr
 100 105 110
 Leu Val Thr Val Ser Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
 115 120 125
 Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Ile Thr Cys Pro Pro

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130					135					140					
Pro 145	Met	Ser	Val	Glu	His 150	Ala	Asp	Ile	Trp	Val 155	Lys	Ser	Tyr	Ser	Leu 160
Tyr	Ser	Arg	Glu	Arg 165	Tyr	Ile	Cys	Asn	Ser 170	Gly	Phe	Lys	Arg	Lys 175	Ala
Gly	Thr	Ser	Ser	Leu 180	Thr	Glu	Cys	Val 185	Leu	Asn	Lys	Ala	Thr 190	Asn	Val
Ala	His	Trp	Thr	Thr 195	Pro	Ser	Leu 200	Lys	Cys	Ile	Arg	Asp 205	Pro	Ala	Leu
Val	His	Gln	Arg	Pro 210	Ala	Pro 215	Pro	Ser	Thr	Val	Thr 220	Thr	Ala	Gly	Val
Thr 225	Pro	Gln	Pro	Glu 230	Ser	Leu	Ser	Pro	Ser	Gly 235	Lys	Glu	Pro	Ala	Ala 240
Ser	Ser	Pro	Ser	Ser 245	Asn	Asn	Thr	Ala	Ala 250	Thr	Thr	Ala	Ala	Ile 255	Val
Pro	Gly	Ser	Gln	Leu 260	Met	Pro	Ser	Lys 265	Ser	Pro	Ser	Thr	Gly 270	Thr	Thr
Glu	Ile	Ser	Ser	His 275	Glu	Ser	Ser	His 280	Gly	Thr	Pro	Ser 285	Gln	Thr	Thr
Ala	Lys	Asn	Trp	Glu 290	Leu	Thr 295	Ala	Ser	Ala	Ser	His 300	Gln	Pro	Pro	Gly
Val 305	Tyr	Pro	Gln	Gly 310	His	Ser	Asp	Thr	Thr 315	Glu	Pro	Lys	Ser	Ser	Asp 320
Lys	Thr	His	Thr	Cys 325	Pro	Pro	Cys	Pro	Ala 330	Pro	Glu	Leu	Leu	Gly 335	Gly
Pro	Ser	Val	Phe	Leu 340	Phe	Pro	Pro	Lys 345	Pro	Lys	Asp	Thr	Leu 350	Met	Ile
Ser	Arg	Thr	Pro	Glu 355	Val	Thr	Cys 360	Val	Val	Val	Asp	Val 365	Ser	His	Glu
Asp	Pro	Glu	Val	Lys 370	Phe	Asn 375	Trp	Tyr	Val	Asp	Gly 380	Val	Glu	Val	His
Asn 385	Ala	Lys	Thr	Lys 390	Pro	Arg	Glu	Glu	Gln 395	Tyr	Asn	Ser	Thr	Tyr	Arg 400
Val	Val	Ser	Val	Leu 405	Thr	Val	Leu	His	Gln 410	Asp	Trp	Leu	Asn	Gly 415	Lys
Glu	Tyr	Lys	Cys	Lys 420	Val	Ser	Asn	Lys 425	Ala	Leu	Pro	Ala 430	Pro	Ile	Glu
Lys	Thr	Ile	Ser	Lys 435	Ala	Lys	Gly 440	Gln	Pro	Arg	Glu	Pro 445	Gln	Val	Tyr
Thr 450	Leu	Pro	Pro	Ser 455	Arg	Glu	Glu	Met	Thr	Lys	Asn 460	Gln	Val	Ser	Leu
Thr 465	Cys	Arg	Val	Lys 470	Gly	Phe	Tyr	Pro	Ser	Asp	Ile 475	Ala	Val	Glu	Trp 480
Glu	Ser	Asn	Gly	Gln 485	Pro	Glu	Asn	Asn	Tyr 490	Lys	Thr	Thr	Pro	Pro	Val 495
Leu	Lys	Ser	Asp	Gly 500	Ser	Phe	Phe	Leu 505	Ala	Ser	Lys	Leu 510	Thr	Val	Asp
Lys	Ser	Arg	Trp	Gln 515	Gln	Gly	Asn 520	Val	Phe	Ser	Cys 525	Ser	Val	Met	His
Glu 530	Ala	Leu	His	Asn 535	His	Tyr	Thr	Gln	Lys	Ser	Leu 540	Ser	Leu	Ser	Pro

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Gly Lys
545

<210> SEQ ID NO 58
<211> LENGTH: 242
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 58

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1 5 10 15
Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Glu Asn Ile Tyr Ser Asn
20 25 30
Leu Ala Trp Tyr Gln Gln Lys Pro Gly Lys Ser Pro Lys Leu Leu Val
35 40 45
Tyr Ala Ala Ser His Leu Pro Asp Gly Val Pro Ser Arg Phe Ser Gly
50 55 60
Ser Gly Ser Gly Thr Asp Tyr Ser Leu Thr Ile Ser Ser Leu Gln Pro
65 70 75 80
Glu Asp Phe Ala Thr Tyr Tyr Cys Gln His Phe Trp Gly Thr Pro Arg
85 90 95
Thr Cys Gly Gln Gly Thr Lys Leu Glu Ile Lys Gly Gly Gly Ser
100 105 110
Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Ser Gly
115 120 125
Asn Trp Val Asn Val Ile Ser Asp Leu Lys Lys Ile Glu Asp Leu Ile
130 135 140
Gln Ser Met His Ile Asp Ala Thr Leu Tyr Thr Glu Ser Asp Val His
145 150 155 160
Pro Ser Cys Lys Val Thr Ala Met Lys Cys Phe Leu Leu Glu Leu Gln
165 170 175
Val Ile Ser Leu Glu Ser Gly Asp Ala Ser Ile His Asn Thr Val Gln
180 185 190
Asp Leu Ile Ile Leu Ala Asn Asn Ser Leu Ser Ser Asn Gly Asn Val
195 200 205
Thr Glu Ser Gly Cys Lys Glu Cys Glu Glu Leu Glu Lys Asn Ile
210 215 220
Lys Glu Phe Leu Gln Ser Phe Val His Ile Val Gln Met Phe Ile Asn
225 230 235 240
Thr Ser

<210> SEQ ID NO 59
<211> LENGTH: 547
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 59

Gln Val Gln Leu Gln Glu Ser Gly Pro Gly Leu Val Lys Pro Ser Glu
1 5 10 15
Thr Leu Ser Leu Thr Cys Ala Val Tyr Gly Asp Ser Ile Thr Ser Gly
20 25 30

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Tyr	Trp	Asn	Trp	Ile	Arg	Lys	Pro	Pro	Gly	Lys	Cys	Leu	Glu	Tyr	Ile
354045															
Gly	Tyr	Ile	Ser	Tyr	Thr	Gly	Ser	Thr	Tyr	Gln	Asn	Pro	Ser	Leu	Lys
505560															
Ser	Arg	Ile	Thr	Met	Ser	Arg	Asp	Thr	Ser	Lys	Asn	Gln	Tyr	Tyr	Leu
65707580															
Lys	Leu	Ser	Ser	Val	Thr	Ala	Ala	Asp	Thr	Ala	Val	Tyr	Tyr	Cys	Ala
859095															
Arg	Ser	Arg	Ala	Trp	Ile	Arg	Thr	Tyr	Phe	Asp	Tyr	Trp	Gly	Gln	Gly
100105110															
Thr	Leu	Val	Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly
115120125															
Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Ile	Thr	Cys	Pro
130135140															
Pro	Pro	Met	Ser	Val	Glu	His	Ala	Asp	Ile	Trp	Val	Lys	Ser	Tyr	Ser
145150155160															
Leu	Tyr	Ser	Arg	Glu	Arg	Tyr	Ile	Cys	Asn	Ser	Gly	Phe	Lys	Arg	Lys
165170175															
Ala	Gly	Thr	Ser	Ser	Leu	Thr	Glu	Cys	Val	Leu	Asn	Lys	Ala	Thr	Asn
180185190															
Val	Ala	His	Trp	Thr	Thr	Pro	Ser	Leu	Lys	Cys	Ile	Arg	Asp	Pro	Ala
195200205															
Leu	Val	His	Gln	Arg	Pro	Ala	Pro	Pro	Ser	Thr	Val	Thr	Thr	Ala	Gly
210215220															
Val	Thr	Pro	Gln	Pro	Glu	Ser	Leu	Ser	Pro	Ser	Gly	Lys	Glu	Pro	Ala
225230235240															
Ala	Ser	Ser	Pro	Ser	Ser	Asn	Asn	Thr	Ala	Ala	Thr	Thr	Ala	Ala	Ile
245250255															
Val	Pro	Gly	Ser	Gln	Leu	Met	Pro	Ser	Lys	Ser	Pro	Ser	Thr	Gly	Thr
260265270															
Thr	Glu	Ile	Ser	Ser	His	Glu	Ser	Ser	His	Gly	Thr	Pro	Ser	Gln	Thr
275280285															
Thr	Ala	Lys	Asn	Trp	Glu	Leu	Thr	Ala	Ser	Ala	Ser	His	Gln	Pro	Pro
290295300															
Gly	Val	Tyr	Pro	Gln	Gly	His	Ser	Asp	Thr	Thr	Glu	Pro	Lys	Ser	Ser
305310315320															
Asp	Lys	Thr	His	Thr	Cys	Pro	Pro	Cys	Pro	Ala	Pro	Glu	Leu	Leu	Gly
325330335															
Gly	Pro	Ser	Val	Phe	Leu	Phe	Pro	Pro	Lys	Pro	Lys	Asp	Thr	Leu	Met
340345350															
Ile	Ser	Arg	Thr	Pro	Glu	Val	Thr	Cys	Val	Val	Val	Asp	Val	Ser	His
355360365															
Glu	Asp	Pro	Glu	Val	Lys	Phe	Asn	Trp	Tyr	Val	Asp	Gly	Val	Glu	Val
370375380															
His	Asn	Ala	Lys	Thr	Lys	Pro	Arg	Glu	Glu	Gln	Tyr	Asn	Ser	Thr	Tyr
385390395400															
Arg	Val	Val	Ser	Val	Leu	Thr	Val	Leu	His	Gln	Asp	Trp	Leu	Asn	Gly
405410415															
Lys	Glu	Tyr	Lys	Cys	Lys	Val	Ser	Asn	Lys	Ala	Leu	Pro	Ala	Pro	Ile
420425430															

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Glu	Lys	Thr	Ile	Ser	Lys	Ala	Lys	Gly	Gln	Pro	Arg	Glu	Pro	Gln	Val
	435						440					445			
Tyr	Thr	Leu	Pro	Pro	Ser	Arg	Glu	Glu	Met	Thr	Lys	Asn	Gln	Val	Ser
	450					455					460				
Leu	Thr	Cys	Arg	Val	Lys	Gly	Phe	Tyr	Pro	Ser	Asp	Ile	Ala	Val	Glu
465					470					475					480
Trp	Glu	Ser	Asn	Gly	Gln	Pro	Glu	Asn	Asn	Tyr	Lys	Thr	Thr	Pro	Pro
			485						490					495	
Val	Leu	Lys	Ser	Asp	Gly	Ser	Phe	Phe	Leu	Ala	Ser	Lys	Leu	Thr	Val
			500					505					510		
Asp	Lys	Ser	Arg	Trp	Gln	Gln	Gly	Asn	Val	Phe	Ser	Cys	Ser	Val	Met
	515						520					525			
His	Glu	Ala	Leu	His	Asn	His	Tyr	Thr	Gln	Lys	Ser	Leu	Ser	Leu	Ser
	530				535						540				
Pro	Gly	Lys													
545															

<210> SEQ ID NO 60
 <211> LENGTH: 243
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 60

Asp	Ile	Gln	Met	Thr	Gln	Ser	Pro	Ser	Ser	Leu	Ser	Ala	Ser	Val	Gly
1				5					10					15	
Asp	Arg	Val	Thr	Ile	Thr	Cys	Ser	Val	Ser	Ser	Ser	Ile	Ser	Ser	Ser
		20						25					30		
Asn	Leu	His	Trp	Tyr	Gln	Gln	Lys	Pro	Gly	Lys	Ala	Pro	Lys	Pro	Trp
		35					40					45			
Ile	Tyr	Gly	Thr	Ser	Asn	Leu	Ala	Ser	Gly	Val	Pro	Ser	Arg	Phe	Ser
	50				55					60					
Gly	Ser	Gly	Ser	Gly	Thr	Asp	Tyr	Thr	Leu	Thr	Ile	Ser	Ser	Leu	Gln
65					70					75					80
Pro	Glu	Asp	Phe	Ala	Thr	Tyr	Tyr	Cys	Gln	Gln	Trp	Ser	Ser	Tyr	Pro
			85					90						95	
Leu	Thr	Phe	Gly	Cys	Gly	Thr	Lys	Leu	Glu	Ile	Lys	Gly	Gly	Gly	Gly
		100						105					110		
Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser
		115				120					125				
Gly	Asn	Trp	Val	Asn	Val	Ile	Ser	Asp	Leu	Lys	Lys	Ile	Glu	Asp	Leu
	130				135					140					
Ile	Gln	Ser	Met	His	Ile	Asp	Ala	Thr	Leu	Tyr	Thr	Glu	Ser	Asp	Val
145				150						155					160
His	Pro	Ser	Cys	Lys	Val	Thr	Ala	Met	Lys	Cys	Phe	Leu	Leu	Glu	Leu
			165					170						175	
Gln	Val	Ile	Ser	Leu	Glu	Ser	Gly	Asp	Ala	Ser	Ile	His	Asn	Thr	Val
		180						185					190		
Gln	Asp	Leu	Ile	Ile	Leu	Ala	Asn	Asn	Ser	Leu	Ser	Ser	Asn	Gly	Asn
	195					200						205			
Val	Thr	Glu	Ser	Gly	Cys	Lys	Glu	Cys	Glu	Glu	Leu	Glu	Glu	Lys	Asn
	210					215					220				

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Ile	Lys	Glu	Phe	Leu	Gln	Ser	Phe	Val	His	Ile	Val	Gln	Met	Phe	Ile
225					230					235					240

Asn Thr Ser

<210> SEQ ID NO 61
 <211> LENGTH: 448
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 61

Gln	Val	Gln	Leu	Val	Gln	Ser	Gly	Ala	Glu	Val	Lys	Lys	Pro	Gly	Ser
1			5						10					15	
Ser	Val	Lys	Val	Ser	Cys	Lys	Ala	Ser	Gly	Tyr	Ser	Phe	Thr	Ser	Tyr
			20					25					30		
Trp	Met	Asn	Trp	Val	Arg	Gln	Ala	Pro	Gly	Gln	Gly	Leu	Glu	Trp	Ile
		35					40					45			
Gly	Met	Ile	Arg	Pro	Ser	Asp	Ser	Glu	Thr	Arg	Leu	Asn	Gln	Met	Phe
	50					55					60				
Lys	Asp	Arg	Val	Thr	Ile	Thr	Val	Asp	Lys	Ser	Thr	Ser	Thr	Ala	Tyr
65					70					75					80
Met	Glu	Leu	Ser	Ser	Leu	Arg	Ser	Glu	Asp	Thr	Ala	Val	Tyr	Tyr	Cys
			85						90					95	
Ala	Gly	Ile	His	Asp	Tyr	Gly	His	Gly	Ala	Tyr	Trp	Gly	Gln	Gly	Thr
			100					105					110		
Leu	Val	Thr	Val	Ser	Ser	Ala	Ser	Thr	Lys	Gly	Pro	Ser	Val	Phe	Pro
		115						120					125		
Leu	Ala	Pro	Ser	Ser	Lys	Ser	Thr	Ser	Gly	Gly	Thr	Ala	Ala	Leu	Gly
	130					135					140				
Cys	Leu	Val	Lys	Asp	Tyr	Phe	Pro	Glu	Pro	Val	Thr	Val	Ser	Trp	Asn
145					150					155					160
Ser	Gly	Ala	Leu	Thr	Ser	Gly	Val	His	Thr	Phe	Pro	Ala	Val	Leu	Gln
			165					170						175	
Ser	Ser	Gly	Leu	Tyr	Ser	Leu	Ser	Ser	Val	Val	Thr	Val	Pro	Ser	Ser
			180					185					190		
Ser	Leu	Gly	Thr	Gln	Thr	Tyr	Ile	Cys	Asn	Val	Asn	His	Lys	Pro	Ser
	195						200					205			
Asn	Thr	Lys	Val	Asp	Lys	Arg	Val	Glu	Pro	Lys	Ser	Cys	Asp	Lys	Thr
	210					215					220				
His	Thr	Cys	Pro	Pro	Cys	Pro	Ala	Pro	Glu	Leu	Leu	Gly	Gly	Pro	Ser
225					230					235					240
Val	Phe	Leu	Phe	Pro	Pro	Lys	Pro	Lys	Asp	Thr	Leu	Met	Ile	Ser	Arg
			245						250					255	
Thr	Pro	Glu	Val	Thr	Cys	Val	Val	Val	Asp	Val	Ser	His	Glu	Asp	Pro
		260						265					270		
Glu	Val	Lys	Phe	Asn	Trp	Tyr	Val	Asp	Gly	Val	Glu	Val	His	Asn	Ala
		275					280					285			
Lys	Thr	Lys	Pro	Arg	Glu	Glu	Gln	Tyr	Asn	Ser	Thr	Tyr	Arg	Val	Val
	290					295					300				
Ser	Val	Leu	Thr	Val	Leu	His	Gln	Asp	Trp	Leu	Asn	Gly	Lys	Glu	Tyr
305					310					315					320
Lys	Cys	Lys	Val	Ser	Asn	Lys	Ala	Leu	Pro	Ala	Pro	Ile	Glu	Lys	Thr

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      325              330              335
Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr Thr Leu
      340              345              350

Pro Pro Ser Arg Glu Glu Met Thr Lys Asn Gln Val Ser Leu Trp Cys
      355              360              365

Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu Ser
      370              375              380

Asn Gly Gln Pro Glu Asn Asn Tyr Asp Thr Thr Pro Pro Val Leu Asp
      385              390              395              400

Ser Asp Gly Ser Phe Phe Leu Tyr Ser Asp Leu Thr Val Asp Lys Ser
      405              410              415

Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met His Glu Ala
      420              425              430

Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro Gly Lys
      435              440              445


<210> SEQ ID NO 62
<211> LENGTH: 214
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthesized


<400> SEQUENCE: 62

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1          5          10          15

Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Glu Asn Ile Tyr Ser Asn
      20          25          30

Leu Ala Trp Tyr Gln Gln Lys Pro Gly Lys Ser Pro Lys Leu Leu Val
      35          40          45

Tyr Ala Ala Ser His Leu Pro Asp Gly Val Pro Ser Arg Phe Ser Gly
      50          55          60

Ser Gly Ser Gly Thr Asp Tyr Ser Leu Thr Ile Ser Ser Leu Gln Pro
      65          70          75          80

Glu Asp Phe Ala Thr Tyr Tyr Cys Gln His Phe Trp Gly Thr Pro Arg
      85          90          95

Thr Phe Gly Gln Gly Thr Lys Leu Glu Ile Lys Arg Thr Val Ala Ala
      100         105         110

Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu Gln Leu Lys Ser Gly
      115         120         125

Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe Tyr Pro Arg Glu Ala
      130         135         140

Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln Ser Gly Asn Ser Gln
      145         150         155         160

Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser Thr Tyr Ser Leu Ser
      165         170         175

Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu Lys His Lys Val Tyr
      180         185         190

Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser Pro Val Thr Lys Ser
      195         200         205

Phe Asn Arg Gly Glu Cys
      210

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<210> SEQ ID NO 63
<211> LENGTH: 546
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 63

Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ser
1          5          10          15

Ser Val Lys Val Ser Cys Lys Ala Ser Gly Tyr Ser Phe Thr Ser Tyr
20          25          30

Trp Met Asn Trp Val Arg Gln Ala Pro Gly Gln Gly Leu Glu Trp Ile
35          40          45

Gly Met Ile Arg Pro Ser Asp Ser Glu Thr Arg Leu Asn Gln Met Phe
50          55          60

Lys Asp Arg Val Thr Ile Thr Val Asp Lys Ser Thr Ser Thr Ala Tyr
65          70          75          80

Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Val Tyr Tyr Cys
85          90          95

Ala Gly Ile His Asp Tyr Gly His Gly Ala Tyr Trp Gly Gln Gly Thr
100         105         110

Leu Val Thr Val Ser Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
115         120         125

Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Ile Thr Cys Pro Pro
130         135         140

Pro Met Ser Val Glu His Ala Asp Ile Trp Val Lys Ser Tyr Ser Leu
145         150         155         160

Tyr Ser Arg Glu Arg Tyr Ile Cys Asn Ser Gly Phe Lys Arg Lys Ala
165         170         175

Gly Thr Ser Ser Leu Thr Glu Cys Val Leu Asn Lys Ala Thr Asn Val
180         185         190

Ala His Trp Thr Thr Pro Ser Leu Lys Cys Ile Arg Asp Pro Ala Leu
195         200         205

Val His Gln Arg Pro Ala Pro Pro Ser Thr Val Thr Thr Ala Gly Val
210         215         220

Thr Pro Gln Pro Glu Ser Leu Ser Pro Ser Gly Lys Glu Pro Ala Ala
225         230         235         240

Ser Ser Pro Ser Ser Asn Asn Thr Ala Ala Thr Thr Ala Ala Ile Val
245         250         255

Pro Gly Ser Gln Leu Met Pro Ser Lys Ser Pro Ser Thr Gly Thr Thr
260         265         270

Glu Ile Ser Ser His Glu Ser Ser His Gly Thr Pro Ser Gln Thr Thr
275         280         285

Ala Lys Asn Trp Glu Leu Thr Ala Ser Ala Ser His Gln Pro Pro Gly
290         295         300

Val Tyr Pro Gln Gly His Ser Asp Thr Thr Glu Pro Lys Ser Ser Asp
305         310         315         320

Lys Thr His Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu Leu Gly Gly
325         330         335

Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile
340         345         350

Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser His Glu

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355					360					365					
Asp	Pro	Glu	Val	Lys	Phe	Asn	Trp	Tyr	Val	Asp	Gly	Val	Glu	Val	His
370						375					380				
Asn	Ala	Lys	Thr	Lys	Pro	Arg	Glu	Glu	Gln	Tyr	Asn	Ser	Thr	Tyr	Arg
385						390					395				400
Val	Val	Ser	Val	Leu	Thr	Val	Leu	His	Gln	Asp	Trp	Leu	Asn	Gly	Lys
				405					410					415	
Glu	Tyr	Lys	Cys	Lys	Val	Ser	Asn	Lys	Ala	Leu	Pro	Ala	Pro	Ile	Glu
			420					425					430		
Lys	Thr	Ile	Ser	Lys	Ala	Lys	Gly	Gln	Pro	Arg	Glu	Pro	Gln	Val	Tyr
		435					440					445			
Thr	Leu	Pro	Pro	Ser	Arg	Glu	Glu	Met	Thr	Lys	Asn	Gln	Val	Ser	Leu
	450					455					460				
Thr	Cys	Arg	Val	Lys	Gly	Phe	Tyr	Pro	Ser	Asp	Ile	Ala	Val	Glu	Trp
465				470						475					480
Glu	Ser	Asn	Gly	Gln	Pro	Glu	Asn	Asn	Tyr	Lys	Thr	Thr	Pro	Pro	Val
			485						490					495	
Leu	Lys	Ser	Asp	Gly	Ser	Phe	Phe	Leu	Ala	Ser	Lys	Leu	Thr	Val	Asp
			500					505					510		
Lys	Ser	Arg	Trp	Gln	Gln	Gly	Asn	Val	Phe	Ser	Cys	Ser	Val	Met	His
		515					520					525			
Glu	Ala	Leu	His	Asn	His	Tyr	Thr	Gln	Lys	Ser	Leu	Ser	Leu	Ser	Pro
	530					535					540				
Gly	Lys														
545															
<210> SEQ ID NO 64															
<211> LENGTH: 242															
<212> TYPE: PRT															
<213> ORGANISM: Artificial Sequence															
<220> FEATURE:															
<223> OTHER INFORMATION: Synthesized															
<400> SEQUENCE: 64															
Asp	Ile	Gln	Met	Thr	Gln	Ser	Pro	Ser	Ser	Leu	Ser	Ala	Ser	Val	Gly
1				5					10					15	
Asp	Arg	Val	Thr	Ile	Thr	Cys	Arg	Ala	Ser	Glu	Asn	Ile	Tyr	Ser	Asn
		20					25						30		
Leu	Ala	Trp	Tyr	Gln	Gln	Lys	Pro	Gly	Lys	Ser	Pro	Lys	Leu	Leu	Val
	35					40						45			
Tyr	Ala	Ala	Ser	His	Leu	Pro	Asp	Gly	Val	Pro	Ser	Arg	Phe	Ser	Gly
	50					55					60				
Ser	Gly	Ser	Gly	Thr	Asp	Tyr	Ser	Leu	Thr	Ile	Ser	Ser	Leu	Gln	Pro
65				70					75					80	
Glu	Asp	Phe	Ala	Thr	Tyr	Tyr	Cys	Gln	His	Phe	Trp	Gly	Thr	Pro	Arg
		85						90					95		
Thr	Phe	Gly	Gln	Gly	Thr	Lys	Leu	Glu	Ile	Lys	Gly	Gly	Gly	Ser	
		100					105						110		
Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	
	115					120					125				
Asn	Trp	Val	Asn	Val	Ile	Ser	Asp	Leu	Lys	Lys	Ile	Glu	Asp	Leu	Ile
	130					135					140				
Gln	Ser	Met	His	Ile	Asp	Ala	Thr	Leu	Tyr	Thr	Glu	Ser	Asp	Val	His

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145	150	155	160
Pro Ser Cys Lys Val Thr Ala Met Lys Cys Phe Leu Leu Glu Leu Gln			
	165	170	175
Val Ile Ser Leu Glu Ser Gly Asp Ala Ser Ile His Asn Thr Val Gln			
	180	185	190
Asp Leu Ile Ile Leu Ala Asn Asn Ser Leu Ser Ser Asn Gly Asn Val			
	195	200	205
Thr Glu Ser Gly Cys Lys Glu Cys Glu Glu Leu Glu Glu Lys Asn Ile			
	210	215	220
Lys Glu Phe Leu Gln Ser Phe Val His Ile Val Gln Met Phe Ile Asn			
225	230	235	240
Thr Ser			

<210> SEQ ID NO 65
 <211> LENGTH: 546
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 65

Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ser			
1	5	10	15
Ser Val Lys Val Ser Cys Lys Ala Ser Gly Tyr Ser Phe Thr Ser Tyr			
	20	25	30
Trp Met Asn Trp Val Arg Gln Ala Pro Gly Gln Gly Leu Glu Trp Ile			
	35	40	45
Gly Met Ile Arg Pro Ser Asp Ser Glu Thr Arg Leu Asn Gln Met Phe			
	50	55	60
Lys Asp Arg Val Thr Ile Thr Val Asp Lys Ser Thr Ser Thr Ala Tyr			
65	70	75	80
Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Val Tyr Tyr Cys			
	85	90	95
Ala Gly Ile His Asp Tyr Gly His Gly Ala Tyr Trp Gly Gln Gly Thr			
	100	105	110
Leu Val Thr Val Ser Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser			
	115	120	125
Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Ile Thr Cys Pro Pro			
	130	135	140
Pro Met Ser Val Glu His Ala Asp Ile Trp Val Lys Ser Tyr Ser Leu			
145	150	155	160
Tyr Ser Arg Glu Arg Tyr Ile Cys Asn Ser Gly Phe Lys Cys Lys Ala			
	165	170	175
Gly Thr Ser Ser Leu Thr Glu Cys Val Leu Asn Lys Ala Thr Asn Val			
	180	185	190
Ala His Trp Thr Thr Pro Ser Leu Lys Cys Ile Arg Asp Pro Ala Leu			
	195	200	205
Val His Gln Arg Pro Ala Pro Pro Ser Thr Val Thr Thr Ala Gly Val			
	210	215	220
Thr Pro Gln Pro Glu Ser Leu Ser Pro Ser Gly Lys Glu Pro Ala Ala			
225	230	235	240
Ser Ser Pro Ser Ser Asn Asn Thr Ala Ala Thr Thr Ala Ala Ile Val			
	245	250	255

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Pro Gly Ser Gln Leu Met Pro Ser Lys Ser Pro Ser Thr Gly Thr Thr
    260                265                270

Glu Ile Ser Ser His Glu Ser Ser His Gly Thr Pro Ser Gln Thr Thr
    275                280                285

Ala Lys Asn Trp Glu Leu Thr Ala Ser Ala Ser His Gln Pro Pro Gly
    290                295                300

Val Tyr Pro Gln Gly His Ser Asp Thr Thr Glu Pro Lys Ser Ser Asp
    305                310                315                320

Lys Thr His Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu Leu Gly Gly
    325                330                335

Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile
    340                345                350

Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser His Glu
    355                360                365

Asp Pro Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val Glu Val His
    370                375                380

Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Tyr Asn Ser Thr Tyr Arg
    385                390                395                400

Val Val Ser Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly Lys
    405                410                415

Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu Pro Ala Pro Ile Glu
    420                425                430

Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr
    435                440                445

Thr Leu Pro Pro Ser Arg Glu Glu Met Thr Lys Asn Gln Val Ser Leu
    450                455                460

Thr Cys Arg Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp
    465                470                475                480

Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val
    485                490                495

Leu Lys Ser Asp Gly Ser Phe Phe Leu Ala Ser Lys Leu Thr Val Asp
    500                505                510

Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met His
    515                520                525

Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro
    530                535                540

Gly Lys
545

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<210> SEQ ID NO 66
<211> LENGTH: 436
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthesized

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<400> SEQUENCE: 66

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Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ser
1          5          10          15

Ser Val Lys Val Ser Cys Lys Ala Ser Gly Tyr Ser Phe Thr Ser Tyr
20          25          30

Trp Met Asn Trp Val Arg Gln Ala Pro Gly Gln Gly Leu Glu Trp Ile
35          40          45

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Gly	Met	Ile	Arg	Pro	Ser	Asp	Ser	Glu	Thr	Arg	Leu	Asn	Gln	Met	Phe	50	55	60
Lys	Asp	Arg	Val	Thr	Ile	Thr	Val	Asp	Lys	Ser	Thr	Ser	Thr	Ala	Tyr	65	70	75
Met	Glu	Leu	Ser	Ser	Leu	Arg	Ser	Glu	Asp	Thr	Ala	Val	Tyr	Tyr	Cys	85	90	95
Ala	Gly	Ile	His	Asp	Tyr	Gly	His	Gly	Ala	Tyr	Trp	Gly	Gln	Gly	Thr	100	105	110
Leu	Val	Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	115	120	125
Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Ile	Thr	Cys	Pro	Pro	130	135	140
Pro	Met	Ser	Val	Glu	His	Ala	Asp	Ile	Trp	Val	Lys	Ser	Tyr	Ser	Leu	145	150	155
Tyr	Ser	Arg	Glu	Arg	Tyr	Ile	Cys	Asn	Ser	Gly	Phe	Lys	Arg	Lys	Ala	165	170	175
Gly	Thr	Ser	Ser	Leu	Thr	Glu	Cys	Val	Leu	Asn	Lys	Ala	Thr	Asn	Val	180	185	190
Ala	His	Trp	Thr	Thr	Pro	Ser	Leu	Lys	Cys	Ile	Arg	Glu	Pro	Lys	Ser	195	200	205
Ser	Asp	Lys	Thr	His	Thr	Cys	Pro	Pro	Cys	Pro	Ala	Pro	Glu	Leu	Leu	210	215	220
Gly	Gly	Pro	Ser	Val	Phe	Leu	Phe	Pro	Pro	Lys	Pro	Lys	Asp	Thr	Leu	225	230	235
Met	Ile	Ser	Arg	Thr	Pro	Glu	Val	Thr	Cys	Val	Val	Val	Asp	Val	Ser	245	250	255
His	Glu	Asp	Pro	Glu	Val	Lys	Phe	Asn	Trp	Tyr	Val	Asp	Gly	Val	Glu	260	265	270
Val	His	Asn	Ala	Lys	Thr	Lys	Pro	Arg	Glu	Glu	Gln	Tyr	Asn	Ser	Thr	275	280	285
Tyr	Arg	Val	Val	Ser	Val	Leu	Thr	Val	Leu	His	Gln	Asp	Trp	Leu	Asn	290	295	300
Gly	Lys	Glu	Tyr	Lys	Cys	Lys	Val	Ser	Asn	Lys	Ala	Leu	Pro	Ala	Pro	305	310	315
Ile	Glu	Lys	Thr	Ile	Ser	Lys	Ala	Lys	Gly	Gln	Pro	Arg	Glu	Pro	Gln	325	330	335
Val	Tyr	Thr	Leu	Pro	Pro	Ser	Arg	Glu	Glu	Met	Thr	Lys	Asn	Gln	Val	340	345	350
Ser	Leu	Thr	Cys	Arg	Val	Lys	Gly	Phe	Tyr	Pro	Ser	Asp	Ile	Ala	Val	355	360	365
Glu	Trp	Glu	Ser	Asn	Gly	Gln	Pro	Glu	Asn	Asn	Tyr	Lys	Thr	Thr	Pro	370	375	380
Pro	Val	Leu	Lys	Ser	Asp	Gly	Ser	Phe	Phe	Leu	Ala	Ser	Lys	Leu	Thr	385	390	395
Val	Asp	Lys	Ser	Arg	Trp	Gln	Gln	Gly	Asn	Val	Phe	Ser	Cys	Ser	Val	405	410	415
Met	His	Glu	Ala	Leu	His	Asn	His	Tyr	Thr	Gln	Lys	Ser	Leu	Ser	Leu	420	425	430
Ser	Pro	Gly	Lys													435		

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<210> SEQ ID NO 67
<211> LENGTH: 436
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 67

Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ser
1          5          10          15
Ser Val Lys Val Ser Cys Lys Ala Ser Gly Tyr Ser Phe Thr Ser Tyr
20          25          30
Trp Met Asn Trp Val Arg Gln Ala Pro Gly Gln Gly Leu Glu Trp Ile
35          40          45
Gly Met Ile Arg Pro Ser Asp Ser Glu Thr Arg Leu Asn Gln Met Phe
50          55          60
Lys Asp Arg Val Thr Ile Thr Val Asp Lys Ser Thr Ser Thr Ala Tyr
65          70          75          80
Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Val Tyr Tyr Cys
85          90          95
Ala Gly Ile His Asp Tyr Gly His Gly Ala Tyr Trp Gly Gln Gly Thr
100         105         110
Leu Val Thr Val Ser Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
115         120         125
Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Ile Ala Cys Pro Pro
130         135         140
Pro Met Ser Val Glu His Ala Asp Ile Trp Val Lys Ser Tyr Ser Leu
145         150         155         160
Tyr Ser Arg Glu Arg Tyr Ile Cys Asn Ser Gly Phe Lys Arg Lys Ala
165         170         175
Gly Thr Ser Ser Leu Thr Glu Cys Val Leu Asn Lys Ala Thr Asn Val
180         185         190
Ala His Trp Thr Thr Pro Ser Leu Lys Cys Ile Arg Glu Pro Lys Ser
195         200         205
Ser Asp Lys Thr His Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu Leu
210         215         220
Gly Gly Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu
225         230         235         240
Met Ile Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser
245         250         255
His Glu Asp Pro Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val Glu
260         265         270
Val His Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Tyr Asn Ser Thr
275         280         285
Tyr Arg Val Val Ser Val Leu Thr Val Leu His Gln Asp Trp Leu Asn
290         295         300
Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu Pro Ala Pro
305         310         315         320
Ile Glu Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln
325         330         335
Val Tyr Thr Leu Pro Pro Ser Arg Glu Glu Met Thr Lys Asn Gln Val
340         345         350

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Ser Leu Thr Cys Arg Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val
 355 360 365

Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro
 370 375 380

Pro Val Leu Lys Ser Asp Gly Ser Phe Phe Leu Ala Ser Lys Leu Thr
 385 390 395 400

Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val
 405 410 415

Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu
 420 425 430

Ser Pro Gly Lys
 435

<210> SEQ ID NO 68
 <211> LENGTH: 485
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 68

Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ser
 1 5 10 15

Ser Val Lys Val Ser Cys Lys Ala Ser Gly Tyr Ser Phe Thr Ser Tyr
 20 25 30

Trp Met Asn Trp Val Arg Gln Ala Pro Gly Gln Gly Leu Glu Trp Ile
 35 40 45

Gly Met Ile Arg Pro Ser Asp Ser Glu Thr Arg Leu Asn Gln Met Phe
 50 55 60

Lys Asp Arg Val Thr Ile Thr Val Asp Lys Ser Thr Ser Thr Ala Tyr
 65 70 75 80

Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95

Ala Gly Ile His Asp Tyr Gly His Gly Ala Tyr Trp Gly Gln Gly Thr
 100 105 110

Leu Val Thr Val Ser Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
 115 120 125

Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Ile Ala Cys Pro Pro
 130 135 140

Pro Met Ser Val Glu His Ala Asp Ile Trp Val Lys Ser Tyr Ser Leu
 145 150 155 160

Tyr Ser Arg Glu Arg Tyr Ile Cys Asn Ser Gly Phe Lys Arg Lys Ala
 165 170 175

Gly Thr Ser Ser Leu Thr Glu Cys Val Leu Asn Lys Ala Thr Asn Val
 180 185 190

Ala His Trp Thr Thr Pro Ser Leu Lys Cys Ile Arg Asp Pro Ala Leu
 195 200 205

Val His Gln Arg Pro Ala Pro Pro Ser Thr Val Ala Thr Ala Gly Val
 210 215 220

Ala Pro Gln Pro Glu Ser Leu Ser Pro Ser Gly Lys Glu Pro Ala Ala
 225 230 235 240

Ser Ser Pro Ser Ser Asn Asn Thr Ala Ala Thr Thr Ala Glu Pro Lys
 245 250 255

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Ser	Ser	Asp	Lys	Thr	His	Thr	Cys	Pro	Pro	Cys	Pro	Ala	Pro	Glu	Leu
			260					265					270		
Leu	Gly	Gly	Pro	Ser	Val	Phe	Leu	Phe	Pro	Pro	Lys	Pro	Lys	Asp	Thr
			275				280					285			
Leu	Met	Ile	Ser	Arg	Thr	Pro	Glu	Val	Thr	Cys	Val	Val	Val	Asp	Val
	290					295					300				
Ser	His	Glu	Asp	Pro	Glu	Val	Lys	Phe	Asn	Trp	Tyr	Val	Asp	Gly	Val
305					310					315					320
Glu	Val	His	Asn	Ala	Lys	Thr	Lys	Pro	Arg	Glu	Glu	Gln	Tyr	Asn	Ser
				325					330					335	
Thr	Tyr	Arg	Val	Val	Ser	Val	Leu	Thr	Val	Leu	His	Gln	Asp	Trp	Leu
			340					345					350		
Asn	Gly	Lys	Glu	Tyr	Lys	Cys	Lys	Val	Ser	Asn	Lys	Ala	Leu	Pro	Ala
		355					360					365			
Pro	Ile	Glu	Lys	Thr	Ile	Ser	Lys	Ala	Lys	Gly	Gln	Pro	Arg	Glu	Pro
	370					375					380				
Gln	Val	Tyr	Thr	Leu	Pro	Pro	Ser	Arg	Glu	Glu	Met	Thr	Lys	Asn	Gln
385					390					395					400
Val	Ser	Leu	Thr	Cys	Arg	Val	Lys	Gly	Phe	Tyr	Pro	Ser	Asp	Ile	Ala
				405					410					415	
Val	Glu	Trp	Glu	Ser	Asn	Gly	Gln	Pro	Glu	Asn	Asn	Tyr	Lys	Thr	Thr
			420				425						430		
Pro	Pro	Val	Leu	Lys	Ser	Asp	Gly	Ser	Phe	Phe	Leu	Ala	Ser	Lys	Leu
		435					440					445			
Thr	Val	Asp	Lys	Ser	Arg	Trp	Gln	Gln	Gly	Asn	Val	Phe	Ser	Cys	Ser
	450					455					460				
Val	Met	His	Glu	Ala	Leu	His	Asn	His	Tyr	Thr	Gln	Lys	Ser	Leu	Ser
465					470					475					480
Leu	Ser	Pro	Gly	Lys											
				485											

<210> SEQ ID NO 69

<211> LENGTH: 584

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 69

Glu	Val	Gln	Leu	Gln	Glu	Ser	Gly	Pro	Gly	Leu	Val	Lys	Pro	Ser	Glu
1				5					10				15		
Thr	Leu	Ser	Leu	Thr	Cys	Ala	Val	Tyr	Gly	Asp	Ser	Ile	Thr	Ser	Gly
			20					25				30			
Tyr	Trp	Asn	Trp	Ile	Arg	Lys	Pro	Pro	Gly	Lys	Gly	Leu	Glu	Tyr	Met
		35				40						45			
Gly	Tyr	Ile	Ser	Tyr	Thr	Gly	Ser	Thr	Tyr	Gln	Asn	Pro	Ser	Leu	Lys
		50				55					60				
Ser	Arg	Ile	Thr	Phe	Ser	Arg	Asp	Thr	Ser	Lys	Asn	Gln	Tyr	Tyr	Leu
65					70				75						80
Lys	Leu	Ser	Ser	Val	Thr	Ala	Ala	Asp	Thr	Ala	Thr	Tyr	Tyr	Cys	Ala
				85				90						95	
Arg	Ser	Arg	Ala	Trp	Ile	Arg	Thr	Tyr	Phe	Asp	Tyr	Trp	Gly	Gln	Gly
			100				105						110		

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Thr	Leu	Val	Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	115	120	125
Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Cys	Pro	Asp	Leu	130	135	140
Val	Cys	Tyr	Thr	Asp	Tyr	Leu	Gln	Thr	Val	Ile	Cys	Ile	Leu	Glu	Met	145	150	155
Trp	Asn	Leu	His	Pro	Ser	Thr	Leu	Thr	Leu	Thr	Trp	Gln	Asp	Gln	Tyr	165	170	175
Glu	Glu	Leu	Lys	Asp	Glu	Ala	Thr	Ser	Cys	Ser	Leu	His	Arg	Ser	Ala	180	185	190
His	Asn	Ala	Thr	His	Ala	Thr	Tyr	Thr	Cys	His	Met	Asp	Val	Phe	His	195	200	205
Phe	Met	Ala	Asp	Asp	Ile	Phe	Ser	Val	Gln	Ile	Thr	Asp	Gln	Ser	Gly	210	215	220
Gln	Tyr	Ser	Gln	Glu	Cys	Gly	Ser	Phe	Leu	Leu	Ala	Glu	Ser	Ile	Lys	225	230	235
Pro	Ala	Pro	Pro	Phe	Asp	Val	Thr	Val	Thr	Phe	Ser	Gly	Gln	Tyr	Gln	245	250	255
Ile	Ser	Trp	Arg	Ser	Asp	Tyr	Glu	Asp	Pro	Ala	Phe	Tyr	Met	Leu	Lys	260	265	270
Gly	Lys	Leu	Gln	Tyr	Glu	Leu	Gln	Tyr	Arg	Asn	Arg	Gly	Asp	Pro	Trp	275	280	285
Ala	Val	Ser	Pro	Arg	Arg	Lys	Leu	Ile	Ser	Val	Asp	Ser	Arg	Ser	Val	290	295	300
Ser	Leu	Leu	Pro	Leu	Glu	Phe	Arg	Lys	Asp	Ser	Ser	Tyr	Glu	Leu	Gln	305	310	315
Val	Arg	Ala	Gly	Pro	Met	Pro	Gly	Ser	Ser	Tyr	Gln	Gly	Thr	Trp	Ser	325	330	335
Glu	Trp	Ser	Asp	Pro	Val	Ile	Phe	Gln	Thr	Gln	Ser	Glu	Glu	Leu	Lys	340	345	350
Glu	Glu	Pro	Lys	Ser	Ser	Asp	Lys	Thr	His	Thr	Cys	Pro	Pro	Cys	Pro	355	360	365
Ala	Pro	Glu	Leu	Leu	Gly	Gly	Pro	Ser	Val	Phe	Leu	Phe	Pro	Pro	Lys	370	375	380
Pro	Lys	Asp	Thr	Leu	Met	Ile	Ser	Arg	Thr	Pro	Glu	Val	Thr	Cys	Val	385	390	395
Val	Val	Asp	Val	Ser	His	Glu	Asp	Pro	Glu	Val	Lys	Phe	Asn	Trp	Tyr	405	410	415
Val	Asp	Gly	Val	Glu	Val	His	Asn	Ala	Lys	Thr	Lys	Pro	Arg	Glu	Glu	420	425	430
Gln	Tyr	Asn	Ser	Thr	Tyr	Arg	Val	Val	Ser	Val	Leu	Thr	Val	Leu	His	435	440	445
Gln	Asp	Trp	Leu	Asn	Gly	Lys	Glu	Tyr	Lys	Cys	Lys	Val	Ser	Asn	Lys	450	455	460
Ala	Leu	Pro	Ala	Pro	Ile	Glu	Lys	Thr	Ile	Ser	Lys	Ala	Lys	Gly	Gln	465	470	475
Pro	Arg	Glu	Pro	Gln	Val	Tyr	Thr	Leu	Pro	Pro	Ser	Arg	Glu	Glu	Met	485	490	495
Thr	Lys	Asn	Gln	Val	Ser	Leu	Thr	Cys	Arg	Val	Lys	Gly	Phe	Tyr	Pro	500	505	510
Ser	Asp	Ile	Ala	Val	Glu	Trp	Glu	Ser	Asn	Gly	Gln	Pro	Glu	Asn	Asn			

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515	520	525
Tyr Lys Thr Thr Pro Pro Val Leu Lys Ser Asp Gly Ser Phe Phe Leu		
530	535	540
Ala Ser Lys Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val		
545	550	555
Phe Ser Cys Ser Val Met His Glu Ala Leu His Asn His Tyr Thr Gln		
	565	570
		575
Lys Ser Leu Ser Leu Ser Pro Gly		
580		

<210> SEQ ID NO 70
 <211> LENGTH: 448
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 70

Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ser		
1	5	10
Ser Val Lys Val Ser Cys Lys Ala Ser Gly Tyr Ser Phe Thr Ser Tyr		
	20	25
Trp Met Asn Trp Val Arg Gln Ala Pro Gly Gln Gly Leu Glu Trp Ile		
	35	40
Gly Met Ile Arg Pro Ser Asp Ser Glu Thr Arg Leu Asn Gln Met Phe		
50	55	60
Lys Asp Arg Val Thr Ile Thr Val Asp Lys Ser Thr Ser Thr Ala Tyr		
65	70	75
Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Val Tyr Tyr Cys		
	85	90
Ala Gly Ile His Asp Tyr Gly His Gly Ala Tyr Trp Gly Gln Gly Thr		
	100	105
Leu Val Thr Val Ser Ser Ala Ser Thr Lys Gly Pro Ser Val Phe Pro		
	115	120
Leu Ala Pro Ser Ser Lys Ser Thr Ser Gly Gly Thr Ala Ala Leu Gly		
130	135	140
Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val Thr Val Ser Trp Asn		
145	150	155
Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro Ala Val Leu Gln		
	165	170
Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val Thr Val Pro Ser Ser		
	180	185
Ser Leu Gly Thr Gln Thr Tyr Ile Cys Asn Val Asn His Lys Pro Ser		
195	200	205
Asn Thr Lys Val Asp Lys Arg Val Glu Pro Lys Ser Cys Asp Lys Thr		
210	215	220
His Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu Leu Gly Gly Pro Ser		
225	230	235
Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile Ser Arg		
	245	250
Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser His Glu Asp Pro		
	260	265
		270
Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val Glu Val His Asn Ala		

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275	280	285
Lys Thr Lys Pro Arg Glu Glu Gln Tyr Asn Ser Thr Tyr Arg Val Val		
290	295	300
Ser Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly Lys Glu Tyr		
305	310	315
Lys Cys Lys Val Ser Asn Lys Ala Leu Pro Ala Pro Ile Glu Lys Thr		
325	330	335
Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr Thr Leu		
340	345	350
Pro Pro Ser Arg Glu Glu Met Thr Lys Asn Gln Val Ser Leu Trp Cys		
355	360	365
Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu Ser		
370	375	380
Asn Gly Gln Pro Glu Asn Asn Tyr Asp Thr Thr Pro Pro Val Leu Asp		
385	390	395
Ser Asp Gly Ser Phe Phe Leu Tyr Ser Asp Leu Thr Val Asp Lys Ser		
405	410	415
Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met His Glu Ala		
420	425	430
Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro Gly Lys		
435	440	445

<210> SEQ ID NO 71
 <211> LENGTH: 119
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 71

Gln Val Gln Leu Gln Glu Ser Gly Pro Gly Leu Val Lys Pro Ser Glu
1 5 10 15
Thr Leu Ser Leu Thr Cys Ala Val Tyr Gly Asp Ser Ile Thr Ser Gly
20 25 30
Tyr Trp Asn Trp Ile Arg Lys Pro Pro Gly Lys Cys Leu Glu Tyr Ile
35 40 45
Gly Tyr Ile Ser Tyr Thr Gly Ser Thr Tyr Gln Asn Pro Ser Leu Lys
50 55 60
Ser Arg Ile Thr Met Ser Arg Asp Thr Ser Lys Asn Gln Tyr Tyr Leu
65 70 75 80
Lys Leu Ser Ser Val Thr Ala Ala Asp Thr Ala Val Tyr Tyr Cys Ala
85 90 95
Arg Ser Arg Ala Trp Ile Arg Thr Tyr Phe Asp Tyr Trp Gly Gln Gly
100 105 110
Thr Leu Val Thr Val Ser Ser
115

<210> SEQ ID NO 72
 <211> LENGTH: 108
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 72

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Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
 1 5 10 15
 Asp Arg Val Thr Ile Thr Cys Ser Val Ser Ser Ser Ile Ser Ser Ser
 20 25 30
 Asn Leu His Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Pro Trp
 35 40 45
 Ile Tyr Gly Thr Ser Asn Leu Ala Ser Gly Val Pro Ser Arg Phe Ser
 50 55 60
 Gly Ser Gly Ser Gly Thr Asp Tyr Thr Leu Thr Ile Ser Ser Leu Gln
 65 70 75 80
 Pro Glu Asp Phe Ala Thr Tyr Tyr Cys Gln Gln Trp Ser Ser Tyr Pro
 85 90 95
 Leu Thr Phe Gly Cys Gly Thr Lys Leu Glu Ile Lys
 100 105

<210> SEQ ID NO 73
 <211> LENGTH: 118
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 73

Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ser
 1 5 10 15
 Ser Val Lys Val Ser Cys Lys Ala Ser Gly Tyr Ser Phe Thr Ser Tyr
 20 25 30
 Trp Met Asn Trp Val Arg Gln Ala Pro Gly Gln Gly Leu Glu Trp Ile
 35 40 45
 Gly Met Ile Arg Pro Ser Asp Ser Glu Thr Arg Leu Asn Gln Met Phe
 50 55 60
 Lys Asp Arg Val Thr Ile Thr Val Asp Lys Ser Thr Ser Thr Ala Tyr
 65 70 75 80
 Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95
 Ala Gly Ile His Asp Tyr Gly His Gly Ala Tyr Trp Gly Gln Gly Thr
 100 105 110
 Leu Val Thr Val Ser Ser
 115

<210> SEQ ID NO 74
 <211> LENGTH: 107
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 74

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
 1 5 10 15
 Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Glu Asn Ile Tyr Ser Asn
 20 25 30
 Leu Ala Trp Tyr Gln Gln Lys Pro Gly Lys Ser Pro Lys Leu Leu Val
 35 40 45
 Tyr Ala Ala Ser His Leu Pro Asp Gly Val Pro Ser Arg Phe Ser Gly
 50 55 60

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Ser Gly Ser Gly Thr Asp Tyr Ser Leu Thr Ile Ser Ser Leu Gln Pro
 65 70 75 80

Glu Asp Phe Ala Thr Tyr Tyr Cys Gln His Phe Trp Gly Thr Pro Arg
 85 90 95

Thr Phe Gly Gln Gly Thr Lys Leu Glu Ile Lys
 100 105

<210> SEQ ID NO 75
 <211> LENGTH: 119
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 75

Glu Val Gln Leu Gln Glu Ser Gly Pro Gly Leu Val Lys Pro Ser Glu
 1 5 10 15

Thr Leu Ser Leu Thr Cys Ala Val Tyr Gly Asp Ser Ile Thr Ser Gly
 20 25 30

Tyr Trp Asn Trp Ile Arg Lys Pro Pro Gly Lys Gly Leu Glu Tyr Met
 35 40 45

Gly Tyr Ile Ser Tyr Thr Gly Ser Thr Tyr Gln Asn Pro Ser Leu Lys
 50 55 60

Ser Arg Ile Thr Phe Ser Arg Asp Thr Ser Lys Asn Gln Tyr Tyr Leu
 65 70 75 80

Lys Leu Ser Ser Val Thr Ala Ala Asp Thr Ala Thr Tyr Tyr Cys Ala
 85 90 95

Arg Ser Arg Ala Trp Ile Arg Thr Tyr Phe Asp Tyr Trp Gly Gln Gly
 100 105 110

Thr Leu Val Thr Val Ser Ser
 115

<210> SEQ ID NO 76
 <211> LENGTH: 108
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 76

Glu Ile Val Leu Thr Gln Ser Pro Asp Phe Gln Ser Val Thr Pro Lys
 1 5 10 15

Glu Lys Val Thr Ile Thr Cys Ser Val Ser Ser Ser Ile Ser Ser Ser
 20 25 30

Asn Leu His Trp Tyr Gln Gln Lys Pro Asp Gln Ser Pro Lys Leu Leu
 35 40 45

Ile Tyr Gly Thr Ser Asn Leu Ala Ser Gly Val Pro Ser Arg Phe Ser
 50 55 60

Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Asn Ser Leu Glu
 65 70 75 80

Ala Glu Asp Ala Ala Thr Tyr Tyr Cys Gln Gln Trp Ser Ser Tyr Pro
 85 90 95

Leu Thr Phe Gly Gln Gly Thr Lys Leu Glu Ile Lys
 100 105

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<210> SEQ ID NO 77
<211> LENGTH: 536
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 77

Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ser
1          5          10          15

Ser Val Lys Val Ser Cys Lys Ala Ser Gly Tyr Ser Phe Thr Ser Tyr
20          25          30

Trp Met Asn Trp Val Arg Gln Ala Pro Gly Gln Gly Leu Glu Trp Ile
35          40          45

Gly Met Ile Arg Pro Ser Asp Ser Glu Thr Arg Leu Asn Gln Met Phe
50          55          60

Lys Asp Arg Val Thr Ile Thr Val Asp Lys Ser Thr Ser Thr Ala Tyr
65          70          75          80

Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Val Tyr Tyr Cys
85          90          95

Ala Gly Ile His Asp Tyr Gly His Gly Ala Tyr Trp Gly Gln Gly Thr
100         105         110

Leu Val Thr Val Ser Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
115         120         125

Gly Gly Gly Gly Ser Gly Gly Gly Gly Cys Gly Glu Leu Cys Asp Asp
130         135         140

Asp Pro Pro Glu Ile Pro His Ala Thr Phe Lys Ala Met Ala Tyr Lys
145         150         155         160

Glu Gly Thr Met Leu Asn Cys Glu Cys Lys Arg Gly Phe Arg Arg Ile
165         170         175

Lys Ser Gly Ser Leu Tyr Met Leu Cys Thr Gly Asn Ser Ser His Ser
180         185         190

Ser Trp Asp Asn Gln Cys Gln Cys Thr Ser Ser Ala Thr Arg Asn Thr
195         200         205

Thr Lys Gln Val Thr Pro Gln Pro Glu Glu Gln Lys Glu Arg Lys Thr
210         215         220

Thr Glu Met Gln Ser Pro Met Gln Pro Val Asp Gln Ala Ser Leu Pro
225         230         235         240

Gly His Cys Arg Glu Pro Pro Pro Trp Glu Asn Glu Ala Thr Glu Arg
245         250         255

Ile Tyr His Phe Val Val Gly Gln Met Val Tyr Tyr Gln Cys Val Gln
260         265         270

Gly Tyr Arg Ala Leu His Arg Gly Pro Ala Glu Ser Val Cys Lys Met
275         280         285

Thr His Gly Lys Thr Arg Trp Thr Gln Pro Gln Leu Ile Cys Thr Gly
290         295         300

Glu Pro Lys Ser Ser Asp Lys Thr His Thr Cys Pro Pro Cys Pro Ala
305         310         315         320

Pro Glu Leu Leu Gly Gly Pro Ser Val Phe Leu Phe Pro Pro Lys Pro
325         330         335

Lys Asp Thr Leu Met Ile Ser Arg Thr Pro Glu Val Thr Cys Val Val
340         345         350

Val Asp Val Ser His Glu Asp Pro Glu Val Lys Phe Asn Trp Tyr Val

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355	360	365
Asp Gly Val Glu Val His Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln		
370	375	380
Tyr Asn Ser Thr Tyr Arg Val Val Ser Val Leu Thr Val Leu His Gln		
385	390	395
400		
Asp Trp Leu Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala		
405	410	415
Leu Pro Ala Pro Ile Glu Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro		
420	425	430
Arg Glu Pro Gln Val Tyr Thr Leu Pro Pro Ser Arg Glu Glu Met Thr		
435	440	445
Lys Asn Gln Val Ser Leu Thr Cys Arg Val Lys Gly Phe Tyr Pro Ser		
450	455	460
Asp Ile Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr		
465	470	475
480		
Lys Thr Thr Pro Pro Val Leu Lys Ser Asp Gly Ser Phe Phe Leu Ala		
485	490	495
Ser Lys Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe		
500	505	510
Ser Cys Ser Val Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys		
515	520	525
Ser Leu Ser Leu Ser Pro Gly Lys		
530	535	

<210> SEQ ID NO 78

<211> LENGTH: 261

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 78

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1 5 10 15
Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Glu Asn Ile Tyr Ser Asn
20 25 30
Leu Ala Trp Tyr Gln Gln Lys Pro Gly Lys Ser Pro Lys Leu Leu Val
35 40 45
Tyr Ala Ala Ser His Leu Pro Asp Gly Val Pro Ser Arg Phe Ser Gly
50 55 60
Ser Gly Ser Gly Thr Asp Tyr Ser Leu Thr Ile Ser Ser Leu Gln Pro
65 70 75 80
Glu Asp Phe Ala Thr Tyr Tyr Cys Gln His Phe Trp Gly Thr Pro Arg
85 90 95
Thr Phe Gly Gln Gly Thr Lys Leu Glu Ile Lys Gly Gly Gly Gly Ser
100 105 110
Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly
115 120 125
Ala Pro Thr Ser Ser Ser Thr Lys Lys Thr Gln Leu Gln Leu Glu His
130 135 140
Leu Leu Leu Asp Leu Gln Met Ile Leu Asn Gly Ile Asn Asn Tyr Lys
145 150 155 160
Asn Pro Lys Leu Thr Arg Met Leu Thr Phe Lys Phe Tyr Met Pro Lys

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	165		170		175										
Lys	Ala	Thr	Glu	Leu	Lys	His	Leu	Gln	Cys	Leu	Glu	Glu	Glu	Leu	Lys
	180							185						190	
Pro	Leu	Glu	Glu	Val	Leu	Asn	Leu	Ala	Gln	Cys	Lys	Asn	Phe	His	Leu
	195						200						205		
Arg	Pro	Arg	Asp	Leu	Ile	Ser	Asn	Ile	Asn	Val	Ile	Val	Leu	Glu	Leu
	210					215					220				
Lys	Gly	Ser	Glu	Thr	Thr	Phe	Met	Cys	Glu	Tyr	Ala	Asp	Glu	Thr	Ala
225					230					235					240
Thr	Ile	Val	Glu	Phe	Leu	Asn	Arg	Trp	Ile	Thr	Phe	Cys	Gln	Ser	Ile
			245						250					255	
Ile	Ser	Thr	Leu	Thr											
			260												

<210> SEQ ID NO 79
 <211> LENGTH: 449
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 79

Glu	Val	Gln	Leu	Gln	Glu	Ser	Gly	Pro	Gly	Leu	Val	Lys	Pro	Ser	Glu
1			5					10					15		
Thr	Leu	Ser	Leu	Thr	Cys	Ala	Val	Tyr	Gly	Asp	Ser	Ile	Thr	Ser	Gly
			20					25				30			
Tyr	Trp	Asn	Trp	Ile	Arg	Lys	Pro	Pro	Gly	Lys	Gly	Leu	Glu	Tyr	Met
		35				40						45			
Gly	Tyr	Ile	Ser	Tyr	Thr	Gly	Ser	Thr	Tyr	Gln	Asn	Pro	Ser	Leu	Lys
	50					55					60				
Ser	Arg	Ile	Thr	Phe	Ser	Arg	Asp	Thr	Ser	Lys	Asn	Gln	Tyr	Tyr	Leu
65					70					75					80
Lys	Leu	Ser	Ser	Val	Thr	Ala	Ala	Asp	Thr	Ala	Thr	Tyr	Tyr	Cys	Ala
				85					90					95	
Arg	Ser	Arg	Ala	Trp	Ile	Arg	Thr	Tyr	Phe	Asp	Tyr	Trp	Gly	Gln	Gly
			100					105					110		
Thr	Leu	Val	Thr	Val	Ser	Ser	Ala	Ser	Thr	Lys	Gly	Pro	Ser	Val	Phe
		115					120					125			
Pro	Leu	Ala	Pro	Ser	Ser	Lys	Ser	Thr	Ser	Gly	Gly	Thr	Ala	Ala	Leu
	130					135					140				
Gly	Cys	Leu	Val	Lys	Asp	Tyr	Phe	Pro	Glu	Pro	Val	Thr	Val	Ser	Trp
145					150					155					160
Asn	Ser	Gly	Ala	Leu	Thr	Ser	Gly	Val	His	Thr	Phe	Pro	Ala	Val	Leu
			165					170						175	
Gln	Ser	Ser	Gly	Leu	Tyr	Ser	Leu	Ser	Ser	Val	Val	Thr	Val	Pro	Ser
			180					185					190		
Ser	Ser	Leu	Gly	Thr	Gln	Thr	Tyr	Ile	Cys	Asn	Val	Asn	His	Lys	Pro
		195					200					205			
Ser	Asn	Thr	Lys	Val	Asp	Lys	Arg	Val	Glu	Pro	Lys	Ser	Cys	Asp	Lys
	210					215					220				
Thr	His	Thr	Cys	Pro	Pro	Cys	Pro	Ala	Pro	Glu	Leu	Leu	Gly	Gly	Pro
225					230					235					240
Ser	Val	Phe	Leu	Phe	Pro	Pro	Lys	Pro	Lys	Asp	Thr	Leu	Met	Ile	Ser

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245					250					255					
Arg	Thr	Pro	Glu	Val	Thr	Cys	Val	Val	Val	Asp	Val	Ser	His	Glu	Asp
			260					265					270		
Pro	Glu	Val	Lys	Phe	Asn	Trp	Tyr	Val	Asp	Gly	Val	Glu	Val	His	Asn
			275				280					285			
Ala	Lys	Thr	Lys	Pro	Arg	Glu	Glu	Gln	Tyr	Asn	Ser	Thr	Tyr	Arg	Val
			290			295					300				
Val	Ser	Val	Leu	Thr	Val	Leu	His	Gln	Asp	Trp	Leu	Asn	Gly	Lys	Glu
305					310					315					320
Tyr	Lys	Cys	Lys	Val	Ser	Asn	Lys	Ala	Leu	Pro	Ala	Pro	Ile	Glu	Lys
				325					330					335	
Thr	Ile	Ser	Lys	Ala	Lys	Gly	Gln	Pro	Arg	Glu	Pro	Gln	Val	Tyr	Thr
			340					345					350		
Leu	Pro	Pro	Ser	Arg	Glu	Glu	Met	Thr	Lys	Asn	Gln	Val	Ser	Leu	Trp
			355				360					365			
Cys	Leu	Val	Lys	Gly	Phe	Tyr	Pro	Ser	Asp	Ile	Ala	Val	Glu	Trp	Glu
	370					375					380				
Ser	Asn	Gly	Gln	Pro	Glu	Asn	Asn	Tyr	Asp	Thr	Thr	Pro	Pro	Val	Leu
385					390					395					400
Asp	Ser	Asp	Gly	Ser	Phe	Phe	Leu	Tyr	Ser	Asp	Leu	Thr	Val	Asp	Lys
			405						410					415	
Ser	Arg	Trp	Gln	Gln	Gly	Asn	Val	Phe	Ser	Cys	Ser	Val	Met	His	Glu
			420					425					430		
Ala	Leu	His	Asn	His	Tyr	Thr	Gln	Lys	Ser	Leu	Ser	Leu	Ser	Pro	Gly
			435				440					445			

Lys

<210> SEQ ID NO 80
 <211> LENGTH: 215
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 80

Glu	Ile	Val	Leu	Thr	Gln	Ser	Pro	Asp	Phe	Gln	Ser	Val	Thr	Pro	Lys
1			5					10						15	
Glu	Lys	Val	Thr	Ile	Thr	Cys	Ser	Val	Ser	Ser	Ser	Ile	Ser	Ser	Ser
		20					25					30			
Asn	Leu	His	Trp	Tyr	Gln	Gln	Lys	Pro	Asp	Gln	Ser	Pro	Lys	Leu	Leu
	35						40					45			
Ile	Tyr	Gly	Thr	Ser	Asn	Leu	Ala	Ser	Gly	Val	Pro	Ser	Arg	Phe	Ser
	50				55					60					
Gly	Ser	Gly	Ser	Gly	Thr	Asp	Phe	Thr	Leu	Thr	Ile	Asn	Ser	Leu	Glu
65				70				75						80	
Ala	Glu	Asp	Ala	Ala	Thr	Tyr	Tyr	Cys	Gln	Gln	Trp	Ser	Ser	Tyr	Pro
			85					90						95	
Leu	Thr	Phe	Gly	Gln	Gly	Thr	Lys	Leu	Glu	Ile	Lys	Arg	Thr	Val	Ala
			100					105						110	
Ala	Pro	Ser	Val	Phe	Ile	Phe	Pro	Pro	Ser	Asp	Glu	Gln	Leu	Lys	Ser
			115				120					125			
Gly	Thr	Ala	Ser	Val	Val	Cys	Leu	Leu	Asn	Asn	Phe	Tyr	Pro	Arg	Glu
	130					135					140				

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Ala Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln Ser Gly Asn Ser
 145 150 155 160

Gln Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser Thr Tyr Ser Leu
 165 170 175

Ser Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu Lys His Lys Val
 180 185 190

Tyr Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser Pro Val Thr Lys
 195 200 205

Ser Phe Asn Arg Gly Glu Cys
 210 215

<210> SEQ ID NO 81
 <211> LENGTH: 504
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 81

Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ser
 1 5 10 15

Ser Val Lys Val Ser Cys Lys Ala Ser Gly Tyr Ser Phe Thr Ser Tyr
 20 25 30

Trp Met Asn Trp Val Arg Gln Ala Pro Gly Gln Gly Leu Glu Trp Ile
 35 40 45

Gly Met Ile Arg Pro Ser Asp Ser Glu Thr Arg Leu Asn Gln Met Phe
 50 55 60

Lys Asp Arg Val Thr Ile Thr Val Asp Lys Ser Thr Ser Thr Ala Tyr
 65 70 75 80

Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95

Ala Gly Ile His Asp Tyr Gly His Gly Ala Tyr Trp Gly Gln Gly Thr
 100 105 110

Leu Val Thr Val Ser Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
 115 120 125

Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Ala Pro Thr Ser Ser
 130 135 140

Ser Thr Lys Lys Thr Gln Leu Gln Leu Glu His Leu Leu Leu Asp Leu
 145 150 155 160

Gln Met Ile Leu Asn Gly Ile Asn Asn Tyr Lys Asn Pro Lys Leu Thr
 165 170 175

Arg Met Leu Thr Phe Lys Phe Tyr Met Pro Lys Lys Ala Thr Glu Leu
 180 185 190

Lys His Leu Gln Cys Leu Glu Glu Glu Leu Lys Pro Leu Glu Glu Val
 195 200 205

Leu Asn Leu Ala Gln Cys Lys Asn Phe His Leu Arg Pro Arg Asp Leu
 210 215 220

Ile Ser Asn Ile Asn Val Ile Val Leu Glu Leu Lys Gly Ser Glu Thr
 225 230 235 240

Thr Phe Met Cys Glu Tyr Ala Asp Glu Thr Ala Thr Ile Val Glu Phe
 245 250 255

Leu Asn Arg Trp Ile Thr Phe Cys Gln Ser Ile Ile Ser Thr Leu Thr
 260 265 270

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Glu Pro Lys Ser Ser Asp Lys Thr His Thr Cys Pro Pro Cys Pro Ala
 275 280 285
 Pro Glu Leu Leu Gly Gly Pro Ser Val Phe Leu Phe Pro Pro Lys Pro
 290 295 300
 Lys Asp Thr Leu Met Ile Ser Arg Thr Pro Glu Val Thr Cys Val Val
 305 310 315 320
 Val Asp Val Ser His Glu Asp Pro Glu Val Lys Phe Asn Trp Tyr Val
 325 330 335
 Asp Gly Val Glu Val His Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln
 340 345 350
 Tyr Asn Ser Thr Tyr Arg Val Val Ser Val Leu Thr Val Leu His Gln
 355 360 365
 Asp Trp Leu Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala
 370 375 380
 Leu Pro Ala Pro Ile Glu Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro
 385 390 395 400
 Arg Glu Pro Gln Val Tyr Thr Leu Pro Pro Ser Arg Glu Glu Met Thr
 405 410 415
 Lys Asn Gln Val Ser Leu Thr Cys Arg Val Lys Gly Phe Tyr Pro Ser
 420 425 430
 Asp Ile Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr
 435 440 445
 Lys Thr Thr Pro Pro Val Leu Lys Ser Asp Gly Ser Phe Phe Leu Ala
 450 455 460
 Ser Lys Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe
 465 470 475 480
 Ser Cys Ser Val Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys
 485 490 495
 Ser Leu Ser Leu Ser Pro Gly Lys
 500

<210> SEQ ID NO 82
 <211> LENGTH: 293
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 82

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
 1 5 10 15
 Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Glu Asn Ile Tyr Ser Asn
 20 25 30
 Leu Ala Trp Tyr Gln Gln Lys Pro Gly Lys Ser Pro Lys Leu Leu Val
 35 40 45
 Tyr Ala Ala Ser His Leu Pro Asp Gly Val Pro Ser Arg Phe Ser Gly
 50 55 60
 Ser Gly Ser Gly Thr Asp Tyr Ser Leu Thr Ile Ser Ser Leu Gln Pro
 65 70 75 80
 Glu Asp Phe Ala Thr Tyr Tyr Cys Gln His Phe Trp Gly Thr Pro Arg
 85 90 95
 Thr Phe Gly Gln Gly Thr Lys Leu Glu Ile Lys Gly Gly Gly Gly Ser
 100 105 110

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Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Cys Gly
 115 120 125
 Glu Leu Cys Asp Asp Asp Pro Pro Glu Ile Pro His Ala Thr Phe Lys
 130 135 140
 Ala Met Ala Tyr Lys Glu Gly Thr Met Leu Asn Cys Glu Cys Lys Arg
 145 150 155 160
 Gly Phe Arg Arg Ile Lys Ser Gly Ser Leu Tyr Met Leu Cys Thr Gly
 165 170 175
 Asn Ser Ser His Ser Ser Trp Asp Asn Gln Cys Gln Cys Thr Ser Ser
 180 185 190
 Ala Thr Arg Asn Thr Thr Lys Gln Val Thr Pro Gln Pro Glu Glu Gln
 195 200 205
 Lys Glu Arg Lys Thr Thr Glu Met Gln Ser Pro Met Gln Pro Val Asp
 210 215 220
 Gln Ala Ser Leu Pro Gly His Cys Arg Glu Pro Pro Pro Trp Glu Asn
 225 230 235 240
 Glu Ala Thr Glu Arg Ile Tyr His Phe Val Val Gly Gln Met Val Tyr
 245 250 255
 Tyr Gln Cys Val Gln Gly Tyr Arg Ala Leu His Arg Gly Pro Ala Glu
 260 265 270
 Ser Val Cys Lys Met Thr His Gly Lys Thr Arg Trp Thr Gln Pro Gln
 275 280 285
 Leu Ile Cys Thr Gly
 290

<210> SEQ ID NO 83
 <211> LENGTH: 536
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 83

Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ser
 1 5 10 15
 Ser Val Lys Val Ser Cys Lys Ala Ser Gly Tyr Ser Phe Thr Ser Tyr
 20 25 30
 Trp Met Asn Trp Val Arg Gln Ala Pro Gly Gln Gly Leu Glu Trp Ile
 35 40 45
 Gly Met Ile Arg Pro Ser Asp Ser Glu Thr Arg Leu Asn Gln Met Phe
 50 55 60
 Lys Asp Arg Val Thr Ile Thr Val Asp Lys Ser Thr Ser Thr Ala Tyr
 65 70 75 80
 Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95
 Ala Gly Ile His Asp Tyr Gly His Gly Ala Tyr Trp Gly Gln Gly Thr
 100 105 110
 Leu Val Thr Val Ser Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
 115 120 125
 Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Glu Leu Cys Asp Asp
 130 135 140
 Asp Pro Pro Glu Ile Pro His Ala Thr Phe Lys Ala Met Ala Tyr Lys
 145 150 155 160

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Glu Gly Thr Met Leu Asn Cys Glu Cys Lys Arg Gly Phe Arg Arg Ile
 165 170 175
 Lys Ser Gly Ser Cys Tyr Met Leu Cys Thr Gly Asn Ser Ser His Ser
 180 185 190
 Ser Trp Asp Asn Gln Cys Gln Cys Thr Ser Ser Ala Thr Arg Asn Thr
 195 200 205
 Thr Lys Gln Val Thr Pro Gln Pro Glu Glu Gln Lys Glu Arg Lys Thr
 210 215 220
 Thr Glu Met Gln Ser Pro Met Gln Pro Val Asp Gln Ala Ser Leu Pro
 225 230 235 240
 Gly His Cys Arg Glu Pro Pro Pro Trp Glu Asn Glu Ala Thr Glu Arg
 245 250 255
 Ile Tyr His Phe Val Val Gly Gln Met Val Tyr Tyr Gln Cys Val Gln
 260 265 270
 Gly Tyr Arg Ala Leu His Arg Gly Pro Ala Glu Ser Val Cys Lys Met
 275 280 285
 Thr His Gly Lys Thr Arg Trp Thr Gln Pro Gln Leu Ile Cys Thr Gly
 290 295 300
 Glu Pro Lys Ser Ser Asp Lys Thr His Thr Cys Pro Pro Cys Pro Ala
 305 310 315 320
 Pro Glu Leu Leu Gly Gly Pro Ser Val Phe Leu Phe Pro Pro Lys Pro
 325 330 335
 Lys Asp Thr Leu Met Ile Ser Arg Thr Pro Glu Val Thr Cys Val Val
 340 345 350
 Val Asp Val Ser His Glu Asp Pro Glu Val Lys Phe Asn Trp Tyr Val
 355 360 365
 Asp Gly Val Glu Val His Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln
 370 375 380
 Tyr Asn Ser Thr Tyr Arg Val Val Ser Val Leu Thr Val Leu His Gln
 385 390 395 400
 Asp Trp Leu Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala
 405 410 415
 Leu Pro Ala Pro Ile Glu Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro
 420 425 430
 Arg Glu Pro Gln Val Tyr Thr Leu Pro Pro Ser Arg Glu Glu Met Thr
 435 440 445
 Lys Asn Gln Val Ser Leu Thr Cys Arg Val Lys Gly Phe Tyr Pro Ser
 450 455 460
 Asp Ile Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr
 465 470 475 480
 Lys Thr Thr Pro Pro Val Leu Lys Ser Asp Gly Ser Phe Phe Leu Ala
 485 490 495
 Ser Lys Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe
 500 505 510
 Ser Cys Ser Val Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys
 515 520 525
 Ser Leu Ser Leu Ser Pro Gly Lys
 530 535

<210> SEQ ID NO 84

<211> LENGTH: 261

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<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 84
Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1             5             10             15
Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Glu Asn Ile Tyr Ser Asn
20            25            30
Leu Ala Trp Tyr Gln Gln Lys Pro Gly Lys Ser Pro Lys Leu Leu Val
35            40            45
Tyr Ala Ala Ser His Leu Pro Asp Gly Val Pro Ser Arg Phe Ser Gly
50            55            60
Ser Gly Ser Gly Thr Asp Tyr Ser Leu Thr Ile Ser Ser Leu Gln Pro
65            70            75            80
Glu Asp Phe Ala Thr Tyr Tyr Cys Gln His Phe Trp Gly Thr Pro Arg
85            90            95
Thr Phe Gly Gln Gly Thr Lys Leu Glu Ile Lys Gly Gly Gly Gly Ser
100           105           110
Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly
115           120           125
Ala Pro Thr Ser Ser Ser Thr Lys Lys Thr Gln Leu Gln Leu Glu His
130           135           140
Leu Leu Leu Asp Leu Gln Met Ile Leu Asn Gly Ile Asn Asn Tyr Lys
145           150           155           160
Asn Pro Lys Leu Thr Arg Met Leu Thr Cys Lys Phe Tyr Met Pro Lys
165           170           175
Lys Ala Thr Glu Leu Lys His Leu Gln Cys Leu Glu Glu Glu Leu Lys
180           185           190
Pro Leu Glu Glu Val Leu Asn Leu Ala Gln Ser Lys Asn Phe His Leu
195           200           205
Arg Pro Arg Asp Leu Ile Ser Asn Ile Asn Val Ile Val Leu Glu Leu
210           215           220
Lys Gly Ser Glu Thr Thr Phe Met Cys Glu Tyr Ala Asp Glu Thr Ala
225           230           235           240
Thr Ile Val Glu Phe Leu Asn Arg Trp Ile Thr Phe Cys Gln Ser Ile
245           250           255
Ile Ser Thr Leu Thr
260

```

```

<210> SEQ ID NO 85
<211> LENGTH: 536
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthesized

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```

<400> SEQUENCE: 85
Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ser
1             5             10             15
Ser Val Lys Val Ser Cys Lys Ala Ser Gly Tyr Ser Phe Thr Ser Tyr
20            25            30
Trp Met Asn Trp Val Arg Gln Ala Pro Gly Gln Gly Leu Glu Trp Ile
35            40            45

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Gly	Met	Ile	Arg	Pro	Ser	Asp	Ser	Glu	Thr	Arg	Leu	Asn	Gln	Met	Phe	50	55	60
Lys	Asp	Arg	Val	Thr	Ile	Thr	Val	Asp	Lys	Ser	Thr	Ser	Thr	Ala	Tyr	65	70	75
Met	Glu	Leu	Ser	Ser	Leu	Arg	Ser	Glu	Asp	Thr	Ala	Val	Tyr	Tyr	Cys	85	90	95
Ala	Gly	Ile	His	Asp	Tyr	Gly	His	Gly	Ala	Tyr	Trp	Gly	Gln	Gly	Thr	100	105	110
Leu	Val	Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	115	120	125
Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Phe	Cys	Gly	Glu	Leu	Cys	Asp	Asp	130	135	140
Asp	Pro	Pro	Glu	Ile	Pro	His	Ala	Thr	Phe	Lys	Ala	Met	Ala	Tyr	Lys	145	150	155
Glu	Gly	Thr	Met	Leu	Asn	Cys	Glu	Cys	Lys	Arg	Gly	Phe	Arg	Arg	Ile	165	170	175
Lys	Ser	Gly	Ser	Leu	Tyr	Met	Leu	Cys	Thr	Gly	Asn	Ser	Ser	His	Ser	180	185	190
Ser	Trp	Asp	Asn	Gln	Cys	Gln	Cys	Thr	Ser	Ser	Ala	Thr	Arg	Asn	Thr	195	200	205
Thr	Lys	Gln	Val	Thr	Pro	Gln	Pro	Glu	Glu	Gln	Lys	Glu	Arg	Lys	Thr	210	215	220
Thr	Glu	Met	Gln	Ser	Pro	Met	Gln	Pro	Val	Asp	Gln	Ala	Ser	Leu	Pro	225	230	235
Gly	His	Cys	Arg	Glu	Pro	Pro	Pro	Trp	Glu	Asn	Glu	Ala	Thr	Glu	Arg	245	250	255
Ile	Tyr	His	Phe	Val	Val	Gly	Gln	Met	Val	Tyr	Tyr	Gln	Cys	Val	Gln	260	265	270
Gly	Tyr	Arg	Ala	Leu	His	Arg	Gly	Pro	Ala	Glu	Ser	Val	Cys	Lys	Met	275	280	285
Thr	His	Gly	Lys	Thr	Arg	Trp	Thr	Gln	Pro	Gln	Leu	Ile	Cys	Thr	Gly	290	295	300
Glu	Pro	Lys	Ser	Ser	Asp	Lys	Thr	His	Thr	Cys	Pro	Pro	Cys	Pro	Ala	305	310	315
Pro	Glu	Leu	Leu	Gly	Gly	Pro	Ser	Val	Phe	Leu	Phe	Pro	Pro	Lys	Pro	325	330	335
Lys	Asp	Thr	Leu	Met	Ile	Ser	Arg	Thr	Pro	Glu	Val	Thr	Cys	Val	Val	340	345	350
Val	Asp	Val	Ser	His	Glu	Asp	Pro	Glu	Val	Lys	Phe	Asn	Trp	Tyr	Val	355	360	365
Asp	Gly	Val	Glu	Val	His	Asn	Ala	Lys	Thr	Lys	Pro	Arg	Glu	Glu	Gln	370	375	380
Tyr	Asn	Ser	Thr	Tyr	Arg	Val	Val	Ser	Val	Leu	Thr	Val	Leu	His	Gln	385	390	395
Asp	Trp	Leu	Asn	Gly	Lys	Glu	Tyr	Lys	Cys	Lys	Val	Ser	Asn	Lys	Ala	405	410	415
Leu	Pro	Ala	Pro	Ile	Glu	Lys	Thr	Ile	Ser	Lys	Ala	Lys	Gly	Gln	Pro	420	425	430
Arg	Glu	Pro	Gln	Val	Tyr	Thr	Leu	Pro	Pro	Ser	Arg	Glu	Glu	Met	Thr	435	440	445

-continued

Lys	Asn	Gln	Val	Ser	Leu	Thr	Cys	Arg	Val	Lys	Gly	Phe	Tyr	Pro	Ser
450						455					460				
Asp	Ile	Ala	Val	Glu	Trp	Glu	Ser	Asn	Gly	Gln	Pro	Glu	Asn	Asn	Tyr
465					470					475					480
Lys	Thr	Thr	Pro	Pro	Val	Leu	Lys	Ser	Asp	Gly	Ser	Phe	Phe	Leu	Ala
			485						490					495	
Ser	Lys	Leu	Thr	Val	Asp	Lys	Ser	Arg	Trp	Gln	Gln	Gly	Asn	Val	Phe
		500						505					510		
Ser	Cys	Ser	Val	Met	His	Glu	Ala	Leu	His	Asn	His	Tyr	Thr	Gln	Lys
		515					520					525			
Ser	Leu	Ser	Leu	Ser	Pro	Gly	Lys								
530						535									

<210> SEQ ID NO 86
 <211> LENGTH: 536
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 86

Gln	Val	Gln	Leu	Val	Gln	Ser	Gly	Ala	Glu	Val	Lys	Lys	Pro	Gly	Ser
1			5						10					15	
Ser	Val	Lys	Val	Ser	Cys	Lys	Ala	Ser	Gly	Tyr	Ser	Phe	Thr	Ser	Tyr
		20						25					30		
Trp	Met	Asn	Trp	Val	Arg	Gln	Ala	Pro	Gly	Gln	Gly	Leu	Glu	Trp	Ile
		35					40					45			
Gly	Met	Ile	Arg	Pro	Ser	Asp	Ser	Glu	Thr	Arg	Leu	Asn	Gln	Met	Phe
	50					55					60				
Lys	Asp	Arg	Val	Thr	Ile	Thr	Val	Asp	Lys	Ser	Thr	Ser	Thr	Ala	Tyr
65					70					75					80
Met	Glu	Leu	Ser	Ser	Leu	Arg	Ser	Glu	Asp	Thr	Ala	Val	Tyr	Tyr	Cys
			85						90					95	
Ala	Gly	Ile	His	Asp	Tyr	Gly	His	Gly	Ala	Tyr	Trp	Gly	Gln	Gly	Thr
		100						105					110		
Leu	Val	Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser
		115					120					125			
Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Ser	Cys	Gly	Glu	Leu	Cys	Asp	Asp
	130					135					140				
Asp	Pro	Pro	Glu	Ile	Pro	His	Ala	Thr	Phe	Lys	Ala	Met	Ala	Tyr	Lys
145					150					155					160
Glu	Gly	Thr	Met	Leu	Asn	Cys	Glu	Cys	Lys	Arg	Gly	Phe	Arg	Arg	Ile
			165						170					175	
Lys	Ser	Gly	Ser	Leu	Tyr	Met	Leu	Cys	Thr	Gly	Asn	Ser	Ser	His	Ser
			180					185					190		
Ser	Trp	Asp	Asn	Gln	Cys	Gln	Cys	Thr	Ser	Ser	Ala	Thr	Arg	Asn	Thr
		195					200					205			
Thr	Lys	Gln	Val	Thr	Pro	Gln	Pro	Glu	Glu	Gln	Lys	Glu	Arg	Lys	Thr
	210					215					220				
Thr	Glu	Met	Gln	Ser	Pro	Met	Gln	Pro	Val	Asp	Gln	Ala	Ser	Leu	Pro
225					230					235					240
Gly	His	Cys	Arg	Glu	Pro	Pro	Pro	Trp	Glu	Asn	Glu	Ala	Thr	Glu	Arg
			245						250					255	

<400> SEQUENCE: 87

Gln	Val	Gln	Leu	Val	Gln	Ser	Gly	Ala	Glu	Val	Lys	Lys	Pro	Gly	Ser
1			5						10					15	
Ser	Val	Lys	Val	Ser	Cys	Lys	Ala	Ser	Gly	Tyr	Ser	Phe	Thr	Ser	Tyr
			20					25					30		
Trp	Met	Asn	Trp	Val	Arg	Gln	Ala	Pro	Gly	Gln	Gly	Leu	Glu	Trp	Ile
		35					40					45			
Gly	Met	Ile	Arg	Pro	Ser	Asp	Ser	Glu	Thr	Arg	Leu	Asn	Gln	Met	Phe
	50					55					60				

-continued

Lys 65	Asp	Arg	Val	Thr	Ile 70	Thr	Val	Asp	Lys 75	Ser	Thr	Ser	Thr	Ala	Tyr 80
Met	Glu	Leu	Ser	Ser	Leu	Arg	Ser	Glu	Asp 90	Thr	Ala	Val	Tyr	Tyr	Cys 95
Ala	Gly	Ile	His	Asp	Tyr	Gly	His	Gly	Ala	Tyr	Trp	Gly	Gln	Gly	Thr 110
Leu	Val	Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser 115
Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Lys	Cys	Gly	Glu	Leu	Cys	Asp	Asp 130
Asp	Pro	Pro	Glu	Ile	Pro	His	Ala	Thr	Phe	Lys	Ala	Met	Ala	Tyr	Lys 145
Glu	Gly	Thr	Met	Leu	Asn	Cys	Glu	Cys	Lys	Arg	Gly	Phe	Arg	Arg	Ile 165
Lys	Ser	Gly	Ser	Leu	Tyr	Met	Leu	Cys	Thr	Gly	Asn	Ser	Ser	His	Ser 180
Ser	Trp	Asp	Asn	Gln	Cys	Gln	Cys	Thr	Ser	Ser	Ala	Thr	Arg	Asn	Thr 195
Thr	Lys	Gln	Val	Thr	Pro	Gln	Pro	Glu	Glu	Gln	Lys	Glu	Arg	Lys	Thr 210
Thr	Glu	Met	Gln	Ser	Pro	Met	Gln	Pro	Val	Asp	Gln	Ala	Ser	Leu	Pro 225
Gly	His	Cys	Arg	Glu	Pro	Pro	Pro	Trp	Glu	Asn	Glu	Ala	Thr	Glu	Arg 245
Ile	Tyr	His	Phe	Val	Val	Gly	Gln	Met	Val	Tyr	Tyr	Gln	Cys	Val	Gln 260
Gly	Tyr	Arg	Ala	Leu	His	Arg	Gly	Pro	Ala	Glu	Ser	Val	Cys	Lys	Met 275
Thr	His	Gly	Lys	Thr	Arg	Trp	Thr	Gln	Pro	Gln	Leu	Ile	Cys	Thr	Gly 290
Glu	Pro	Lys	Ser	Ser	Asp	Lys	Thr	His	Thr	Cys	Pro	Pro	Cys	Pro	Ala 305
Pro	Glu	Leu	Leu	Gly	Gly	Pro	Ser	Val	Phe	Leu	Phe	Pro	Pro	Lys	Pro 325
Lys	Asp	Thr	Leu	Met	Ile	Ser	Arg	Thr	Pro	Glu	Val	Thr	Cys	Val	Val 340
Val	Asp	Val	Ser	His	Glu	Asp	Pro	Glu	Val	Lys	Phe	Asn	Trp	Tyr	Val 355
Asp	Gly	Val	Glu	Val	His	Asn	Ala	Lys	Thr	Lys	Pro	Arg	Glu	Glu	Gln 370
Tyr	Asn	Ser	Thr	Tyr	Arg	Val	Val	Ser	Val	Leu	Thr	Val	Leu	His	Gln 385
Asp	Trp	Leu	Asn	Gly	Lys	Glu	Tyr	Lys	Cys	Lys	Val	Ser	Asn	Lys	Ala 405
Leu	Pro	Ala	Pro	Ile	Glu	Lys	Thr	Ile	Ser	Lys	Ala	Lys	Gly	Gln	Pro 420
Arg	Glu	Pro	Gln	Val	Tyr	Thr	Leu	Pro	Pro	Ser	Arg	Glu	Glu	Met	Thr 435
Lys	Asn	Gln	Val	Ser	Leu	Thr	Cys	Arg	Val	Lys	Gly	Phe	Tyr	Pro	Ser 450
Asp	Ile	Ala	Val	Glu	Trp	Glu	Ser	Asn	Gly	Gln	Pro	Glu	Asn	Asn	Tyr 455

-continued

465	470	475	480
Lys Thr Thr Pro Pro Val Leu Lys Ser Asp Gly Ser Phe Phe Leu Ala			
	485	490	495
Ser Lys Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe			
	500	505	510
Ser Cys Ser Val Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys			
	515	520	525
Ser Leu Ser Leu Ser Pro Gly Lys			
	530	535	

<210> SEQ ID NO 88
 <211> LENGTH: 536
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 88

Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ser			
1	5	10	15
Ser Val Lys Val Ser Cys Lys Ala Ser Gly Tyr Ser Phe Thr Ser Tyr			
	20	25	30
Trp Met Asn Trp Val Arg Gln Ala Pro Gly Gln Gly Leu Glu Trp Ile			
	35	40	45
Gly Met Ile Arg Pro Ser Asp Ser Glu Thr Arg Leu Asn Gln Met Phe			
	50	55	60
Lys Asp Arg Val Thr Ile Thr Val Asp Lys Ser Thr Ser Thr Ala Tyr			
	65	70	75
Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Val Tyr Tyr Cys			
	85	90	95
Ala Gly Ile His Asp Tyr Gly His Gly Ala Tyr Trp Gly Gln Gly Thr			
	100	105	110
Leu Val Thr Val Ser Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser			
	115	120	125
Gly Gly Gly Gly Ser Gly Gly Gly Asp Cys Gly Glu Leu Cys Asp Asp			
	130	135	140
Asp Pro Pro Glu Ile Pro His Ala Thr Phe Lys Ala Met Ala Tyr Lys			
	145	150	155
Glu Gly Thr Met Leu Asn Cys Glu Cys Lys Arg Gly Phe Arg Arg Ile			
	165	170	175
Lys Ser Gly Ser Leu Tyr Met Leu Cys Thr Gly Asn Ser Ser His Ser			
	180	185	190
Ser Trp Asp Asn Gln Cys Gln Cys Thr Ser Ser Ala Thr Arg Asn Thr			
	195	200	205
Thr Lys Gln Val Thr Pro Gln Pro Glu Glu Gln Lys Glu Arg Lys Thr			
	210	215	220
Thr Glu Met Gln Ser Pro Met Gln Pro Val Asp Gln Ala Ser Leu Pro			
	225	230	235
Gly His Cys Arg Glu Pro Pro Pro Trp Glu Asn Glu Ala Thr Glu Arg			
	245	250	255
Ile Tyr His Phe Val Val Gly Gln Met Val Tyr Tyr Gln Cys Val Gln			
	260	265	270
Gly Tyr Arg Ala Leu His Arg Gly Pro Ala Glu Ser Val Cys Lys Met			

-continued

275					280					285					
Thr	His	Gly	Lys	Thr	Arg	Trp	Thr	Gln	Pro	Gln	Leu	Ile	Cys	Thr	Gly
290						295					300				
Glu	Pro	Lys	Ser	Ser	Asp	Lys	Thr	His	Thr	Cys	Pro	Pro	Cys	Pro	Ala
305					310					315					320
Pro	Glu	Leu	Leu	Gly	Gly	Pro	Ser	Val	Phe	Leu	Phe	Pro	Pro	Lys	Pro
				325					330					335	
Lys	Asp	Thr	Leu	Met	Ile	Ser	Arg	Thr	Pro	Glu	Val	Thr	Cys	Val	Val
			340					345					350		
Val	Asp	Val	Ser	His	Glu	Asp	Pro	Glu	Val	Lys	Phe	Asn	Trp	Tyr	Val
		355					360					365			
Asp	Gly	Val	Glu	Val	His	Asn	Ala	Lys	Thr	Lys	Pro	Arg	Glu	Glu	Gln
	370					375					380				
Tyr	Asn	Ser	Thr	Tyr	Arg	Val	Val	Ser	Val	Leu	Thr	Val	Leu	His	Gln
385					390					395					400
Asp	Trp	Leu	Asn	Gly	Lys	Glu	Tyr	Lys	Cys	Lys	Val	Ser	Asn	Lys	Ala
			405						410					415	
Leu	Pro	Ala	Pro	Ile	Glu	Lys	Thr	Ile	Ser	Lys	Ala	Lys	Gly	Gln	Pro
			420					425					430		
Arg	Glu	Pro	Gln	Val	Tyr	Thr	Leu	Pro	Pro	Ser	Arg	Glu	Glu	Met	Thr
		435					440					445			
Lys	Asn	Gln	Val	Ser	Leu	Thr	Cys	Arg	Val	Lys	Gly	Phe	Tyr	Pro	Ser
	450					455					460				
Asp	Ile	Ala	Val	Glu	Trp	Glu	Ser	Asn	Gly	Gln	Pro	Glu	Asn	Asn	Tyr
465				470					475					480	
Lys	Thr	Thr	Pro	Pro	Val	Leu	Lys	Ser	Asp	Gly	Ser	Phe	Phe	Leu	Ala
			485						490					495	
Ser	Lys	Leu	Thr	Val	Asp	Lys	Ser	Arg	Trp	Gln	Gln	Gly	Asn	Val	Phe
		500						505					510		
Ser	Cys	Ser	Val	Met	His	Glu	Ala	Leu	His	Asn	His	Tyr	Thr	Gln	Lys
		515					520					525			
Ser	Leu	Ser	Leu	Ser	Pro	Gly	Lys								
	530					535									

<210> SEQ ID NO 89

<211> LENGTH: 536

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 89

Gln	Val	Gln	Leu	Val	Gln	Ser	Gly	Ala	Glu	Val	Lys	Lys	Pro	Gly	Ser
1				5					10					15	
Ser	Val	Lys	Val	Ser	Cys	Lys	Ala	Ser	Gly	Tyr	Ser	Phe	Thr	Ser	Tyr
		20						25				30			
Trp	Met	Asn	Trp	Val	Arg	Gln	Ala	Pro	Gly	Gln	Gly	Leu	Glu	Trp	Ile
	35					40						45			
Gly	Met	Ile	Arg	Pro	Ser	Asp	Ser	Glu	Thr	Arg	Leu	Asn	Gln	Met	Phe
	50					55					60				
Lys	Asp	Arg	Val	Thr	Ile	Thr	Val	Asp	Lys	Ser	Thr	Ser	Thr	Ala	Tyr
65				70					75					80	
Met	Glu	Leu	Ser	Ser	Leu	Arg	Ser	Glu	Asp	Thr	Ala	Val	Tyr	Tyr	Cys

-continued

85					90					95					
Ala	Gly	Ile	His	Asp	Tyr	Gly	His	Gly	Ala	Tyr	Trp	Gly	Gln	Gly	Thr
			100					105					110		
Leu	Val	Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser
			115				120					125			
Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Phe	Cys	Ser	Glu	Leu	Cys	Asp	Asp
						135					140				
Asp	Pro	Pro	Glu	Ile	Pro	His	Ala	Thr	Phe	Lys	Ala	Met	Ala	Tyr	Lys
					150					155					160
Glu	Gly	Thr	Met	Leu	Asn	Cys	Glu	Cys	Lys	Arg	Gly	Phe	Arg	Arg	Ile
				165					170					175	
Lys	Ser	Gly	Ser	Leu	Tyr	Met	Leu	Cys	Thr	Gly	Asn	Ser	Ser	His	Ser
				180				185					190		
Ser	Trp	Asp	Asn	Gln	Cys	Gln	Cys	Thr	Ser	Ser	Ala	Thr	Arg	Asn	Thr
			195				200					205			
Thr	Lys	Gln	Val	Thr	Pro	Gln	Pro	Glu	Glu	Gln	Lys	Glu	Arg	Lys	Thr
						215					220				
Thr	Glu	Met	Gln	Ser	Pro	Met	Gln	Pro	Val	Asp	Gln	Ala	Ser	Leu	Pro
					230					235					240
Gly	His	Cys	Arg	Glu	Pro	Pro	Pro	Trp	Glu	Asn	Glu	Ala	Thr	Glu	Arg
				245					250					255	
Ile	Tyr	His	Phe	Val	Val	Gly	Gln	Met	Val	Tyr	Tyr	Gln	Cys	Val	Gln
				260				265						270	
Gly	Tyr	Arg	Ala	Leu	His	Arg	Gly	Pro	Ala	Glu	Ser	Val	Cys	Lys	Met
							280					285			
Thr	His	Gly	Lys	Thr	Arg	Trp	Thr	Gln	Pro	Gln	Leu	Ile	Cys	Thr	Gly
						295					300				
Glu	Pro	Lys	Ser	Ser	Asp	Lys	Thr	His	Thr	Cys	Pro	Pro	Cys	Pro	Ala
					310					315					320
Pro	Glu	Leu	Leu	Gly	Gly	Pro	Ser	Val	Phe	Leu	Phe	Pro	Pro	Lys	Pro
				325					330					335	
Lys	Asp	Thr	Leu	Met	Ile	Ser	Arg	Thr	Pro	Glu	Val	Thr	Cys	Val	Val
				340				345					350		
Val	Asp	Val	Ser	His	Glu	Asp	Pro	Glu	Val	Lys	Phe	Asn	Trp	Tyr	Val
							360					365			
Asp	Gly	Val	Glu	Val	His	Asn	Ala	Lys	Thr	Lys	Pro	Arg	Glu	Glu	Gln
						375					380				
Tyr	Asn	Ser	Thr	Tyr	Arg	Val	Val	Ser	Val	Leu	Thr	Val	Leu	His	Gln
					390					395					400
Asp	Trp	Leu	Asn	Gly	Lys	Glu	Tyr	Lys	Cys	Lys	Val	Ser	Asn	Lys	Ala
				405					410					415	
Leu	Pro	Ala	Pro	Ile	Glu	Lys	Thr	Ile	Ser	Lys	Ala	Lys	Gly	Gln	Pro
				420				425					430		
Arg	Glu	Pro	Gln	Val	Tyr	Thr	Leu	Pro	Pro	Ser	Arg	Glu	Glu	Met	Thr
							440					445			
Lys	Asn	Gln	Val	Ser	Leu	Thr	Cys	Arg	Val	Lys	Gly	Phe	Tyr	Pro	Ser
						455					460				
Asp	Ile	Ala	Val	Glu	Trp	Glu	Ser	Asn	Gly	Gln	Pro	Glu	Asn	Asn	Tyr
					470					475					480
Lys	Thr	Thr	Pro	Pro	Val	Leu	Lys	Ser	Asp	Gly	Ser	Phe	Phe	Leu	Ala
				485					490					495	

-continued

Ser Lys Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe
500 505 510

Ser Cys Ser Val Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys
515 520 525

Ser Leu Ser Leu Ser Pro Gly Lys
530 535

<210> SEQ ID NO 90

<211> LENGTH: 536

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 90

Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ser
1 5 10 15

Ser Val Lys Val Ser Cys Lys Ala Ser Gly Tyr Ser Phe Thr Ser Tyr
20 25 30

Trp Met Asn Trp Val Arg Gln Ala Pro Gly Gln Gly Leu Glu Trp Ile
35 40 45

Gly Met Ile Arg Pro Ser Asp Ser Glu Thr Arg Leu Asn Gln Met Phe
50 55 60

Lys Asp Arg Val Thr Ile Thr Val Asp Lys Ser Thr Ser Thr Ala Tyr
65 70 75 80

Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Val Tyr Tyr Cys
85 90 95

Ala Gly Ile His Asp Tyr Gly His Gly Ala Tyr Trp Gly Gln Gly Thr
100 105 110

Leu Val Thr Val Ser Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
115 120 125

Gly Gly Gly Gly Ser Gly Gly Gly Lys Cys Asp Glu Leu Cys Asp Asp
130 135 140

Asp Pro Pro Glu Ile Pro His Ala Thr Phe Lys Ala Met Ala Tyr Lys
145 150 155 160

Glu Gly Thr Met Leu Asn Cys Glu Cys Lys Arg Gly Phe Arg Arg Ile
165 170 175

Lys Ser Gly Ser Leu Tyr Met Leu Cys Thr Gly Asn Ser Ser His Ser
180 185 190

Ser Trp Asp Asn Gln Cys Gln Cys Thr Ser Ser Ala Thr Arg Asn Thr
195 200 205

Thr Lys Gln Val Thr Pro Gln Pro Glu Glu Gln Lys Glu Arg Lys Thr
210 215 220

Thr Glu Met Gln Ser Pro Met Gln Pro Val Asp Gln Ala Ser Leu Pro
225 230 235 240

Gly His Cys Arg Glu Pro Pro Pro Trp Glu Asn Glu Ala Thr Glu Arg
245 250 255

Ile Tyr His Phe Val Val Gly Gln Met Val Tyr Tyr Gln Cys Val Gln
260 265 270

Gly Tyr Arg Ala Leu His Arg Gly Pro Ala Glu Ser Val Cys Lys Met
275 280 285

Thr His Gly Lys Thr Arg Trp Thr Gln Pro Gln Leu Ile Cys Thr Gly
290 295 300

-continued

Glu Pro Lys Ser Ser Asp Lys Thr His Thr Cys Pro Pro Cys Pro Ala
 305 310 315 320
 Pro Glu Leu Leu Gly Gly Pro Ser Val Phe Leu Phe Pro Pro Lys Pro
 325 330 335
 Lys Asp Thr Leu Met Ile Ser Arg Thr Pro Glu Val Thr Cys Val Val
 340 345 350
 Val Asp Val Ser His Glu Asp Pro Glu Val Lys Phe Asn Trp Tyr Val
 355 360 365
 Asp Gly Val Glu Val His Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln
 370 375 380
 Tyr Asn Ser Thr Tyr Arg Val Val Ser Val Leu Thr Val Leu His Gln
 385 390 395 400
 Asp Trp Leu Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala
 405 410 415
 Leu Pro Ala Pro Ile Glu Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro
 420 425 430
 Arg Glu Pro Gln Val Tyr Thr Leu Pro Pro Ser Arg Glu Glu Met Thr
 435 440 445
 Lys Asn Gln Val Ser Leu Thr Cys Arg Val Lys Gly Phe Tyr Pro Ser
 450 455 460
 Asp Ile Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr
 465 470 475 480
 Lys Thr Thr Pro Pro Val Leu Lys Ser Asp Gly Ser Phe Phe Leu Ala
 485 490 495
 Ser Lys Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe
 500 505 510
 Ser Cys Ser Val Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys
 515 520 525
 Ser Leu Ser Leu Ser Pro Gly Lys
 530 535

<210> SEQ ID NO 91
 <211> LENGTH: 536
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 91

Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ser
 1 5 10 15
 Ser Val Lys Val Ser Cys Lys Ala Ser Gly Tyr Ser Phe Thr Ser Tyr
 20 25 30
 Trp Met Asn Trp Val Arg Gln Ala Pro Gly Gln Gly Leu Glu Trp Ile
 35 40 45
 Gly Met Ile Arg Pro Ser Asp Ser Glu Thr Arg Leu Asn Gln Met Phe
 50 55 60
 Lys Asp Arg Val Thr Ile Thr Val Asp Lys Ser Thr Ser Thr Ala Tyr
 65 70 75 80
 Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95
 Ala Gly Ile His Asp Tyr Gly His Gly Ala Tyr Trp Gly Gln Gly Thr
 100 105 110

Leu	Val	Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Ser
115														
Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Ser	Cys	Asp	Glu	Leu	Cys	Asp
130														
Asp	Pro	Pro	Glu	Ile	Pro	His	Ala	Thr	Phe	Lys	Ala	Met	Ala	Lys
145														
Glu	Gly	Thr	Met	Leu	Asn	Cys	Glu	Cys	Lys	Arg	Gly	Phe	Arg	Ile
165														
Lys	Ser	Gly	Ser	Leu	Tyr	Met	Leu	Cys	Thr	Gly	Asn	Ser	Ser	His
180														
Ser	Trp	Asp	Asn	Gln	Cys	Gln	Cys	Thr	Ser	Ser	Ala	Thr	Arg	Thr
195														
Thr	Lys	Gln	Val	Thr	Pro	Gln	Pro	Glu	Glu	Gln	Lys	Glu	Arg	Thr
210														
Thr	Glu	Met	Gln	Ser	Pro	Met	Gln	Pro	Val	Asp	Gln	Ala	Ser	Pro
225														
Gly	His	Cys	Arg	Glu	Pro	Pro	Pro	Trp	Glu	Asn	Glu	Ala	Thr	Arg
245														
Ile	Tyr	His	Phe	Val	Val	Gly	Gln	Met	Val	Tyr	Tyr	Gln	Cys	Gln
260														
Gly	Tyr	Arg	Ala	Leu	His	Arg	Gly	Pro	Ala	Glu	Ser	Val	Cys	Met
275														
Thr	His	Gly	Lys	Thr	Arg	Trp	Thr	Gln	Pro	Gln	Leu	Ile	Cys	Gly
290														
Glu	Pro	Lys	Ser	Ser	Asp	Lys	Thr	His	Thr	Cys	Pro	Pro	Cys	Ala
305														
Pro	Glu	Leu	Leu	Gly	Gly	Pro	Ser	Val	Phe	Leu	Phe	Pro	Pro	Pro
325														
Lys	Asp	Thr	Leu	Met	Ile	Ser	Arg	Thr	Pro	Glu	Val	Thr	Cys	Val
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Val	Asp	Val	Ser	His	Glu	Asp	Pro	Glu	Val	Lys	Phe	Asn	Trp	Val
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Asp	Gly	Val	Glu	Val	His	Asn	Ala	Lys	Thr	Lys	Pro	Arg	Glu	Gln
370														
Tyr	Asn	Ser	Thr	Tyr	Arg	Val	Val	Ser	Val	Leu	Thr	Val	Leu	Gln
385														
Asp	Trp	Leu	Asn	Gly	Lys	Glu	Tyr	Lys	Cys	Lys	Val	Ser	Asn	Ala
405														
Leu	Pro	Ala	Pro	Ile	Glu	Lys	Thr	Ile	Ser	Lys	Ala	Lys	Gly	Pro
420														
Arg	Glu	Pro	Gln	Val	Tyr	Thr	Leu	Pro	Pro	Ser	Arg	Glu	Glu	Thr
435														
Lys	Asn	Gln	Val	Ser	Leu	Thr	Cys	Arg	Val	Lys	Gly	Phe	Tyr	Ser
450														
Asp	Ile	Ala	Val	Glu	Trp	Glu	Ser	Asn	Gly	Gln	Pro	Glu	Asn	Tyr
465														
Lys	Thr	Thr	Pro	Pro	Val	Leu	Lys	Ser	Asp	Gly	Ser	Phe	Phe	Ala
485														
Ser	Lys	Leu	Thr	Val	Asp	Lys	Ser	Arg	Trp	Gln	Gln	Gly	Asn	Phe
500														

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Ser Cys Ser Val Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys
515 520 525

Ser Leu Ser Leu Ser Pro Gly Lys
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<210> SEQ ID NO 92
<211> LENGTH: 536
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 92

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Ser Val Lys Val Ser Cys Lys Ala Ser Gly Tyr Ser Phe Thr Ser Tyr
20 25 30

Trp Met Asn Trp Val Arg Gln Ala Pro Gly Gln Gly Leu Glu Trp Ile
35 40 45

Gly Met Ile Arg Pro Ser Asp Ser Glu Thr Arg Leu Asn Gln Met Phe
50 55 60

Lys Asp Arg Val Thr Ile Thr Val Asp Lys Ser Thr Ser Thr Ala Tyr
65 70 75 80

Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Val Tyr Tyr Cys
85 90 95

Ala Gly Ile His Asp Tyr Gly His Gly Ala Tyr Trp Gly Gln Gly Thr
100 105 110

Leu Val Thr Val Ser Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
115 120 125

Gly Gly Gly Gly Ser Gly Gly Gly Asp Cys Phe Glu Leu Cys Asp Asp
130 135 140

Asp Pro Pro Glu Ile Pro His Ala Thr Phe Lys Ala Met Ala Tyr Lys
145 150 155 160

Glu Gly Thr Met Leu Asn Cys Glu Cys Lys Arg Gly Phe Arg Arg Ile
165 170 175

Lys Ser Gly Ser Leu Tyr Met Leu Cys Thr Gly Asn Ser Ser His Ser
180 185 190

Ser Trp Asp Asn Gln Cys Gln Cys Thr Ser Ser Ala Thr Arg Asn Thr
195 200 205

Thr Lys Gln Val Thr Pro Gln Pro Glu Glu Gln Lys Glu Arg Lys Thr
210 215 220

Thr Glu Met Gln Ser Pro Met Gln Pro Val Asp Gln Ala Ser Leu Pro
225 230 235 240

Gly His Cys Arg Glu Pro Pro Pro Trp Glu Asn Glu Ala Thr Glu Arg
245 250 255

Ile Tyr His Phe Val Val Gly Gln Met Val Tyr Tyr Gln Cys Val Gln
260 265 270

Gly Tyr Arg Ala Leu His Arg Gly Pro Ala Glu Ser Val Cys Lys Met
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Thr His Gly Lys Thr Arg Trp Thr Gln Pro Gln Leu Ile Cys Thr Gly
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Glu Pro Lys Ser Ser Asp Lys Thr His Thr Cys Pro Pro Cys Pro Ala
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<210> SEQ ID NO 93
<211> LENGTH: 536
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthesized
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<400> SEQUENCE: 93

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		20					25						30		
Trp	Met	Asn	Trp	Val	Arg	Gln	Ala	Pro	Gly	Gln	Gly	Leu	Glu	Trp	Ile
	35					40						45			
Gly	Met	Ile	Arg	Pro	Ser	Asp	Ser	Glu	Thr	Arg	Leu	Asn	Gln	Met	Phe
	50					55					60				
Lys	Asp	Arg	Val	Thr	Ile	Thr	Val	Asp	Lys	Ser	Thr	Ser	Thr	Ala	Tyr
65					70					75				80	
Met	Glu	Leu	Ser	Ser	Leu	Arg	Ser	Glu	Asp	Thr	Ala	Val	Tyr	Tyr	Cys
			85						90					95	
Ala	Gly	Ile	His	Asp	Tyr	Gly	His	Gly	Ala	Tyr	Trp	Gly	Gln	Gly	Thr
		100					105						110		
Leu	Val	Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser
	115					120					125				

-continued

Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Ser	Cys	Lys	Glu	Leu	Cys	Asp	Asp
130						135					140				
Asp	Pro	Pro	Glu	Ile	Pro	His	Ala	Thr	Phe	Lys	Ala	Met	Ala	Tyr	Lys
145					150					155					160
Glu	Gly	Thr	Met	Leu	Asn	Cys	Glu	Cys	Lys	Arg	Gly	Phe	Arg	Arg	Ile
			165						170					175	
Lys	Ser	Gly	Ser	Leu	Tyr	Met	Leu	Cys	Thr	Gly	Asn	Ser	Ser	His	Ser
		180						185					190		
Ser	Trp	Asp	Asn	Gln	Cys	Gln	Cys	Thr	Ser	Ser	Ala	Thr	Arg	Asn	Thr
		195					200					205			
Thr	Lys	Gln	Val	Thr	Pro	Gln	Pro	Glu	Glu	Gln	Lys	Glu	Arg	Lys	Thr
	210					215					220				
Thr	Glu	Met	Gln	Ser	Pro	Met	Gln	Pro	Val	Asp	Gln	Ala	Ser	Leu	Pro
225					230					235					240
Gly	His	Cys	Arg	Glu	Pro	Pro	Pro	Trp	Glu	Asn	Glu	Ala	Thr	Glu	Arg
			245						250					255	
Ile	Tyr	His	Phe	Val	Val	Gly	Gln	Met	Val	Tyr	Tyr	Gln	Cys	Val	Gln
		260						265					270		
Gly	Tyr	Arg	Ala	Leu	His	Arg	Gly	Pro	Ala	Glu	Ser	Val	Cys	Lys	Met
		275					280					285			
Thr	His	Gly	Lys	Thr	Arg	Trp	Thr	Gln	Pro	Gln	Leu	Ile	Cys	Thr	Gly
	290					295					300				
Glu	Pro	Lys	Ser	Ser	Asp	Lys	Thr	His	Thr	Cys	Pro	Pro	Cys	Pro	Ala
305					310					315					320
Pro	Glu	Leu	Leu	Gly	Gly	Pro	Ser	Val	Phe	Leu	Phe	Pro	Pro	Lys	Pro
			325						330					335	
Lys	Asp	Thr	Leu	Met	Ile	Ser	Arg	Thr	Pro	Glu	Val	Thr	Cys	Val	Val
		340						345					350		
Val	Asp	Val	Ser	His	Glu	Asp	Pro	Glu	Val	Lys	Phe	Asn	Trp	Tyr	Val
		355					360					365			
Asp	Gly	Val	Glu	Val	His	Asn	Ala	Lys	Thr	Lys	Pro	Arg	Glu	Glu	Gln
	370					375					380				
Tyr	Asn	Ser	Thr	Tyr	Arg	Val	Val	Ser	Val	Leu	Thr	Val	Leu	His	Gln
385					390					395					400
Asp	Trp	Leu	Asn	Gly	Lys	Glu	Tyr	Lys	Cys	Lys	Val	Ser	Asn	Lys	Ala
			405						410					415	
Leu	Pro	Ala	Pro	Ile	Glu	Lys	Thr	Ile	Ser	Lys	Ala	Lys	Gly	Gln	Pro
		420						425					430		
Arg	Glu	Pro	Gln	Val	Tyr	Thr	Leu	Pro	Pro	Ser	Arg	Glu	Glu	Met	Thr
	435						440					445			
Lys	Asn	Gln	Val	Ser	Leu	Thr	Cys	Arg	Val	Lys	Gly	Phe	Tyr	Pro	Ser
	450					455					460				
Asp	Ile	Ala	Val	Glu	Trp	Glu	Ser	Asn	Gly	Gln	Pro	Glu	Asn	Asn	Tyr
465					470					475					480
Lys	Thr	Thr	Pro	Pro	Val	Leu	Lys	Ser	Asp	Gly	Ser	Phe	Phe	Leu	Ala
			485						490					495	
Ser	Lys	Leu	Thr	Val	Asp	Lys	Ser	Arg	Trp	Gln	Gln	Gly	Asn	Val	Phe
			500					505						510	

-continued

Ser Cys Ser Val Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys
515 520 525

Ser Leu Ser Leu Ser Pro Gly Lys
530 535

What is claimed is:

1. A bispecific fusion polypeptide, comprising a first antigen-binding moiety which comprises:

a first polypeptide comprising from a first heavy chain variable domain VH1 of a first antibody from the N terminus to the C terminus, which is operably linked to a first conjugate fragment; and

a second polypeptide comprising a first light chain variable domain VL1 of the first antibody from the N terminus to the C terminus, which is operably linked to a second conjugate fragment, wherein the first conjugate fragment and the second conjugate fragment are capable of specifically binding; and

the first conjugate fragment is a receptor and the second conjugate fragment is a ligand; or the first conjugate fragment is a ligand and the second conjugate fragment is a receptor.

2. The bispecific fusion polypeptide according to claim 1, further comprising a second antigen-binding moiety, wherein the second antigen-binding moiety is different from the first antigen-binding moiety, and

the second antigen-binding moiety comprises:

a third polypeptide comprising a second heavy chain variable domain VH2 of a second antibody from the N terminus to the C terminus, which is operably linked to a third conjugate fragment; and

a fourth polypeptide comprising a second light chain variable domain VL2 of the second antibody from the N terminus to the C terminus, which is operably linked to a fourth conjugate fragment, wherein the third conjugate fragment and the fourth conjugate fragment are capable of specifically binding; the third conjugate fragment is a receptor and the fourth conjugate fragment is a ligand; or the third conjugate fragment is a ligand and the fourth conjugate fragment is a receptor; and

the third conjugate fragment and/or the fourth conjugate fragment, and the first conjugate fragment and/or the second conjugate fragment are selected from different receptors and ligands; or

the second antigen-binding moiety comprises:

a third polypeptide comprising from the N terminus to the C terminus a second heavy chain variable domain VH2 of a second antibody from the N terminus to the C terminus, which is operably linked to an antibody heavy chain constant region CH1; and

a fourth polypeptide comprising from the N terminus to the C terminus a second light chain variable domain VL2 of the second antibody from the N terminus to the C terminus, which is operably linked to an antibody light chain constant region CL.

3. (canceled)

4. The bispecific fusion polypeptide according to claim 2, wherein at least one non-natural interchain bond is comprised between the receptor and the ligand or between the first heavy chain variable domain VH1 and the first light

chain variable domain VL1, and the non-natural interchain bond is capable of enhancing a specific binding force between the receptor and the ligand.

5. The bispecific fusion polypeptide according to claim 4, wherein the non-natural interchain bond is formed between a first mutant residue comprised by the receptor and a second mutant residue comprised by the ligand; or

wherein the non-natural interchain bond is formed between a first mutant residue comprised by the first heavy chain variable domain VH1 and a second mutant residue comprised by the first light chain variable domain VL1.

6. (canceled)

7. The bispecific fusion polypeptide according to claim 5, wherein at least one of the first mutant residue and the second mutant residue is a cysteine residue;

preferably, wherein the non-natural interchain bond is a disulfide bond.

8. (canceled)

9. The bispecific fusion polypeptide according to claim 1, wherein at least one natural glycosylation site is not present in the receptor and/or the ligand.

10. The bispecific fusion polypeptide according to claim 1, wherein the receptor and the ligand thereof are selected from an interleukin and a receptor thereof.

11. The bispecific fusion polypeptide according to claim 7, wherein the interleukin and the receptor thereof have a lift-type steric conformation and are selected from IL15/IL15R, IL2/IL2R, IL4/IL-4R α +R γ , IL-6/IL-6R, IL-11/IL-11R, IL-13/IL-13R1, IL-20/IL20R α +IL20R β , and/or IL24/IL20R α +IL20R β ; or

wherein the interleukin and the receptor thereof have a pincer-type steric conformation and are selected from IL7/IL7R, IL21/IL21R, and IL23A/IL12B.

12. (canceled)

13. The bispecific fusion polypeptide according to claim 11,

wherein the ligand and the receptor are selected from IL15 and IL15R α ;

preferably, wherein the E at position 90 of the IL15 is mutated into C and the P at position 67 of the IL15R α is mutated into C; or

wherein a non-natural disulfide bond is present between the first heavy chain variable domain VH1 and the first light chain variable domain VL1; and preferably, the first heavy chain variable domain VH1 and the first light chain variable domain VL1 comprise any one of the following mutation combinations:

Combination	VH	VL
1	37C	95C
2	44C	100C
3	44C	101C
4	44C	105C

-continued

Combination	VH	VL
5	45C	87C
6	45C	98C
7	100C	50C
8	100bC	49C
9	98C	46C
10	101C	46C
11	105C	43C
12	106C	57C
13	108C	43C.

14. (canceled)
15. (canceled)
16. The bispecific fusion polypeptide according to claim 13, wherein the D at position 61 of the IL15 is mutated into N, the E at position 64 is mutated into Q, and/or the N at position 65 is mutated into D; or
- wherein at least one N-glycosylation site of the IL15 is not present; preferably, the N-glycosylation site is selected from N71, N79, and/or N112; and preferably, the IL15 comprises the following amino acid mutations: N71Q, N79Q, and/or N112Q; or
- wherein at least one O-glycosylation site of the IL15R α is not present; preferably, the O-glycosylation site is selected from T2, T81, and/or T86; and preferably, the IL15R α comprises the following amino acid mutations: T2A, T81A, and/or T86A.
17. (canceled)
18. (canceled)
19. The bispecific fusion polypeptide according to claim 8, wherein the ligand and the receptor are selected from IL2 and IL2R α ;
- preferably, wherein S at position 75 of the IL2 is mutated into C and the N terminus of the IL2R α is extended by two or three amino acids;
- when the N terminus of the IL2R α is extended by two amino acids, the second extended amino acid is cysteine, and the first extended amino acid is any one of a non-polar fatty acid amino acid, an aromatic amino acid, an amino acid with an uncharged R group, an amino acid with a positively charged R group or an amino acid with a negatively charged R group; and
- when the N terminus of the IL2R α is extended by three amino acids, the second extended amino acid is cysteine, and the first and third extended amino acids are any one of a non-polar fatty acid amino acid, an aromatic amino acid, an amino acid with an uncharged R group, an amino acid with a positively charged R group or an amino acid with a negatively charged R group.
20. (canceled)
21. The bispecific fusion polypeptide according to claim 1, comprising an antibody Fc constant region; preferably, wherein the antibody Fc constant region is a heterodimer;

preferably, wherein the antibody Fc constant region is a heterodimer associated by KiH, a hydrophobic interaction, an electrostatic interaction, a hydrophilic interaction, and/or an increased flexibility;

preferably, wherein the antibody Fc constant region comprises CH2, CH3, and optionally CH4, and the CH2, the CH3, and/or the optionally CH4 are replaced by the receptor and the ligand thereof.

22. (canceled)

23. (canceled)

24. (canceled)

25. The bispecific fusion polypeptide according to claim 1, wherein the first antigen-binding moiety and the second antigen-binding moiety bind to different antigens or bind to different epitopes of the same antigen;

preferably, the first antigen-binding moiety targets an immune cell and the second antigen-binding moiety targets a tumor cell;

preferably, the first antigen-binding moiety and the second antigen-binding moiety both target a tumor cell; and

preferably, the first antigen-binding moiety and the second antigen-binding moiety both target an immune cell.

26. An isolated nucleic acid, encoding the bispecific fusion polypeptide according to claim 1.

27. A vector comprising the nucleic acid according to claim 16.

28. A host cell comprising the nucleic acid according to claim 16.

29. A pharmaceutical composition, comprising the bispecific fusion polypeptide according to claim 1, and a pharmaceutically acceptable carrier, an excipient, or a stabilizer.

30. Use of the bispecific fusion polypeptide according to claim 1 in the preparation of drugs for the treatment of diseases.

31. A method of treating or preventing disease, wherein the method comprises administering at least one of the following to a subject having or suspected of having disease: the bispecific fusion polypeptide of claim 1;

an isolated nucleic acid encoding the bispecific fusion polypeptide according to claim 1;

a vector comprising an isolated nucleic acid encoding the bispecific fusion polypeptide according to claim 1;

a host cell comprising an isolated nucleic acid encoding the bispecific fusion polypeptide according to claim 1; or

a pharmaceutical composition, comprising the bispecific fusion polypeptide according to claim 1, and a pharmaceutically acceptable carrier, an excipient, or a stabilizer.

32. Use of the bispecific fusion polypeptide of claim 1 or a pharmaceutical composition comprising the bispecific fusion polypeptide according to claim 1, and a pharmaceutically acceptable carrier, an excipient, or a stabilizer in the prevention and/or treatment of cancer.

* * * * *